Optimization of a Seed and Blanket Thorium-Uranium Fuel Cycle for Pressurized Water Reactors

by

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Submitted to the Department of Nuclear Engineering
in partial fulfillment of the requirements for the degree of

DOCTOR OF PHILOSOPHY

at the

MASSACHUSETTS INSTITUTE OF TECHNOLOGY

June 2003

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Abstract

A heterogeneous LWR core design, which employs a thorium/uranium once through fuel cycle, is optimized for good economics, wide safety margins, minimal waste burden and high proliferation resistance. The focus is on the **W**hole **A**ssembly **S**eed and **B**lanket (**WASB**) concept, in which the individual seed and blanket regions each occupy one full-size PWR assembly in a checkerboard core configuration. A Westinghouse 4-loop 1150 MWe PWR was selected as the reference plant design. The optimized heterogeneous core, after several iterations, employs 84 seed assemblies and 109 blanket assemblies. Each assembly has the characteristic 17x17 rod array. The seed fuel is composed of 20 w/o enriched annular UO_2 pellets. Erbium is used in the fresh seed to help regulate local power peaking and reduce soluble boron concentrations. Erbium was evenly distributed into all pin central holes except for the peripheral pins and four corner pins of each assembly where more erbium was used due to their higher power level. The blanket fuel is a mixture of 87% ThO_2 – 13% UO_2 by volume, where the uranium is enriched to 10 w/o. The blanket fuel pin diameter is larger than the seed fuel pin diameter. There are two separate fuel management flows: a standard three-batch scheme is adopted for the seed (18 month cycle length) and a single-batch for the blanket, which is to stay in the core for up to 9 seed cycles.

The WASB core design was analyzed by well known tools in the nuclear industry. The neutronic analysis was performed using the Studsvik Core Management System (CMS), which consists of three codes: CASMO-4, TABLES-3 and SIMULATE-3. Thermal-hydraulic analysis was performed using EPRI's VIPRE-01. Fuel performance was analyzed using FRAPCON. The radioactivity and decay heat from the spent seed and blanket fuel were studied using MIT's MCODE (which couples MCNP and ORIGEN) to do depletion calculations, and ORIGEN to analyze the spent fuel characteristics after discharge.

The analyses show that the WASB core can satisfy the requirements of fuel cycle length and safety margins of conventional PWRs. The coefficients of reactivity are comparable to currently operating PWRs. However, the reduction in effective delayed neutron fraction (β_{eff}) requires careful review of the control systems because of its importance to short term power transients. Whole core analyses show that the total control rod worth of the WASB core is about 1/3 less than those of a typical PWR for a standard arrangement of Ag-In-Cd control

rods in the core. The use of enriched boron in the control rods can effectively improve the control rod worth. The control rods have higher worth in the seed than in the blanket. Therefore, a new loading pattern has been designed so that almost all the control rods will be located in seed assemblies. However, the new pattern requires a redesign of the vessel head of the reactor, which is an added cost in case of retrofitting in existing PWRs. Though the WASB core has high power peaking factors, acceptable MDNBR in the core can be achieved under conservative assumptions by using grids with large local pressure loss coefficient in the blanket. However, the core pressure drop will increase by 70%. The very high burnup reached by the seed pins will require larger free gas volume, and both the seed and blanket will require advanced cladding materials.

Compared with conventional UO_2 designs, the WASB core produces 40% less spent fuel assemblies and 50% less spent fuel heavy metal per year. The radioactivity and decay heat per unit heavy metal in the spent fuel are higher because of the higher burnup (the average discharge burnup of the seed is 145 and of the blanket is 88 MWd/kgHM). For the first 20,000 years, WASB has lower decay heat than the conventional UO_2 fuel in Watts/GWe-yr. Thereafter, the decay heat of the WASB design is higher, but only at a level of 200 Watts/GWe-yr or lower. The proliferation resistance attributes of the WASB fuel cycle are significantly better because of reduced plutonium production (by a factor of approximately 3), degraded plutonium isotope mix in the discharged fuel, and enhanced radioactivity barrier per assembly of both the seed and the blanket.

A simplified WASB fuel cycle economic analysis model was developed based on OECD methodology. The major components are found to be the uranium material cost and the enrichment cost. The WASB design has lower fabrication cost, and spent fuel storage and disposal costs when compared with conventional PWRs. However, the front end fuel cycle cost of the WASB is 11.7% more expensive because of its greater SWU requirements. The blanket fuel cycle cost is only about 13% of the total cost. The WASB costs will have a net economic benefit of 2% if the cost of disposal is 500 $/kgHM, which is equivalent to the disposal fee of 1.0 mill/kWe-hr for a discharge burnup of about 30 MWd/kgHM. However, the small economic benefits (within the range of uncertainty in predictions), the potentially high cost of transition cycles, and the lack of established mechanisms for rewarding utilities for reduction in spent fuel mass and plutonium production suggest that the WASB and similar core designs will have difficulty in penetrating the LWR refueling market.

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Acknowledgements

I begin by thanking my advisors, Professor Mujid S. Kazimi and Professor Michael J. Driscoll, for their support throughout my four years at MIT. They have been great sources of knowledge, encouragement, and guidance.

I must also sincerely thank my colleague, Dr. Edward E. Pilat, for his invaluable help on my thesis work. Ed always has sincere confidence in me, and without his encouragement, I would not be where I am today.

I am grateful to all the members of the Advanced Fuel Cycle Group at the MIT Center for Advanced Nuclear Energy Systems (CANES). Special thanks are due to my colleagues Dr. Pavel Hejzlar, Dr. Pradip Saha, Dr. Xianfeng Zhao, Dr. Yun Long, Dr. Zhiwen Xu, Mr. Eugene Shwageraus, and Mr. Dandong Feng who provided help on various aspects of the work. Special thanks are also extended to Dr. Tamer Bahadir of STUDSVIK of America for his assistance using the CMS code package.

This research was funded under the U.S. Department of Energy's NERI program. Collaboration with Professor Alex Galperin at Ben-Gurion University of the Negev, Israel and Dr. Michael Todosow of BNL are gratefully acknowledged.

Finally, I wish to thank my wife, Xiaoqian Ma, for her love, support, and patience.

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Nomenclature

ARI	All Control Rods In
ARO	All Control Rods Out
BOC	Beginning of Cycle
BOL	Beginning of Life
BP	Burnable Poison
BWR	Boiling Water Reactor
CANDU	Canada Deuterium-Uranium (Reactor)
CBC	Critical Boron Concentration
CHF	Critical Heat Flux
CMS	Core Management System
CRD	control rod
CZP	Cold Zero Power
DNB	Departure from Nucleate Boiling
DNBR	Departure from Nucleate Boiling Ratio
DUPIC	direct use of PWR spent fuel in CANDU
EFPM	Effective Full Power Months
EOC	End of Cycle
EOL	End of Life
HFP	Hot Full Power
HTGR	High Temperature Gas-cooled Reactor
HTR	High Temperature Reactor
HZP	Hot Zero Power
iHM	Initial Heavy Metal
LER	Low Enrichment Region
LMFBR	Liquid Metal Fast Breeder Reactor
LWBR	Light Water Breeder Reactor
LWR	Light Water Reactor
MCNP	Monte Carlo N-Particle Transport Code
MDNBR	Minimum Departure from Nucleate Boiling Ratio

Mills/kWe-hr	0.001 dollar per kilowatt hour electricity
MTC	Moderator Temperature Coefficient
MOC	Middle of Cycle
MWd/kgHM	MWd per kg HM, equivalent to GWD per MTHM
ORIGEN	Oak Ridge Isotope GENeration and Depletion Code
PCI	Pellet-Cladding Interaction
pcm	per cent mille (10^{-5})
ppm	part per million (10^{-6})
PWR	Pressurized Water Reactor
PWSCC	Primary Water Stress Corrosion Cracking
RCCA	Rod Cluster Control Assembly
RTF	Radkowsky Thorium Fuel
SBU	Seed-Blanket Unit
SWU	Separative Work Unit
VIPRE	Versatile Internals and Component Program for Reactors; EPRI
v/o	volume percent
VVER	Russian designed pressurized water reactor
WASB	Whole Assembly Seed and Blanket
w/o	weight percent
wt%	weight percent

Chapter 1

INTRODUCTION

1.1 Motivation and Objectives

More than 430 nuclear power plants are currently operating throughout the world, supplying about 17 percent of the world's electricity. These plants perform safely and reliably, and they help satisfy the objectives of diversity, independence and security of energy supply. Unlike fossil fuel plants, nuclear plants do not produce carbon dioxide, sulfur or nitrogen oxides which are released into the environment. Growing concerns for the environment will favor energy sources that can satisfy the need for electricity and other energy-intensive products on a sustainable basis with minimal environmental impact and competitive economics. Nuclear energy will continue to provide a vital part of a long-term, diversified energy supply if the challenging goals facing nuclear energy systems are realized. These goals have been defined in the recent effort of so called GEN-IV reactors as falling in the broad areas of sustainability, economics, safety and reliability, and proliferation resistance and physical protection [U-3]. Sustainability goals focus on fuel utilization, and waste management. Economics goals focus on competitive life cycle and energy production costs and financial risk. Safety and reliability goals focus on reliable operation, improved accident management and minimization of consequences, investment protection, and essentially eliminating the technical need for emergency response. The proliferation resistance and physical protection goals focus on controlling and securing nuclear material and nuclear facilities.

Most operating nuclear plants use several-percent-enriched uranium fuel. Plutonium, generated from U-238 neutron capture, is the main source of proliferation potential and radiotoxicity. Limits on low cost uranium resources, waste, and proliferation raise concerns about the long-term sustainability of nuclear energy. The use of thorium as a fertile material in nuclear fuel has been of interest since the dawn of nuclear power technology due to the superior neutronic properties of U-233, and to the abundance of thorium in nature compared to uranium

Several once-through thorium-based fuel cycle options have been proposed for PWRs in recent years. Two main design variants have been considered: homogeneous and heterogeneous. The homogeneous design uses a mixture of ThO_2 and UO_2 [H-2][Z-1]. It was concluded that homogeneously mixed ThO_2-UO_2 fuel will not provide economically superior performance to conventional all-U fuel. The heterogeneous approach introduced a seed and blanket assembly design, where the uranium and thorium regions are separated spatially. The principal example is the Radkowsky Thorium Fuel (RTF) concept [G-1]. The present work analyzes an alternative to the RTF design concept for thorium utilization in PWRs: the Whole-Assembly Seed and Blanket (WASB) core, which can simplify fuel assembly fabrication and in-core fuel management compared to the RTF concept. The seed and blanket regions each occupy one full size PWR assembly. The moderator to fuel ratios in the seed and blanket assemblies are optimized to reduce plutonium production, increase the thermal margin in the seed and enhance the conversion ratio and U-233 burning in the blanket. Different refueling schemes are adopted for the seed and blanket: a standard 3-batch scheme for the seed and single batch for the blanket. The blanket will sit in the core for up to 9 seed refueling cycles (about 13 years) [W-2].

The goal of this effort is to design a heterogeneous seed and blanket PWR core which employs a once-through thorium/uranium fuel cycle that meets the requirements of next generation nuclear systems: good economics, high safety, minimal waste and high proliferation resistance, and to define clearly the advantages and disadvantages of the new design versus the conventional all uranium-fueled PWR cores. The detailed objectives are as follows:

- Optimize the seed and blanket fuel design and core configuration in order to keep the fuel and the core within the current PWR operating and safety envelopes. Therefore the S&B array can be fully retrofit into current PWR cores without significant hardware changes.

- Examine the thermal-hydraulic and fuel performance issues. Thermal-hydraulic limitations must be carefully checked because of the high power peaking factors of the WASB core. The very high burnups of the seed and blanket raise concerns about fuel performance; in addition the long residence of the blanket in the core requires evaluation of cladding material qualifications.
- Identify measures to evaluate proliferation resistance and examine the intrinsic proliferation resistance of the WASB cores.
- Investigate the economic performance of WASB cores. Economic competitiveness would provide a key incentive for the deployment of the thorium fuel cycle.

1.2 Characteristics of Thorium Fuel

1.2.1 Neutronics of Th-232/U-233

Thorium (Thor, Scandinavian god of war) was discovered by Berzelius in 1828 [L-3]. Thorium is now thought to be about three times as abundant as uranium. The use of thorium fuel cycles has been studied since the very beginning of the nuclear energy era, but on a smaller scale than uranium or uranium/plutonium cycles.

The thermal capture cross-section of thorium is around three times that of U-238, as shown in Table 1.1, which would indicate enhanced conversion in a highly thermal reactor. However, the increase of absorption in the fertile element requires a compensatory increase of fissile material, which affects the overall conversion ratio, while reducing capture in the moderator and structures. On the other hand, if the spectrum is hardened the effect is reversed. U-238 captures more than Th-232 in an epithermal flux but the resonance self-shielding will reduce the consequences. In a fast flux, the captures of the two isotopes are similar, however the UO_2 has a slight advantage as it has 10% higher density than ThO_2. The fast fission cross-section of U-238 is three to five times that of Th-232. In a conventional PWR, U-238 fission produces 7 to 8% of the total energy, whereas Th-232 fission only accounts for 2%.

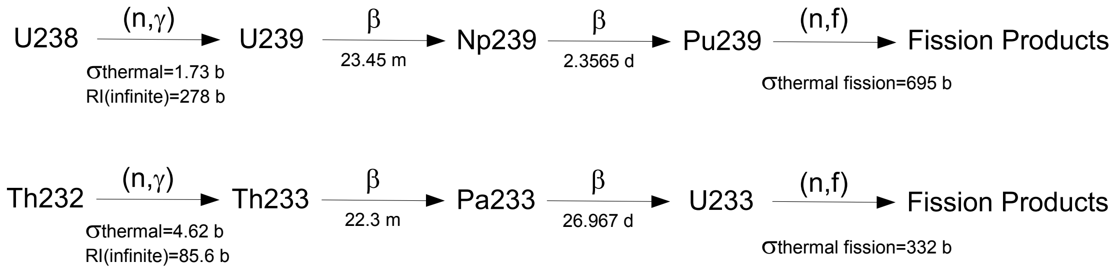


Figure 1.1 Transmutation of U-238 into Pu-239 and Th-232 into U-233

There is no U-233 in nature; it is produced through neutron capture in Th-232 as shown in Figure 1.1. The neutron yields of U-223 in the thermal and epithermal regions are the highest among all fissile isotopes, as shown in Table 1.2. In addition, its low α value limits the accumulation of even mass nuclei (U-234) which favors reaching high burnup.

Pa-233 has a half-life of 27 days and decays to U-233, while Np-239 leads to Pu-239 with a half-life of 2.4 days. In the same neutron flux, the concentration of Pa-233 would be about 10 times that of Np-239, resulting in greater losses, but this effect is limited. However, the larger quantity of Pa-233 poses a problem after reactor shutdown, i.e. reactivity increase due to the accumulation of U-233 from the decay of Pa-233. The steady state concentration of Pa-233 depends on the flux level.

Table 1.1 Fertile Neutronic Properties [K-1]

Parameter	Th-232	U-238	U-234	Pu-240
Thermal σ_a	4.62	1.73	63	203
Epithermal (∞ Dilution) RI_a	85.6	278	660	8500
Self Shielded RI	17	24		

Table 1.2 Fissile Neutronic Properties [K-1]

Parameter		U-233	U-235	Pu-239	Pu-241
Thermal* (barns)	σ_a	364	405	1045	1121
	σ_f	332	346	695	842
$\alpha = \sigma_c / \sigma_f$		0.096	0.171	0.504	0.331
Neutrons per absorption η_{th}		2.26	2.08	1.91	2.23
Epithermal Resonance Integral (∞ Dilution)	RI_a	882	405	474	740
	RI_f	748	272	293	571
$\alpha = RI_c / RI_f$		0.182	0.489	0.618	0.296
Neutrons per absorption η_{epi}		2.10	1.63	1.77	2.29
Neutron Yield ν		2.48	2.43	2.87	2.97
Delayed Neutron Yield β		0.0031	0.0069	0.0026	0.0050

* Average over Maxwellian spectrum at 300 ^{O}C (0.05eV)

All cross-section values in barns

1.2.2 Advantages and Drawbacks of Thorium Fuel Cycles

Even though thorium has long been considered as a supplement to or replacement for uranium, the use of thorium-based fuel cycles has been studied on a much smaller scale as compared to uranium or uranium/plutonium cycles. Recently, there has been a revival of interest in thorium fuel. The main reasons are that:

- The abundance of thorium in the earth's crust is about three times that of uranium.
- Neutron yields from fission of U-233 (produced through neutron capture in Th-232) are higher than those of U-235 and Pu-239.
- The fission product absorption, which is the main contributor to reactor poisoning, is about 25% less for U-233 than for U-235 or Pu-239 [I-1].
- Th-232 produces fewer minor actinides than U-238 does. The resultant lower radiotoxicity of spent thorium fuel is claimed by some to be a benefit in waste management [K-2].
- The thermal conductivity of ThO_2 is about 10% higher than that of UO_2 over a large temperature range, and its melting temperature is about 500 ^{O}C higher than UO_2 (3300 ^{O}C instead of 2800 ^{O}C). As a consequence, fuel-operating temperatures will be

lower than those of UO_2, and all thermally activated processes such as diffusion of fission gas from the fuel will be decreased [L-2].

- ThO_2 is chemically very stable, and it does not oxidize, a benefit for normal operation, postulated accidents, and in waste management.

However, countervailing the advantages, there are some problems in the practical implementation of a thorium fuel cycle.

- The main disadvantage of the thorium cycle is the lack of U-233 in nature. It is necessary to initiate the cycle by using a fissile material such as U-235 or Pu-239, which can be mixed with the thorium fuel or spatially separated as a driver.
- From a neutronic point of view, the epithermal resonance absorption in Th-232 is lower than that in U-238. This may reduce the negative Doppler reactivity feedback in overpower transients.
- U-233 has a smaller delayed neutron fraction, β , than that of U-235 but comparable to that of Pu-239, thus creating a need for faster response of control systems to transients should U-235 not be sufficiently present, since the average worth measured in dollars is increased.
- The energy yield per U-233 fission is a few percent less than that of U-235 and Pu-239. Hence, more fissions are needed per unit energy production. Also more fission gases are produced per U-233 fission. But measurements and simulation calculations have shown that less of that gas is released due to the lower fission gas diffusion coefficient of the thorium fuel [L-2].
- An important complication of the thorium cycle is that a small but significant proportion of U-233 undergoes a (n, 2n) reaction to form U-232, also formed by a sequence starting with the (n, 2n) reaction of Th-232. U-232 decays with a 70 year half-life to Th-228 (half-life 1.9 years), then by way of Ra-224 and other short-lived intermediates with a minor branch to Tl-208 which emits energetic gamma rays. This implies a need for heavier shielding than for the uranium fuel during handling.

There are also concerns about the proliferation implications of thorium fuel cycles. Although pure U-233 appears to be usable for nuclear weapons, the isotope mix produced by a thorium fuel cycle is sometimes claimed to be proliferation resistant because the decay products of the concomitant U-232 would create a radiation hazard sufficient to require remote handling within a short time after chemical separation. In addition, denaturing of U-233 by addition of U-238 to the thorium fuel is also an effective measure. Whether U-233 would be much less attractive to proliferators than plutonium is thus at least open to question, supporting the international view that U-233 requires the same level of safeguards oversight and physical protection as does plutonium. Forsberg, et al, have developed a combined proliferation limit of 12% fissile uranium (U-233 plus 0.6 times U-235) for the case such as the blanket when both U-233 and U-235 are present [F-1].

1.3 Background

1.3.1 Historic Status of Thorium Fuel Development

The use of thorium based fuel cycles has been studied for about 40 years. Many incentives have been identified for its use including the fact that public concerns have increasingly focused on the high radiotoxicity of the long-lived waste produced during reactor operation. In addition, the large stockpiles of plutonium produced in civil and military reactors have raised concerns about weapons proliferation. Basic research and development on the thorium fuel cycle has been conducted in Germany, India, Japan, the Russian Federation, the United Kingdom, and the USA. Studies included the determination of material data, fabrication tests on a laboratory scale, irradiation of thorium-based fuel in material test reactors with post-irradiation examinations, and investigation into the use of thorium-based fuel for LWRs, LMFBRs., and HTRs/HTGRs [I-1].

In the 1960s and 1970s whole-core demonstrations of thorium-uranium oxide fuels in LWRs were explored in the US in two types of fuel arrangements:

i) Homogenously mixing thorium with highly enriched uranium oxide in the BORAX-IV, Indian Point I PWR, and Elk River BWR reactors [K-1].

ii) Using a heterogeneous arrangement of seed and blanket regions. The light water breeder reactor (LWBR) concept was tested in the Shippingport reactor using thorium and U-233 fuel. The core was operated from 1977 to 1982 and successfully demonstrated breeding of U-233 using the "seed/blanket" concept [B-1].

There are two characteristics of the early use of thorium in LWRs. First, highly enriched fuel was used to prime the cycle. For example, the seed of the LWBR was ThO_2-UO_2 fuel, where the UO_2 was 95% enriched U-233. However a 20% uranium enrichment limit is currently defined for non-proliferation purposes. Hence all future fuel designs should keep the enrichment under this limit. Secondly, reprocessing to recover fissile and fertile materials from irradiated fuel was considered a necessary step in the thorium based fuel cycle. But in October 1976, President Ford announced that reprocessing and recycling of uranium and plutonium should not proceed in the U.S. President Carter followed the same line and went further by deferring reprocessing and recycling indefinitely. In 1981, the Reagan administration lifted the ban on reprocessing, but since there were no commercial breeder reactors and recycling of uranium and plutonium in LWRs did not offer any economic advantages, no effort has been made by any industrial firm in the U.S. to engage in reprocessing ventures.

In past attempts to significantly improve LWR fuel cycle efficiency, it has been recognized that advanced converter concepts attain superior fuel utilization by one of three methods: adoption of on-line refueling (CANDU, molten salt and pebble bed reactors), special geometry to capture leakage neutrons (Seed-blanket), or moderation control of neutron capture in fertile material (H_2O/D_2O spectrum shift). At this stage only CANDU reactors have been successfully deployed worldwide. The high neutron economy of the CANDU reactor, its ability to be refueled while operating at full power, its compact fuel channel design, and its fuel bundle provide a promising path for full exploitation of the energy potential of thorium fuel cycles in existing reactors. AECL has done considerable work on many aspects of the thorium fuel cycle. Both recycling and once-through thorium cycles have been investigated. The results confirm the expected improvement in fuel use efficiencies [I-2].

Designs of several reactor types other than LWRs have experimented with the use of thorium. The most notable among these is the first gas cooled, graphite moderated reactor in the U.S. (Peach Bottom, 40 MWe, 1967-69) and the first pebble bed reactor in Germany (AVR, 15 MWe, 1966-72). These experiments included post-irradiation examination of spent Th-based micro sphere particle fuel from the reactors, and fabrication of Th-based fuel, both on a pilot and semi-industrial scale. The AVR was operated for more than 2 decades using HEU-Th fuel, and the fuel achieved burnups of more than 140 MWd/kgHM. The Fort St. Vrain reactor in the U.S. and the THTR in Germany were industrial scale follow-on reactors. FSV was a high-temperature, graphite-moderated, helium-cooled reactor with thorium and high-enriched uranium designed to operate at 842 MWth. The fuel was in microspheres of thorium carbide and Th/U-235 carbide (4/1) coated with silicon carbide and pyrolytic carbon to retain fission products. Excellent agreement was obtained between reactor physics measurements and calculations for control rod worth, temperature coefficient of reactivity, and reactivity behavior with burnup. The THTR was also operated with HEU-Thorium mixed oxide fuel [I-1].

1.3.2 Current Status of Thorium Fuel Development

In the past several years, there has been renewed interest in thorium fuel as an option for next generation fuel cycles, driven by the incentives of reducing plutonium production and the long-term radioactivity of waste. The U.S. Department of Energy has conducted four projects involving use of the thorium fuel cycle. All four projects are based on a once-through, proliferation resistant, high burnup, long refueling cycle use of thorium in a light water reactor [I-2]. Three of these projects are part of the Nuclear Energy Research Initiative (NERI) program. These are: "Advanced Proliferation Resistant, Lower Cost Uranium-Thorium Dioxide Fuels for Light Water Reactors", with INEEL as the lead organization and MIT as a participant; "Fuel for a Once-Through Cycle (Th, U)O_2 in a Metal Matrix", with ANL as the lead; and "A Proliferation Resistant Hexagonal Tight Lattice BWR Fuel Core Design for Increased Burnup and Reduced Fuel Storage Requirements", with BNL as the lead. The fourth project is "The Radkowsky Thorium Fuel (RTF) Project", examining application of this concept in the Russian VVER with DOE and private support, with BNL as the coordinating laboratory.

Generation of nuclear power through use of thorium is the ultimate goal of India's nuclear power strategy. Bhabha Atomic Research Center (BARC) is actively pursuing research, development, fabrication, characterization and irradiating testing of ThO_2, ThO_2-PuO_2, and ThO_2-UO_2 fuels in test and power reactors. Use of ThO_2 fuel for initial flux flattening in PHWRs; Introduction of ThO_2 as axial and radial blankets in FBTR, proposed ThO_2-PuO_2 and ThO_2-$^{233}UO_2$ fuel for AHWR, are the main steps towards use of thorium in India [I-2].

The "Advanced Proliferation Resistant, Lower Cost Uranium-Thorium Dioxide Fuels for Light Water Reactors" project was completed in 2002 [H-2][Z-1][M-1][K-5]. It was found that homogeneously mixed ThO_2-UO_2 fuel in general has better physics behavior than all-UO_2 fuel, e.g. more negative moderator temperature coefficient and less reactivity swing. The hydrogen to heavy metal ratio can be optimized to extend the burnup of both ThO_2-UO_2 and all-UO_2 fuel about 10% and 5% respectively. However, economic evaluations show that the homogeneously mixed ThO_2-UO_2 fuel is 20%~30% more expensive than the equivalent all-UO_2 fuel [Z-1].

The Radkowsky Thorium Fuel (RTF) concept was proposed by Dr. Alvin Radkowsky and co-worker at Ben-Gurion University, based in part on the ideas and experience of the Bettis Atomic Power Laboratory's Light Water Breeder Reactor (LWBR) program as implemented and sucessfully demonstrated at the Shippingport reactor in the late 1970s [G-1]. The RTF is a new core fuel concept, not a new reactor. However, in contrast to the LWBR, the RTF concept assumes a once-through fuel cycle with no reprocessing. The U-233 is burnt in situ, and the fuel rods that contain the U-233 are then disposed of. The main idea of the proposed concept is the utilization of the seed-blanket unit (SBU) fuel assembly geometry having separate uranium (U) and thorium (Th) fuel zones. The central region of the assembly (seed) contains about 20% enriched U, while the peripheral region of the assembly (blanket) contains natural Th spiked by a small amount of 10 to 20% enriched U. This arrangement provides the necessary flexibility for designing the seed as an efficient supplier of well thermalized neutrons to a subcritical blanket which, in turn, is designed for efficient generation and in-situ burning of U-233. The spatial separation of the seed and blanket sub-assemblies permits local optimization of the lattice: an over-moderated seed region

(V_m/V_f=3.3) and an under-moderated blanket region (V_m/V_f=1.8). One of the novel features of the RTF concept is its in-core fuel management scheme. The standard multi-batch fuel management of a PWR is replaced by a scheme based on two separate (seed and blanket) fuel flow routes. Basically, seeds are treated similarly to the standard PWR assemblies, i.e. approximately one-third of the seeds are replaced periodically by "fresh" seeds, and the remaining, partially depleted, seeds are reshuffled together with partially depleted blankets to form a reload configuration for the next cycle. For reasons of fuel economy the Th-blanket in-core residence time is quite long (about 10 years). Such a long residence time is required to achieve a large accumulated burnup for the Th part of the fuel, about 100MWd/kgHM, or 10 MWd/kgHM for each cycle. The design objectives, constraints, and guidelines discussed above were applied to a design of a PWRT (T stands for Th) fuel assembly compatible with a PWR plant of current Western technology. A similar design option for a Russian PWR design, using hexagonal assemblies, designated as VVER, has been developed by the Kurchatov Institute in cooperation with Russian fuel vendors and licensing authorities [G-1]. Originally the SBU concept used U/Zr as the seed fuel; later on this was replaced by annular UO_2 fuel [T-1][T-2]. Based on a thermal-hydraulic analysis [B-5][B-6], the use of UO_2 presents no problem from a thermal point of view and should simplify the licensing process. In addition, the neutronics and economic performance is slightly better than metallic uranium, as will be shown in Chapter 3. Recently, S&B cores of the WASB type have also been studied in Korea with an original focus on breeding and a 2:1 ratio of blanket to seed assemblies [K-3][K-4].

A drawback of the SBU concept is that a fuel assembly has two different regions, which makes design, fabrication and refueling procedures more complex. In this thesis research, a clearer separation of seed and blanket was adopted to simplify refueling and manufacture. Therefore, the reactor core is loaded with separate seed and blanket assemblies as shown in Figure 1.2.

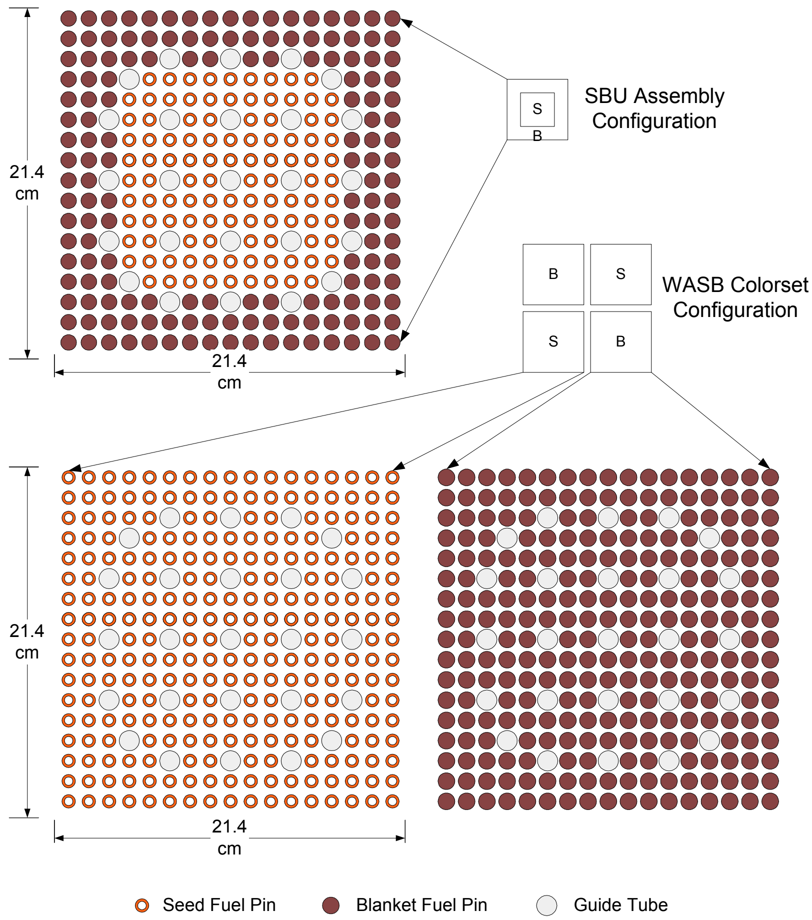


Figure 1.2 SBU and WASB Assembly Configuration

Rubbia et al., of CERN have proposed the use of thorium as the feed material in an accelerator driven system [A-1][R-3]. In this system, a beam of high-energy protons from an accelerator impinges on a target of lead, lead-bismuth or other material. High-energy neutrons are produced by the spallation reaction between the high-energy protons and the target nuclei and can be directed to a subcritical reactor containing thorium. The energy of the neutrons emanating from the spallation reaction is high enough to cause fission of the thorium nuclide itself. These neutrons can be multiplied using a material like beryllium, which has an (n, 2n) reaction. Use of an external source of neutrons to maintain the neutron chain reaction opens up the possibility of achieving very high burnup of the fuel. In addition, accelerator breeders have been investigated by ORNL in the U.S. and KFA/KFK in Germany [I-1]. In this concept, protons are acclerated and directed onto a heavy metal target, producing a cascade of secondary particles and neutrons. The neutrons are absorbed in a surrounding thorium blanket, thus breeding U-233.

Plutonium burning has recently become important due to the release of excess quantities of high-grade plutonium in the nuclear warheads of weapons and to the realization that large quantities of civil plutonium are stockpiled in countries that have had significant nuclear power programs. The efficiency of incinerating plutonium by utilization of mixed oxide fuel (MOX) is significantly reduced by the production of new plutonium. Hence the use of thorium to burn plutonium is an alternative option [I-2].

1.4 Analysis Tools

All calculations in this thesis were performed using state of the art computer codes. The Studsvik Core Management System (CMS) code package was used to do fuel/core neutronic analysis calculations for conventional LWRs (PWRs and BWRs). The CASMO-4 code incorporates the thorium neutron data from the J2/E6 data libraries, so the CMS package has the ability to analyze the thorium fuel cycle. Thermal-hydraulic design is very challenging due to the high power of the seed of the WASB. To meet the thermal-hydraulic limits, a thermal analysis was performed using VIPRE to demonstrate the viability of the physics design. In addition, problems of fission gas release and cladding corrosion were analyzed using FRAPCON. The codes used in this work are briefly discussed in the following paragraphs.

Studsvik Core Management System (CMS) Code Package

The MIT-version of CMS consists of CASMO-4 [E-1], TABLES-3 [U-1], and SIMULATE-3 [U-2]. The CMS has been used by many nuclear utilities to perform in-core fuel management studies, core design calculations, and calculations of safety parameters. The flow of CMS calculations is shown in Figure 1.3.

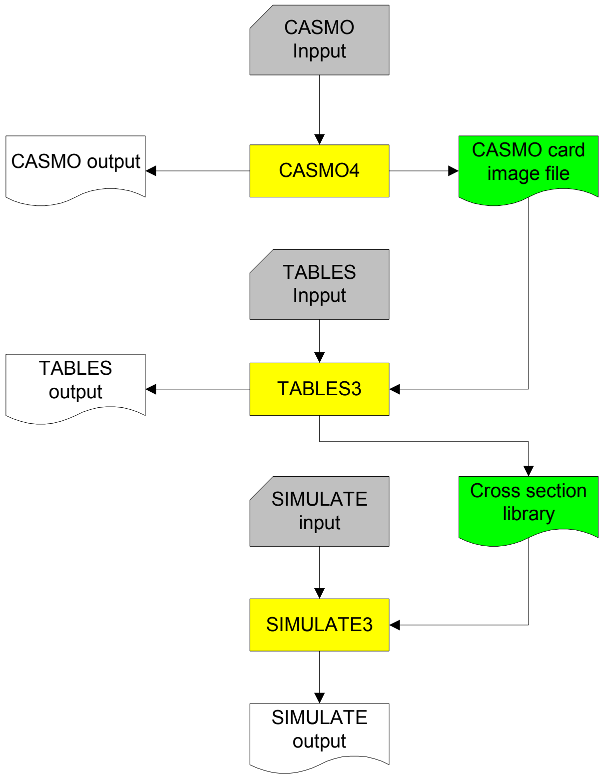


Figure 1.3 Flow of CMS Calculations

The CASMO-4 code is a two dimensional, neutron transport lattice code for generating homogeneous cross sections. Seventy energy groups are used for neutron calculations and eighteen or ten energy groups are used for gamma detector calculations. Heterogeneous pin cell modeling is used for individual assembly types. Individual pin data is collapsed to two neutron groups and two gamma groups and then homogenized into assembly average lattice cross sections. The output also contains flux discontinuity factors for bundle interfaces and reflector regions. Discontinuity factors are used by nodal codes such as SIMULATE-3 in order to preserve net currents calculated by the CASMO multi-group transport solution in two-group diffusion theory.

The TABLES-3 code is used to fit, tabulate and format assembly cross sections into a master binary cross section library. The resulting library holds "look up" tables of cross sections versus core conditions for all lattice types found in the core. This data is then used within the SIMULATE-3 code on a node by node basis.

The SIMULATE-3 code is an advanced two-group nodal code for the analysis of both PWRs and BWRs. This code is used to perform in-core fuel management studies, core design calculations, and calculation of safety parameters.

The Studsvik Core Management System has been proven accurate for performing the core analysis of conventional LWRs. Benchmarks done by Studsvik show that the nodal diffusion model in SIMULATE has achieved the goal of producing very accurate solutions to the two-group diffusion equation even for a very heterogeneous core configuration by the introduction of discontinuity factors.

MCNP-4C

MCNP [B-4], developed at the Los Alamos National Laboratory (LANL), is a general purpose, generalized-geometry, continuous energy, and time-dependent, coupled neutron/photon/electron Monte Carlo transport code that solves transport problems in any arbitrary three-dimensional geometry defined by first-, second-, and some special fourth-degree surfaces. The neutron energy range is from 10^{-5} eV to 20 MeV, the photon energy range is from 1 keV to 100 MeV, and the electron energy is from 1 keV to 1000 MeV, with the upper limit reduced to 100 MeV for coupled photon/electron problems. The capability to calculate the effective multiplication factor (k_{eff}) is also included in the code. Quantities that may be tallied include particle or energy current and flux across a surface, flux or energy deposition averaged over a volume, and flux at a point for a point detector. A wide variety of multipliers are available to augment any of these tallies, ranging from dose conversion factors to any cross section defined for ENDF/B. Sources can be defined in an arbitrary fashion for all particle types. Cross section data are available for most isotopes used in nuclear applications.

ORIGEN2

ORIGEN2.1 [C-3] is used for calculating the generation, decay, and processing of radioactive isotopes. Several libraries of cross section, decay, and production data are available with the code to allow the data used to be tailored to fit the application. Options are included to allow for material reprocessing, continuous isotope feed streams, modification of libraries to incorporate new reactions or isotopes, and a host of output formats and data.

ORIGEN2 performs its calculations in one neutron group, and depletion can be controlled either by total power, or by the flux level within the material. The depletion equation is solved using the matrix exponential technique. ORIGEN2 uses a unique method for storing the necessary coefficients for solution of the coupled equations, and for solving the system. These techniques are described fully in the literature, and will not be further explained here.

MCODE

The MCNP-ORIGEN DEpletion program (MCODE) [X-1] is a MCNP-ORIGEN linkage code developed at MIT, which combines the continuous-energy Monte Carlo code, MCNP-4C, and the one-group depletion code, ORIGEN2.1, to perform burnup calculations. MCNP is capable of providing the neutron flux and effective one-group cross sections for different MCNP-defined regions. ORIGEN, in turn, carries out multi-nuclide depletion calculations for each region and provides material compositions to beginning-of-timestep MCNP input.

VIPRE-01

The VIPRE [C-4] (Versatile Internals and Component Program for Reactors) was developed for nuclear power utility thermal-hydraulic analysis. The code can evaluate LWR core safety limits including minimum departure from nucleate boiling ratio (MDNBR), critical power ratio (CPR), fuel and clad temperatures, and coolant state in normal operation and assumed transient conditions by solving the finite-difference equations for mass, energy and momentum conservation for an interconnected array of channels assuming incompressible thermally expandable homogeneous flow. Like most other core thermal-hydraulic codes, the VIPRE-01 modeling structure is based on a subchannel analysis approach. The core or section of symmetry is defined as an array of parallel flow channels with lateral connections between adjacent channels. The shape and size of the channels and their interconnections are essentially arbitrary. The user has a great deal of flexibility for modeling reactor cores or any other fluid flow geometry.

FRAPCON-3/ FRAPCON3-ThHB

FRAPCON-3 [B-2] is the main code used by the U.S. Nuclear Regulatory Commission (NRC) for fuel performance modeling under conditions of normal operation. This code is designed to calculate the thermal-mechanical and chemical behavior of LWR fuel rods during normal operation. FRAPCON3-ThHB, a revised version of FRAPCON-3, is used for the analysis of thoria fuel since the models currently in the FRAPCON-3 code are applicable only to UO_2 fuel rods [L-2].

1.5 Thesis Organization

This thesis is subdivided into eight chapters. Computational method validation is discussed in Chapter 2. All neutronic calculations in the thesis work have been done using the Studsvik CMS code package, which is developed for all-UO_2 fueled LWRs. In addition, the WASB seed and blanket arrangement is more heterogeneous than usual, with significantly different neutron spectra and flux magnitudes in the seed and blanket assemblies. Thus it is necessary to confirm that CMS is capable of analyzing the thorium fuel.

Chapter 3 describes the WASB seed and blanket fuel design. The fuel material compositions of the seed and blanket are quite different. 20% enriched UO_2 is used for the seed fuel, and the fuel pellet is annular. In order to limit the buildup of Pu-239, a wetter lattice design is employed, which affects both thermal-hydraulics and safety. The blanket is ThO_2-UO_2 mixed fuel, where UO_2 is only about 10-15 wt% of total heavy metal. A dry lattice design is used to improve the conversion ratio in the blanket. A small amount of UO_2 is adopted for two objectives: one is to denature U-233 with U-238, the other is to share some power in order to improve power peaking at BOC in the core.

The core neutronic performance and safety parameters are presented in Chapter 4. Different fuel management schemes are adopted for the seed and blanket fuel. A three-batch 18-month cycle is used for the seed assemblies, while one-batch is used for the blanket, with the blanket residing in the core for up to 9 seed cycles (~13.5 years). The Westinghouse 4-loop PWR was selected as the reference core. Neutronic performance and safety parameters were made comparable with the reference core - a conventional all-UO_2 type - to facilitate retrofittability. Reactivity control issues are discussed in this chapter. Reactivity control of the WASB is more complex than conventional PWRs due to its heterogeneous core design and harder neutron spectra. Short-term reactivity control is still provided by control rods. The WASB control rod configuration for the 17×17 lattice is the same as for the reference PWR. The poison material for the reference PWR control rods is an alloy of Ag-In-Cd. The total control rod worth is about 30% less than that of the reference core if the same control rod design is used. An improved control design is presented in this chapter. Intermediate and long-term reactivity control is provided through the use of soluble boron and burnable poison. Burnable poison is employed to reduce the requirements for soluble boron and to reduce the power peaking factor of the core. All fresh seed fuel pins are poisoned in the WASB design.

Thermal-hydraulic analysis is presented in Chapter 5. The objective of this work is to optimize the thermal-hydraulic performance so as to achieve a reasonable safety margin during normal operation and anticipated transients and accidents. The detailed analysis of core thermal-hydraulics is presented in this chapter. In addition the fuel performance of the seed and blanket at high burnups is also discussed in this chapter.

The proliferation resistance feature of the WASB fuel cycle is a major advantage, and the reduced waste production is also a very attractive feature compared to conventional LWRs. Chapter 6 presents a thorough analysis of spent fuel characteristics and waste management consequences. There is an incentive to recycle the spent fuel from a WASB core due to the high fissile material content in the discharged fuel. About 4.5 wt% U-235 is left in the seed fuel after discharge. So if this amount of fissile can be simply reprocessed and refabricated into PWR fuel using a dry process such as AIROX, it will increase significantly the utilization of natural uranium and improve the economics of the fuel cycle. This topic is covered in Chapter 6.

Fuel Cycle Cost Modeling of the WASB fuel cycle is described in Chapter 7. Existing conventional fuel cycle cost analysis models are not valid for analysis of the WASB fuel cycle because of the complication of having separate fuel management schemes for the seed and blanket. Thus a fuel cycle cost analysis model has been developed for the WASB. A fuel cycle cost comparison with conventional LWRs is also discussed in this chapter.

The last chapter, Chapter 8 provides an overall summary of the effort, and some recommendations for future work.

Chapter 2

COMPUTER CODE VALIDATION

2.1 Introduction

Numerical simulation has been extensively used in nuclear reactor physics design and analysis. The ability to perform such reactor physics calculations depends critically on models employed to predict the neutron density in space, direction and energy. The complexity inherent in explicit modeling of every core component, such as fuel pins, control rods, and water channels, in a light water reactor limits the direct methods of solving the three-dimensional transport equation. Although tools such as three-dimensional continuous energy Monte Carlo models are available, the magnitude of the computational problem posed by modeling is such that the most sophisticated computers are incapable of simulating an entire core over its burnup history with the required degree of accuracy. Deterministic neutron transport methods (e.g. multi-group S_N, integral transport, or collision probability methods) are similarly overwhelmed by the complexity of the computational problem of explicit geometrical modeling on a core-wide basis. Alternative nodal diffusion methods have been developed and improved since the 1970s. The advances in coarse-mesh nodal methods have improved nodal reactor analysis to the point that accurate three-dimensional nodal depletion calculations that allow the determination of local pin power peaking can be performed routinely and economically [S-3].

The WASB core neutronic design and analysis was performed using the Studsvik Core Management System (CMS), which consists of three codes: CASMO-4, TABLES-3 and SIMULATE-3. CASMO-4 can be used to generate 2-group nodal data for SIMULATE-3. The accuracy of the nodal data is dependent upon the accuracy of the CASMO flux distribution, which is used to homogenize and collapse the cross-sections. The most accurate solution may be obtained by modeling the true heterogeneous nature of the lattice in the energy group structure of the nuclear data library.

Some issues should be addressed before the use of the CMS for WASB core analysis. These are discussed in the sections which follows.

Nuclear data for thorium fuel

Cross section libraries based on the evaluated data files JEFF-2.2 and ENDF/B-6 have been processed for CASMO-4. The nuclear data libraries of CAMSO-4 contain data for 35 heavy nuclides in the uranium chain and 44 heavy nuclides in the thorium chain and 189 individual nuclides in the fission product chains. The actinides are represented up to Cf-252 and fission products are represented from Ge-76 through Tb-159. Along with a vast amount of structural material and neutron absorber data, the total number of nuclides contained in the libraries is well in excess of 300 [E-1]. The heavy metal buildup chains which include thorium are shown in Figure 2.1.

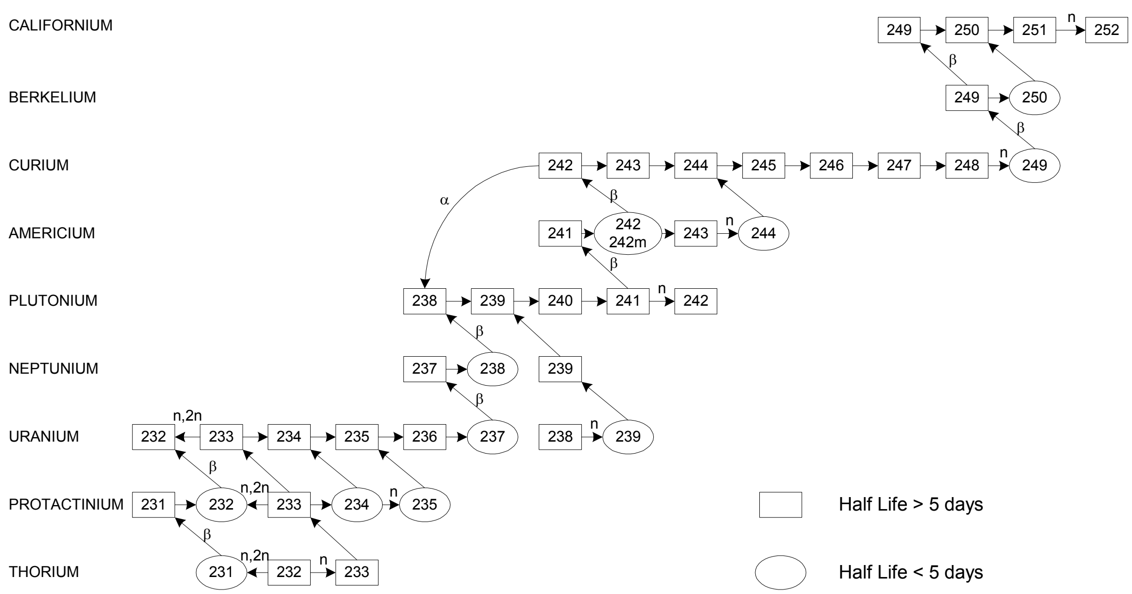


Figure 2.1 Actinide Buildup Chains in CASMO Depletion Calculations

Nuclear data for the thorium fuel chain are available in the nuclear data library, however, since the thorium cycle has not been as intensively studied and reviewed as uranium, the reliability of these data is not at the same level as for uranium. In addition, all present day multi-group cross-sections use a group structure optimized for the uranium cycle. Thus, group boundaries are chosen in part so the resonances of U-238 are close to the midpoint of the energy groups and no resonance straddles two groups. Using the same group structure for the thorium could have undesirable consequences.

The microscopic cross-sections of CASMO-4 are tabulated in 70 energy groups. The resonance region of CASMO-4 is defined to lie between 4 eV and 9118 eV. Resonance absorption above 9118 eV is regarded as being unshielded. The 1 ev resonance in Pu-240 and the 0.3 eV resonance in Pu-239, and the 2 eV resonance in U-233 are adequately covered by the concentration of thermal groups around these resonances and are consequently excluded from the special resonance treatment [E-1]. The resonance region energy structure is shown in Figure 2.2. It can be seen that the resonances of Th-232 are well-situated with respect to the resonance group structure set by U-238, hence the energy structure of CASMO-4 would be expected not to have a negative effect on thorium fuel calculations.

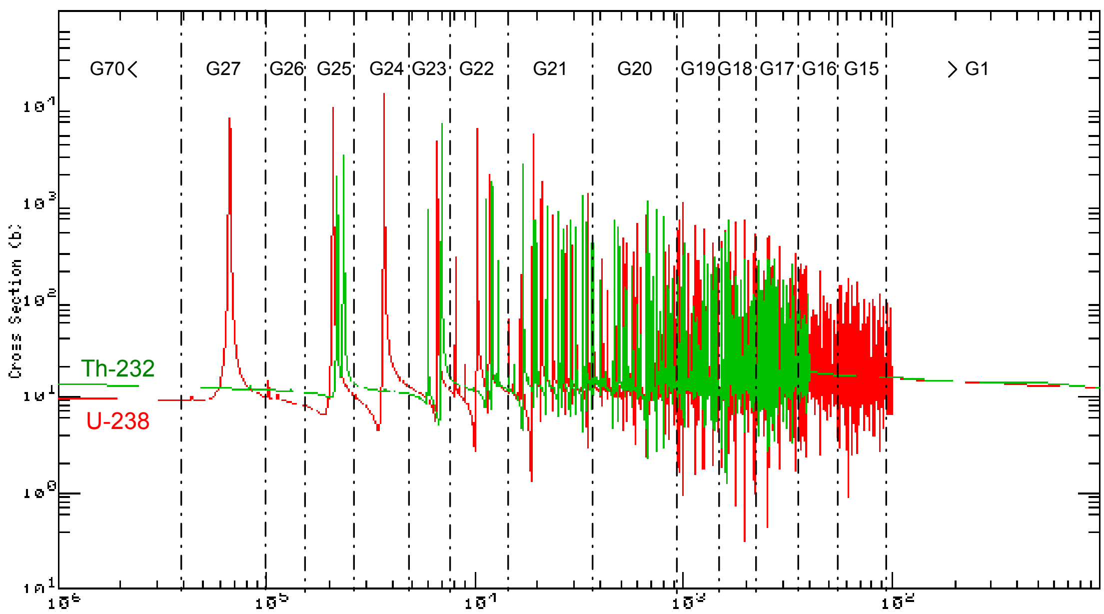


Figure 2.2 Energy Group Structure of CASMO-4

Fission Neutron Spectra

The fission neutron spectrum for the various fissionable isotopes differs somewhat and this difference is of importance in the calculation of leakage and fast fissions. The data library of CASMO-4 contains two fission spectra, one for U-235 and one for Pu-239, and an appropriate average is used in the transport calculations. The fission neutron spectrum of U-233 is very similar to that of U-235, so that the effect on the transport calculation of the fission spectrum is negligible.

High enrichment fuel -- Seed

The CMS has been developed for conventional UO_2 fueled LWR cores and proven to predict reactor parameters very accurately. The enrichment of uranium fuel usually falls into the range of 3% to 5%. However, the seed fuel is enriched to 20%, and the CMS has not been verified for such high enrichment.

High burnups

The seed discharge burnup is about 145 MWd/kgHM and the blanket at 88 MWd/kgHM. The CMS has been proved to be very accurate for ≤65 MWd/kgHM for conventional LWRs. An extension of the validation base of the CMS to higher burnups, especially with respect to the quality of the prediction of the isotopic composition of the exposed fuel, appears to be an issue of importance.

Heterogeneous core

The seed and blanket assemblies are organized in a checkerboard pattern in the WASB core. The complex heterogeneity raises concerns about the modeling capability of the CMS.

Because of the use of spatially homogenized cross-sections, nodal models are capable of providing only smoothly varying flux distributions within each assembly. Consequently, detailed pin-by-pin flux and power distributions are not immediately available from a nodal solution. However many reactor safety analyses require detailed knowledge of pin power distributions. Most pin power reconstruction methods assume that detailed pin-by-pin distributions within an assembly can be approximated by the product of a global homogenized intranodal distribution and a local heterogeneous form function [R-1]:

$$\phi_g(x,y)_{reactor} = \phi_g(x,y)_{\mathrm{hom}\,ogeneous} \times \phi_g(x,y)_{form\ fuction}\,. \tag{2.1}$$

The form function must account for assembly heterogeneities caused by water holes, burnable absorber pins, enrichment variations, etc. The differences between various pin power reconstruction methods arise from (a) the approximations used in calculating the intranodal flux distribution and (b) the spectrum/depletion geometry employed to compute the form functions. The first successful method of obtaining accurate intranodal flux distributions was introduced by Koebke et al.[K-8], by assuming that intranodal flux shapes could be approximated by non-separable polynomial expansions. Such polynomial flux shapes accurately model the behavior of the flux in the interior of an assembly; however, low-order polynomials cannot accurately model the large localized gradients, which are induced by the spectral interaction occurring at assembly interfaces. This difficulty can be overcome by defining the heterogeneous form function from multi-assembly (colorset) calculations. In such an approach, any inaccuracy in the intranodal flux shapes at assembly interfaces can be absorbed in the form function, which is defined by

$$\phi_g(x,y)_{form\ fuction} = \frac{\phi_g(x,y)_{colorset}}{\phi_g(x,y)_{colorset\ \mathrm{hom}ogeneous}}. \tag{2.2}$$

There is no doubt that predictions of heterogeneous fluxes are very accurate using this approach. However there are two obvious disadvantages of this approach. One is the expense involved in performing colorset calculations. The other more significant disadvantage is that accurate pin power can be reconstructed only if colorset depletion calculations have been performed for all important combinations of assemblies.

A new pin power reconstruction method is used in SIMULATE that requires only single-assembly form functions by introducing quantities called discontinuity factors defined by:

$$DF_g^s = \frac{Heterogeneous\ Surface\ Flux}{Homogeneous\ Surface\ Flux}. \tag{2.3}$$

Smith demonstrated that discontinuity factors can be accurately approximated by the ratios of the heterogeneous and homogeneous surface fluxes from isolated assembly spectrum/depletion calculations. Benchmarking versus CASMO colorset calculations and experimental measured pin power distributions showed excellent agreement [R-1].

2.2 Reference PWR Core

The Westinghouse 4-loop PWR was selected as the reference plant for this study because of its widespread use. The core parameters are shown in Table 2.1. All dimensions given are cold dimensions.

It is assumed that the WASB core would be retrofitted into this reference plant. All operation and safety parameters are analyzed and assessed for the WASB core in this reference plant.

Table 2.1 Operating Parameters for a Westinghouse 4-loop PWR [G-2]

Operating Parameter	Value
1. Plant	
Number of primary loops	4
Total heat output of the core (MWth)	3411
Total plant thermal efficiency (%)	34
Electrical output of plant (MWe)	1150
Energy deposited in the fuel (%)	97.4
Energy deposited in the moderator (%)	2.6
2. Core	
Core barrel inside diameter/outside diameter (m)	3.76/3.78
Mass of fuel as UO_2 (MT)	101.0
Mass of fuel as U (MTU)	88.2
Mass of cladding material (MT)	23.1
Rated power density (kW/L)	104.5
Average linear heat generation rate (kW/m)	18.37
Core volume (m^3)	32.6
Design axial enthalpy rise peaking factor ($F_{\Delta H}$)	1.65
Allowable core total peaking factor (F_Q)	2.5
3. Primary coolant	
System pressure (MPa)	15.51
Total core flow rate (Mg/sec)	18.63
Rated coolant mass flux (kg/m^2-sec) in core	2087.6
Core inlet temperature (^oC)	292.7
4. Fuel Rods	
Total number	50,952
Fuel density (% of theoretical)	94

Pellet diameter (mm)	8.194
Pellet height (mm)	13.4
Diametral gap (μm)	165
Cladding material	Zircaloy-4
Cladding thickness (mm)	0.57
Cladding outer diameter (mm)	9.5
Active fuel height (m)	3.66
5. Fuel Assembly	
Number of assemblies	193
Assembly array	17×17
Array geometry	Square
Number of fuel rods per assembly	264
Number of grids per assembly	7
Rod pitch (mm)	12.6
Overall dimensions (mm×mm)	214×214
6. Rod Cluster Control Assemblies	
Neutron absorbing material	Ag-In-Cd
Cladding material	Type 304 SS
Cladding thickness (mm)	0.46
Number of clusters	53
Number of absorber rods per cluster	24

2.3 Benchmarking Calculations

Two modes have been used for benchmark calculations: a single pin model, and a colorset (2 × 2 assembly configuration) model. The single pin mode is used to benchmark CASMO-4 against MCODE. The colorset model is used to benchmark SIMULATE against MCODE and CASMO. The comparison calculations performed are shown in Table 2.2.

Table 2.2 Benchmark Calculations among Codes

	MCODE	CASMO	SIMULATE
Pin cell	×	×	
Colorset	×	×	×

2.3.1 Single Pin Mode

The single pin mode has been employed to verify that CASMO can be used to perform neutronic calculations for the seed and blanket fuel. High burnups of the seed and blanket (seed: 145 MWd/kgHM, blanket: 88 MWd/kgHM) are well above the current licensing limit of 60 MWd/kgHM. In addition, the thorium-based blanket is also a new application for CASMO because CASMO has been developed and used for conventional low enrichment UO_2 fuel. The single pin models for the seed and blanket are shown in Figure 2.3. The schematic illustration is only the cross section of the fuel unit cell. The seed fuel is annular,

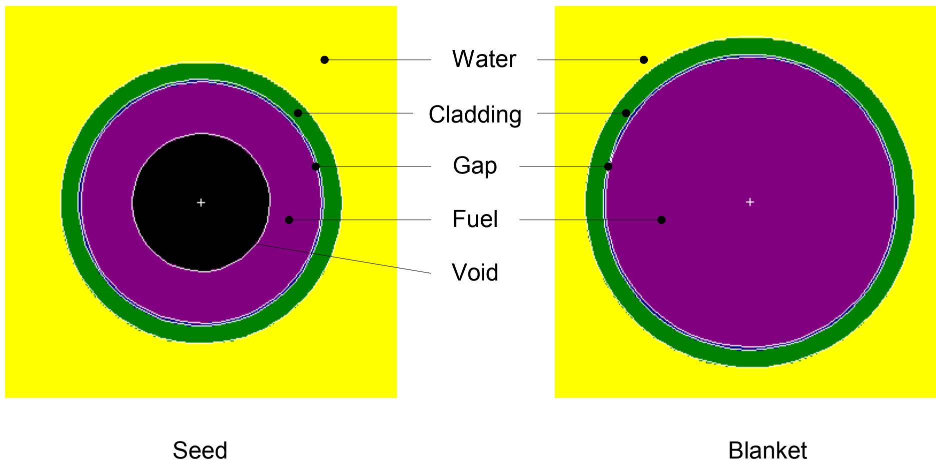


Figure 2.3 Pin Cell Models of Seed and Blanket

The detailed pin cell dimensions and initial fuel compositions are given in Table 2.3 and Table 2.4. The power densities of the seed and blanket used here are the cycle average values from whole core calculations. The boundary condition applied in the calculation is reflective.

Table 2.3 Pin Cell Model Parameters (at 300 K)

Parameter	Value	
	Seed	Blanket
Fuel Pellet Radius, cm	0.22-0.385	0.4646
Cladding Inner Radius, cm	0.3932	0.4728
Pin Pitch, cm	1.26	1.26
Fuel Density, g/cm^3	10.3024	9.3624
Coolant Density, g/cm^3	0.977	
Cladding Density, g/cm^3	6.55	
Power Density, kW/liter	129.58	79.42

Table 2.4 Initial Isotopic Compositions (at 300 K)

		Nuclide	Weight Percent w/o	Number Density 1/ cm^3
Fuel	Seed	U-234	0.1410	3.73911E+19
		U-235	17.6300	4.65393E+21
		U-238	70.3790	1.83438E+22
		O-16	11.8500	4.59793E+22
	Blanket	Th-232	79.0955	1.92206E+22
		U-234	0.0071	1.71054E+18
		U-235	0.8814	2.11441E+20
		U-238	7.9326	1.87893E+21
		O-16	12.0834	4.26071E+22
Coolant, H_2O		H-1	11.19	6.66295E+22
		O-16	88.81	3.33339E+22
Cladding Zircaloy-4		O	0.125	3.08281E+20
		Cr	0.100	7.58663E+19
		Fe	0.210	1.48338E+20
		Zr	98.115	4.24275E+22
		Sn	1.450	4.81835E+20

Figure 2.4 shows comparisons of infinite medium multiplication constants (Kinf) of the seed fuel between CASMO and MCODE. It can be seen CASMO has good agreement with MCODE, even at very high burnups. The CASMO values are a little below those of MCODE. The Kinf difference increases with burnup, and it exceeds 0.015 after 120 MWd/kgHM. When the assembly average discharge burnup reaches 150 MWd/kgHM, the difference becomes 0.017.

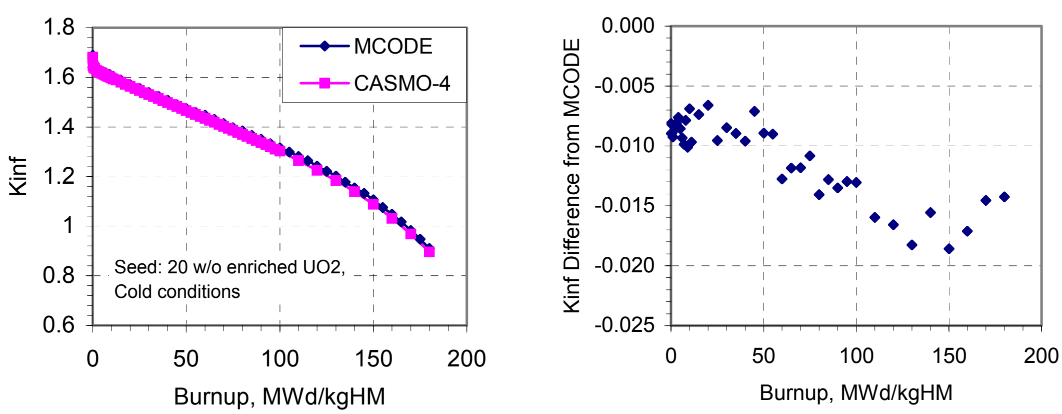


Figure 2.4 Kinf Predictions by CASMO and MCODE for a Seed Pin

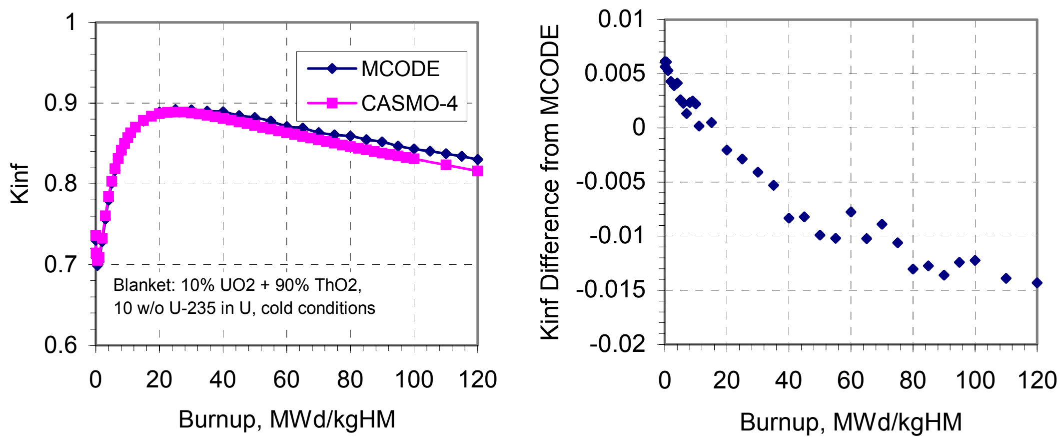


Figure 2.5 Kinf Predictions by CASMO and MCODE for a Blanket Pin

The kinf change as a function of burnup for a blanket fuel pin is shown in Figure 2.5. It appears that the agreement between CASMO and MCODE for Kinf calculations is also very good, even at very high burnups. The blanket discharge burnup after 9 seed cycles is about 88 MWd/kgHM. From the figure the absolute value of the Δk difference is about 0.013 at this burnup. Also it can be seen that the difference increases as the burnup increases.

There are many factors affecting the accuracy of depletion calculations. The early difference (< 60 MWd/kgHM) comes from the library differences between CASMO and MCODE [X-1][M-4].

Isotopic composition in the discharged fuel is important to the analyses of spent fuel characteristics. The isotopic number densities at the discharge burnup from CASMO and MCODE are shown in Table 2.5. Only some of the most important isotopes are presented in the table.

Table 2.5 Isotopic Number Density Predicted by CASMO and MCODE

(Seed: 155 MWd/kgHM, Blanket: 80 MWd/kgHM)

Isotope	Seed			Blanket		
	MCODE #/cm^3	CASMO #/cm^3	DIFF. %	MCODE #/cm^3	CASMO #/cm^3	DIFF. %
Th-232				1.74411E+22	1.74803E+22	0.24
Pa-233				1.63576E+19	1.61328E+19	-1.33
U-232				2.61978E+18	2.62192E+18	-0.37
U-233				3.08024E+20	2.89536E+20	-6.07
U-234	1.67573E+19	1.67965E+19	0.23	9.88011E+19	9.70411E+19	-1.78
U-235	8.22599E+20	8.24252E+20	0.20	2.66926E+19	2.68262E+19	-0.90
U-236	6.49150E+20	6.46310E+20	-0.44	2.99498E+19	2.91658E+19	-2.62
U-238	1.74563E+22	1.74767E+22	0.12	1.56061E+21	1.54547E+21	-0.97
Np-237	4.60440E+19	5.03444E+19	9.34	5.82612E+18	6.13310E+18	5.27
Pu-238	2.10405E+19	2.27440E+19	8.10	4.81151E+18	5.03847E+18	4.72
Pu-239	1.48481E+20	1.43825E+20	-3.14	2.60690E+19	2.61324E+19	0.24
Pu-240	5.61633E+19	5.22088E+19	-7.04	7.56405E+18	7.52192E+18	-0.56
Pu-241	4.37540E+19	4.25487E+19	-2.75	8.89503E+18	8.79242E+18	-1.15
Pu-242	2.14265E+19	2.20717E+19	3.01	8.11065E+18	9.29350E+18	14.73
Am-241	2.10472E+18	1.98123E+18	-5.87	5.00289E+17	4.65315E+17	-6.99
Am-242	4.52101E+15	4.78601E+15	5.86	8.65282E+14	9.48386E+14	11.54
Am-242m	5.47807E+16	2.79593E+16	-48.96	1.41308E+15	7.03461E+15	399.4
Am-243	5.87461E+18	4.60682E+18	-21.58	4.44652E+18	4.04273E+18	-9.20

The agreement is good with the exception of Am-242m and Am-243 in the seed, and Pu-242, Am-242, Am-242m in the blanket. Am-242m is the excited state of Am242, and has a 141 year half-life, however Am-242 has a 16.02 hour half-life. Hence Am-243 is more likely generated from neutron capture in Am-242m. It should be noted that there are significant uncertainties about the americium cross sections in MCNP. In addition, it can be seen that there are moderate differences for U-233 and Pu-242 in the blanket. In both cases CASMO appears to produce numbers that reduce the fuel reactivity; i.e. produce more conservation Kinf values.

2.3.2 Colorset Model

The accuracy of the SIMULATE calculations can be quantified directly by comparing multi-assembly MCODE and CASMO calculations to SIMULATE pin powers reconstructed from nodal colorset geometries. The colorset contains two seed assemblies and blanket assemblies, and the seed and blanket assemblies are located diagonally as shown in Figure 2.6. The assemblies employ the standard 17×17 lattice design. The water gap between assemblies is 0.08 cm wide. The pin pitch is 1.26 cm. The fuel geometrical parameters and isotopic compositions are the same as given in Tables 2.4 and 2.5. There are 25 guide tubes in each assembly as shown in the following figure. In the benchmark calculations the same Zircaloy-4 as the cladding is used for the guide tubes. The guide tube thickness is 0.0457 cm, the inner radius is 0.569 cm and the outer radius is 0.6147 cm.

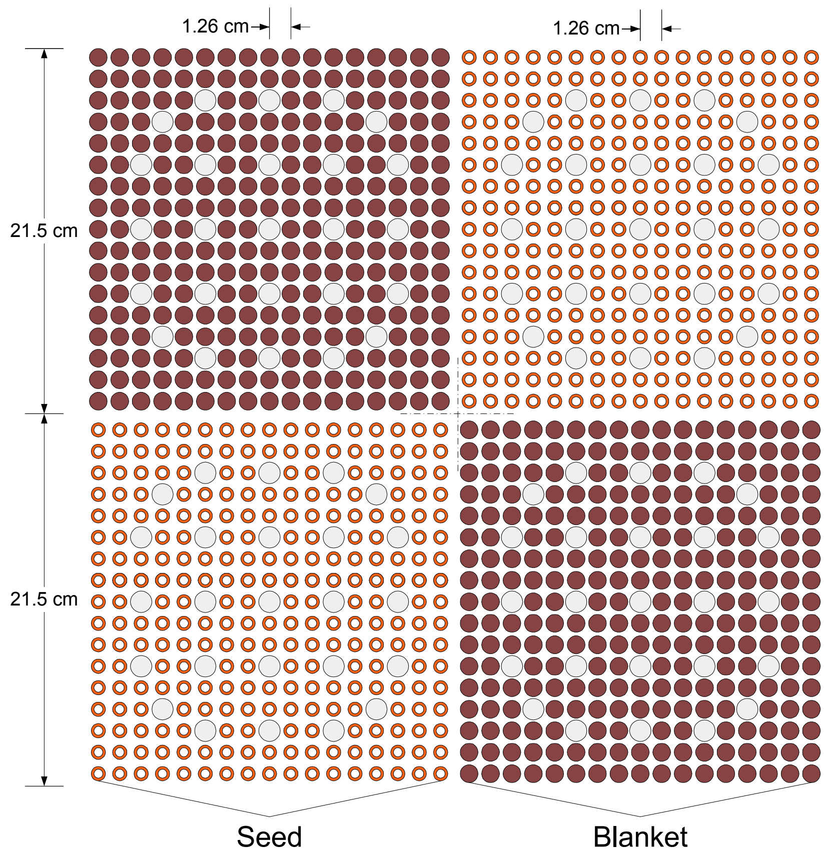


Figure 2.6 A Colorset Configuration

The MCODE calculation uses symmetry to reduce the size of the model as shown on Figure 2.7. The shaded area shows that part of the colorset on which the transport calculation is done with reflective boundary conditions at all boundaries. Every pin cell is modeled in the MCNP input file. The depletion calculations have been performed on a multiple CPU cluster (30 CPU nodes).

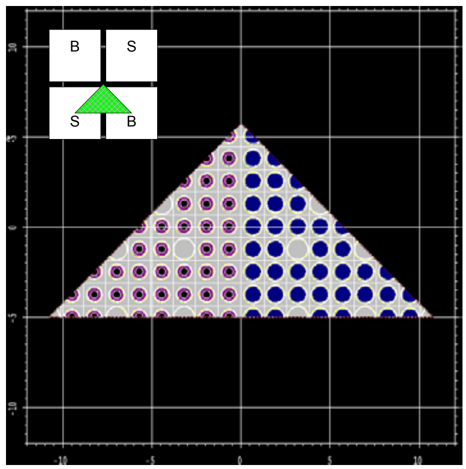


Figure 2.7 MCODE Colorset Calculation Model from MCNP Geometrical Plot

Colorset calculations in CASMO-4 are performed using the HETRAP transmission probability module with homogenized pin cells [E-1], which is different from the code's assembly calculation module. For an assembly, CAMSO-4 performs a two dimensional, heterogeneous calculation on the entire lattice in an 8-group energy structure. Cross sections for each "micro" of the lattice are condensed to 8-group energy structure using the updated spectrum from the "macro" group calculations. The flux distribution throughout the lattice is determined by solving the Boltzmann transport equation.

The Kinf comparisons among MCODE, CASMO and SIMULATE are shown in Figure 2.8. The agreement between CASMO and SIMULATE is very good before 40 MWd/kgHM. After this point the difference increases as the burnup increases. At 80 MWd/kgHM, the seed assembly burnup is about 165 MWd/kgHM, and the blanket is at 36 MWd/kgHM. Good agreement can be seen before 30 MWd/kgHM for the blanket from the single pin-cell comparison. Thus the colorset difference mainly comes from the seed. In addition, both SIMULATE and CASMO underestimate the reactivity compared with MCODE.

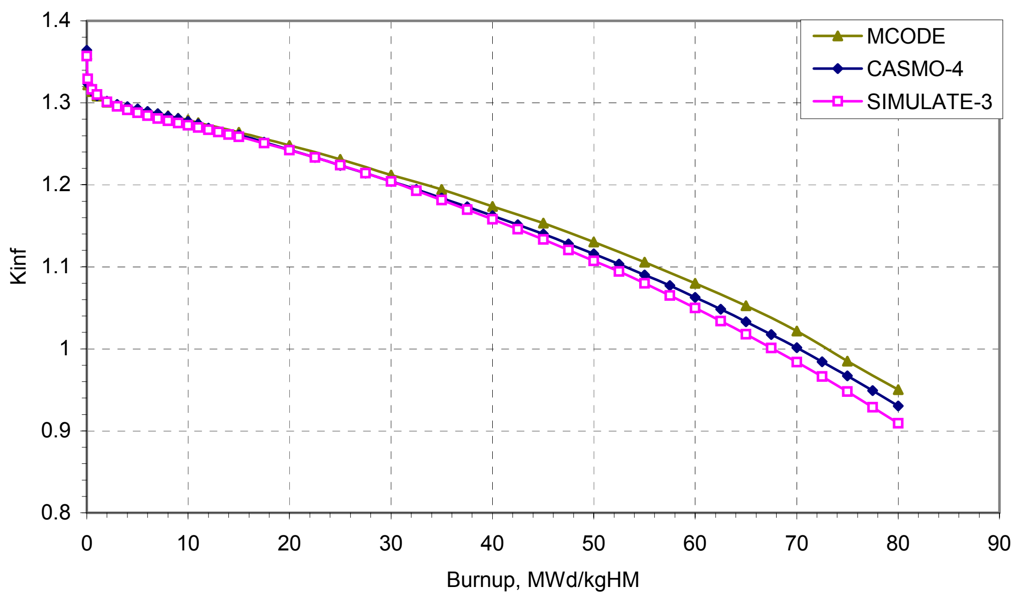


Figure 2.8 Colorset Multiplication Factor Comparison

The major concern in core analyses is the capability of predicting the power distribution, and especially the power peaking factors for WASB using SIMULATE. The use of flux and power form functions computed from single-assembly (zero-leakage) spectrum/depletion calculations in SIMULATE leads to an additional complication because of spectral interactions at assembly interfaces. These spectral interactions produce significantly different plutonium and U-233 buildup and depletion rates at assembly interfaces from those implicit in the single-assembly spectrum depletion. Table 2.6 summarizes the MCODE, CASMO and SIMULATE results for the WASB colorset.

Table 2.6 Pin Power Peaking Results of MCODE, CASMO-4 and SIMULATE-3 (predicted from colorset calculations)

BURNUP	MCODE		CASMO-4		SIMULATE-3	
MWd/kgHM	Peak Pin	RPF – S*	Peak Pin	RPF – S*	Peak Pin	RPF – S*
0.5	1.758	1.613	1.771	1.619	1.919	1.615
20	1.605	1.490	1.605	1.493	1.637	1.516
40	1.483	1.396	1.468	1.397	1.582	1.431
60	1.349	1.297	1.338	1.293	1.522	1.335

* Relative power fraction of the seed assembly

The agreement of the peaking factors between CASMO and MCODE is quite good at different burnup points, and the location of the peak pin is also correctly predicted through depletion of the colorset. But the peaking factors from SIMULATE are about 9% higher than the MCODE values at 0.5 MWd/kgHM, and the difference increases with burnup. There is about 12% error at 60 MWd/kgHM. SIMULATE predicts that the peak pin is located at the corners of the seed assembly at 0.5 and 20 MWd/kgHM because SIMULATE reconstructs the power distribution from the single-assembly power distribution, where the peak pin locations appear at the corners of the assembly at low burnups. However the peak pins in both CASMO and MCODE are close to the center of the seed assembly. Hence SIMULATE is incapable of predicting the peak pin location. The power fraction comparison among MCODE, CASMO and SIMULATE is good. The maximum error in SIMULATE is about 3% at 60 MWd/kgHM. In addition, it can be seen that the maximum power peaking factor is at the beginning of depletion. The poor prediction of power peaking factors might raise concerns about thermal-hydraulic analyses. This issue will be covered in Chapter 5.

The power distribution comparison at 0.5 MWd/kgHM is shown in Figure 2.9. The pin power agreement between MCODE and CASMO is very good for all fuel pins. A relatively higher error can be seen in the blanket due to its lower power level. The comparison between SIMULATE and CASMO results is also good with the exception of the corner pins.

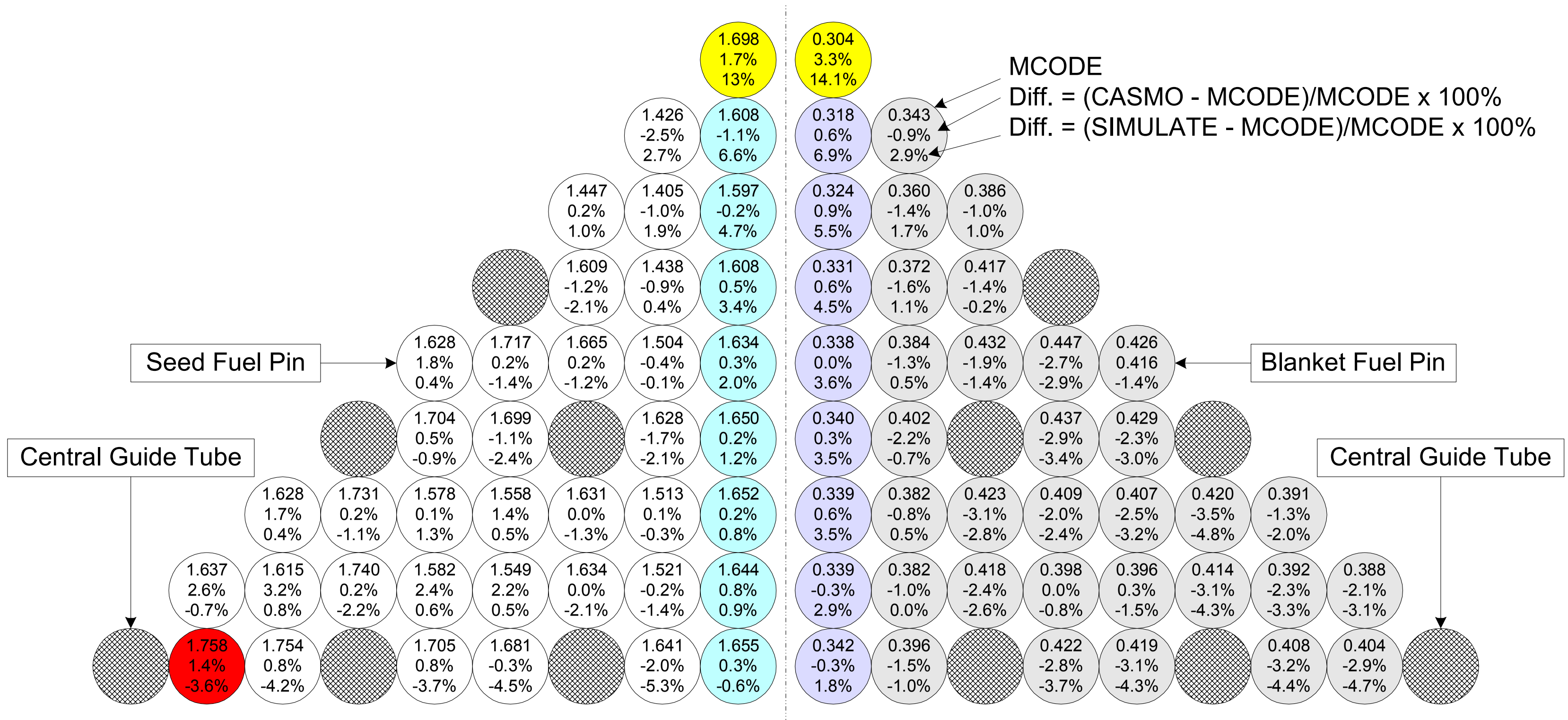


Figure 2.9 Power Distribution Comparison at 0.5 MWd/kgHM

2.4 Chapter Summary

The CMS code package was benchmarked against MCODE in this chapter. Based on the benchmark calculations, the agreement between CASMO-4 and MCODE is good, and more than adequate to confirm that CASMO-4, in particular its "colorset" feature, can be used to calculate thorium-fueled seed and blanket assembly arrays for use in PWR cores. SIMULATE always predicts the lowest reactivity values among the three codes. SIMULATE is inconsistent with MCODE at high burnup. Thus the cycle length calculated using SIMULATE will be conservative. In addition, the power distribution comparison shows good agreement between SIMULATE and MCODE with the exception of corner pins.

Chapter 3

SEED AND BLANKET FUEL DESIGN

3.1 Introduction

Most of the world's nuclear reactors use uranium fuel which is enriched to less than 5% U-235, the only naturally occurring fissile isotope. Neutron capture by the fertile material U-238 produces plutonium. Concerns have been raised about proliferation aspects of nuclear power due to the large, and still increasing stockpiles of plutonium produced in civil and military reactors. However, there has long been an interest in using thorium as a fertile material to produce U-233, an isotope of uranium with the best neutronic properties to enlarge the resources of thermal reactor fuels.

In most prior work, improvement in natural uranium utilization using thorium is achieved only if the self-generated U-233 fissile material is separated and recycled into a closed fuel cycle. However, in the U.S. once-through fuel cycles are exclusively used by all nuclear utilities and no recent effort has been made by any industrial firm to enter into the reprocessing business due to the projected high cost, though the ban on reprocessing was lifted in 1981. There are R&D programs worldwide on the use of once-through thorium fuel cycles in LWRs [K-2][K-3][H-2]. The most recent project completed in the U.S. is the evaluation of homogeneous Th/U dioxide for LWRs, DOE-supported and led by INEEL. It has been conclude that there is no economic advantage compared with conventional uranium fueled LWRs [Z-1]. Another on-going project involving the thorium fuel is the heterogeneous utilization of thorium in PWRs, of which this effort is one part. It is also supported by a DOE-NERI program. BNL is the lead and MIT, Ben-Gurion University of Israel, and others from Russia and Korea are the collaborators.

The heterogeneous thorium project is focused on the seed and blanket concept, where the seed is 20% enriched UO_2 fuel or U/Zr metal fuel, and the blanket is ThO_2 with a small amount of UO_2. The significant difference from the homogeneous fuel is that the seed and blanket fuel are separated spatially, and the blanket remains in the core for a relatively long time (about 10 years). Hence the bred U-233 can be burned in situ. There are two design options: SBU and WASB. as introduced in Chapter 1. The WASB concept is under study at MIT, and this thesis focuses on this design.

3.2 Seed and Blanket Fuel

3.2.1 Seed Fuel Design

The seed fuel in the WASB core is a "driver", which is used to keep the reactor critical and achieve the desired cycle length. UO_2 is chosen as the seed fuel. The SBU initially was designed to use U/Zr as the seed fuel, which could also be used for the WASB concept. However the thermal performance of UO_2 annular seed fuel pins was found acceptable for the WASB design to accommodate a higher linear power [B-6]. The use of UO_2 should simplify the licensing process. In addition, the neutronics and economic performance is slightly better than metallic uranium because of parasitic neutron absorption by zirconium as shown in Figure 3.1. Note that the two seed fuel designs have the same U-235 and total uranium contents. In addition, UO_2 has been widely and successfully used in nuclear power plants, and there is no industrial application of other fuels such as metal fuel in the U.S (although many research rectors use aluminum based fuel alloys).

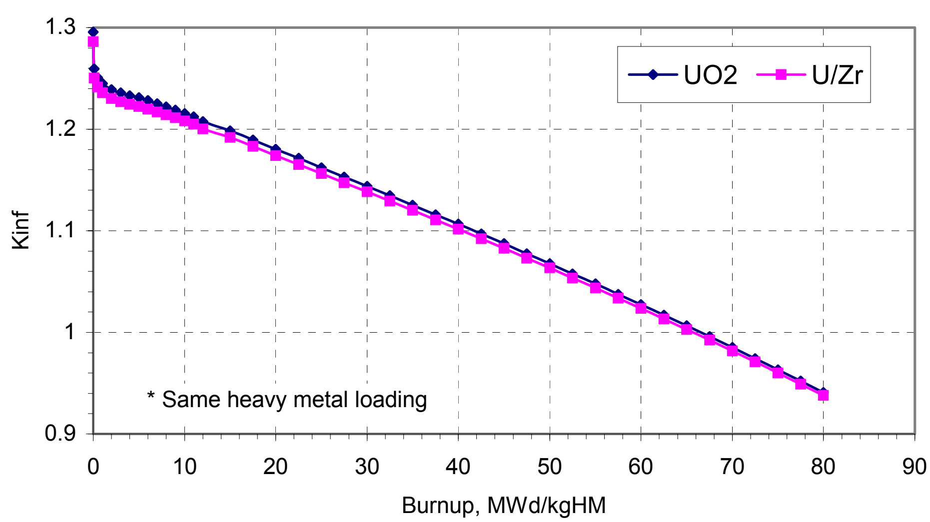


Figure 3.1 CASMO Burnup History for a UO_2 and a U/Zr Fuel Seed Assembly

In the WASB design, the seed fuel enrichment is just below 20%, a percentage generally accepted as being nonproliferative. The relatively high enrichment is necessary to compensate for the smaller volume of the uranium present in the core and to compensate for the high thorium capture rate. The purpose of the seed is to supply neutrons to the blanket in the most efficient way by thermalizing the neutron spectrum (high water content) so as to minimize plutonium formation. However, this high enrichment level is disadvantageous with respect to front-end fuel cycle costs, as will be discussed in Chapter 7. Care is required for such high enrichment because the physics of criticality begins to change as enrichments reach 10% and beyond, where single moderated assemblies can go critical and criticality of weakly moderated or unmoderated systems becomes possible. Enrichments above 5% will require redesign of some fuel fabrication and handling equipment and fuel transportation packages. The possibility of reactivity addition during accidents, in particular in severe accident core melt sequences should also be addressed as this could alter the progression of such accidents.

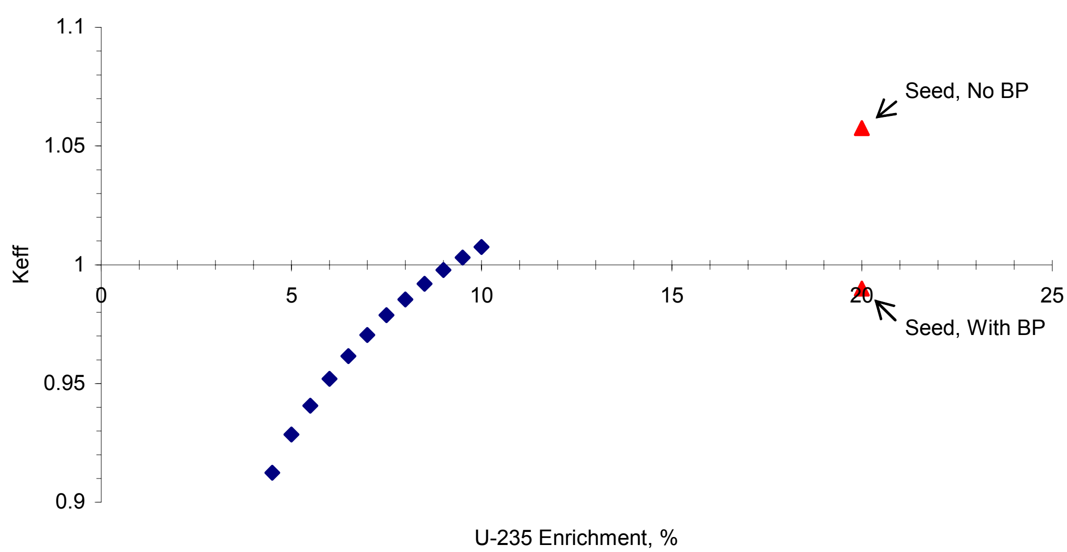


Figure 3.2 Criticality of a Single Assembly in a Water Pool

Figure 3.2 shows the Keff values as a function of enrichment for a standard 17×17 UO_2 lattice in an infinite water pool at a temperature of 300 ^{0}K. The single assembly becomes critical when the fuel enrichment reaches 9.5%. Note that Keff of a fresh unpoisoned seed assembly is quite a bit higher than 1.0 (Keff is about 1.06); however Keff of a poisoned seed assembly with 20% enrichment is subcritical (< 1). Hence the storage area for fresh fuel should be reassessed to reaffirm the validity of the criticality analysis. For a wet fresh fuel storage area the water conditions and rack layout should be maintained within specified limits to ensure adequate subcriticality margins.

The top view of the seed fuel pin is presented in Figure 3.3. Higher enrichment leads to a lower fuel loading in the seed. Annular fuel pellets are employed in the seed rods of WASB, which helps maximize the H/U ratio in the seed. In addition, this also offers space to accommodate the needed burnable poison. Mechanical and thermal benefits also stem from the use of annular fuel pellets instead of solid pellets. These benefits include:

- Increased space for accumulation of fission gas, and, hence reduced rod internal pressure;
- Delayed and reduced pellet-clad-interaction (PCI);
- Lower maximum fuel temperature;
- Lower stored energy in the fuel pins, and therefore benefits for ECCS/LOCA limits.

However, the center void fraction of the fuel is about 32.6% by volume, which is beyond the 10% value which is representative of significant current experience (LWR end blankets and VVER-1000 fuel), hence a center void plug might be needed to support the fuel and prevent relocation of fuel fragments into the inner region of each fuel pellet. B&W has a patent on file which calls for the use of a low density ceramic filler for this purpose [P-1].

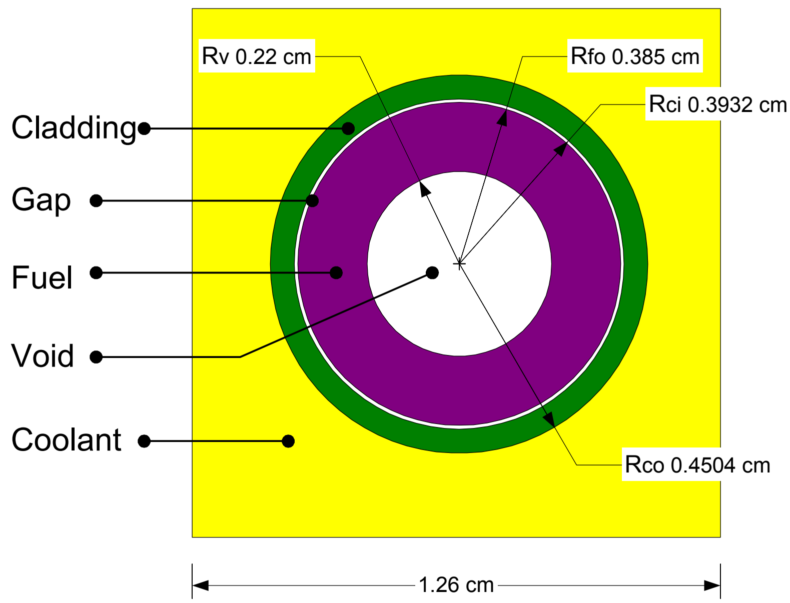


Figure 3.3 Top View of a Seed Fuel Pin

The center-voided fuel has a lower temperature profile than solid fuel at the same linear power rate. The ratio of temperature rise in the center-voided fuel to solid fuel at the same linear power rate is [T-4]:

$$\frac{\Delta T_{annular}}{\Delta T_{solid}} = 1 - \frac{\ln\left(\frac{R_{fo}}{R_v}\right)^2}{\left(\frac{R_{fo}}{R_v}\right)^2 - 1}, \tag{3.1}$$

where R_{fo} is the outer radius of the fuel and R_v is the void radius. Note that the formula only applies with the assumption of uniform power density and conductivity in the fuel. A 32.6% void by volume in the seed fuel leads to a reduction of temperature rise by about 55%.

3.2.2 Blanket Fuel Design

The blanket fuel assemblies in the core are interdispersed among the seed assemblies. The blanket fuel composition is a mixture of ThO_2 and UO_2, where UO_2 is only about 10% of the fuel volume. A pin cell model is shown in Figure 3.4.

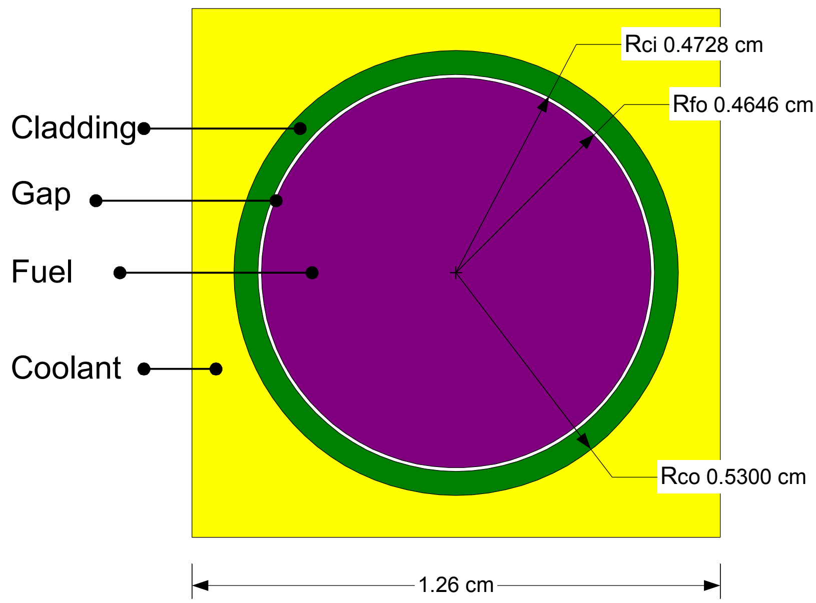


Figure 3.4 Top View of a Blanket Fuel Pin

Under irradiation, thorium undergoes a rapid increase in U-233 concentration and attains a value higher than that of Pu-239 in uranium lattices. Because of the very large fission cross section of Pu-239, it saturates and declines much sooner than U-233 in thorium. This lower equilibrium concentration of Pu-239 is due to its high fission cross section and because of the high capture to fission ratio: for thermal neutrons, it is 0.096 for U-233, and 0.504 for Pu-239. Thus, the main challenge of efficient utilization of thorium in LWRs in a once-through fuel cycle is reduced to the problem of achieving very large accumulated burnup in the thorium.

The WASB fuel cycle is based on extensive utilization of thorium, which produces through a nuclear reaction the fissile isotope U-233. U-233 may be used as a weapon material. Therefore, a special effort was invested in the blanket fuel design to create an effective barrier to diversion of U-233. In order to eliminate the U-233 concern, the U-233 created in the blanket is to be denatured by inclusion of uranium having a high U-238 content in the fuel. It should be noted that the acceptable concentration of mixtures of U-233 and U-235 has not been established in law or in international agreements. The acceptable limit for U-235 in U-238 is 20%. A recent study by Forsberg, et al. [F-1] shows that 12% U-

233 is "critically equivalent" to the 20% U-235 limit (in mixture with U-238), and that the "criticality equivalency" of the limits on LEU for mixtures of (U-233 + U-235) in U-238 can be expressed by the relationship:

$$\frac{^{233}U + 0.6\,^{235}U}{U^{tot}} \leq 12\% \,. \tag{3.2}$$

It should be noted that as U-233 builds up in the thorium, U-235 is consumed in a blanket containing natural or slightly enriched uranium. Thus it is necessary to adjust the initial uranium content or U-235 enrichment of the blanket fuel in order to meet proliferation requirements. Additional uranium isotopes created during the long in-core residence time of the blanket are U-232, U-234, U-235, and U-236, which present another major natural barrier to diversion of U-233. In principle, uranium may be chemically separated from the blanket spent fuel and further enriched by standard industrial methods. However, there are several barriers to the diversion of U-233. As discussed above an addition of natural or slightly enriched uranium provides an appreciable fission rate during this period and allows reasonable power sharing between the seed and blanket, which, in turn, limits the seed power density to an acceptable level.

Parametric studies on blanket fuel compositions are summarized in Table 3.1. Calculations were made using CASMO-4. It can be seen that the U-233 weight percentage in total uranium at the burnup of 80 MWd/kgHM decreases as the UO_2 content increases in the fuel. The enrichment of U-235 in UO_2 has no significant effect on the U-233 content. Thus it is an effective strategy to adjust the UO_2 volume ratio to ThO_2 in the fuel so that the U-233 weight percentage is below 12% in the discharged fuel. Finally 13% UO_2 - 87%ThO_2 has been used for the blanket fuel composition, where the enrichment of U-235 is 10%.

Table 3.1 U-233 Weight Percentage in Total Uranium at 80 MWd/kgHM

Enrich.[b] \ v/o[a]	5%	10%	13%	15%
5%	25.5%	14.7%	11.5%	10.0%
10%	26.4%	15.3%	12.0%	10.5%
15%	27.3%	15.9%	12.5%	10.9%

a. UO_2 volume ratio in UO_2-ThO_2 fuel

b. U-235 enrichment in U

U-233 builds up in the blanket fuel as a function of burnup as shown in Figure 3.5. The isotopic weight percentages are related to the initial heavy metal. At the beginning of life U-233 increases rapidly, then more slowly after 20 MWd/kgHM. Finally the U-233 content remains at an approximately constant level of 1.6%. The Fissile plutonium content is much lower.

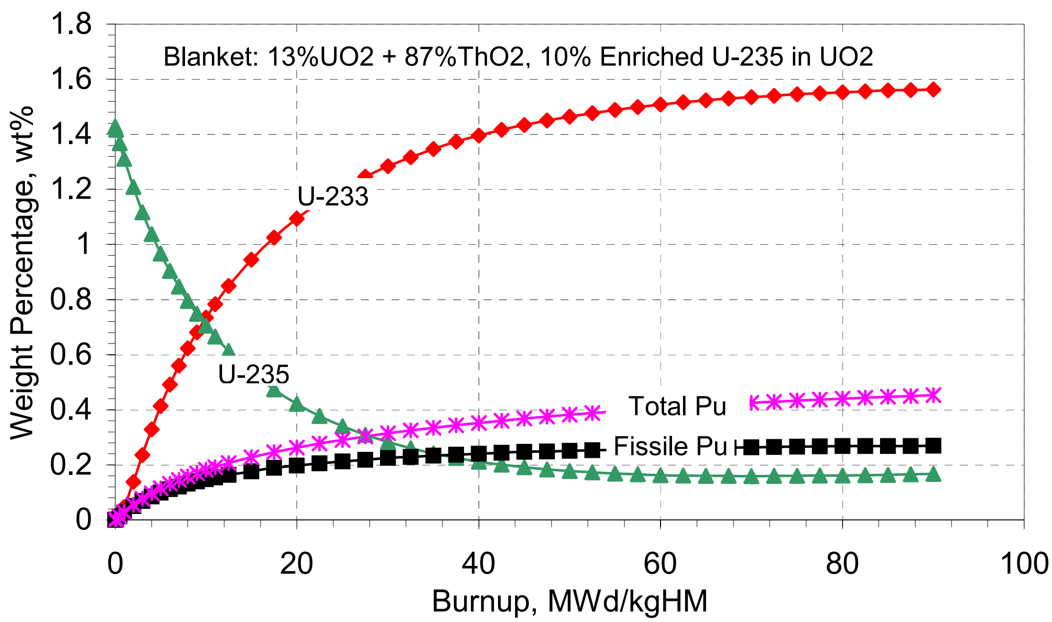


Figure 3.5 Changes in Blanket Fuel Composition with Burnup (wt% is relative to initial heavy metal)

3.3 Moderation Effects

Moderation effects on reactivity behavior and safety parameters are of significant importance for fuel and core design. In terms of LWR safety criteria, there is a general requirement that the moderator temperature coefficient be negative when the core is critical. Moderation in the seed and blanket of the WASB can be adjusted to achieve the optimum neutronic and thermal-hydraulic performance. Generally speaking, a high moderator to fuel volume ratio is desirable in the seed to reduce plutonium generation, and a low moderator to fuel volume ratio in the blanket to give a higher conversion ratio. There are several means to change the moderator to fuel ratio. Varying the fuel rod diameter is used here to investigate moderation effects on seed and blanket fuel design.

A colorset model is employed to analyze the moderation effects on the WASB core. First, the moderator to fuel ratio of the seed fuel is varied to investigate its effects on the colorset neutronic behavior without varying the blanket fuel. The total heavy metal content of the seed fuel is conserved while varying the seed fuel rod inner void diameter and fuel outer diameter. The geometrical parameters and material compositions of the seed and blanket fuel are presented in Table 3.2.

Table 3.2 Fuel Characteristics for the Seed Moderation Study

PARAMETERS	VALUES			
	Seed 1	Seed 2	Seed 3	Blanket
Fuel Material Composition	UO_2, 20 w/o U235			ThO_2+UO_2 [b]
Fuel Assembly Pitch, cm	21.5			
Fuel Pellet Radius, cm	0.10-0.3314	0.22-0.385	0.35-0.4715	0.4646
Gas Gap, cm	0.0082			
Cladding Thickness, cm	0.0572			
Pin Pitch, cm	1.26			
Moderator to Fuel Volume Ratio [a]	3.95	3.50	2.65	1.26

a. Assembly based values; compare to conventional PWRs @ 1.95

b. 87% ThO_2 + 13% UO_2, and 10% Enrichment in UO_2

The impacts of varying the moderator to fuel volume ratio in the seed are shown in Figures 3.6, 3.7, 3.8 and 3.9. Lower values of V_{water}/V_{fuel} lead to poorer neutronic performance as shown in Figure 3.6. It is shown in Figure 3.7 that the conversion ratio increases as the V_{water}/V_{fuel} value increases in the seed, however the effects on the conversion ratio of the blanket are negligible when only varying the moderator to fuel ratio of the seed. In addition, fissile and total plutonium contents in the seed fuel are also shown in Figures 3.8 and 3.9. It is obvious that a wetter seed lattice design is preferred in terms of neutronic performance and plutonium content. It should be noted that fuel rod design is not only determined by neutronic considerations, but also limited by thermal-hydraulic and mechanical requirements.

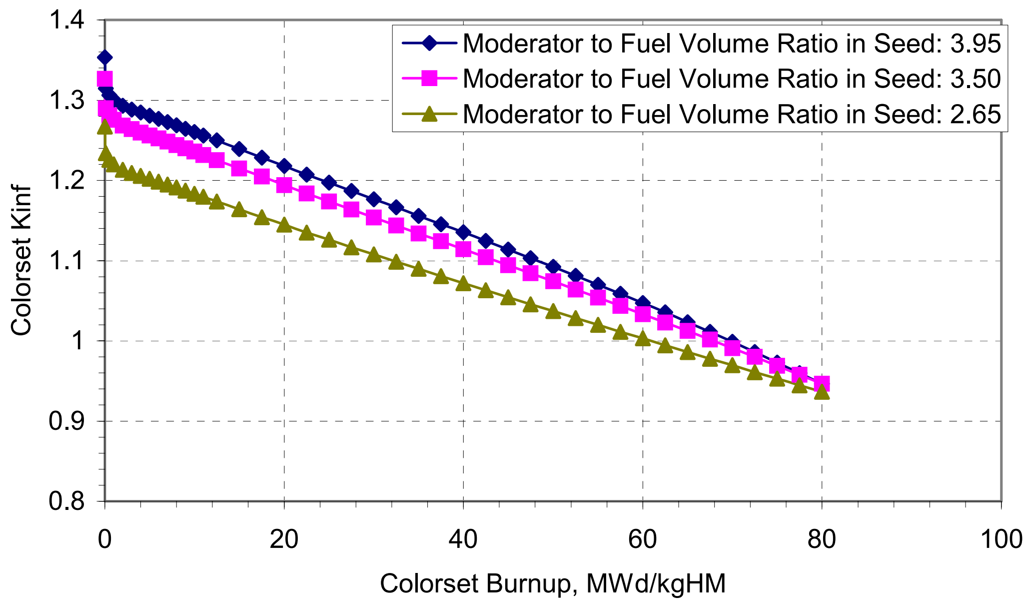


Figure 3.6 Impacts of Varying V_{water}/V_{fuel} on Kinf of a Colorset

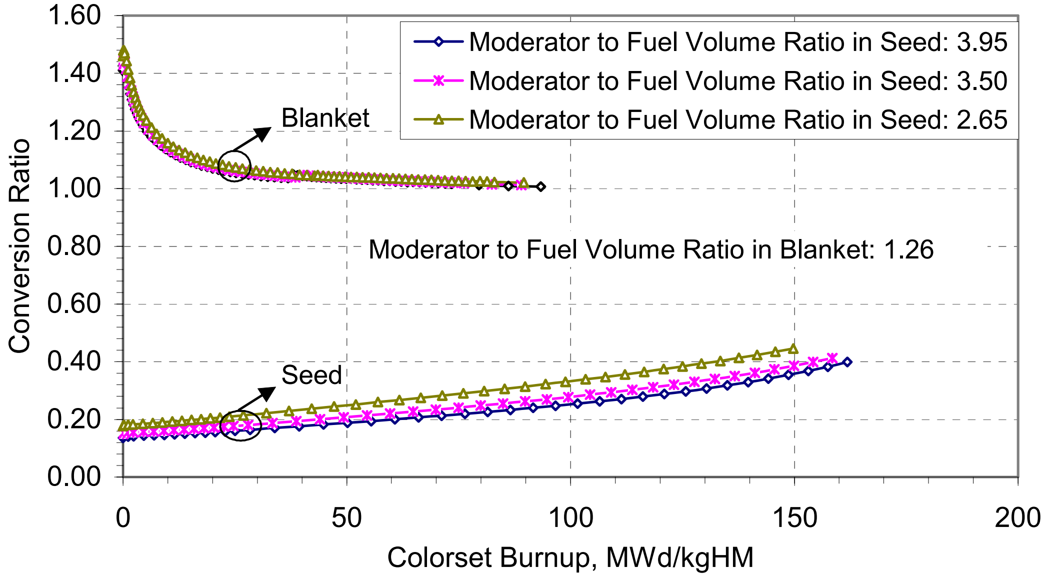


Figure 3.7 Impact of Varying V_{water}/V_{fuel} on CR in Seed

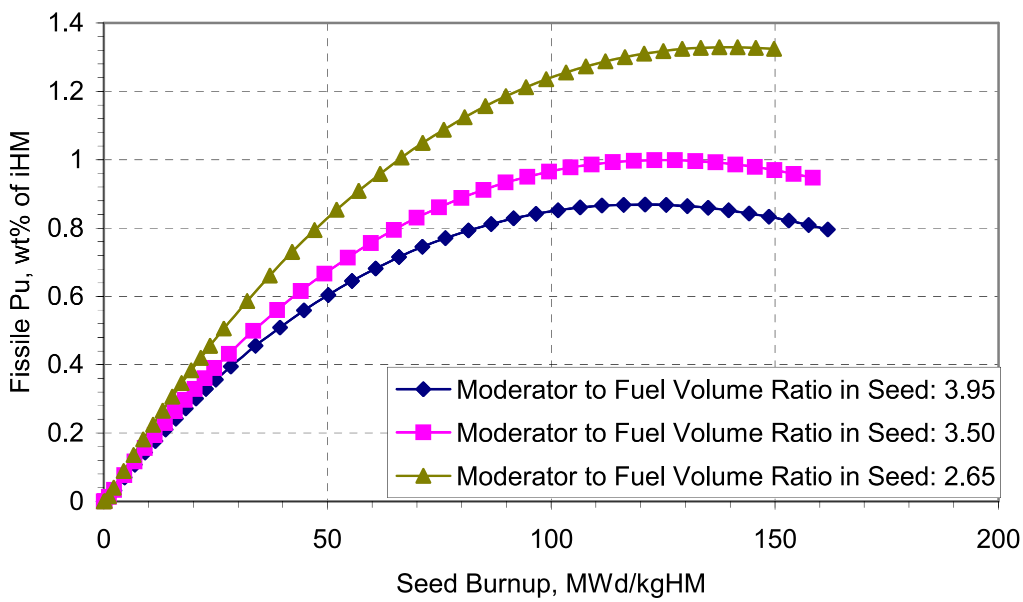


Figure 3.8 Impact of Varying V_{water}/V_{fuel} on Fissile Pu Content in Seed

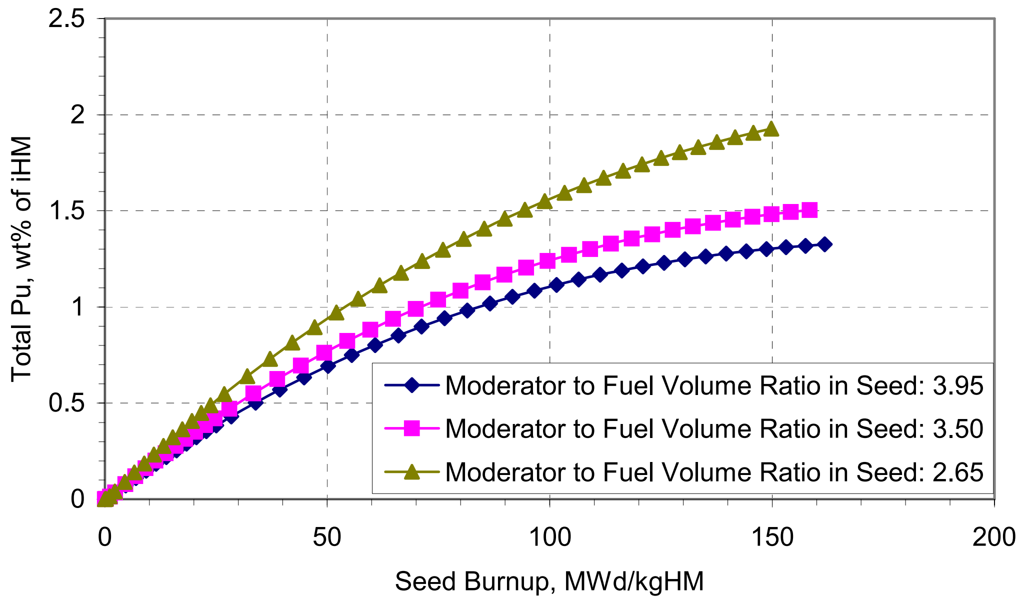


Figure 3.9 Impacts of Varying V_{water}/V_{fuel} on Total Pu Content in Seed

Calculations were also performed to analyze the impacts of varying the moderator to fuel ratio of the blanket fuel. Varying V_{water}/V_{fue} values is realized by changing the blanket fuel diameter. Hence heavy metal loading in the blanket is not preserved when varying the moderator to fuel ratio in the blanket. Three different moderator to fuel ratios have been compared as shown in Table 3.3.

Table 3.3 Fuel Characteristics for the Blanket Moderation Study

PARAMETERS	VALUES			
	Blanket 1	Blanket 2	Blanket 3	Seed
Fuel Material Composition	ThO_2+UO_2 [b]			UO_2 [c]
Fuel Assembly Pitch, cm	21.5			
Fuel Pellet Radius, cm	0.3500	0.4096	0.4646	0.22-0.3850
Gas Gap, cm	0.0082			
Cladding Thickness, cm	0.0572			
Pin Pitch, cm	1.26			
Moderator to Fuel Volume Ratio[a]	3.09	1.95	1.26	3.50

a. Assembly based values; compare to conventional PWRs @ 1.95

b. 87% ThO_2 + 13% UO_2, and 10% Enrichment in UO_2

c. 20% enriched U-235 in UO_2

The Nu-Fission/Absorption values of the seed and blanket fuel in a colorset are presented in Figure 3.10, and comparisons with single assembly results are also shown in the figure. Actually the Nu-Fission/Absorption values from single assembly calculations are just the multiplication factors, Kinf. It can be seen that the Nu-Fission/Absorption values from single assembly calculations are very close to those from colorset calculations. Hence reactivity is approximately a state function of burnup for a PWR assembly - in other words, a unique single-valued function of the specified burnup, and not of the depletion history or the manner in which the burnup was accumulated [D-1].

In order to take into account the influence of neighboring assemblies a colorset was also employed to investigate the moderation effects of the blanket fuel. All results are shown in Figures 3.11, 3.12 and 3.13.

Figure 3.11 shows the impacts on Nu-Fission/Absorption values when varying the moderator to fuel volume ratio of the blanket assemblies in a colorset. It is concluded that a tight lattice is preferred due to higher reactivity and flatter reactivity swing during depletion. However, Figure 3.12 shows that there are only minor effects on the conversion ratio at the early stage of depletion when varying the moderator to fuel volume ratio of the blanket fuel. It is interesting that all three curves of the blanket fuel almost converge to the same conversion ratio after 30 MWd/kgHM, and that ratio remains constant until 90 MWd/kgHM. Note that the conversion ratio of the blanket is always above one, which means that the blanket fuel undergoes net breeding. U-233 builds up with depletion as shown in Figure 3.13, and once again a tight or dry lattice design is desirable due to its high U-233 content in the fuel.

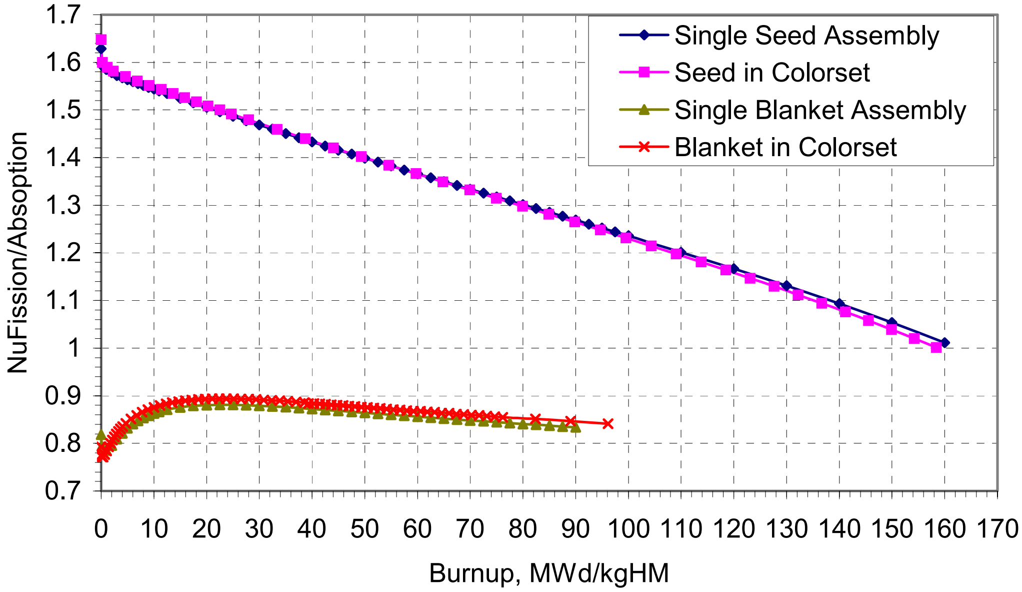


Figure 3.10 Nu-Fission/Absorption Ratio in Seed and Blanket

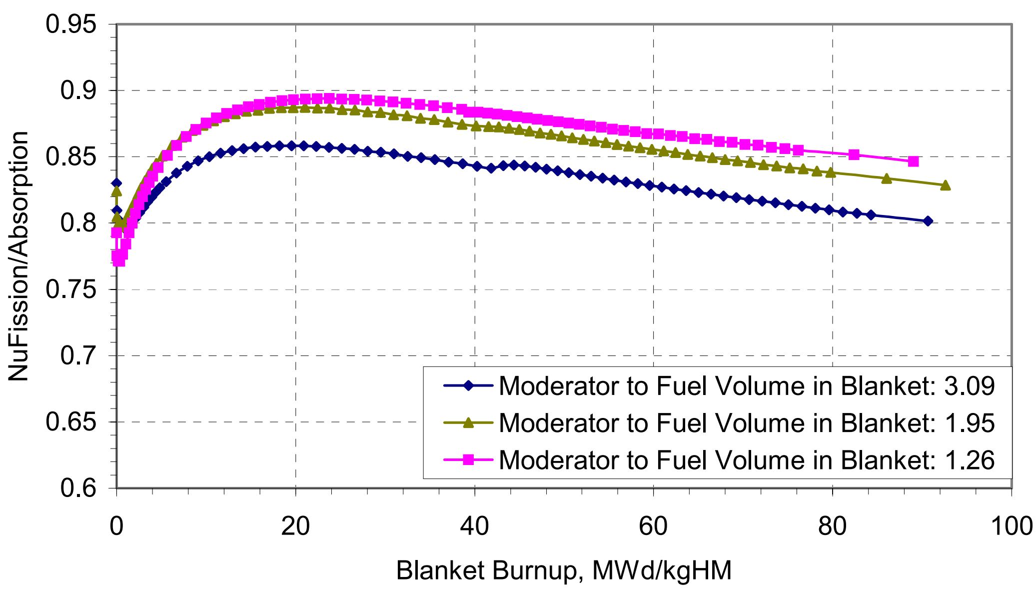


Figure 3.11 Impact of Varying V_{water}/V_{fuel} on Nu-Fission/Absorption in the Blanket

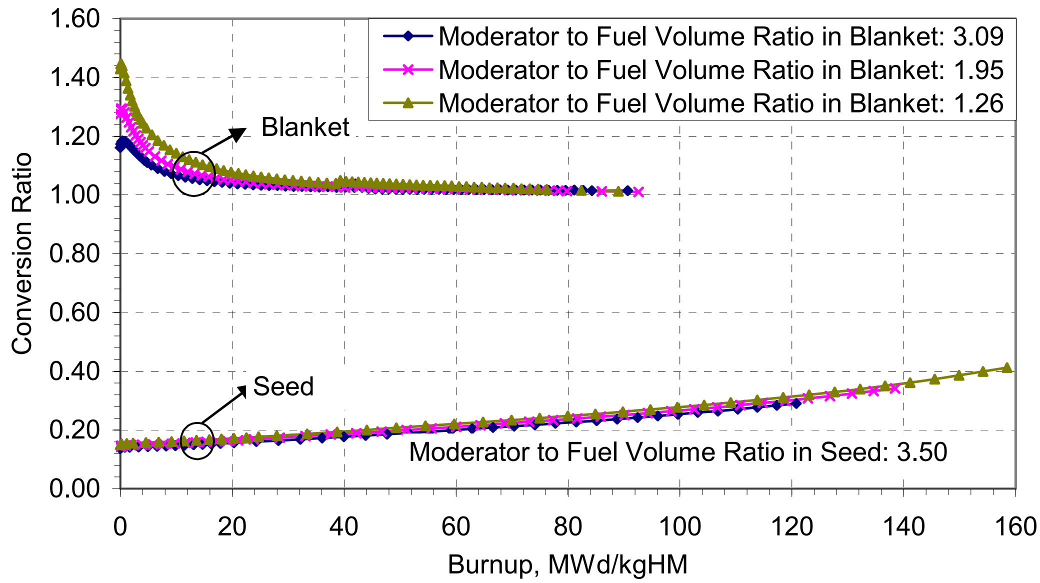


Figure 3.12 Impact of Varying V_{water}/V_{fuel} on Conversion Ratio in the Seed and Blanket

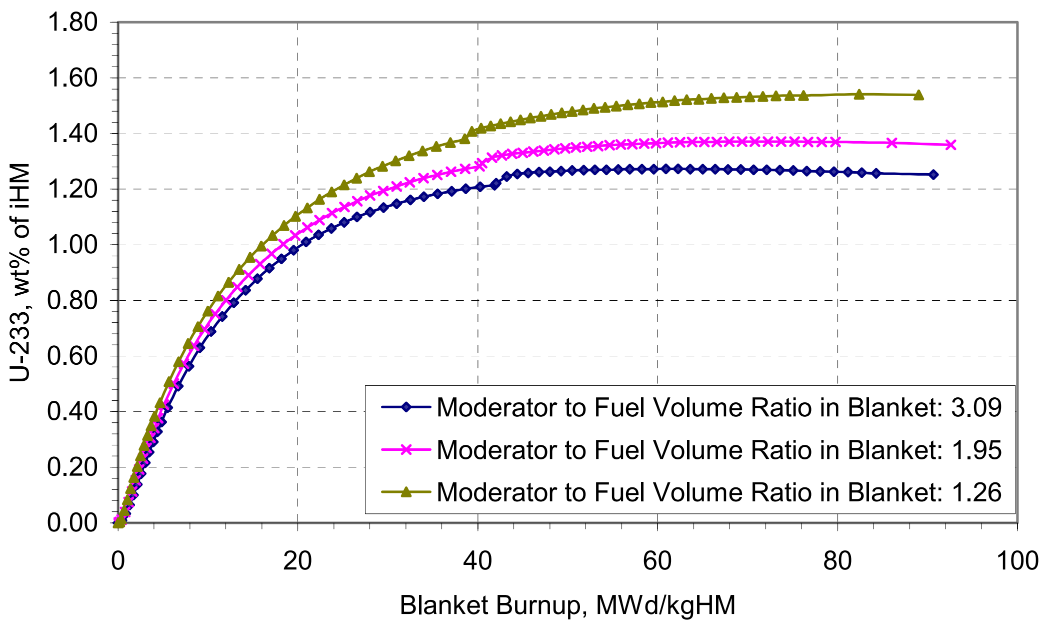


Figure 3.13 Impact of Varying V_{water}/V_{fuel} on U-233 Content in the Blanket

Based on the preceding detailed analyses of the seed and blanket fuel assemblies, an "optimum" reference design of the WASB fuel, in terms of neutronic performance, has been proposed (and subjected subsequently to whole core analyses) as shown in Table 3.4. More fuel design issues (thermal-hydraulic performance, etc.) will be addressed and assessed in the following chapters.

Table 3.4 Seed and Blanket Assembly Optimized Parameters

PARAMETERS	VALUES	
	Seed	Blanket
Fuel Material Composition	UO_2 20 w/o U-235 in U	87 v/o ThO_2 + 13 v/o UO_2 10 w/o U-235 in U
Fuel Density, g/cm^3	10.302	9.403
Fuel Assembly Pitch, cm	21.5	
Fuel Pellet Radius, cm	Void: 0.22, Fuel: 0.385	0.4646
Gas Gap, cm	0.0082	
Cladding Thickness, cm	0.0572	
Pin Pitch, cm	1.26	
Moderator to Fuel Volume Ratio[a]	3.50	1.26

a. Conventional PWR $V_m / V_f = 1.95$

3.4 Burnable Poison Design

Burnable poisons are used to hold down beginning of cycle core reactivity and to shape the power distribution in the core so that physics design limits are met and hot spots are avoided. These limits are set to insure that the coolant has the capacity of extracting all the energy locally generated in the fuel rods, and the fuel and clad temperatures do not exceed the design limits. Some fuel performance parameters such as fission gas release or internal pin pressure are closely related to the temperature attained in the fuel. One shortcoming of burnable poisons is the residual reactivity penalty. The concentration of burnable poison is reduced as it captures neutrons, in which process it competes with fissile atoms. Towards the last part of the cycle, burnable poisons are less important or not needed, since the excess reactivity can be held down by an acceptable amount of soluble boron, and the less reactive core has lower peaking factors. However, absorbing isotopes are generated in the neutron capture process (e.g. Gd-157 from Gd-156), producing a residual poisoning. The core reactivity is hence reduced and the core life shortened. However, burnable poisons tend to contribute to hardening of the spectrum, which enhances the conversion of fertile materials to fissile materials. This effect tends to enhance reactivity. Ideally, a burnable poison should:

- Have a low residual penalty;
- Yield low intra-assembly power peaking;
- Create a fairly flat reactivity shape throughout core life, thus avoiding undesirable shifts in core power distribution;
- Not adversely affect oxide thermal conductivity.

3.4.1 Erbium vs. Gadolinium and Boron

Introduction of burnable poison into the central void of the seed fuel pellets provides substantial benefits in the WASB design. This concept was originally proposed in a B&W patent [P-1] but to our knowledge never applied in practice. Several burnable poisons are analyzed and evaluated here.

Gadolinium: Gadolinium is a rare earth element with atomic number 64. It has several natural isotopes, with two main thermal absorbers: Gd-155 and Gd-157. The oxide form Gd_2O_3 has been extensively used as burnable absorber in both PWRs and BWRs. The thermal neutron absorption cross sections of Gd-157 and natural gadolinium are 250,000 and 49,000 barns respectively.

Boron: The absorbing material B-10 does not generate residual absorbing daughter products. Therefore, the residual penalty is greatly reduced. On the other hand, the boron creates helium after neutron absorption, and this gas accumulates in the rod void space. As a result, internal pin pressure significantly increases through the cycle, raising concerns about fuel integrity.

Erbium: In the WASB design, Er_2O_3 is used in the seed fuel rod design. The obvious benefits will be shown in the following paragraphs.

Erbium is a rare earth element with atomic number 68, similar in chemical and metallurgical properties to gadolinium. Like gadolinium, erbium is comprised of several natural occurring isotopes, of which Er-167 is the primary neutron absorber. The energy-dependent neutron absorption properties of Er-167 include a large double resonance in the vicinity of 0.5eV, as shown in Figure 3.14.

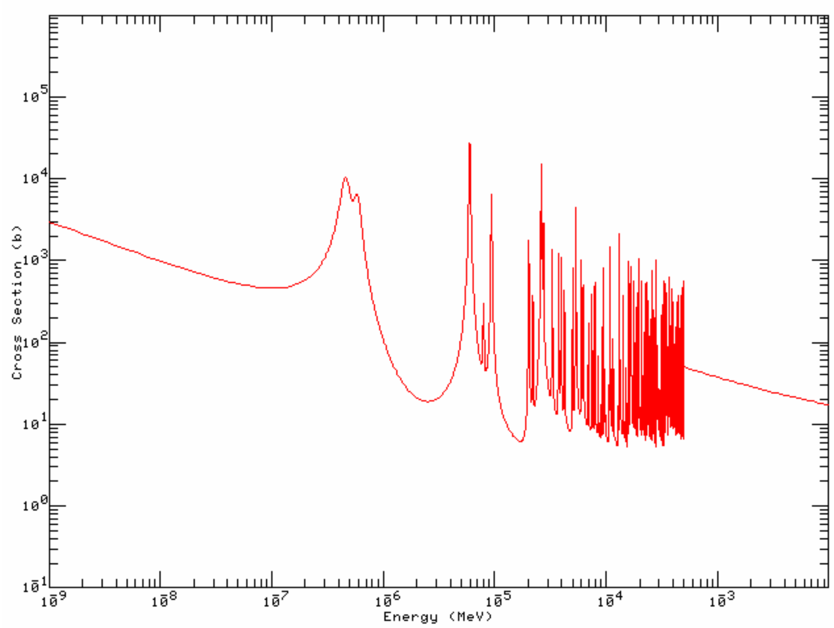


Figure 3.14 Neutron Cross Section of Er-167

This enhances the thermal neutron absorption of erbium (i.e., providing a non-1/v absorption characteristic), and provides the additional characteristic of improving the negative fuel temperature coefficient (FTC) and moderator temperature coefficient (MTC) due to the location of the resonance at the high end of the thermal energy spectrum near the resonance in Pu-239 absorption. In contrast to gadolinium, erbium has a slower depletion characteristic as a burnable poison, releasing reactivity gradually over a longer period of fuel burnup.

Erbium has been used in TRIGA cores to provide a more negative fuel temperature coefficient for high enrichment uranium fuel. ABB-CE (now Westinghouse) has developed the application of erbium as a burnable poison for PWRs, in the form of Er_2O_3 admixed with enriched UO_2 [J-2]. This application was developed as an optimized burnable poison design for 18- and 24-month UO_2 fuel cycles (i.e., the range of cycle lengths currently in use for all U.S. ABB-CE plants). For extended UO_2 cycle lengths the erbium burnable poison design shows major advantages of improving thermal margins (reducing power peaking over long cycle lengths by distribution of the required burnable poison over a large number of fuel rod locations) and providing a more negative moderator coefficient at beginning-of-cycle with higher UO_2 enrichment loadings. The ABB-CE erbium burnable poison design has completed irradiation demonstration in two operating ABB-CE plants and generic licensing approval by the NRC, and is scheduled for full batch reload implementation [J-2].

The application of erbium burnable poison offers key benefits for the WASB design. These include:

- The addition of Er_2O_3 in seed fuel rods provides the capability to accommodate high fissile uranium loading. The effectiveness of erbium for controlling excess reactivity and providing a more negative FTC and MTC is pronounced at the beginning-of-cycle (BOL), and diminishes by end-of-life (EOL). Since the erbium poison is added in the central voids of seed fuel, it provides the ability to control a large amount of excess reactivity, while precluding the possibility of loss of this reactivity control by any mechanism, including misoperation or mechanical disassembly of the fuel;
- The 0.5 eV neutron absorption resonance of Er-167 enhances the neutron absorption worth of burnable poison in comparison to use of purely 1/v absorbers, such as B-10, which have significantly diminished reactivity worth in the presence of a high enrichment loading of U-235;
- The long-term reactivity control characteristics and ability to vary the distribution of the erbium concentration over the fuel lattice provide flexibility for control of power distribution over lifetime, in order to minimize peaking factors and maintain a high degree of thermal operating margin;
- Location of the erbium in the central void avoids its detrimental effect on UO_2 thermal conductivity: a significant benefit because of the high seed power density.

3.4.2 Comparison Calculations of Various Absorbers

Reactivity versus Burnup

All calculations were performed using CASMO-4. The poison is evenly loaded in the central void of each seed pin. The poison content is shown in Table 3.5.

Table 3.5 Burnable Poison Loading

	Fuel U	Er_2O_3 Nat. Er	Er_2O_3 Er-167	ZrB_2 B-10	Gd_2O_3 Nat. Gd
Mass per cm in a pin, g/cm	2.848	0.0332	0.0226	0.0011	0.0049
Bp w/o relative to Fuel		1.166%	0.794%	0.039%	0.172%

Figures 3.15 shows the effect that different burnable poisons have on a seed assembly's neutronic behavior, and Figure 3.16 demonstrates the effect that different burnable poisons have on a colorset's neutronic behavior. The burnup used in Figure 3.16 is the average over both the seed and blanket.

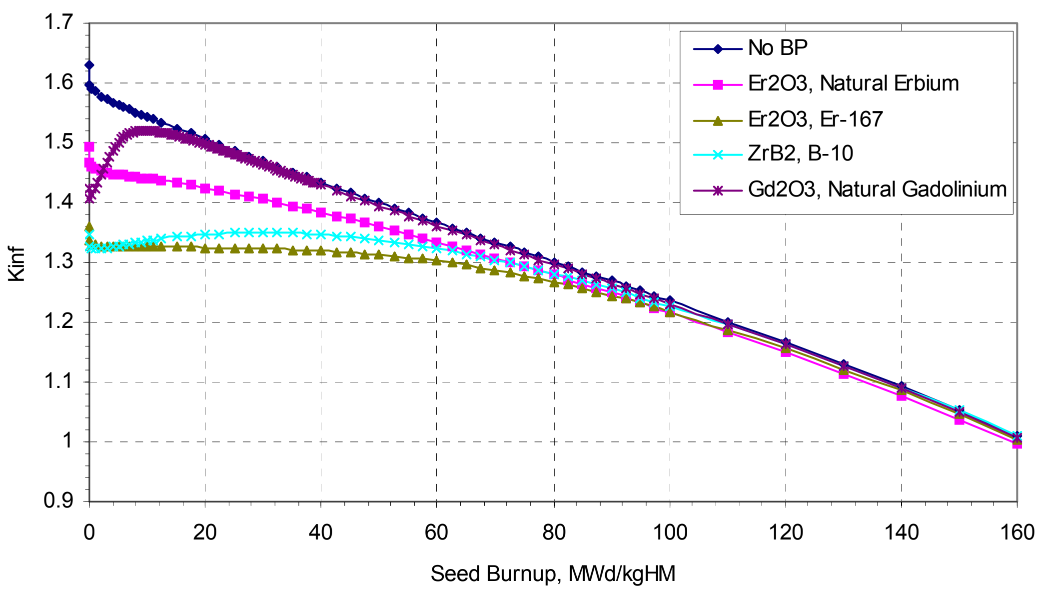


Figure 3.15 Seed Reactivity Response to Burnable Poisons

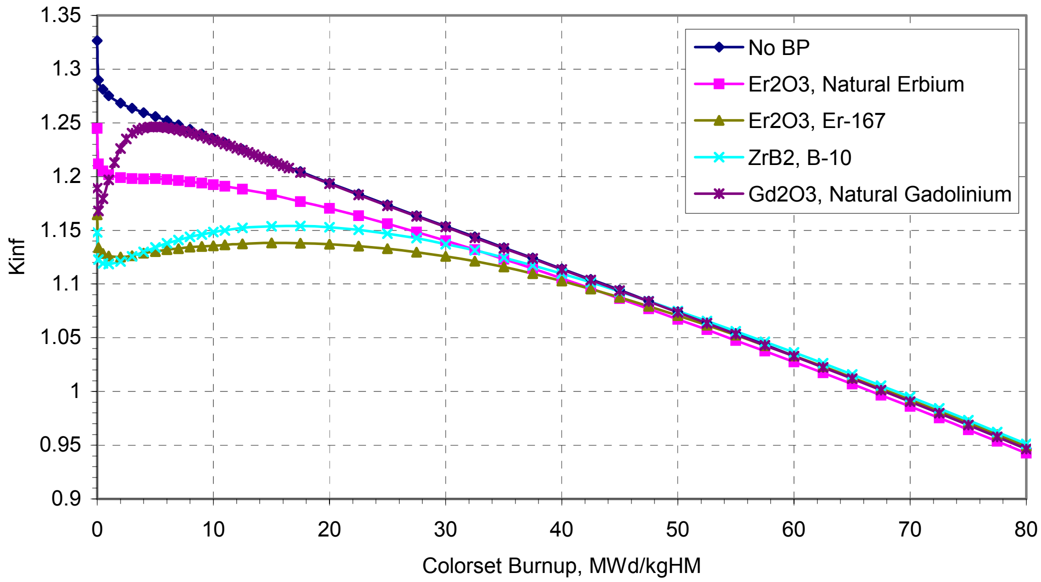


Figure 3.16 Colorset Reactivity Response to Burnable Poisons

It should be noted that the coolant is taken as free of soluble boron for all cases. The behavior of an unpoisoned case is also shown for reference. The rapid burnout of gadolinium produces a sharp rise in assembly reactivity compared to the other poison schemes. The best poison is Er-167, which keeps the reactivity at a significantly lower level than the other poisons. It is also interesting for the WASB configuration that no obvious reactivity penalty exists at high burnup for all burnable poisons with the exception of the natural Er_2O_3 case.

Flattening Power

The power distribution at BOC based on colorset calculations shows that the seed has a much higher power density than the blanket. As a general rule, the seed rods near the blanket have higher power than other seed rods. The peripheral seed rods have higher power than the inner rods. The highest power is found for a seed rod at the corner of the Colorset. In order to reduce the power peaking and flatten the power profile burnable poison was used in the seed fuel rods. Er_2O_3 (Er-167) was loaded in the central void of each seed fuel pin. The burnable poison loading is as shown in Figure 3.17.

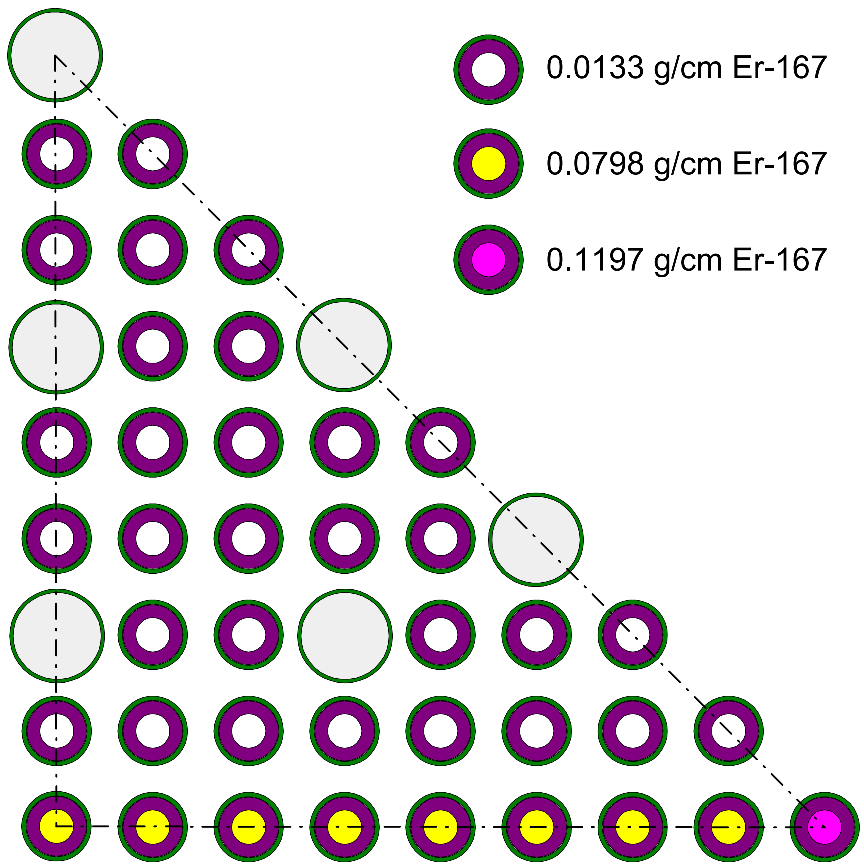


Figure 3.17 Burnable Poison Loading Pattern

Table 3.6 shows the effect of erbium on the power peaking. Using burnable poison in this manner the average peak pin power is decreased about 8.5% compared to an unpoisoned pin.

Table 3.6 Linear Heat Generation Rate Comparison

	Seed Assembly	
	No BP	With Er_2O_3
Avg. Pin Power, w/cm	281.7	272.8
Peak Pin Power, w/cm	314.65	287.8

The addition of BP effectively improves the power sharing between the seed and blanket and decreases the peak power. The introduction of BP in the WASB leads to an increase in the blanket power output, and a reduction in the maximum power of the seed assembly. The relative power of the seed versus exposure is shown in Figure 3.18. During the life of the core, the power sharing between the seed and blanket improves, as expected.

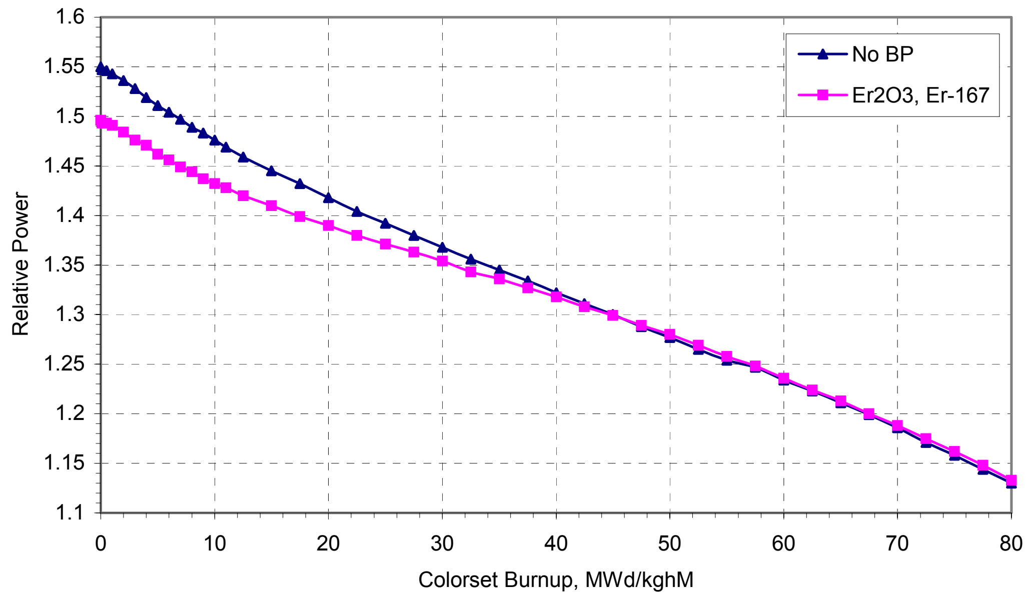


Figure 3.18 Relative Power of Seed over Cycle Exposure

3.5 Neutron Spectrum

The neutron spectrum of the WASB core was investigated using MCNP-4C based on a colorset model. Figure 3.19 shows the neutron spectrum in the seed and blanket at the early depletion stage, where the burnups of the seed and blanket are very low, all fluxes shown are constant-lethargy-width group values. A conventional UO_2 fueled core neutron spectrum is also presented for comparison. The neutron spectrum of the WASB core has some specific characteristics; however it is very similar to that of a conventional PWR core. The thermal neutron flux in the blanket is about two orders of magnitude higher than in the seed, and lower for the fast neutron flux. Thus thermal neutrons flow from the blanket to the seed and fast neutrons flow from the seed to the blanket. The neutron spectra show characteristic dips in the resonance region because of the resonance absorption of U-238 and Th-232.

It can be seen in Figures 3.20 and 3.21 that the thermal flux in the seed increases as the seed burnup increases due to the loss of fissile material. Figure 3.20 shows that the thermal flux in the blanket has a small drop if compared with Figure 3.19, however there is hardly any difference between Figures 3.20 and 3.21, since U-233 increases with burnup as shown in Figure 3.13, hence a relatively flat reactivity has been preserved during depletion. In addition, the resonance absorption around 1 eV of Pu-240 has a significant effect on the neutron spectrum when it builds up in the seed as shown in Figures 3.20 and 3.21.

For a whole core case, seed and blanket assemblies are located in a checkerboard pattern. The highly heterogeneous neutron spectrum in the seed and blanket brings complication in predicting power peaking using nodal methods as discussed in Chapter 2.

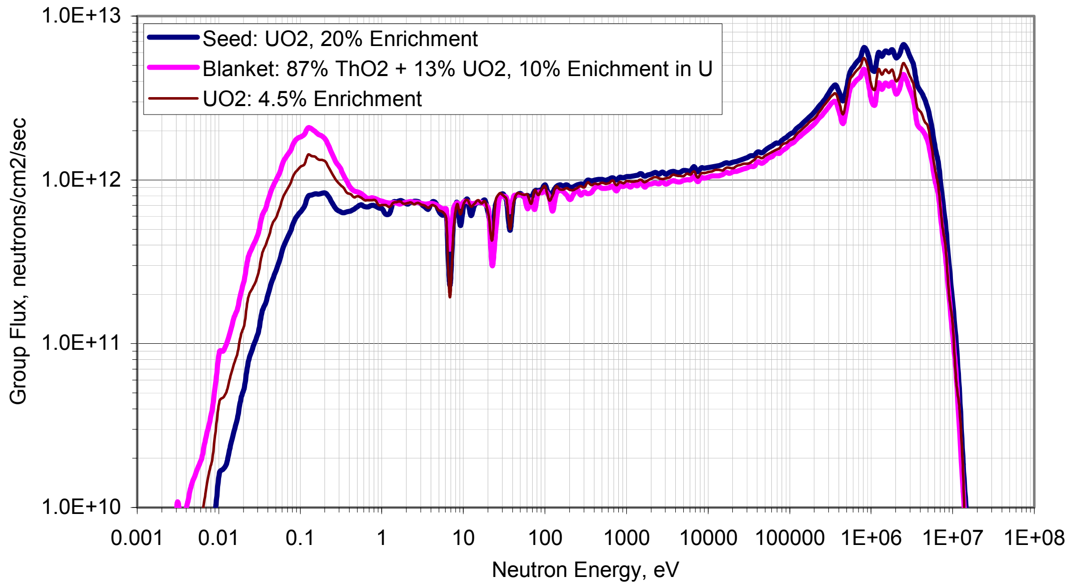


Figure 3.19 Neutron Spectra in WASB S&B Regions at BOL

(Seed: 9.1 MWd/kgHM, Blanket: 1.4 MWd/kgHM)

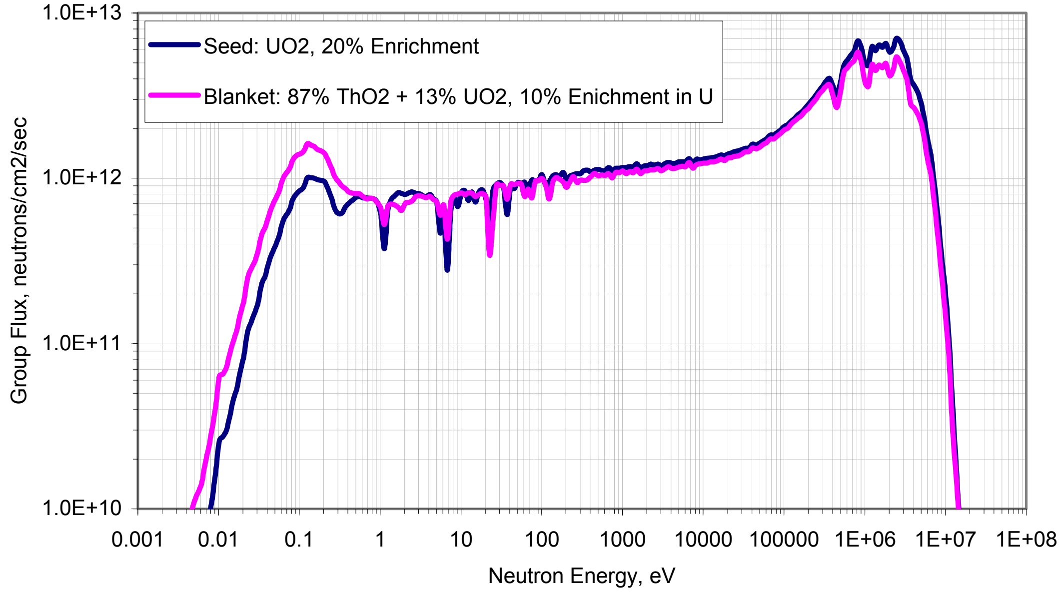


Figure 3.20 Neutron Spectra in WASB S&B Regions

(Seed: 74.9 MWd/kgHM, Blanket: 14.7 MWd/kgHM)

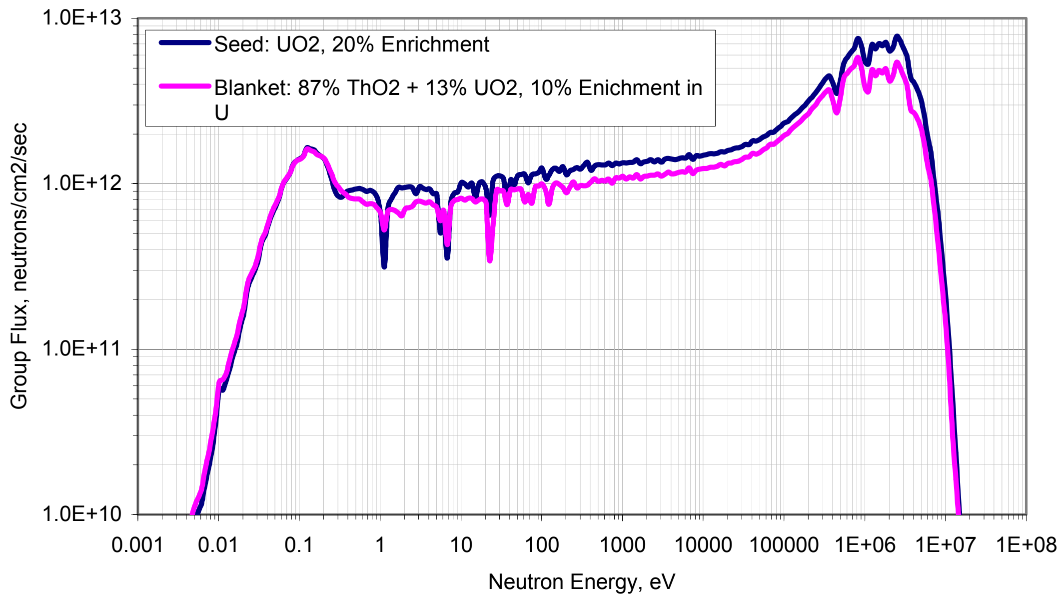


Figure 3.21 Neutron Spectra in WASB S&B Regions

(Seed: 141.1 MWd/kgHM, Blanket: 33.9 MWd/kgHM)

3.6 Chapter Summary

The seed and blanket fuel design has been extensively discussed in this chapter. 20% enriched annular UO_2 fuel is adopted for the seed fuel. It is concluded that a wet lattice improves neutronic performance and limits the fissile plutonium content. Even higher moderator to fuel volume ratios can be achieved by using thinner fuel rods, but a more practical seed fuel pin diameter was used for the seed fuel because of mechanical and thermal-hydraulic concerns.

The blanket fuel uses a mixture of ThO_2 and UO_2, in which a small amount of UO_2 serves to denature U-233 bred in the fuel and to share thermal power at the BOL. Detailed analyses were performed to determine the appropriate mixing ratio of ThO_2 and UO_2. The smallest UO_2 ratio needs to be about 13% (volume percent) in order to meet non-proliferation requirements in the blanket fuel. In addition, a tight blanket lattice (fatter fuel rod) design leads to better neutronic performance, that is, higher reactivity and a flatter core reactivity curve.

Analyses show that an even distribution of erbium in the seed fuel rods may yield a fairly constant reactivity for the cycle, and improve inner assembly power peaking factors. Erbium must be mixed with another material which acts as a void plug, as in the B&W patent cited earlier. By making the plug highly porous it can still serve the function of providing added volume to accommodate gaseous fission products.

Chapter 4

CORE DESIGN AND PERFORMANCE ASSESSMENT

4.1 Introduction

The objective of this chapter is to examine the core design for implementation of the WASB thorium/uranium fuel cycle in PWRs and identify the core and fuel management strategies which will maximize the benefits from inclusion of thorium in fuel. The focus is on once-through fuel cycles that do not involve reprocessing of the spent fuel. A 193 assembly Westinghouse PWR utilizing a17×17 lattice is taken as the model core.

WASB fuel management has its specific characteristics, which are significantly different from conventional PWR cores. A conventional PWR has only one type of fuel with minor variations in fuel composition such as enrichment and burnable poison loading. However a WASB core has two types of fuel: seed and blanket. For the seed a three-batch loading pattern is adopted; the blanket is used in single-batch loading and sits in the core for up to 9 seed cycles. The cycle length is 18 months. One design objective is to retrofit this type of heterogeneous seed and blanket core into existing PWRs in the future. But challenges in core management due to this unconventional heterogeneous core design should be clearly addressed. The objective of core management is to ensure the safe and reliable use of the fuel in the reactor as well as to minimize the fuel costs, with consideration of the limits imposed on the design of the fuel, on the basis of plant safety analyses. Thus the key parameters of the WASB core must be kept within its acceptable operational limits.

The performance of the core must be analyzed by comparison of neutronics and fuel performance parameters to a set of design limits. Some important options relating to core design and management are as follows:

- Ratio of the number of seed and blanket assemblies and their locations in the core to maximize the use of thorium;

- Choice of fuel to moderator ratio in each type of assembly;
- Level of enrichments of uranium in reload fuel;
- Distribution of burnable poison;
- The adequacy of reactivity to achieve the designed cycle length;
- Reactivity coefficients of temperature, and reactivity worth of boron in coolant;
- Power distribution in the core and within the fuel assemblies and their control by appropriate movement of control rods or burnable poison;
- Control rod worth and shutdown margin;
- Thermal-hydraulic parameters such as MDNBR, maximum fuel temperature;
- High burnup fuel performance and cladding corrosion due to long residence of the blanket.

The Studsvik' CMS code package was used to perform core physics analysis. Its capability of simulating the WASB core has been discussed in Chapter 2. Thermal-hydraulic performance will be discussed in Chapter 5.

4.2 Core Modeling

The model set up for analysis of the WASB core with SIMULATE-3 is described here. The core was modeled in 3 dimensions with 24 axial nodes, and used the full core model. Each assembly had 4 radial nodes. The core representation included radial and axial reflectors (top and bottom) that account for the coolant and structural materials. The core was analyzed at a condition of steady-state Hot Full Power (HFP) with all the control rods completely withdrawn, and was depleted to End Of Cycle (EOC).

4.2.1 Loading Pattern

The WASB core of choice consists of 84 seed assemblies and 109 blanket assemblies. Both the seed and blanket assemblies use the standard Westinghouse 17×17 lattice design. The seed and blanket assembly design parameters are given in Table 3.4. The overall length of a fuel assembly is 365.76cm. The active fuel length is 335.28cm. The core loading pattern is devised to meet the requirements of in-core fuel management. There are two separate fuel management streams: a three-batch stream for the seed (18 month cycle length) and a single-batch stream for the blanket, which is to stay in the core for up to 9 seed cycles. The WASB core loading pattern is based on the general rule that more energy is extracted from the fuel if the ratio of peak-to-average power in the core is as low as possible. In order to reduce the neutron radiation damage to the pressure vessel and improve neutron economy, a low-leakage core strategy is applied to the WASB core. Figure 4.1 shows the pattern which reduces the flux by placing blanket assemblies at the periphery of the core. The seed assemblies and other blanket assemblies are scattered inside the core in a "checkerboard" scheme. It should be noted that the fresh seed assemblies are located in the core interior as far as possible. The main advantage of this scheme is that it produces a fairly uniform power distribution throughout the core, and therefore leads to lower power peaking and a low-leakage core. Calculations show that the whole core leakage is less than 1.4%.

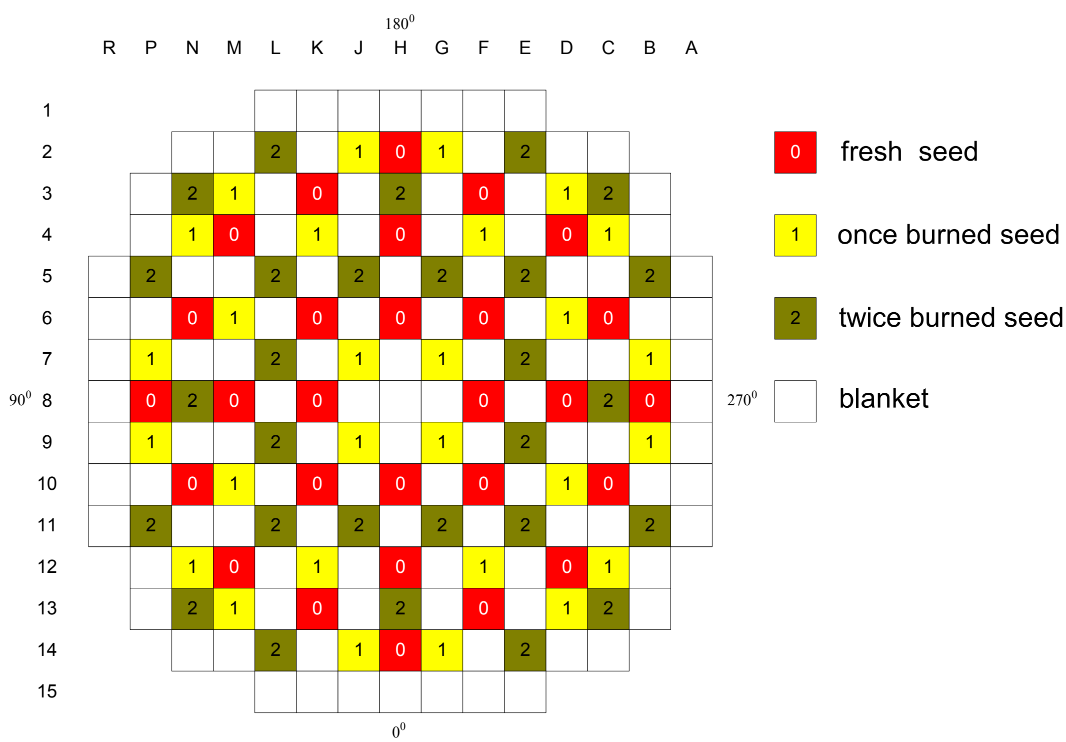


Figure 4.1 The WASB Core Layout and Refueling Scheme

4.2.2 Axial Blankets and Poison Zoning

Power distribution is not uniform in the axial direction. Neutron leakage through the top and bottom of the core reduces the energy extracted from the rod ends. The coolant temperature in the bottom half of the core is lower than in the top half, hence leading to increased moderation and higher thermal flux in this region. As a result, the typical power shape for the initially axially uniform fuel is a cosine type, skewed towards to the bottom. This power shape changes throughout core life as fuel depletes, and peaking shifts to other axial locations.

Optimization of neutron economy leads to minimization of neutron leakage through the top and bottom of the core. Current PWR cores use axial "blankets" of lower enrichment than core average, therefore improving uranium utilization in these Low Enrichment Regions (LER). The optimum length for axial LERs determined in a published Westinghouse analysis is 15.24 cm [M-3]. The WASB core design adopts a mixture of seed and blanket fuel assemblies. High enrichment in the seed (20%) calls for implementation of strategies that reduce axial leakage. Axial LERs for the blanket fuel are not necessary because of its low fissile content. For the WASB core, axial LERs for the seed were analyzed with enrichment ranging from 0.711 to 10%. The axial LERs incorporated in this core are composed of unpoisoned annular fuel pellets with the same dimension as the seed fuel pellets. The length of the axial LERs is 15.24 cm. The annular axial LER fuel not only reduces fuel cost, but also provides some extra space to accommodate fission gases.

The analysis is performed with 3-dimensional models in SIMULATE-3. All cases have no burnable poison in the fuel. The performance is assessed by comparison of the values from the equilibrium cycles. Table 4.1 compares the performance of the core for the cases analyzed.

Table 4.1 Comparative Performance of Axial LERs

Axial LER	Cycle Length (EFPM)	Max. $F_{\Delta H}$	Max. F_Q
20%	17.4	2.716	3.432
10%	17.1	2.691	3.508
8%	16.9	2.690	3.531
6%	16.8	2.689	3.556
4%	16.7	2.686	3.584
2%	16.5	2.682	2.619
0.711%	16.3	2.680	3.660

It can be observed that reducing the enrichment of the axial LERs leads to shorter cycle length. However, the fuel cost also decreases due to the lower enrichment in these axial regions. The maximum $F_{\Delta H}$ is essentially the same in all cases; since the power of the axial LERs is small relative to that of the whole core. Changes in the total peaking factor are more important. Of all cases in the table, the case with 0.711% enrichment blankets presents the highest peaking factor F_Q. The power generated in the rod ends decreases with decreasing enrichment of the axial LERs. As a result, power is shifted towards the center of the rods, yielding higher peaking as shown in Figure 4.2. The figure presents the power shape of the hottest seed assembly at BOC. Note that the critical boron concentration in the coolant was used for all cases in the calculations.

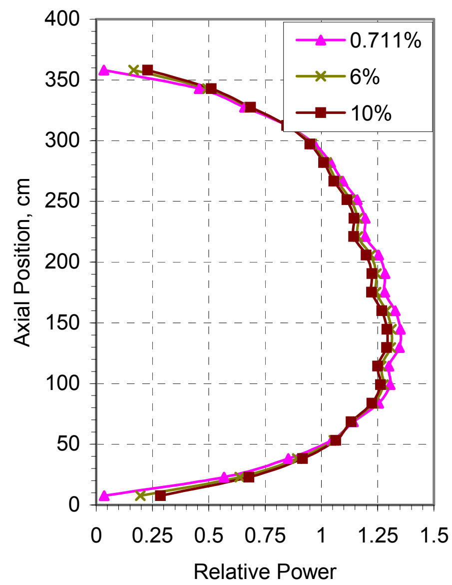


Figure 4.2 Seed Assembly Axial Power Distribution at BOC for Different Enrichment in Axial LERs

Axial zoning of both fuel enrichment and burnable poison can reduce the power peaking. A perfectly tailored distribution of fuel and burnable poison would yield a very flat axial power shape throughout core life. In practice, axial poison zoning is likely to provide a significant reduction in power peaking. Fuel enrichment zoning is another option, but not considered for the WASB design conducted in the present work, since U-235 should be kept within the 20% limit, and lower values would shorten the cycle length.

Figure 4.3 shows the poison zoning in the seed fuel. Actually in the WASB design a single concentration of burnable poison was used in the seed fuel, but the burnable poison loading is closer to the bottom to offset the effect of higher coolant density, hence higher flux and power at BOL.

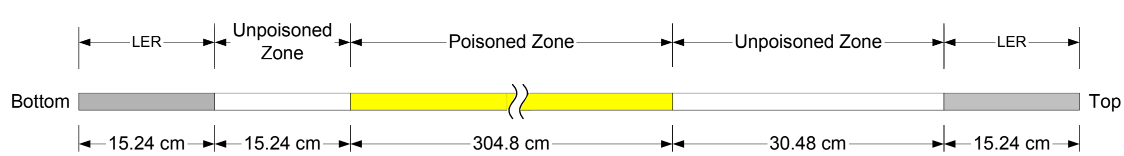


Figure 4.3 Burnable Poison Zoning in the Seed Fuel (not to scale)

Figure 4.4 shows a comparison of the axial power shape of the "unpoisoned" and the "poisoned" seed fuel. Both cases use 0.711% enrichment UO_2 (natural uranium) as the axial blankets. From the figure it can be seen that the axial power peaking is in the bottom half. Burnable poison axial zoning yields a flatter power distribution than the "unpoisoned" case. The performance of the core for both the "unpoisoned" and "poisoned" cases is summarized in Table 4.2. 100% enriched Er_2O_3 (Er-167) is used in the "poisoned" case.

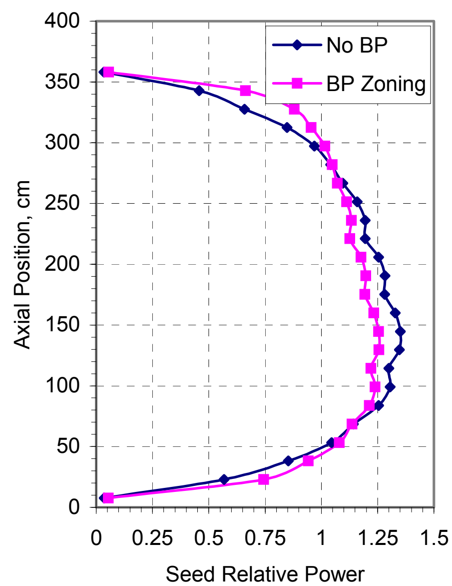


Figure 4.4 Effect of Axial Poison Zoning on the Seed Axial Power shape at BOC

Table 4.2 Core Performance for Cases with and without Axial Poison Zoning

	Cycle Length (EFPM)	Max. $F_{\Delta H}$	Max. F_Q
No BP	16.4	2.678	3.648
BP Zoning	16.3	2.501	3.171

The cycle length is slightly shorter for the "poisoned" case than the "unpoisoned" case. The maximum $F_{\Delta H}$ decreases by about 6.6% for the "poisoned" case, and the maximum F_Q decreases by about 13.1%. A finer tuning of the burnable poison concentration should further improve the power peaking factors. The evolution of the peaking factors throughout core life is shown in Figure 4.5. Overall, the peaking factors of the case with burnable poison are lower than those of the case without burnable poison for the WASB core. The maximum peaking factors are seen at BOC. Compared with conventional PWRs, the power peaking factors are much higher for the WASB core. However, there is more coolant flow in the seed assemblies than the blanket assemblies. Preliminary thermal-hydraulic analysis shows that the seed assemblies can meet the requirements of the MDNBR and maximum fuel temperature limits [B-6]. More analysis will be presented in Chapter 5.

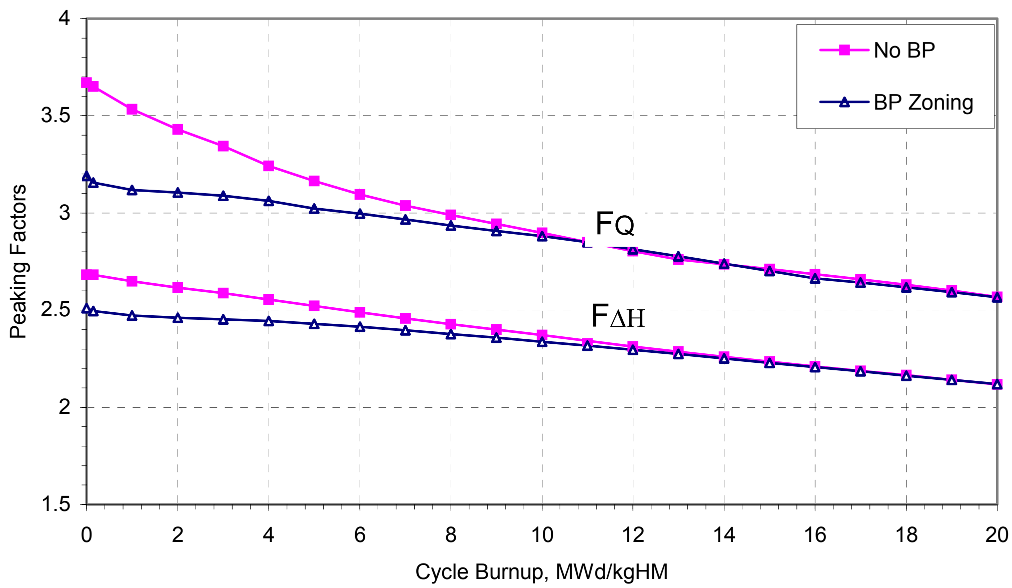


Figure 4.5 Effect of Axial Poison Zoning on Power Peaking Factors

Taking the preceding refinements into account, the final structural dimensions for the WASB fuel design are given in Table 4.3. The blanket fuel has no burnable poison and no axial LERs.

Table 4.3 Mechanical Design Parameters of the Seed and Blanket Fuel

	Seed	Blanket
Fuel Assembly		
Overall Length, cm	406.3	406.3
Assembly Pitch, cm	21.5	21.5
Fuel Rod Pitch, cm	1.26	1.26
Number of Fuel Pins	264	264
Fuel Rod		
Active Fuel Length, cm	365.76	365.76
Clad OD, cm	0.9008	1.06
Clad ID, cm	0.7864	0.9456
Pellet Type	Annular	Solid
Pellet OD, cm	0.77	0.9292
Pellet ID, cm	0.44	NA
Initial Pellet Density, g/cm^3	10.3024	9.4028
Burnable Poison		
Material	Er_2O_3, 100 w/o Er-167	N/A
Content*, mg/cm	Inner Pin: 9.12 Peripheral Pin: 54.74 Corner Pin: 82.11	N/A
Active Length, cm	304.8	
Axial Blanket		
Material	UO_2	N/A
Enrichment, w/o U-235	0.711	N/A
Top Blanket Length, cm	15.24	N/A
Bottom Blanket Length, cm	15.24	N/A
Pellet Type	Annular	N/A

* The burnable poison loading pattern in seed assemblies is shown in Figure 3.17

4.3 Core Performance

4.3.1 Heavy Metal Loading and Cycle Mass Flow

The core heavy metal content in the WASB core is summarized in Table 4.4. The reference case is a conventional 3-batch PWR core. The seed heavy metal loading is about 40% of the blanket. The total amount of heavy metal of the WASB core is about 8.8% less than the reference core, and the fissile content in the WASB fuel is 25% more than the reference core.

The WASB core is made up of three-batch seed and one-batch blanket assemblies. In Cycle 1 there are 28 fresh seed assemblies, 28 seed assemblies burned in one previous cycle, 28 seed assemblies burned in two previous cycles and 109 fresh blanket assemblies in the core; 1/3 of the seed assemblies are refueled at the end of each cycle whereas the blanket will sit in the core for up to 9 seed cycles. After 9 cycles the spent blanket is replaced with fresh blanket assemblies. At cycle 5 the blanket assemblies are shuffled once in order to balance burnups of the assemblies: the outside assemblies are shuffled into the inside, and the inside blanket assemblies are moved to the outside. Figure 4.6 shows the cycle mass flow in the WASB core. The heavy metal mass flow of each cycle is much less than for a conventional UO_2 core, but it does not follow that the fuel cost of the WASB is cheaper than a conventional core because the cost of separative work increases as the enrichment of the fuel increases. A detailed discussion of fuel cycle economics will be covered in Chapter 7.

Table 4.4 Core Heavy Metal Loading, kg

	Seed		Blanket	S+B	Ref. UO_2 core
	Seed Fuel	End Blanket			
U-235	4234	14	842	5090	4050
U-238	16936	1925	7576	26437	85950
Th-232			50579	50579	
Total	23109		58997	82106	90000

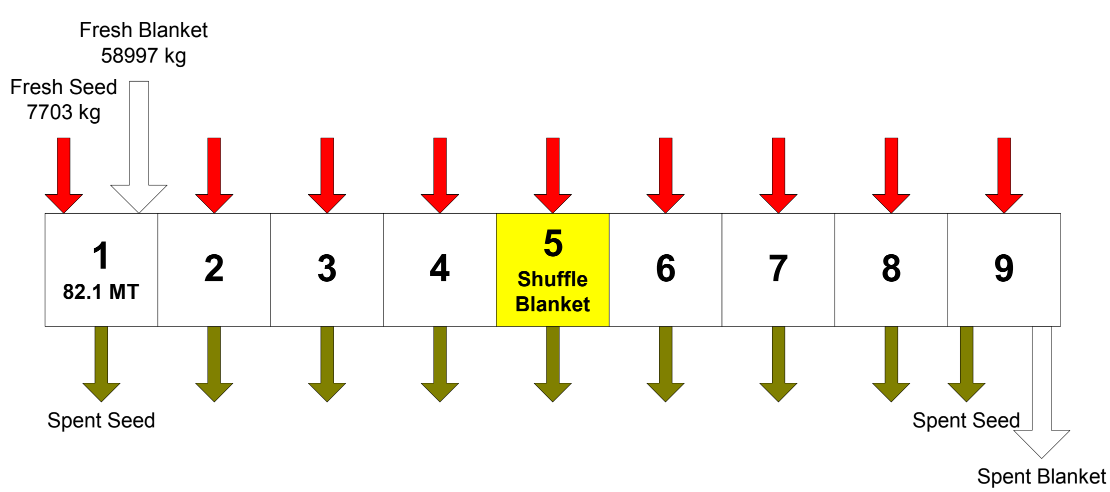


Figure 4.6 Cycle Mass Flow

4.3.2 Core Reactivity and Cycle Length

Generally speaking, the excess reactivity in the core is suppressed by soluble boron and burnable poison to give the target cycle length. As fuel and burnable poison are depleted through normal operation, the concentration of soluble boron is adjusted to keep the core critical. The reactor begins coastdown when the concentration of boron is reduced to zero. The excess reactivity in the core has been used up, and the core cannot maintain full power. The necessary reactivity to keep the core critical is provided by the negative moderator coefficient in conjunction with a reduction in moderator temperature caused by the slow core power decrease in the coastdown mode. The process of coastdown is self limiting, as the core power will continuously decrease to maintain criticality. Eventually the cycle is terminated and fresh fuel is again required.

The excess reactivity in the WASB core is also controlled by soluble boron and burnable poison (Er_2O_3). Boron is used as a soluble poison to control reactivity in PWRs. High boron concentration can have adverse effects on primary water chemistry and on the moderator temperature coefficient (MTC). In most commercial PWRs, LiOH is used to neutralize boric acid in order to achieve the optimum pH for the plant. Figure 4.7 shows lithium-boron operating regions for PWRs [E-2]. Practical evidence has shown that operation within a band corresponding to a pH value higher than 6.9 at 300 ^{0}C leads to significantly less deposits on the fuel than lower pH values (Area A on figure 4.7) and that radiation fields on piping and steam generators build up more slowly. This has been confirmed by a detailed assessment of the chemistry at selected Westinghouse plants, which indicates lower radiation buildup at plants operated with constant pH than at plants operated with low or varying pH. Accordingly, operation in area A should be avoided. In fact, almost all U.S. plants operate in area B or C. Boric acid concentrations greater than 1200 ppm may be required for extended fuel cycles. For these situations, operation in area B' rather than A' is recommended. The Electric Power Research Institute has developed guidelines for maintaining proper Li/B/pH coordinated primary chemistry in commercial PWRs. There are two major principles in the guidelines which would raise concerns about WASB core operation. First, the reactor should operate at or above pH = 6.9 to minimize crud deposition on fuel and Zircaloy corrosion. Second, for operation above 2.2 ppm lithium for extended periods of time (> 3 months) to achieve a pH = 6.9 during an extended fuel cycle, a plant specific fuel and materials review should be performed. Prolonged exposure to elevated concentrations of lithium raises concerns about PWSCC and Zircaloy corrosion.

Figure 4.8a reveals the behavior of critical boron concentration as a function of WASB core exposure. As the figure shows, the WASB core equilibrium xenon CBC stays below 1885 ppm. A lithium concentration above 3.0 ppm is needed for a pH = 6.9 when the boron concentration is 1885 ppm. This is much higher than the 2.2 ppm recommended by EPRI. More burnable poison can be added into the core to reduce the need for soluble boron, however it has adverse effects on neutronic performance. If enriched boron is used for water chemistry control instead of natural boron, the lithium concentration can be reduced to the value of 2.2 ppm recommended by EPRI, or even lower. For example, The 1885 ppm of natural boron (18.3% B-10) can be reduced to a satisfactory value of 1000 ppm by changing to 35% enriched B-10.

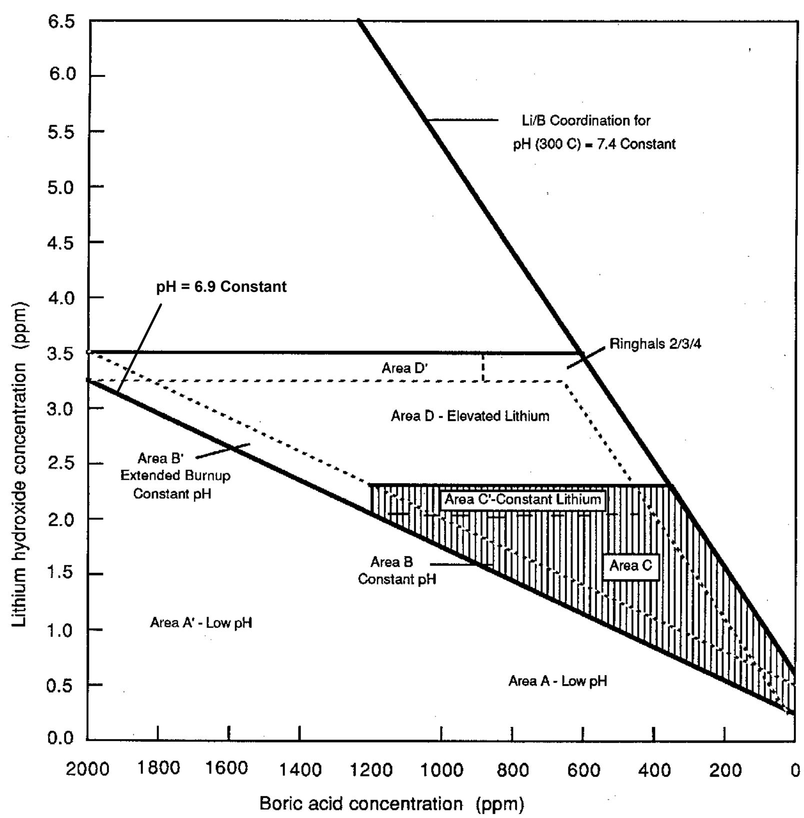


Figure 4.7 Lithium-Boron Operating Regions [E-2]

In addition to affecting primary water chemistry, the soluble boron concentration also influences the moderator temperature coefficient (MTC). In a typical undermoderated PWR core the MTC is negative. However soluble boron contributes to a positive MTC because an increase in moderator temperature results in decrease in the poison density in the coolant. If the soluble boron concentration becomes excessively high, an undesirable positive MTC can result. However, the WASB core is much less susceptible to a positive MTC than a conventional core because of its unique neutronic characteristics. The hardened neutron energy spectrum in the WASB core results in a more negative MTC than a conventional PWR core. So a positive MTC will not occur in the WASB core during normal operation.

The Keff changes as a function of core exposure for HFP (Hot Full Power) conditions are shown in Figure 4.8. The core reactivity swing is found to be less than that of a conventional 18-month cycle PWR due to the relatively high conversion ratio of thorium. In the analysis the same fresh seed assemblies were used to replace 28 twice burned assemblies at each cycle. Thus the reactivity behavior difference among cycles comes from the blanket reactivity change during its life. The outside and inside blanket assemblies are exchanged once at Cycle 5 in order to balance burnups of the blanket assemblies, because the inside assemblies burn faster than the outside ones.

Figures 4.8a and 4.8b show that the cycle lengths decrease gradually due to the reactivity loss of the blanket at the end of its life. The blanket reactivity increases initially with the buildup of U-233 and reaches a maximum after about 3 seed cycles, then decreases slowly because of the accumulation of fission products as shown in Figure 4.9 In order to compensate for the blanket reactivity decline either a gradual increase in the fuel loading of the seed or refueling of the blanket earlier than 9 seed cycles is necessary. A better strategy would be to adopt a multi-batch refueling scheme. For the blanket this will average-out blanket reactivity effects, at the expense of more refueling work. It should be noted that all fresh seed assemblies have burnable poison in the fuel.

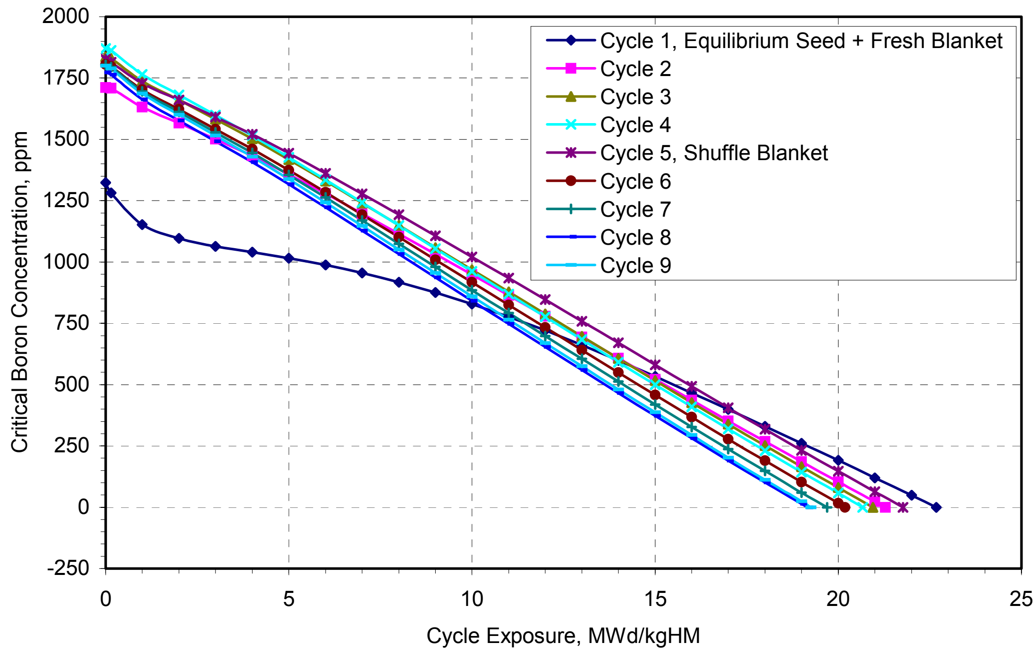


Figure 4.8a Critical Boron Concentration versus Cycle Burnup

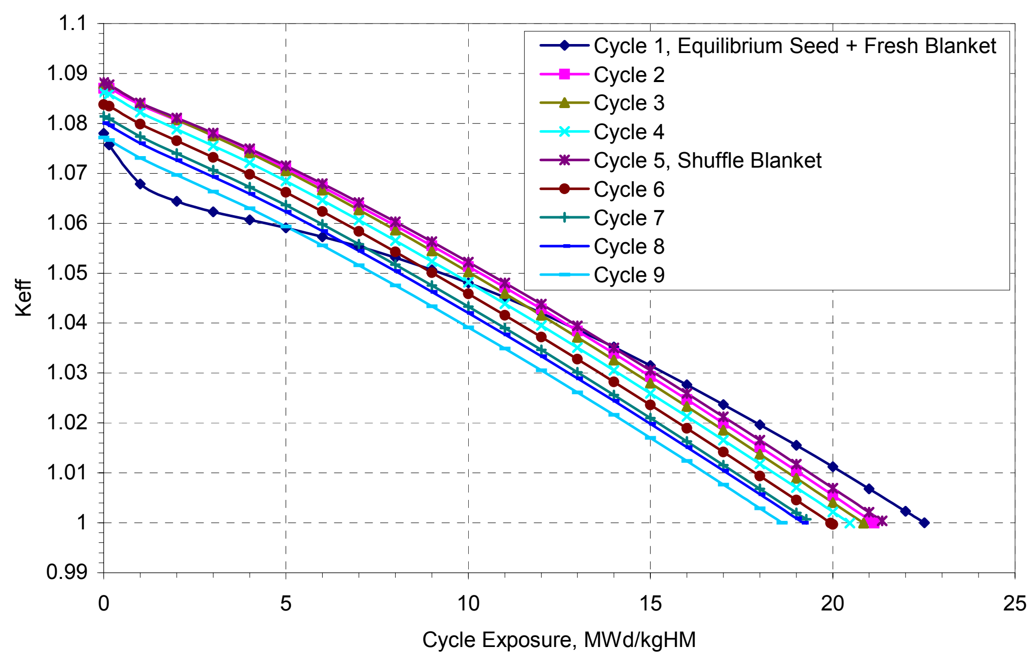


Figure 4.8b Core Keff versus Cycle Burnup

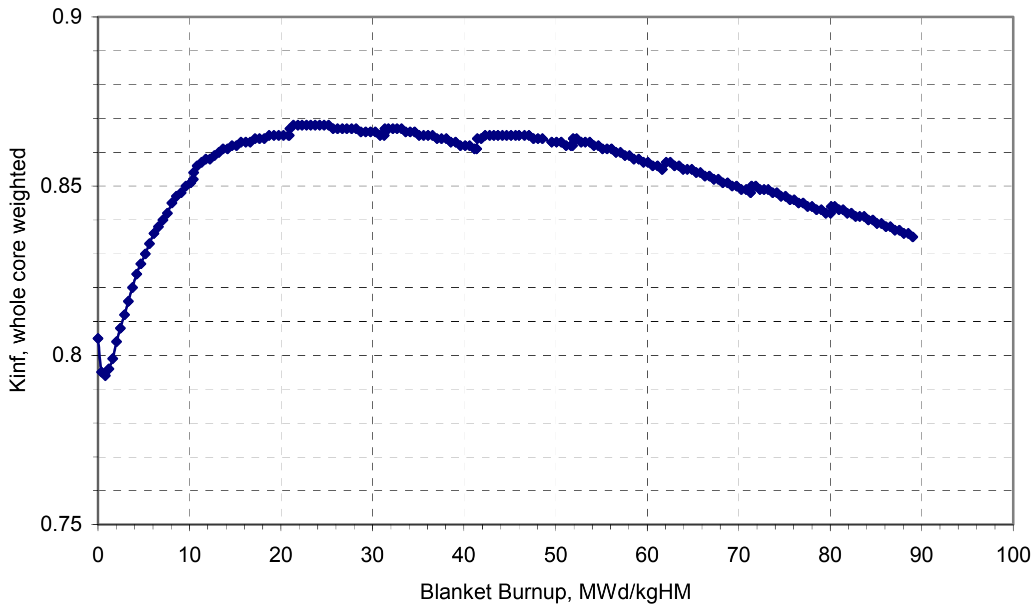


Figure 4.9 Blanket Kinf versus Burnup

The cycle lengths of 9 successive cycles are presented in Table 4.5. The cycle length of Cycle 1 is the longest because all blanket assemblies are fresh. In addition it can be seen that there is an increase in cycle length after shuffling the blanket at cycle 5. In order to analyze the reactivity penalty of burnable poison a case without burnable poison is also given in Table 4.5. It is interesting that the case with burnable poison does not show adverse effects on cycle length. One reason is that 100% Er-167 enriched Er_2O_3 is used as burnable poison. The other possible reason is that burnable poison in the seed fuel increases the conversion ratio of neighboring blanket fuel so that U-233 increases in the blanket. The average capacity factor (CF) of the WASB core is about 91%: hence any cycle length in calendar months is about 10% longer than the EFPM values tabulated.

Table 4.5 Cycle Length, EFPM

	Cycle 1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**Avg.**
No BP	18.39	16.78	16.27	16.52	17.00	15.91	15.52	15.30	15.03	16.30
With BP	17.94	16.84	16.57	16.35	17.22	15.97	15.58	15.16	15.23	16.32

4.3.3 Reactivity Parameters

The transient behavior of the reactor system is dependent on reactivity feedback effects. The physics design of reactors must guarantee that the moderator and fuel reactivity coefficients meet the requirements of plant operations. The coefficients are determined for variations of core moderator temperature, core fuel temperature, boron concentration, core power and isothermal temperature. In this study all reactivity coefficients were calculated without control rods in the core.

The moderator temperature coefficient (MTC) is defined as the reactivity change divided by the average moderator temperature change. The units of pcm (10^{-5}) per degree Fahrenheit are used. The reactivity coefficients due to moderator temperature change are sensitive to boron concentration. They are also sensitive to burnable poison in the fresh fuel. Figure 4.10a shows the dependence of MTC on cycle exposure for both HZP and HFP conditions. The MTC of a typical PWR is also given in the figure for comparison. Light-water reactor fuel assemblies, in general, are designed to be slightly undermoderated such that decreased neutron thermalization reduces reactivity and thereby enhances stability. The figure shows the MTC increases as burnup increases due to the buildup of plutonium (especially Pu-240). The MTC values of the WASB core are more negative than those of a typical PWR. The negative MTC is beneficial to normal operation. As we know, thermal reactors may experience spatial power oscillations because of the presence of Xe-135. In a reactor core with strongly negative temperature feedbacks, Xe-135 oscillations can be damped more readily. However high MTC raises concerns about accidents due to a decrease in coolant temperature such as cold water reflooding. The moderator temperature coefficients for HFP operation at different boron concentrations for a series of cycle exposure points are shown in Figure 4.10b. The figure shows that the MTC values decrease as boron concentration increases, and increases as the moderator temperature increases.

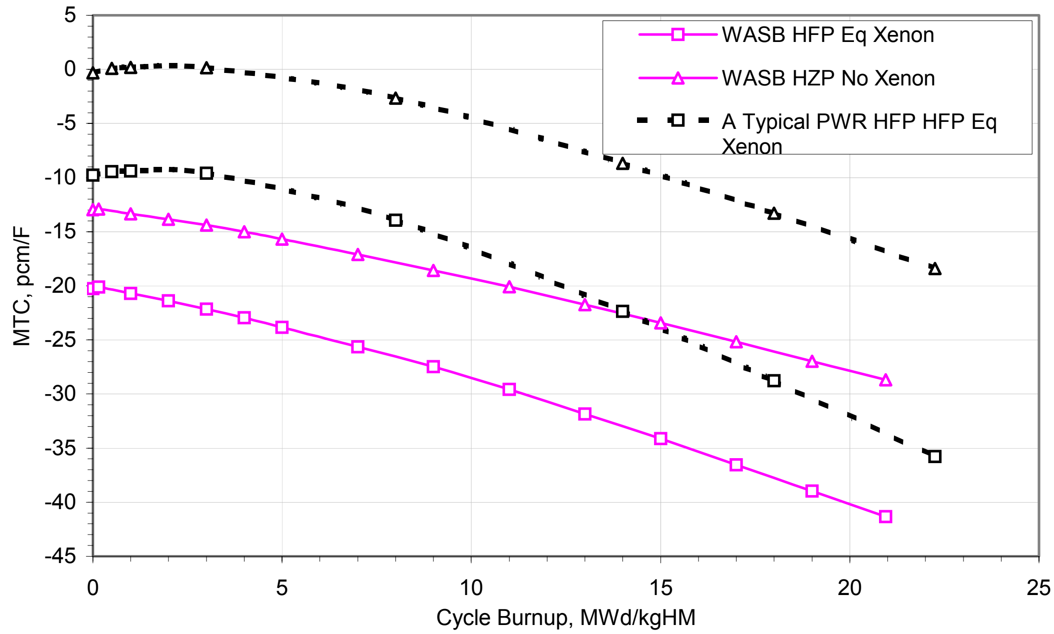


Figure 4.10a Moderator Temperature Coefficients vs. Core Exposure (Conditions: HZP ARO No Xenon Critical and HFP ARO Eq Xenon Critical)

* Data for a typical PWR is from a reference PWR core simulation using SIMULATE

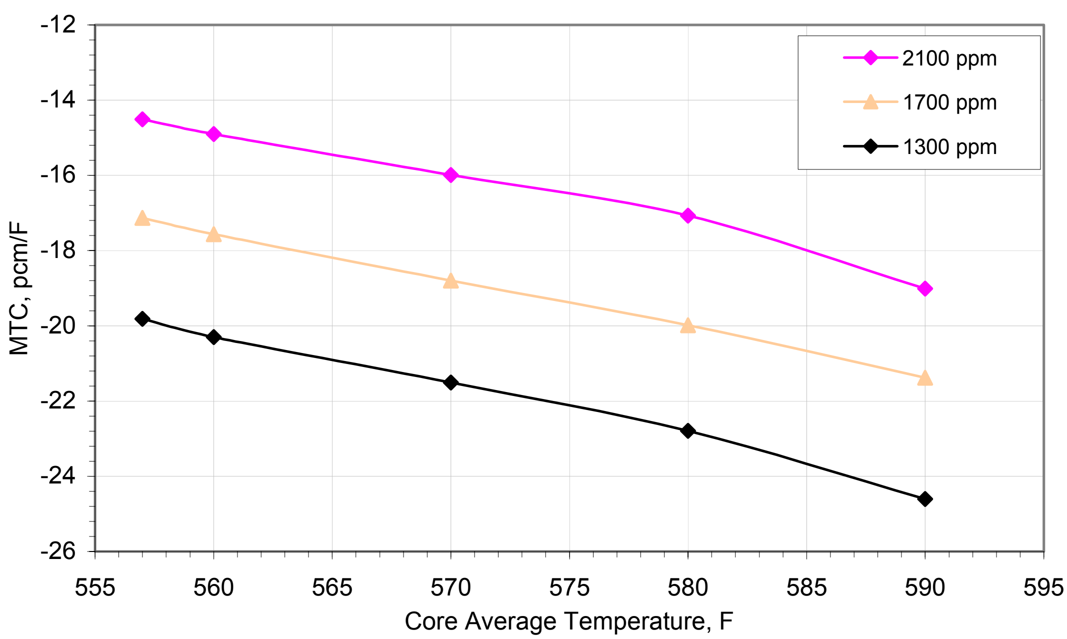


Figure 4.10b MTC vs. Core Average Moderator Temperature at BOC (Conditions: HFP ARO Eq Xenon)

The reactivity change of the WASB due to fuel temperature mainly depends on the Doppler effect. The Doppler effect is a major contributor to the safety and operational stability of thermal reactors. The fuel reactivity is given in terms of reactivity per unit power level change. The Doppler-only power coefficient (DOC) is defined as the reactivity not due to the moderator over a small power change. The Doppler-only power coefficients at BOC, MOC and EOC under critical conditions with equilibrium xenon are given in Figure 4.10c. In practice, the fuel will change due to depletion, because of fuel swelling and cladding creep, which will affect the fuel temperature. SIMULATE does not consider this effect. Figure 4.11 presents the total power defect from different power levels to HZP. The power defect is the reactivity defect associated with a change in power level from the reference power to the perturbed power. Both moderator and fuel temperatures are to be changed based on the perturbed power. The total power defect in the WASB core is higher than a typical PWR core because of its higher MTC and DOC.

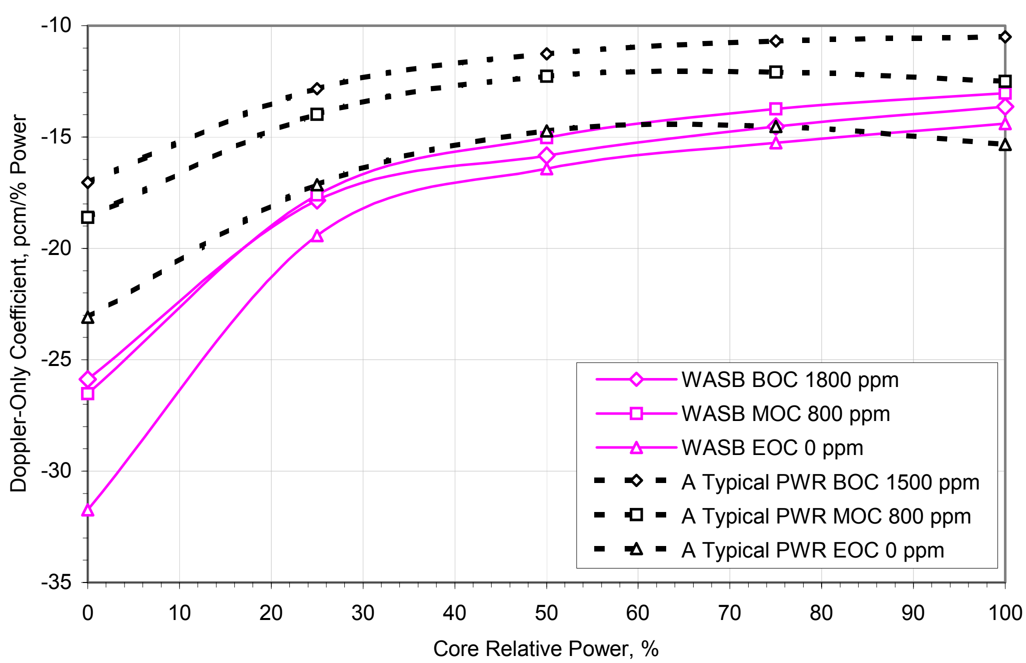


Figure 4.10c Doppler-Only Power Coefficients vs. Power Level

(Conditions: BOC, MOC, EOC HFP ARO Eq Xenon)

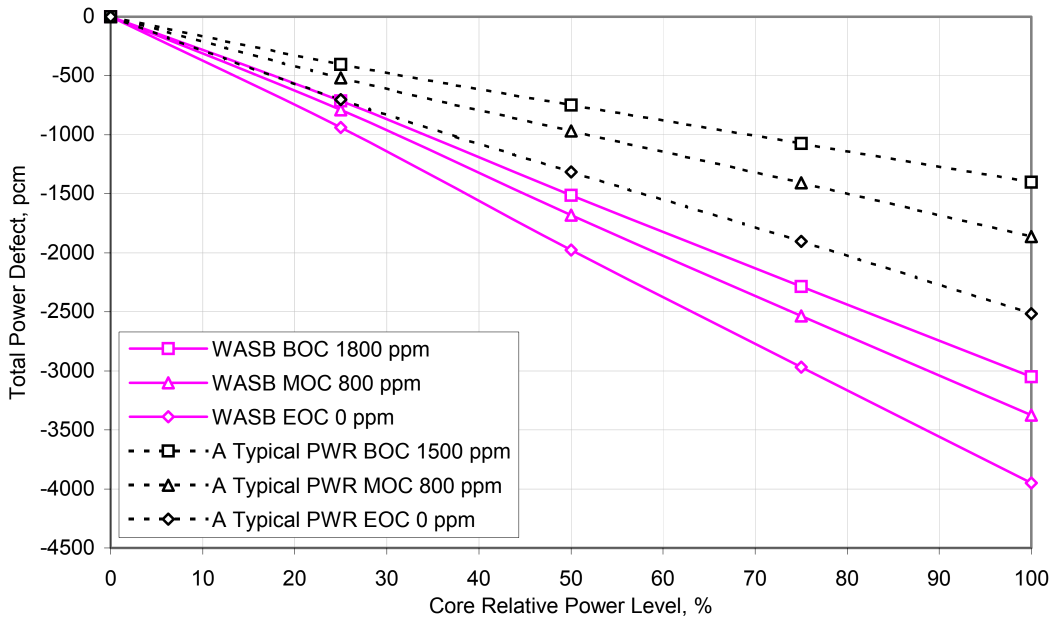


Figure 4.11 Total Power Defect vs. Power Level

(Conditions: BOC, MOC, EOC HFP ARO Eq Xenon)

The worth of soluble boron in the reactor changes as a function of boron concentration and cycle exposure. The boron worth coefficient is in terms of reactivity per unit boron concentration. The sensitivity to exposure and power level is given in Figure 4.12. The worth of soluble boron is less than that of a conventional PWR due to the harder neutron spectrum in the WASB core. Thus for a lower excess reactivity at BOC, higher boron concentration is needed as shown in Figures 4.8a and 4.8b.

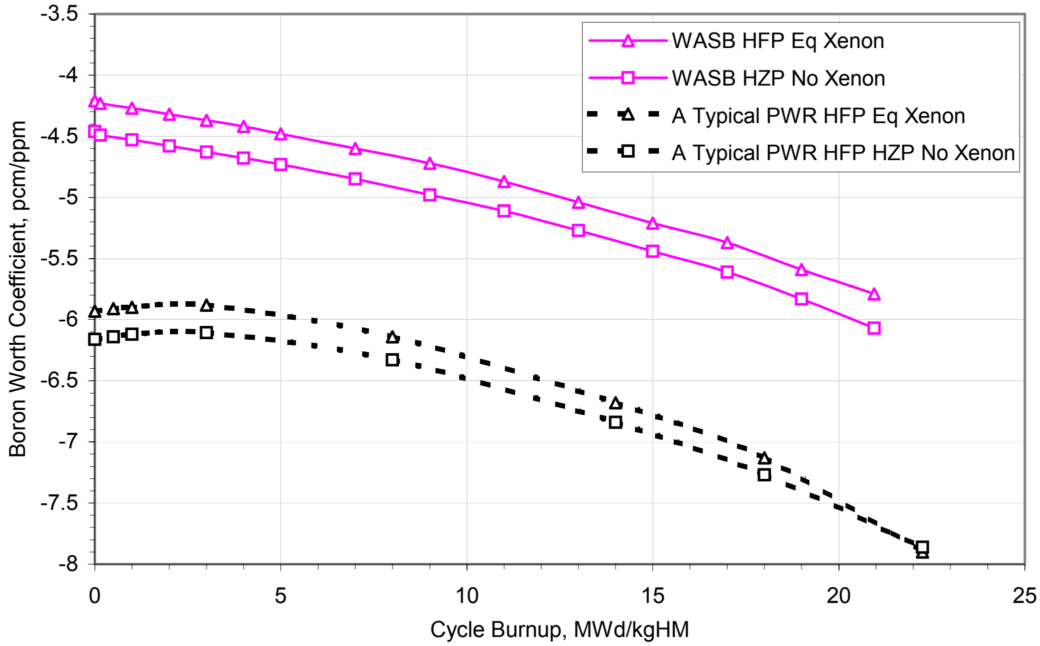


Figure 4.12 Boron Worth Coefficients vs. Core Exposure

(Conditions: HZP ARO No Xenon Critical and HFP ARO Eq Xenon Critical)

4.3.4 Kinetic Parameters

The kinetic parameters are very important to reactivity transients. A reactor usually operates at a steady power level based on equilibrium of both the prompt and delayed neutrons. When reactivity is inserted, the power level will change in a time-dependent manner in accordance with the point kinetics equations [H-1]. The two key parameters of the point kinetics equations are the prompt neutron lifetime l^* and delayed neutron fraction β. Based on the point kinetics equations, higher β values are desirable in terms of reactivity feedback response. The 6-group delayed neutron fractions and decay constants of the WASB core are summarized in Table 4.6. Table 4.7 shows a comparison of the effective delayed neutron fraction and prompt neutron lifetime of the WASB and a typical PWR. It can be seen that the effective β values at BOC and EOC are slightly lower than those of a typical PWR because part of the fissions is contributed by U-223, and β of U-233 is lower than that of U-235. However the values are comparable with the effective β values of MOX LWRs, which are around 0.004 [K-6]. In addition the prompt neutron lifetime is also slightly lower in the WASB than in a typical PWR.

Dynamic responses of the core to any event are consequences of changes in the Doppler and moderator reactivity feedback, and the delayed neutron fraction. A reduction in delayed neutron fraction decreases the required response time in reactivity transients. Hence these dynamic characteristics pose both safety and control issues that will require careful review of the neutronics analysis of the WASB core.

Table 4.6 6-group Delayed Neutron Fractions and Decay Constants

Group	BOC		EOC	
	β	λ	β	λ
1	0.000181	0.012766	0.000148	0.012785
2	0.001095	0.031559	0.000951	0.031426
3	0.000974	0.120578	0.000836	0.122361
4	0.002095	0.320704	0.001170	0.324522
5	0.000722	1.410915	0.000628	1.418405
6	0.000161	3.835709	0.000142	3.794119

Table 4.7 Effective Delayed Neutron Fraction β and Prompt Neutron Lifetime l*

	WASB		A Typical PWR	
	BOC	EOC	BOC	EOC
β	0.0052285	0.0044754	0.006356	0.005218
l^*, second	1.0513×10^{-5}	1.4412×10^{-5}	1.4351×10^{-5}	1.8935×10^{-5}

4.3.5 Power Distribution and Peaking Factors

Reactor core design depends on the peaking factors that represent the limiting conditions for operation of the system. Power peaking is a challenge for this heterogeneous seed and blanket arrangement. An optimized loading pattern can significantly reduce the power peaking factor as well as improve the whole core neutronic performance. Figure 4.13 shows the 1/4 core power distributions of the WASB at BOC, MOC and EOC. Burnable poison is used in the fresh seed to help regulate local power peaking and reduce soluble boron concentrations. Erbium has been shown to be a desirable burnable poison. In this study Erbium was evenly distributed into all pin central holes except for the peripheral pins and the four corner pins, which were more heavily loaded due to their higher power level than in other pins.

The assembly with maximum relative power is indicated with a triangle in the figure. The analysis results show that maximum power always appears in the fresh seed assemblies at BOC. The relative power sharing between the seed and blanket is also shown in the figure. Changes in the power sharing are very small during core life, although a decrease in the seed assemblies and accordingly an increase in the blanket are observed. The average power share of the blanket in a cycle is about 35%, which means the seed share is around 65%. It has also been found that an assembly with higher reactivity has higher power. This can be allowed for in-core fuel management. In order to improve power peaking, fresh seed assemblies should be separated as far apart as possible, and should be surrounded by more blanket assemblies instead of once burned and twice burned seed assemblies. Putting fresh seed assemblies at the periphery of the core is also an option to reduce power peaking, but not an economical option. Introduction of more burnable poison in the assemblies with higher power is widely used by nuclear utilities currently.

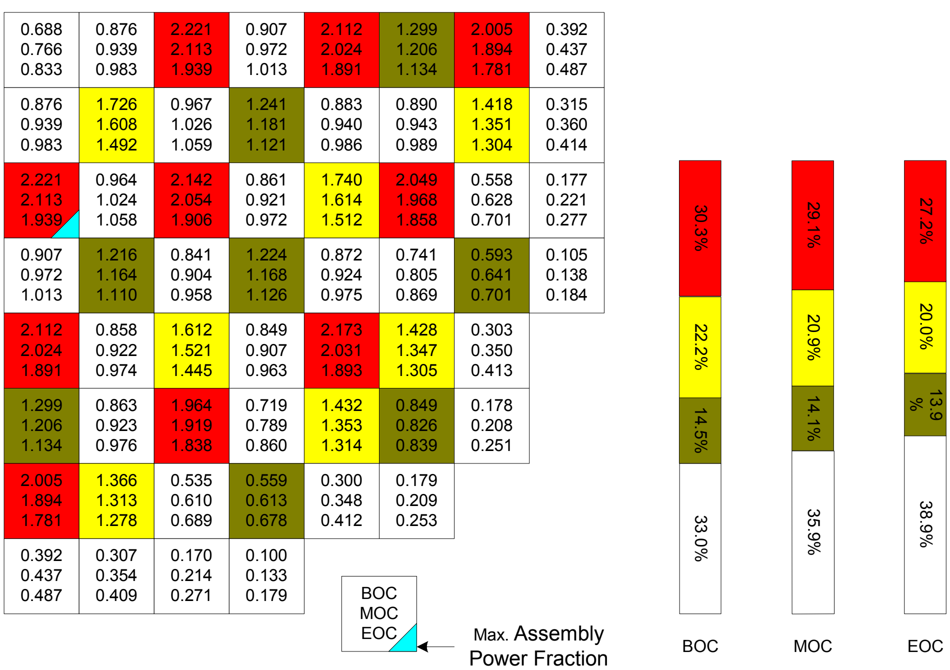


Figure 4.13 Radial Relative Assembly Power Fraction and Power Sharing Between Seed and Blanket at BOC, MOC and EOC

The maximum $F_{\Delta H}$ in each assembly is shown in Figure 4.14 and changes of peaking factors over cycle duration are shown in Figure 4.15. Both maximum $F_{\Delta H}$ and F_Q appear in the fresh seed assemblies at BOC. The maximum $F_{\Delta H}$ is bout 2.5, which is far beyond the limit of 1.65 for current PWRs, and the maximum F_Q is about 3.15, which is also above the design criteria of 2.5. However it should be noted that the seed fuel design is quite different from conventional UO_2 fuel. The seed fuel uses smaller annular pellets, therefore the coolant flows at a higher velocity in the seed assemblies and the fuel temperature remains low. The peaking factors raise concerns about the safety margin for cladding integrity. The DNBR limit ensures that no fuel cladding would be statistically expected to fail during anticipated operational occurrences, and that adequate fuel performance margin during steady-state operation is maintained. Thus proper thermal-hydraulic modeling of the WASB core and fuel performance assessment needs to be done to address this issue. This will be covered in the next chapter.

The axial power shapes of the hottest seed assembly and the whole core are shown in Figure 4.16. As discussed previously axial poison zoning provides a relatively flat power shape throughout cycle duration by reducing axial peaking.

0.728 0.802 0.874	1.032 1.090 1.117	2.501 2.324 2.099	0.990 1.054 1.085	2.379 2.224 2.043	1.428 1.335 1.281	2.330 2.149 1.986	0.700 0.757 0.814
1.032 1.090 1.117	1.862 1.717 1.591	1.060 1.114 1.137	1.326 1.292 1.265	0.957 1.017 1.058	0.976 1.034 1.077	1.746 1.617 1.522	0.633 0.687 0.743
2.501 2.324 2.099	1.058 1.113 1.136	2.414 2.264 2.071	0.953 1.017 1.057	1.966 1.805 1.685	2.396 2.238 2.066	0.812 0.884 0.951	0.387 0.442 0.507
0.990 1.054 1.085	1.305 1.281 1.260	0.943 1.010 1.053	1.335 1.286 1.270	0.959 1.010 1.055	0.904 0.969 1.024	0.870 0.875 0.905	0.230 0.288 0.362
2.379 2.224 2.043	0.941 1.007 1.052	1.806 1.692 1.597	0.951 1.004 1.051	2.476 2.252 2.057	1.791 1.638 1.564	0.522 0.580 0.669	
1.428 1.335 1.281	0.935 1.004 1.059	2.287 2.177 2.041	0.867 0.941 1.007	1.795 1.644 1.564	1.276 1.218 1.209	0.428 0.472 0.536	
2.330 2.149 1.986	1.689 1.575 1.490	0.780 0.860 0.937	0.795 0.840 0.909	0.517 0.578 0.672	0.433 0.479 0.547		
0.700 0.757 0.814	0.622 0.677 0.735	0.372 0.430 0.497	0.219 0.279 0.356				

BOC
MOC
EOC

Max. $F_{\Delta H}$

Figure 4.14 Maximum Enthalpy Rise $F_{\Delta H}$

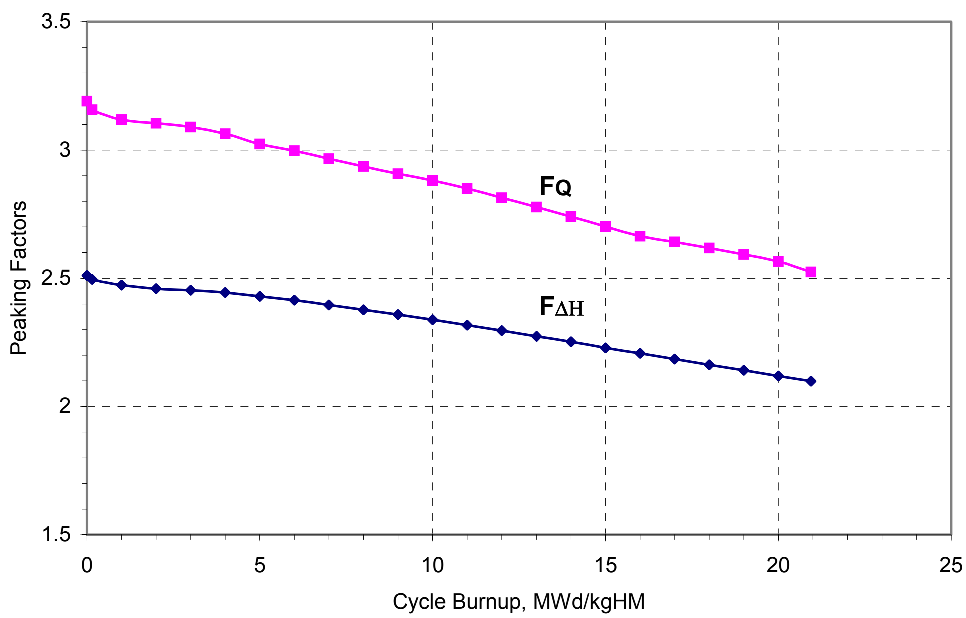


Figure 4.15 Changes of Peaking Factors Over Cycle Duration

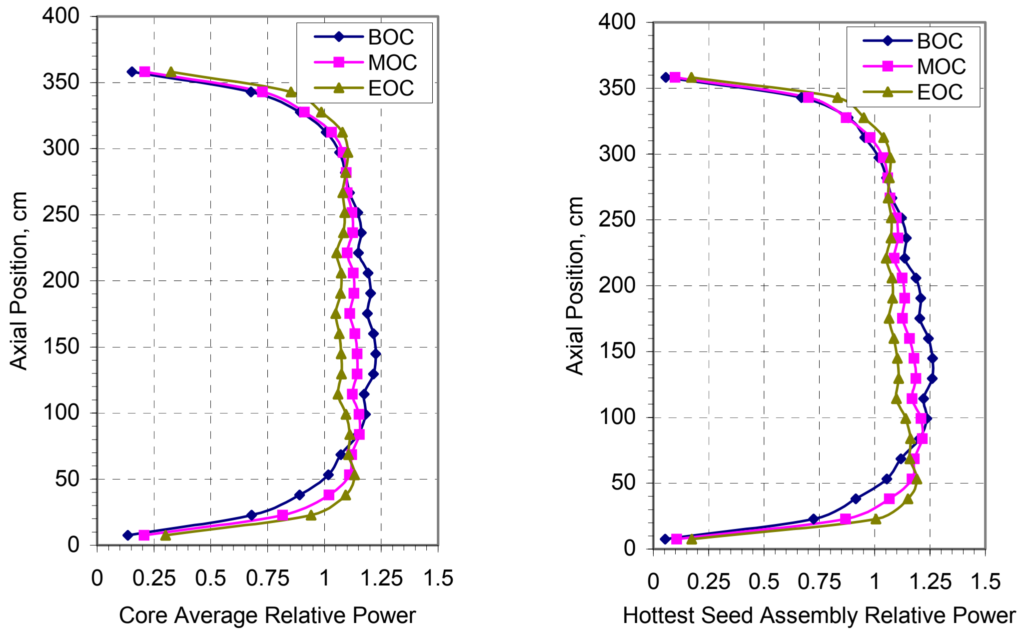


Figure 4.16 Axial Power Distribution

(Conditions: HFP ARO)

4.4 WASB Core Reactivity Control

Reactivity control of the WASB core is more complex than conventional PWRs due to its heterogeneous core design and harder neutron spectra. Direct and short-term reactivity control is still provided by control rods. Immediate and long-term reactivity control is provided through the use of soluble boron and burnable poison. Burnable poison is employed to reduce the requirements of soluble boron and to reduce the power peaking factor of the core. All fresh seed fuel pins are poisoned in the WASB design. All analyses were done based on whole core calculations using SIMULATE-3.

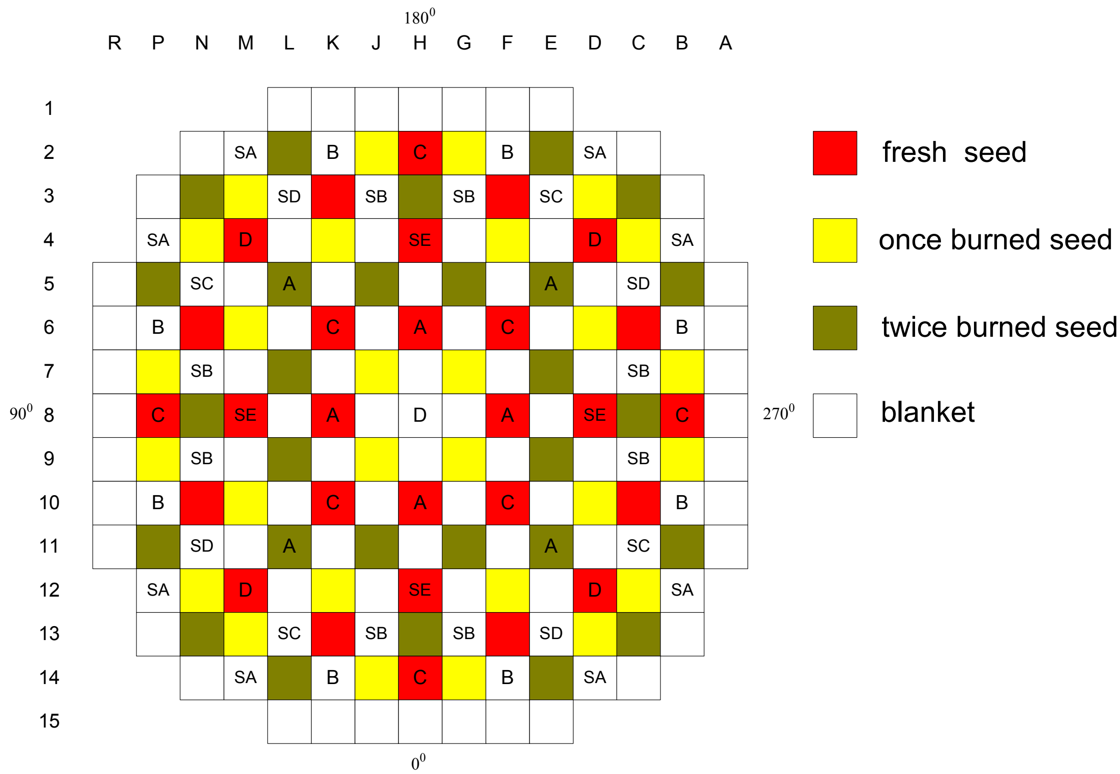
4.4.1 Control Rod Worth

The standard Control rod configuration for the 17×17 lattice is a 24 finger silver-indium-cadmium (80 w/o Ag - 15 w/o In - 5 w/o Cd) Rod Control Cluster Assembly (RCCA). Two different types of neutron absorber material were analyzed for use in the RCCAs of the WASB core. The two cases modeled are listed below:

- Standard Ag-In-Cd RCCAs;

- B_4C RCCAs, 36.6 w/o enriched B-10, which has higher worth than Ag-In-Cd RCCAs.

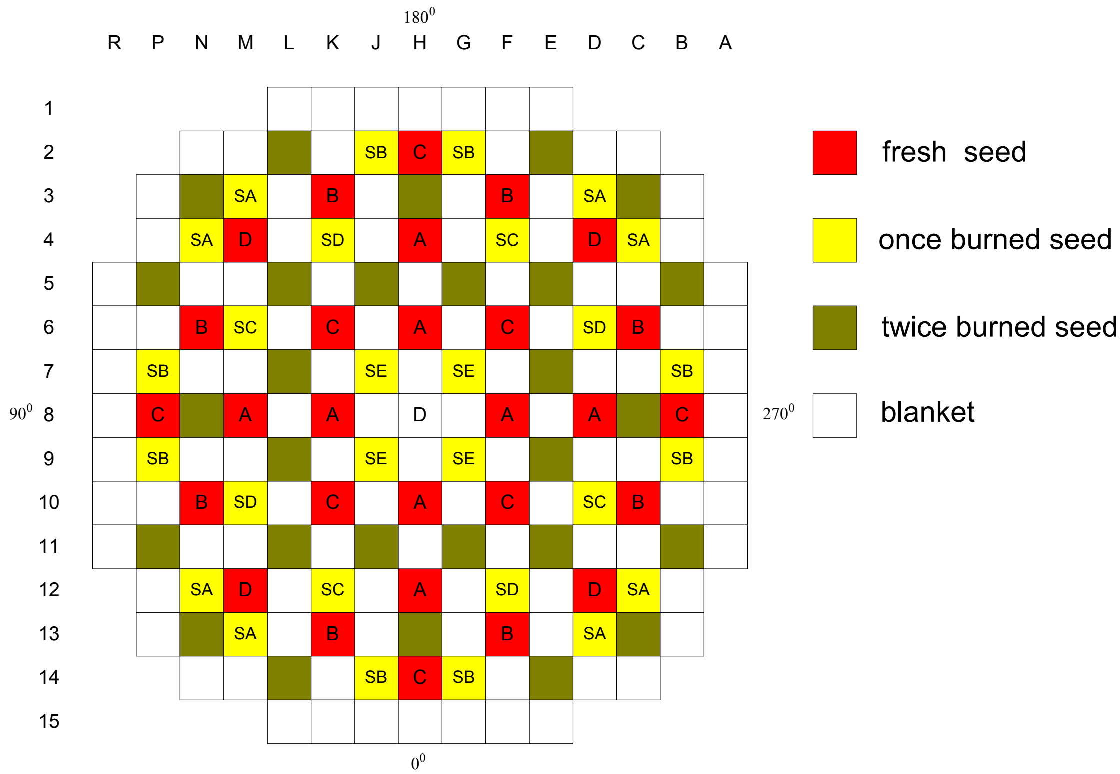
The standard arrangement of RCCAs in the core and their assignment to the various control and shutdown banks is shown in Figure 4.17 together with the typical WASB assembly loading pattern. There are 57 control clusters: 29 are power control rod clusters, and 28 are shutdown control rod cluster. Whole core analyses show that the control rods have higher worth in the seed than those in the blanket. Therefore a new control loading pattern, shown in Figure 4.18, has been designed specifically so that almost all the control rods will be inserted in seed assemblies. Control rod worth comparisons for the different cases are summarized in Table 4.9.



CONTROL BANK	NUMBER OF RCCAs
A	8
B	8
C	8
D	5
TOTAL	29

SHUTDOWN BANK	NUMBER OF RCCAs
SA	8
SB	8
SC	4
SD	4
SE	4
TOTAL	28

Figure 4.17 Standard RCCA loading Pattern



CONTROL BANK	NUMBER OF RCCAs
A	8
B	8
C	8
D	5
TOTAL	29

SHUTDOWN BANK	NUMBER OF RCCAs
SA	8
SB	8
SC	4
SD	4
SE	4
TOTAL	28

Figure 4.18 New RCCA loading Pattern

Generally speaking control rods with B_4C have 20%-30% higher worth than control rods with Ag-In-Cd as shown in Table 4.8. It can be seen the new control rod pattern also results in a significant improvement in control rod worth. Neither the use of enriched boron alone nor use of the altered control rod location pattern alone provides the WASB as much reactivity control as the standard PWR achieves. However, use of both enriched boron and the altered control rod pattern provides the WASB with somewhat more reactivity control than is present in the typical standard PWR. The new pattern requires a redesign of the vessel head of the reactor, which is an added cost in case of retrofitting in existing PWRs. The control rod worth could be further improved if higher enrichment B-10 is used for the poison material.

Table 4.8 Control Rod Bank Worth, pcm

Control bank	A Typical PWR		Standard Pattern				New Pattern			
			AIC*		B_4C**		AIC		B_4C	
	BOC	EOC	BOC	EOC	BOC	EOC	BOC	EOC	BOC	EOC
D	401	456	548	591	683	734	548	591	683	734
C	942	999	1166	1209	1500	1526	1166	1209	1500	1526
B	951	1077	263	367	406	495	1019	1100	1301	1359
A	842	1113	862	958	1103	1181	1042	1091	1278	1309
SE	514	N/A	610	638	805	838	368	396	459	467
SD	690	N/A	210	272	300	366	451	508	608	654
SC	958	N/A	283	284	326	383	517	548	693	704
SB	1312	N/A	631	738	880	978	546	588	676	703
SA	589	N/A	83	128	125	170	445	477	522	551
Total	7199	N/A	4656	5185	6128	6671	6102	6508	7720	8007

* 80 w/o Ag + 15 w/o In + 5 w/o Cd

** 36.6 w/o B-10 + 63.4 w/o B-11

4.4.2 Boron Concentration Requirement

The excess reactivity of the cycle is controlled by use of soluble boron in the coolant and of burnable poison in the fuel. The burnable poison loading and its placement within the fuel assemblies is fixed during fuel fabrication and by initial arrangement of the assemblies. The soluble boron concentration is adjusted during the cycle. The boron concentration requirements during refueling and reactor startup are summarized in Table 4.9.

More soluble boron is necessary for the WASB core for shutdown and refueling because of reduction in control rod worth. Also, the new control rod pattern design reduces the soluble boron required in the WASB core.

A Minimum shutdown boron concentration comparison is shown in Table 4.10. It is obvious that the WASB core with the standard loading pattern of control rods requires more soluble boron for shutdown than the new WASB core design. However, the new design requires more soluble boron than the conventional PWR.

Table 4.9 Boron Concentration Requirements for Refueling and Startup

Conditions		Boron Concentration, ppm				
		A Typical PWR	WASB Standard Pattern		WASB New Pattern	
		AIC*	AIC	B_4C**	AIC	B_4C
Refueling (Keff = 0.95), ARI, Cold		1968	3275	2925	2220	1310
Shutdown (Keff = 0.987), ARI, Cold		1502	2563	2212	1588	783
Shutdown (Keff = 0.981), ARI, HZP		1268	1857	1043	910	< 0
Shutdown (Keff = 0.987), ARO, Cold		2190	3543			
Shutdown (Keff = 0.981), ARO, HZP		2429	3505			
HZP, Keff = 1.0, ARO		2112	3047			
HFP Keff = 1.0 ARO	No Xe, Peak Sm 0 MWd/kgHM	1894	2350			
	Eq Xe and Sm 0.15 MWd/kgHM	1473	1821			

* 80 w/o Ag + 15 w/o In + 5 w/o Cd

** 36.6 w/o B-10 + 63.4 w/o B-11

Table 4.10 Minimum Shutdown Boron Concentrations

(Conditions: Keff = 0.987, CZP, No Xenon, ARI without the most reactive rod)

	A Typical PWR	WASB Standard Pattern		WASB New Pattern	
		AIC	B_4C	AIC	B_4C
BOC	1684	2620	2360	2105	1200
EOC	616	1085	905	755	600

4.4.3 Shutdown Margin

The shutdown margin is defined as the instantaneous value of reactivity by which the reactor is sub-critical or would be sub-critical from its present condition assuming all full-length rod cluster assemblies (shutdown and control) are fully inserted except for the single rod cluster assembly of highest reactivity worth which is assumed to be fully withdrawn. The results show that the WASB core shutdown margin is less than that of a typical 18-month cycle PWR core due to the smaller control rod worth for the same control rod design. Table

4.19 shows that enriched boron is a promising alternative poison material for control rods, since it has a higher worth than Ag-In-Cd.

4.5 Chapter Summary

Neutronic design of the WASB core has been assessed in this chapter. The analyses show that the WASB core is technically feasible for retrofitting into conventional PWRs. The coefficients of reactivity are comparable with currently operating PWRs, however the MTC of the WASB is more negative than a typical 18-month cycle PWR. In addition the reduction in effective delayed neutron fraction (β_{eff}) requires careful review of the control systems because of their importance to short term power transients. Whole core analyses show that the total control rod worth of the WASB core is about 1/3 less than those of a typical PWR for the standard arrangement of Ag-In-Cd RCCAs in the core. The use of enriched boron in the control rods can effectively improve the control rod worth. In addition calculations indict that the control rods have higher worth in the seed than those in the blanket. Therefore a new control loading pattern has been designed specifically so that almost all the control rods will be inserted in seed assemblies, however the new pattern requires a redesign of the vessel head of the reactor, which is an added cost in case of retrofitting in existing PWRs.

Obviously all aspects of the WASB concept analysis need further refinement. Here only major issues are addressed:

- Blanket reactivity degradation in later life may require either a gradual increase in the fuel loading of the seed, or earlier refueling of the blanket than 9 seed cycles, or a staggered schedule for blanket replacement.
- The control rod needs to be redesigned because the worth in the WASB core is lower than in a typical PWR due to the hard neutron spectrum in the core, and transient and accident scenarios must be simulated; particular attention should be given to the rod ejection accident.
- Fuel pin performance for very high burnup requires cladding material qualification for the long residence period to withstand water side corrosion and increased fission gas production.

- Thermal hydraulic limitations should be checked due to the high power peaking factors of the WASB. To meet the thermal hydraulics limits the thermal hydraulic design should be coupled more closely with the physics design. This topic will be covered in the next chapter

C h a p t e r 5

THERMAL-HYDRAULICS AND FUEL PERFORMANCE

5.1 Introduction

The unique design of the WASB core has an impact on thermal-hydraulic performance. Whole core neutronic analyses show that the seed fuel has much higher power than the blanket, especially for the fresh seed fuel. The maximum $F_{\Delta H}$ value of 2.5 in the core found at the BOC, is well beyond the values around 1.5 encountered in conventional PWRs. Hence detailed thermal-hydraulic analyses should be made to check if the WASB core has an adequate thermal safety margin during normal or transient conditions. The most widely used criterion for fuel integrity is the departure from nucleate boiling ratio (DNBR) for PWRs. A typical PWR criterion is that the minimum DNBR (MDNBR) is equal to or greater than about 1.3 at an appropriate overpower condition (the limit is dependent on the CHF correlation used). In order to consider normal operating conditions and likely deviation, a conservative approach was used in the analysis: DNBR calculations were performed at an overpower of 12% of rated power, at an inlet coolant temperature that is 2 ^{0}C above the expected value, and at a flow rate that is 5% lower than the nominal flow rate.

Fuel performance must also be assessed due to the higher burnup of the WASB fuel. The discharge burnup of the seed is about 145 MWd/kgHM, and the peak assembly average burnup is about 160 MWd/kgHM. The discharge burnup of the average blanket assemblies is about 88 MWd/kgHM, and the peak assembly burnup is about 100 MWd/kgHM. These values are significantly higher than current fuel burnups (40-60 MWd/kgHM). Factors limiting fuel burnup include: cladding corrosion, internal gas pressure and dimensional changes of both fuel assembly and fuel rods. Fission gas release, due to the resulting fuel rod internal pressure, is an important limiting factor in the fuel thermal-mechanical design. At high burnup (higher than 40-50 MWd/kgHM) fission gas release tends to increase rapidly. The increase in fission gas release can lead to high fuel rod internal pressure and could also lead to a deterioration of the thermal conductivity of the gas and, more importantly, reduction of the heat transfer between the pellets and the cladding due to the resulting gap size modification. Another consideration is that the high rod internal pressure can have an important effect on fuel cladding behavior (ballooning, bursting, etc.) under transients and postulated accidents. The criterion of acceptable internal gas pressure is that the rod internal pressure should be held below normal pressure in the reactor coolant system during normal operation in order to prevent outward creep of the cladding. Cladding corrosion is another concern for the WASB fuel, especially the blanket assemblies which have more than ten year residence in the reactor. Corrosion is caused by the special nature of the environment in the reactor, such as heat flux and radiolysis. Corrosion behavior in reactor and autoclave tests show that with increased exposure in a reactor, higher corrosion rates will be observed. This makes cladding performance a limiting factor for WASB fuel implementation.

In this chapter, the whole core thermal-hydraulic performance has been evaluated using VIPRE-01 [C-4].Fuel performance assessment has also been done using FRAPCON3-ThHB, which is a revised version of FRAPCON3 with the capability of modeling thorium fuel and high burnup fuel [L-2].

5.2 Thermal-Hydraulics

Preliminary work on thermal-hydraulic design and analysis of the WASB core was focused on the assembly/colorset model [B-5][B-6]. Recently, Dandong Feng has developed a whole core model using VIPRE-01 [T-2]. All calculations and analyses in this section are based on this model.

5.2.1 VIPRE Core Model

For a whole core analysis, rod arrays are modeled as sub-channels and lumped sub-channels. In the area around the hottest pin, detailed sub-channel representation is necessary; three or more pitches away, rod arrays can be lumped gradually as lumped rods and sub-channels. Since seed fuel rods and blanket fuel rods have different diameters, it is not advisable to lump them. Therefore, in all calculations, each assembly is treated as one lumped channel.

Figure 5.1 shows the normalized power distribution for a typical equilibrium cycle of the WASB core, at BOC, MOC and EOC, in 1/4th of a whole core model. These power distributions are the results obtained by SIMULATE for the WASB core design with burnable poison. Because the power distribution is essentially symmetric in the 1/4th core, the VIPRE 1/8th core model uses the higher power peaking factors for each symmetric assembly, which should be conservative. Figure 5.2 shows the normalized pin power distribution in the hottest seed assembly. This is the result obtained from the CASMO colorset model. Note that the normalization includes guide tubes with zero power factors. The hottest pin is at the corner of the assembly and its peaking factor within the assembly is 1.153. Based on this power distribution, a rod and channel numbering scheme, known as the "corner" model, was used to investigate the MDNBR. However the calculation results from the corner model showed that the MDNBR does not occur at the corner of the hottest seed assembly because it is surrounded by much cooler sub-channels of the adjacent blanket assemblies [T-2].

To investigate this effect, a "center" model was constructed by D. Dong [T-2]. This includes the details of sub-channels in the area of Channel 10 as shown in Figure 5.3. Figure 5.3 shows that in this model, the two-sub-channels that receive the highest power from the adjacent four pins are numbered as Channels 1 and 2, and explicit sub-channels are modeled around these two. Figure 5.4 shows the channel numbering scheme in a 1/8th core.

Table 5.1 summarizes the key thermal and hydraulic parameters of the VIPRE-01 model. In order to address transient operation conditions, the core mass flow rate is reduced by 5 % from the normal 17.7 to 16.8 Mg/s, coolant inlet temperature is increased by 2 ^{0}C from the nominal 292.8 to 294.8 ^{0}C and core power is increased from the nominal 3411 MWt to the 112% overpower of 3820.3 MWt. A chopped cosine with peak-to-average ratio of 1.55 is used for the axial power profile. The different fuel and blanket geometries lead to several minor issues. Because the gap width between fuel rods in a seed assembly differs from that in a blanket assembly (their pin diameters are different), the gap width for a lumped channel between a seed and a blanket assembly is modeled as the average of the two. Furthermore, turbulent mixing between the seed and blanket is not included, which is conservative from the MDNBR point of view. To drive more flow into the seed assemblies, a grid loss coefficient of 2.0 is used in the blanket compared to 0.6 in the seed in order to represent high resistance grids in the blanket compared to standard grids in the seed.

Table 5.2 shows that the MDNBR in the WASB core at BOC is 1.309, which means that the thermal-hydraulic design of the WASB core is acceptable from a DNBR point of view. The DNBR of 4.439 in Channel 22 shows that the blanket has a very large thermal-hydraulic safety margin (although it is assembly-averaged). Even at the end of cycle, when the blanket share goes from 33% to 39% of power, there remains good margin in that region (3.988 MDNBR). Figure 5.5 shows the BOC detailed DNBR distribution in seed and blanket sub-channels using the W3L correlation [C-4]. It should be noted that the 1.3 MDNBR of the WASB core has been achieved by the use of high resistance grids (modeled by allocating a much higher grid form loss coefficient in the blanket), as a result, more flow is driven into the seed assemblies. The side effect of this method is that the pressure drop in the core becomes higher. For such a core, the pressure drop along the pins, which doesn't include inlet and outlet form loss, is 0.175 MPa compared to about 0.1 MPa in a standard PWR.

The temperature profile for the hottest rod (Rod 7 in Figure 5.3) in the seed at BOC is shown in Figure 5.6. The hottest point in the seed (approximately at 1567 ^{0}C) is well below the UO_2 melting point (2800 ^{0}C), leaving a large margin in transients and power oscillation. The maximum temperature in the blanket fuel is much lower than that in the seed fuel as shown in Table 5.2. Note that the parameters in Table 5.1 have been used in the calculations, hence the results are very conservative. In addition, analyses show that the maximum temperature in the blanket is very low because of low power, so there is no concern of fuel temperature limitation in the blanket.

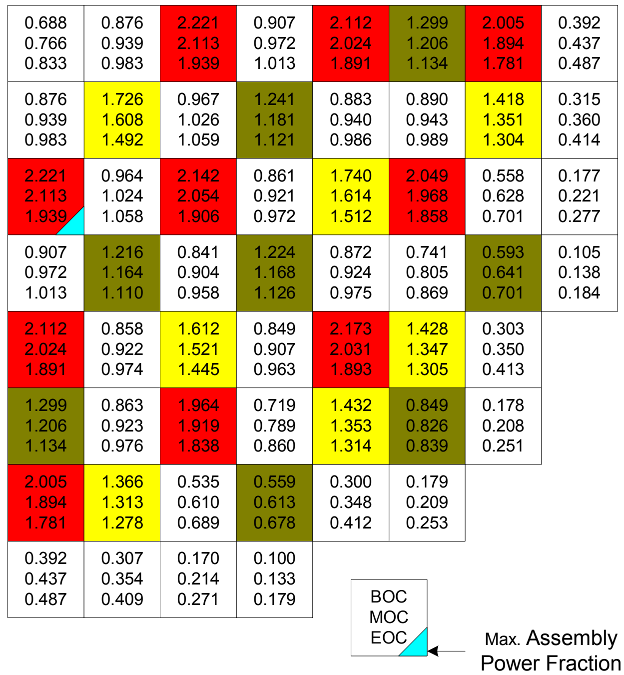


Figure 5.1 Radial Power Distribution at BOC, MOC and EOC for a WASB Core (Cycle 3: Equilibrium Seed, and Blanket in the Third Seed Cycle)

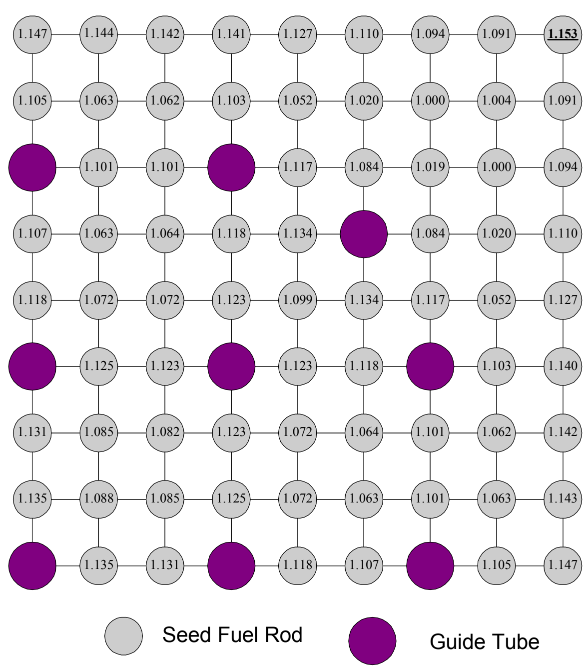


Figure 5.2 Normalized Pin Power Distribution in the 1/4th Hottest Seed Assembly for WASB

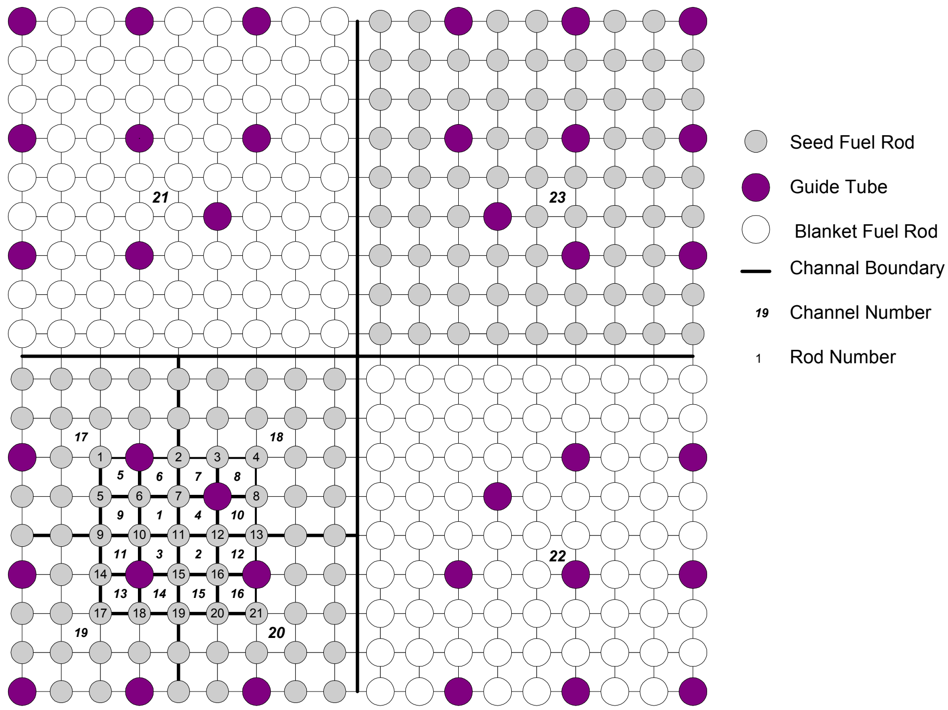


Figure 5.3 Rod and Channel Numbering in Hot Colorset

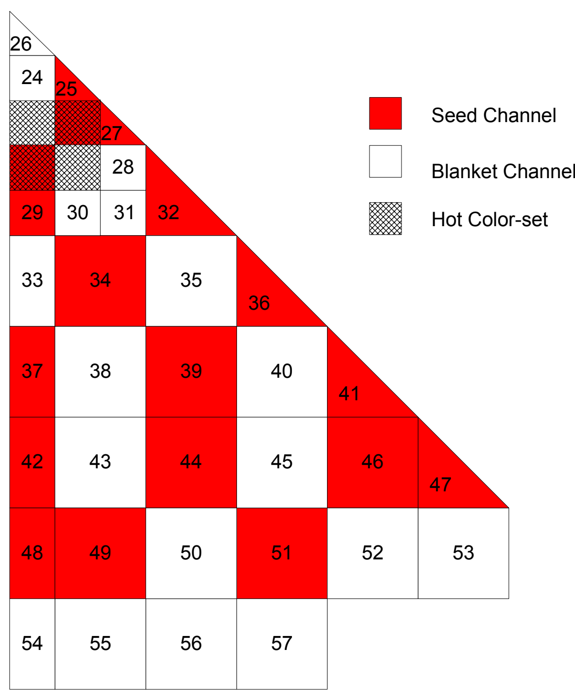


Figure 5.4 Channel Numbering in the 1/8th Core Model of VIPRE-01

Table 5.1 VIPRE-01 Model Key Parameters

Reactor Power	112% overpower (3820 MWt)
Inlet Temperature (^{0}C)	294.8 (Increased by 2 ^{0}C)
Whole Core Coolant Flow Rate (Mg/s)	16.815 (decreased by 5%)
Local Loss Coefficient of Grids	0.6 (Seed), 2.0 (Blanket)
Axial Power Profile	Chopped cosine, peak-to-average ratio = 1.55
Axial Friction Coefficient (Turbulent)	$f_{ax} = 0.32 R_e^{-0.25}$
Turbulent Mixing Model	None
Cross Flow Resistance Coefficient	$K_G = 0.5$
CHF Correlation	W-3L, mixing factor 0.043, grid spacing factor 0.066

Table 5.2 Fuel and Cladding Maximum Temperatures and MDNBR Values

	Seed		Blanket	
	BOC	EOC	BOC	EOC
T_{max} in the Fuel, ^{0}C	1567	1407	966	1023
T_{max} in the Cladding, ^{0}C	436	427	376	378
MDNBR	1.309	1.628	4.439	3.988

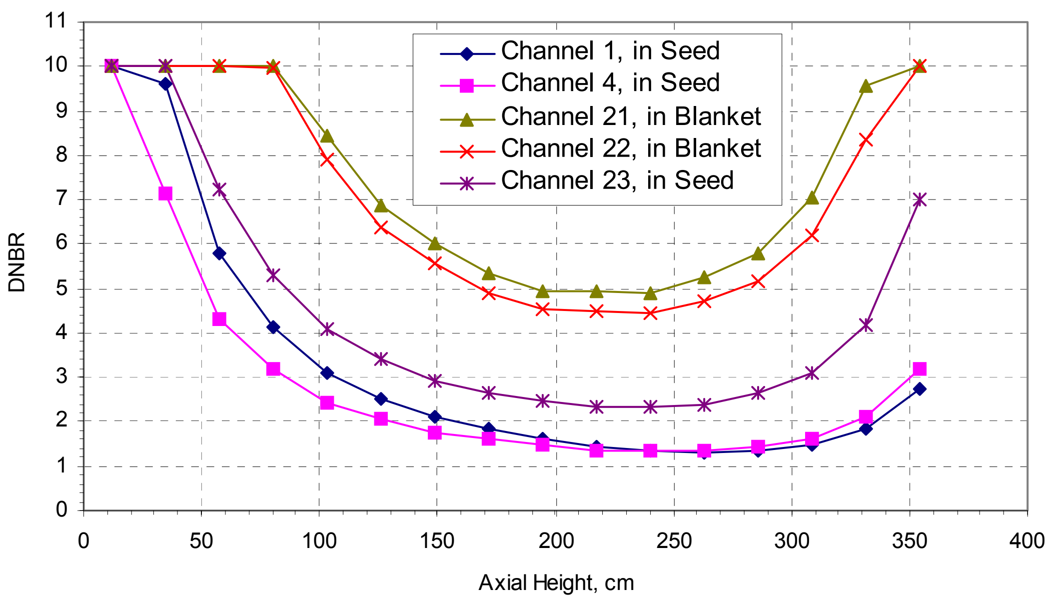


Figure 5.5 DNBR in Seed and Blanket Sub-Channels at BOC (DNBR > 10 are depicted as 10)

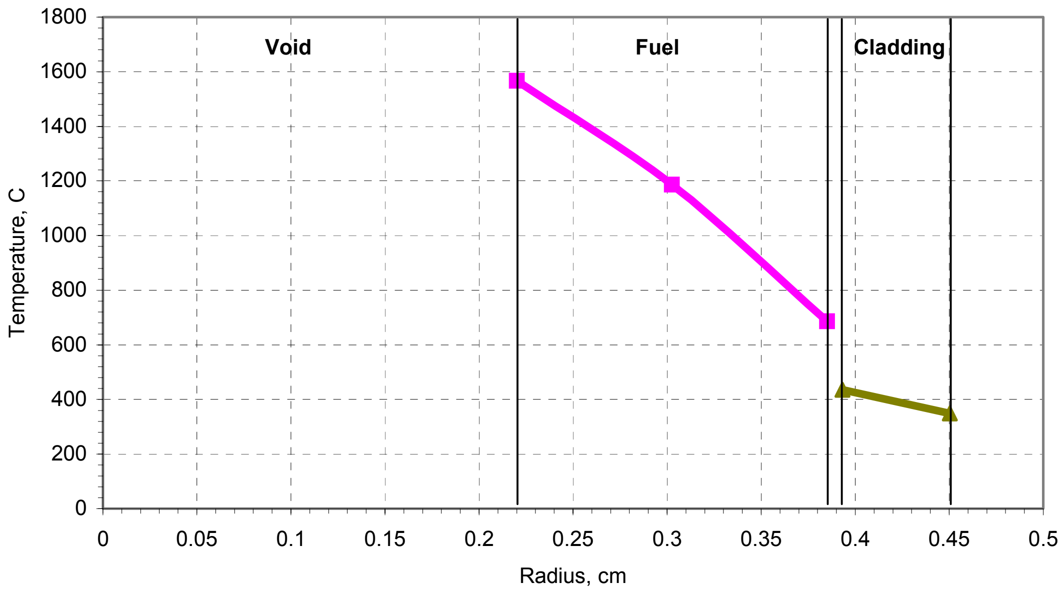


Figure 5.6 Temperature Profile in the Hottest Seed Fuel Rod at BOC

5.2.2 Sensitivity of MDNBR to Various Parameters

The sensitivity of the MDNBR to various parameters such as inlet temperature, power, coolant flow rate and form loss coefficient, etc., has been investigated using the VIPRE "center" model. The parameters in Table 5.1 are used as reference parameters..The power distribution at BOC was use in the calculations. All other parameters are unchanged when varying one parameter.

Core Inlet Temperature

The effect of core inlet coolant temperature on MDNBR is shown in Table 5.3. It shows that the MDNBR is very sensitive to the inlet temperature. The MDNBR increases as the inlet temperature decreases. Note that the normal inlet temperature is 292.8 ^{0}C. If the WASB core is operated at a lower inlet temperature such as 290.8 ^{0}C, the MDNBR margin will be significantly improved.

Table 5.3 Effect of Core Inlet Temperature on MDNBR

ΔT_{in} (^0C)	MDNBR	Channel No.	Rod No.	Axial Position (cm)
-6	1.494	4	12	228.6 - 251.46
-4	1.443	4	12	228.6 - 251.46
-2	1.386	1	7	251.46 - 274.32
Reference, 294.8	1.309	1	7	251.46 - 274.32
+2	1.287	4	12	228.6 - 251.46
+4	1.169	1	7	251.46 - 274.32

Nominal Flow Rate

Table 5.4 shows the impact on the MDNBR of varying the core flow rate. Again, the MDNBR will increase as the flow rate increases. However, higher flow rate leads to a higher pressure drop in the core, hence higher pumping power.

Table 5.4 Effect of Core Flow Rate on MDNBR

% of nominal flow rate	MDNBR	Channel No.	Rod No.	Axial Position (cm)
90%	1.051	1	7	251.46 - 274.32
95%	1.178	1	7	251.46 - 274.32
Reference, 100%	1.309	1	7	251.46 - 274.32
105%	1.441	4	12	228.6 - 251.46
110%	1.552	4	12	228.6 - 251.46

Nominal Core Power

It is obvious that the MDNBR decreases as the core power increases as shown in Table 5.5. It should be noted that the reference case is 12% overpower compared to normal operating power. We can improve the MDNBR by reducing core nominal power, however it is not desirable to run the reactor at a lower power level because of economic considerations.

Table 5.5 Core Power Effect on MDNBR

% of Reference Power	MDNBR	Channel No.	Rod No.	Axial Position (cm)
85%	1.772	4	12	228.6 - 251.46
90%	1.622	4	12	228.6 - 251.46
95%	1.478	4	12	228.6 - 251.46
Reference, 100%*	1.309	1	7	251.46 - 274.32
105%	1.150	1	7	251.46 - 274.32

* Taken as 112% of rated power

Form Loss Coefficient

The seed has a much higher power than the blanket, especially for fresh seed assemblies at BOC. The results show that the most constrained rod (where the MDNBR is found) is always located in a seed assembly. The grid form loss coefficient in the blanket is increased in order to redirect more coolant flow into the seed, hence increase the MDNBR. Table 5.6 shows that an increase in the loss coefficient of the blanket can effectively improve the MDNBR, however it will also increase the pressure drop in the core.

Table 5.6 Effect of Loss Coefficient in Blanket on MDNBR

Loss Coefficient in Blanket*	MDNBR	Channel No.	Rod No.	Axial Position (cm)
0.6	0.985	1	7	251.46 - 274.32
1.0	1.107	1	7	251.46 - 274.32
1.5	1.217	1	7	251.46 - 274.32
Reference, 2.0	1.309	1	7	251.46 - 274.32
2.5	1.394	1	7	251.46 - 274.32

* 0.6 in Seed

Seed Fuel Pin Diameter

The diameters of seed rods and blanket rods, which are 0.9 cm and 1.06 cm respectively, have been optimized from a neutronic rather than a thermal-hydraulic point of view in Chapter 3. Although an analysis of the sensitivity of DNBR to the rod diameters should ideally be done by iterating between thermal-hydraulics and neutronics, a preliminary investigation has been carried out by assuming that the power distribution is invariant when the rod diameters change. Since there are two kinds of fuel rods in the core, a sensitivity analysis of rod diameters can be done by changing either the seed rod diameter or blanket rod diameter. However, the DNBR restrictions always come from the seed rods, so changing the diameter of the seed rods will affect both the heat flux and the flow rate in these seed channels and thus directly affect the MDNBR. The results, shown in Figure 5.7 show that better thermal-hydraulic performance can be achieved by increasing the seed rod radius by about 10% from the present value of 0.45 cm to about 0.495 cm. Although the large seed rod diameter increases the pressure drop in the seed channels, thus driving coolant flow into the blanket, the accompanying decrease in the surface heat flux is the dominant effect and leads to a larger MDNBR until the rod radius exceeds 0.495 cm, at which point the effect of reduced coolant flow in the seed channel becomes dominant, thus the MDNBR decreases.

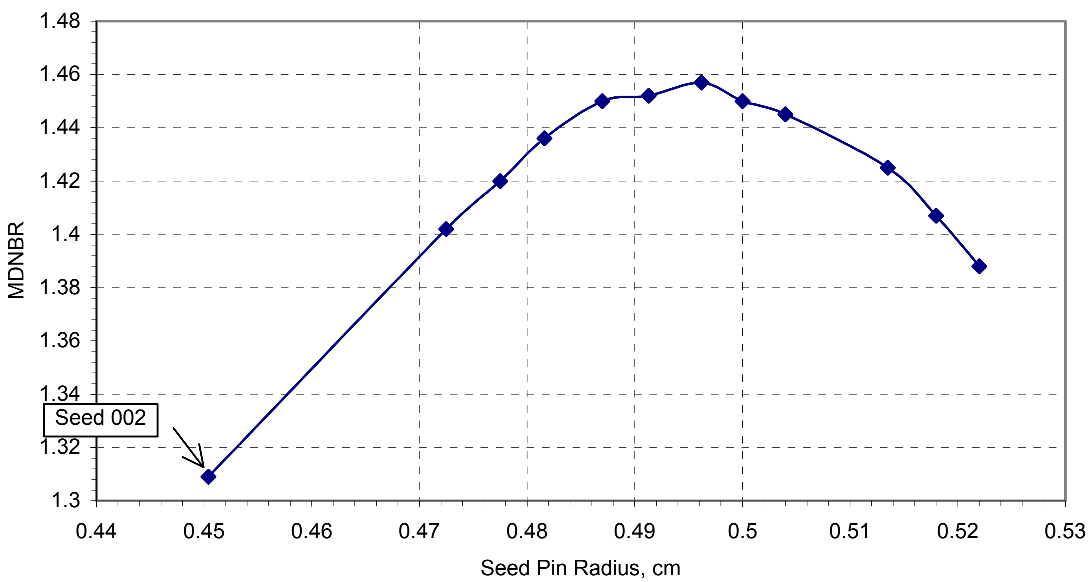


Figure 5.7 DNBR Sensitivity to Seed Fuel Pin Radius

5.3 Seed and Blanket Fuel Performance

Design of reliable WASB fuel presents a challenge for both the seed and blanket assemblies. The seed, operating at a much higher linear power than conventional PWR fuel, has to be designed to prevent excessive release of fission gas which would over-pressurize the fuel rods internally. It must also be designed to prevent excessive corrosion of the outside surface of the fuel rods, although this is expected to be less of a problem since the four and a half year operating time is not greater than that of typical PWR fuel. The blanket, although it operates at a much lower linear power than typical PWR fuel, is expected to remain in-core for up to nine 18-month cycles, equivalently about 13 calendar years, and must therefore be designed to prevent excessive corrosion on the outside surface during its long residence time. Fission gas release may also be a problem because of the high burnup of the blanket, but is expected to be less of a problem than corrosion because of the low linear power and therefore low temperature of the fuel material.

The seed and blanket are two types of fuel, and both reach very high burnups. The UO_2 seed fuel would reach up to a batch average of 145 MWd/kgHM and the ThO_2/UO_2 blanket fuel would reach up to 88 MWd/kgHM. The fuel behavior code FRAPCON3-ThHB [L-2], a modified version of the NRC's FRAPCON-3, which has the capability of modeling high burnup fuel and thoria fuel, was used to analyze fuel performance.

5.3.1 FRAPCON Model

The FRAPCON code is designed to perform steady-state fuel rod calculations. The code uses a single channel coolant enthalpy rise model. Key parameters of the seed and blanket fuel are shown in Table 5.7.

Core analyses show that the power history of each batch of fuel is relatively uniform over each cycle, so that a constant power history approximation was used in fuel behavior analysis for each cycle. The simplified histories reduce the amount of work necessary to input the power shapes in the code and yet represent fairly well the envelope fuel pin. The power histories for the seed and blanket are shown in Figure 5.8. The seed fuel rod experiences a decreasing power during three cycle operation, whereas the blanket rod is approximately constant over its whole lifetime. It can be seen that the seed has much higher power in the first cycle than the second and third cycle during its operation. Three axial power shapes at BOL, MOL and EOL for the seed and blanket have been used in the FRAPCON analysis, which were obtained from SIMULATE for the whole core analyses.

Table 5.7 WASB Fuel Parameters for FRAPCON Model

Parameter	Seed	Blanket
w/o of UO_2 in Fuel	100%	13%
U-235 Enrichment in U	20%	10%
Fuel Rod Radius, cm	0.4504	0.53
Cladding Thickness, cm	0.0572	0.0572
Gap, cm	0.0082	0.0082
Void Radius, cm	0.22	0
Fuel Density	94%	94%
Plenum Length, cm	40	40
Initial Fill Gas Pressure, Pa	10^6	10^6
Pitch, cm	1.26	1.26
Inlet Coolant Temperature, K	565.95	565.95
Coolant Mass Flux, kg/m^{2}s *	4450	3230

* Values from the VIPRE-01 core analysis. Higher coolant mass flux in the seed is because of a higher hydraulic resistance in the blanket.

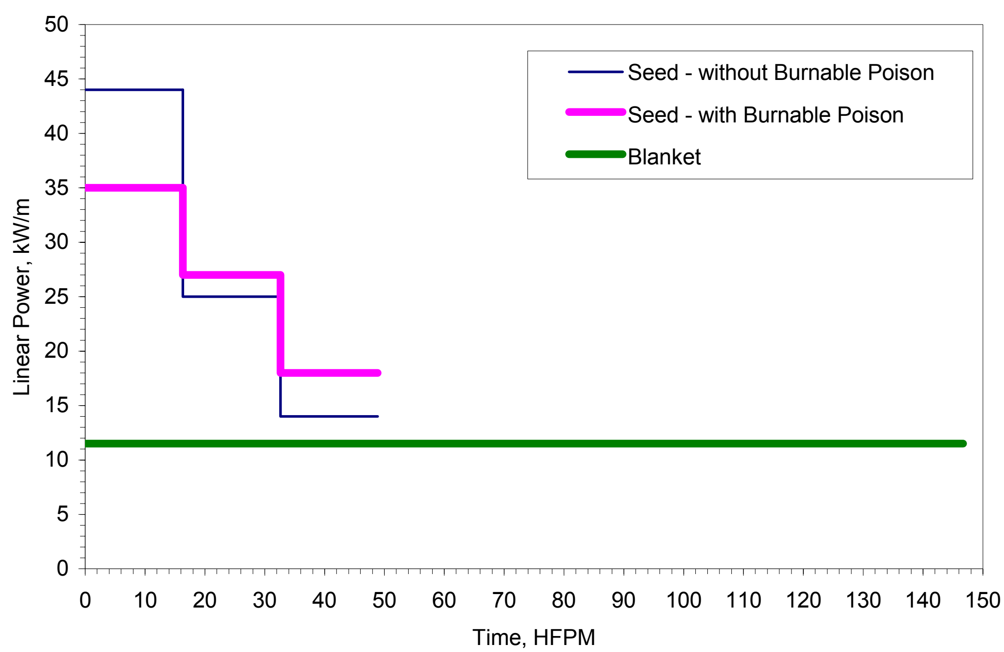


Figure 5.8 Assembly Average Power Histories Used in Fuel Behavior Analysis

5.3.2 Results and Discussion

Figure 5.9 shows the predicted fission gas release for the seed cases with and without burnable poison. Figure 5.10 shows the internal pressure results for both cases. The burnable poison effectively restrains fission gas release at the early stage of fuel irradiation because of the lower power operation. However at high burnup the difference is small. The reason is that as the burnup approaches 145 MWd/kgHM (the peak assembly average burnup is about 160 MWd/kgHM), the fission gas release is mostly a function of burnup and a large portion of fission gas will be released regardless of the power history. Thus with the present neutronic and mechanical fuel rod design and power history, a large fraction of the fission gas would be released in the seed fuel, and the internal pressure would exceed the system pressure of 15.5 MPa.

To investigate how the design might be made more acceptable, a case was calculated assuming a tripled gas plenum volume. Figure 5.11 shows that the triple length plenum can significantly reduce the internal pressure at EOL. However, adding 0.8 m to the fuel pin (20% of current length) will increase the core pressure drop. It may not be easily accommodated in current reactors. The impact of such a change needs to be investigated.

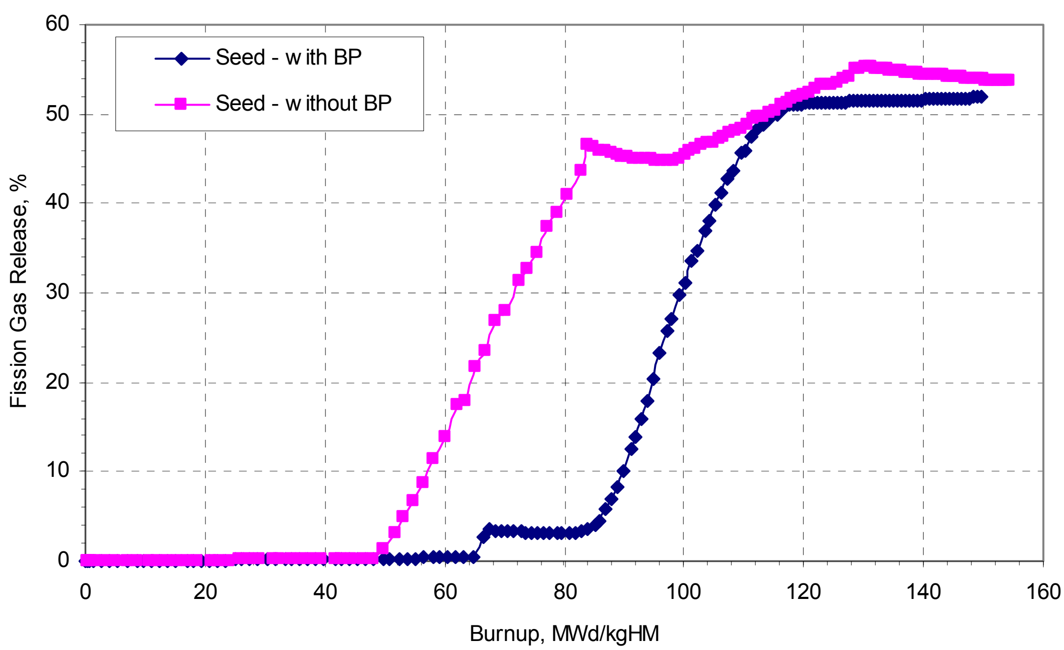


Figure 5.9 Fission Gas Release in Seed Fuel

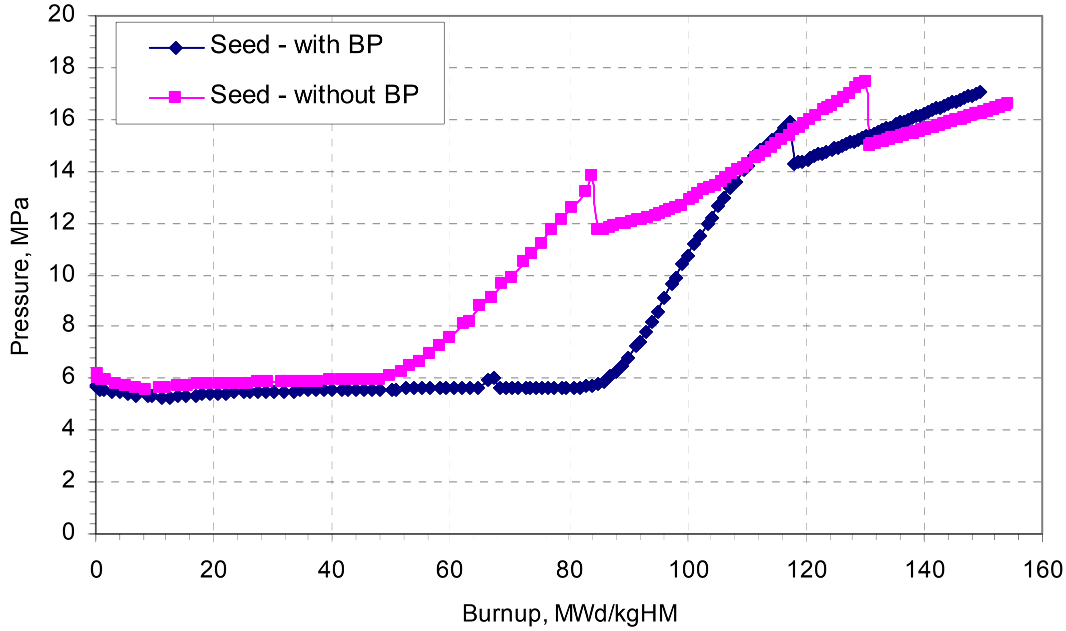


Figure 5.10 Internal Pressure In Seed Fuel

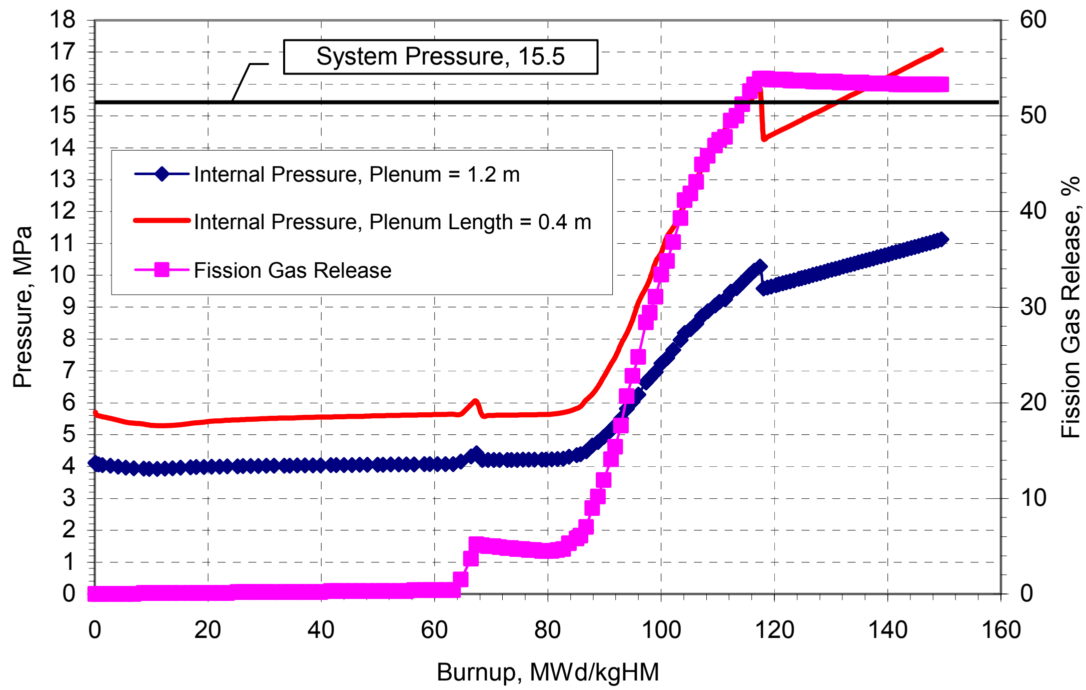


Figure 5.11 Fission Gas Release and Internal Pressure in Seed Fuel

Similarly, the blanket fuel has been analyzed using FRAPCON-ThHB. The results in Figure 5.12 show that the blanket fuel performance is much more favorable, as fission release and internal pressure start to rise significantly only near the end of life. If the residence time of the blanket in the core is reduced to only seven or eight cycles of operation due to neutronic reactivity limitations, the blanket behavior would be even more acceptable. Note that the internal pressure for the case with the 0.4 m plenum is higher than for the case with the 1.2 m plenum at initial irradiation as shown in Figure 5.11, because the average temperature of the internal fill gas for the case with the shorter plenum is higher than that for the case with the longer plenum under hot conditions. The internal pressure drops in the seed fuel between the first cycle and second cycle, and between the second cycle and the third cycle can be seen in Figures 5.10 and 5.11, which is due to the decrease in the power level.

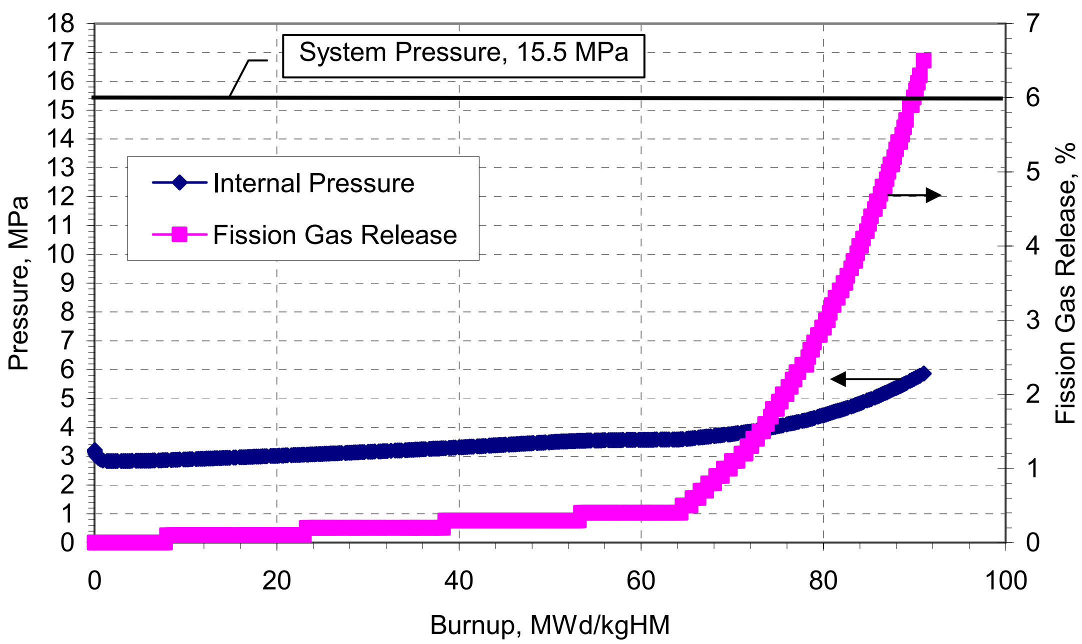


Figure 5.12 Fission Gas Release and Internal Pressure in Blanket Rod

Corrosion of seed and blanket fuel rods is an issue in the WASB design. Corrosion is a function of both surface temperature and time at temperature. The seed has a much higher linear power load during its operation and the blanket has an extremely long residence time in the core. The calculated (FRAPCON-ThHB) oxide thickness is plotted versus burnup in

Figure 5.13. At EOL both the seed and blanket fuel Zr-4 cladding corrosion is well above 150 microns, where oxide thickness greater than 150 microns is cut off in the figure. Figures 5.14 and 5.15 show the axial dependence of the corrosion (oxide) level expected on the outside surface of the seed and blanket fuel rods at EOL. Also, comparisons between the Zircaloy-4 and the more advanced M5 cladding are shown in the figures. The model used for the M5 cladding in FRAPCON-ThHB is a simple one and needs further verification [L-2].

M5 is an advanced cladding type, now coming into use in commercial PWRs. It has less sensitivity to temperature, exhibits less data scatter than Zr-4, and did not show any irradiation or high burnup transition up to 64 MWd/kgHM [W-5]. Figures 5.14 and 5.15 show that M5 can significantly reduce corrosion in the seed and blanket cladding. The blanket has more corrosion than the seed because of the three times longer exposure time. Since a typical corrosion film-thickness limit is on the order of 100 microns, the figures indicate that an advanced cladding like M5 can deliver good corrosion performance, with large margins for the WASB fuel.

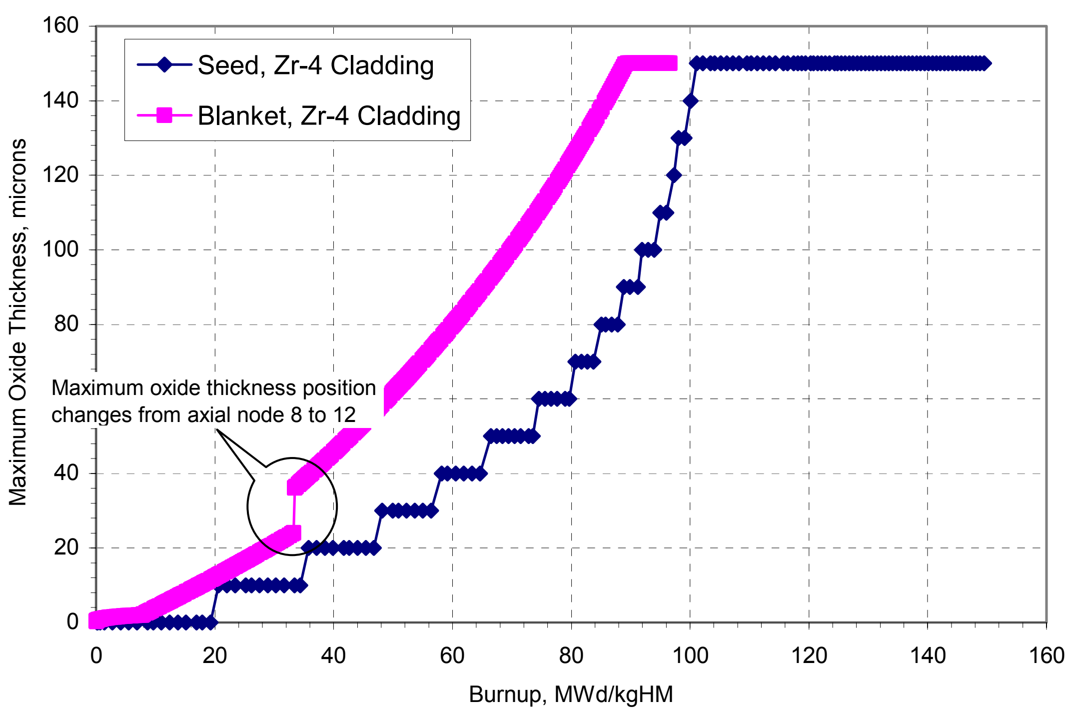


Figure 5.13 Oxide Thickness versus Burnup

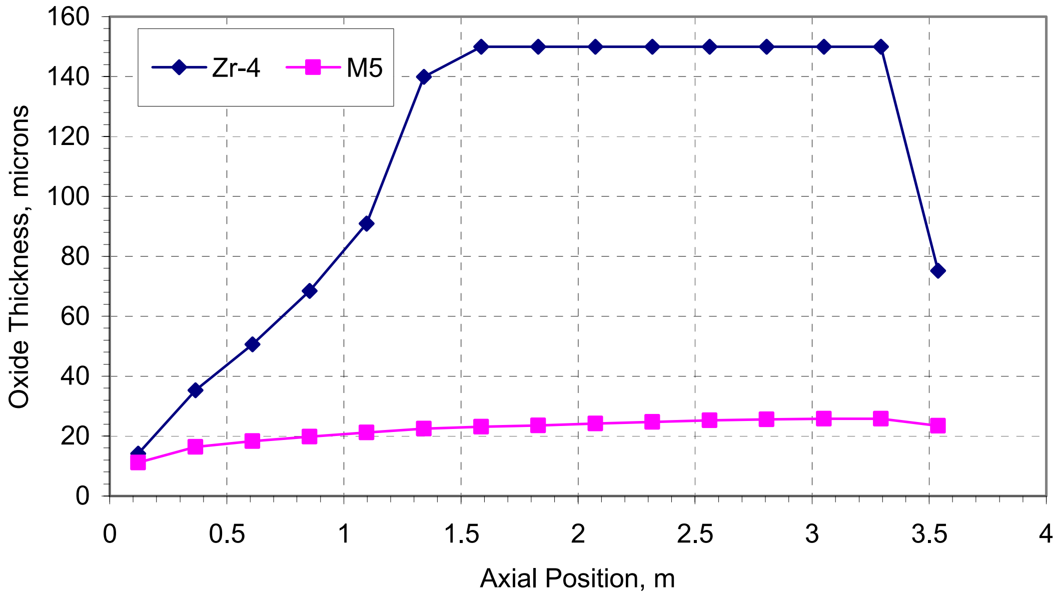


Figure 5.14 External Corrosion on Poisoned Seed Rod at Discharge of 4.5 Years

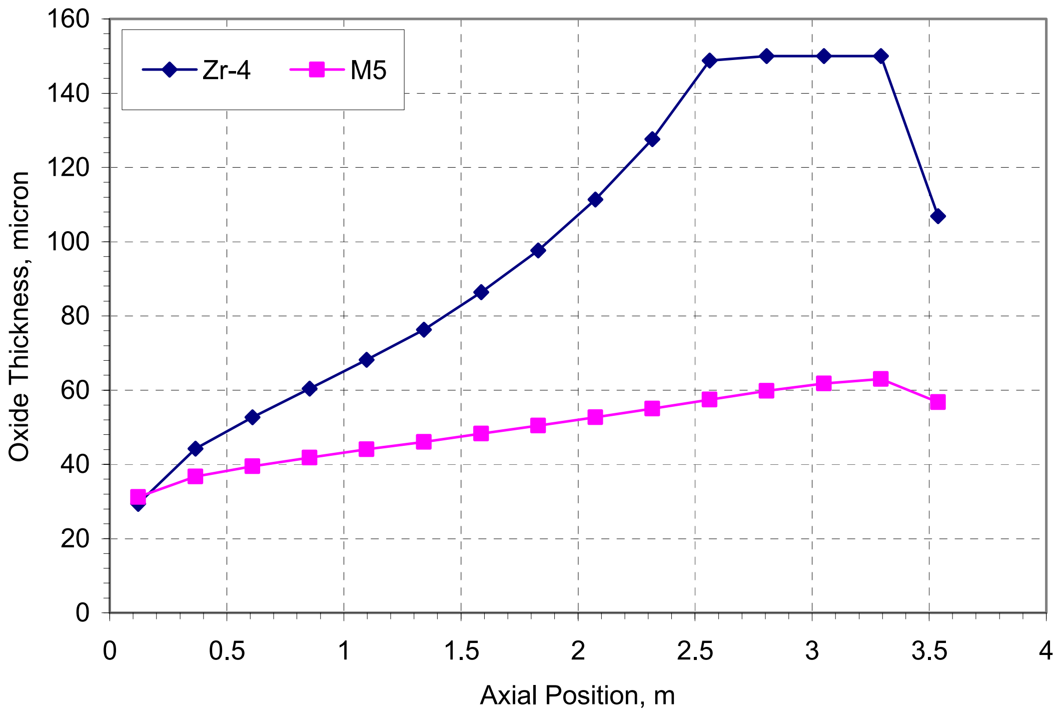


Figure 5.15 External Corrosion on Blanket Rod at Discharge of 13.5 Years

5.4 Chapter Summary

In this chapter, thermal-hydraulic performance and fuel behavior of the WASB core have been studied in detail. The thermal-hydraulic analyses show that acceptable MDNBR in the core can be achieved even assuming 12% overpower, increased inlet coolant temperature, and reduced flow rate, if a large local grid loss coefficient is used for the blanket. This implies that the WASB core will have a higher pressure drop than today's PWRs. Five variables were quantitatively analyzed to check the sensitivity of MDNBR to those variables: core inlet temperature, coolant flow rate, core power, form loss coefficient, and seed rod diameter.

The more demanding power history of the seed and the longer residence time of the blanket bring challenges to the thermal and mechanical performance of fuel rods. The calculated results show the internal pressure of the seed rod is more of a concern than that of the blanket because the large expected fission gas release in the seed fuel leads to an internal pressure of the seed fuel that is greater than the system pressure at EOL. However, further analysis shows that the internal pressure can be effectively reduced if a longer plenum is used for the seed fuel rods. The calculated results indicate that cladding materials with improved corrosion resistance such as M5 are required to prevent excessive cladding corrosion.

Chapter 6

SPENT FUEL CHARACTERISTICS AND PROLIFERATION RESISTANCE

6.1 Introduction

The fuel management scheme of the WASB core will have an impact on spent fuel storage. There are two separate fuel management flows: a standard three-batch scheme has been adopted for the seed (18 month cycle length) whereas a single-batch scheme is used for the blanket which will stay in the core for up to 9 seed cycles. The basis for the back-end of a cycle analysis is the amount (weight) and composition of the fuel discharged from the core. This fuel is initially stored, and finally disposed of in a permanent storage facility. The isotopic composition of the spent fuel stockpile defines its radioactivity and heat emission levels, as well as the overall toxicity of the spent fuel as a function of time. Its fissile content, mainly the amount and composition of plutonium, defines the proliferation potential of the fuel cycle.

For the same energy production, fewer spent fuel assemblies (about 38% less) need to be stored and disposed of for the WASB compared to today's PWRs. The seed assembly discharge burnup is about 145 MWd/kgHM, and the blanket is at 88 MWd/kgHM after 9 seed cycles, whereas the discharge burnup of today's PWR is about 50 MWd/kgHM. Although high burnup fuel reduces the amount of spent fuel, it is important to evaluate the radioactivity and the heat load of such highly burned spent fuel, which would have significant impact on storage.

For the seed fuel, high burnup would improve the proliferation resistance of plutonium by both reducing its amount per unit energy and degrading the isotopic mixture. The proliferation risk of the blanket is mainly due to U-233; the solution is to denature U-233 with a certain amount of U-238. In addition, a high radiation barrier is naturally provided by the gamma emission from U-232 daughters.

6.2 Spent Fuel Characteristics

6.2.1 Spent Fuel Discharge Rate

As discussed previously, the WASB is very different from conventional PWRs in its in-core fuel management. One third of seed assemblies are refueled every cycle, and all the blanket assemblies are refueled every nine cycles. The discharged waste per unit energy is summarized in Table 6.1. The discharged assemblies per GWe-yr of the WASB are nearly 40% less than those of a typical 18 month cycle PWR, and the discharged waste mass is about half that of a typical PWR. Therefore, the WASB has much less waste production than conventional PWRs. In particular, the WASB requires 40% less volume of a storage pool or dry casks. However, decay heat generation may be different. In addition, only half the core assemblies need to be shuffled in an outage. That could reduce the number of operations during a refueling outage.

Table 6.1 Spent Fuel Discharge Rate

	WASB			A Typical PWR
	Seed	Blanket	S + B	
Assemblies/GWe-yr	16.2	7.0	23.2	37.3
MTHM/GWe-yr	4.5	3.8	8.3	17.4

6.2.2 Radioactivity and Decay Heat

Currently there are two storage options for spent fuel used by utilities in the U.S.: wet and dry. Spent fuel assemblies have been successfully stored in water pools at many reactor sites and this wet storage represents a sound demonstrated technology for the safe storage of irradiated fuel for decades with negligible environmental impact. However, the utilities have to increase the on-site spent fuel pool capacity as storage capacity is exhausted. Dry storage is a method of storing the spent fuel in casks that are cooled by natural air circulation. They are designed to shield the fuel properly, while still allowing air circulation. Many spent fuel pools are reaching their maximum capacity and some will reach their limits in the near future; so the expansion of storage capacity is of strong interest to utilities.

There are three general groups of radioisotopes that are formed: fission products, actinides, and activation products. Generally speaking, the fission products (more than 350 nuclides) dominate the near term radioactivity of spent fuel due to their short half lives, and the long-term characteristics are determined by actinides because of their long half lives. For storage design, it is important to know the decay heat level expected from the spent fuel. The heat load and radioactivity of higher burnup spent fuel is higher per unit mass due to increased accumulation of fission products and higher actinides, but is lower per unit power production.

The radioactivity and decay heat from the seed and blanket spent fuel were studied using MCODE[X-1] to do depletion calculations and ORIGEN [C-3] to analyze the spent fuel characteristics after discharge.

Figure 6.1a shows the radioactivity of the WASB spent fuel in Ci/MTHM. The seed radioactivity is obviously higher than that of the blanket due to its higher burnup. In addition, the blanket is mostly composed of ThO_2 (about 90 wt%), hence the actinide content is much less than that of the seed. The radioactivity in Ci/Assembly and Ci/GWe-yr are shown in Figures 6.1b and 6.1c, respectively. The radioactivity per unit energy values in the short term are lower than conventional fuel up to about five years after discharge. In the near term (< 10,000 years) they are comparable with conventional UO_2 fueled LWRs. The blanket fuel is even better than UO_2 fuel. However, between 10,000 and one million years, there is a hump or peak for the blanket radioactivity because of the decay products of U-233, whose decay chain is shown in Figure 6.2. The buildups of Th-229 and its daughters in this time period are the major contributors to the increase in radioactivity. In addition, the structural materials of the blanket will achieve higher activity due to longer core residence time.

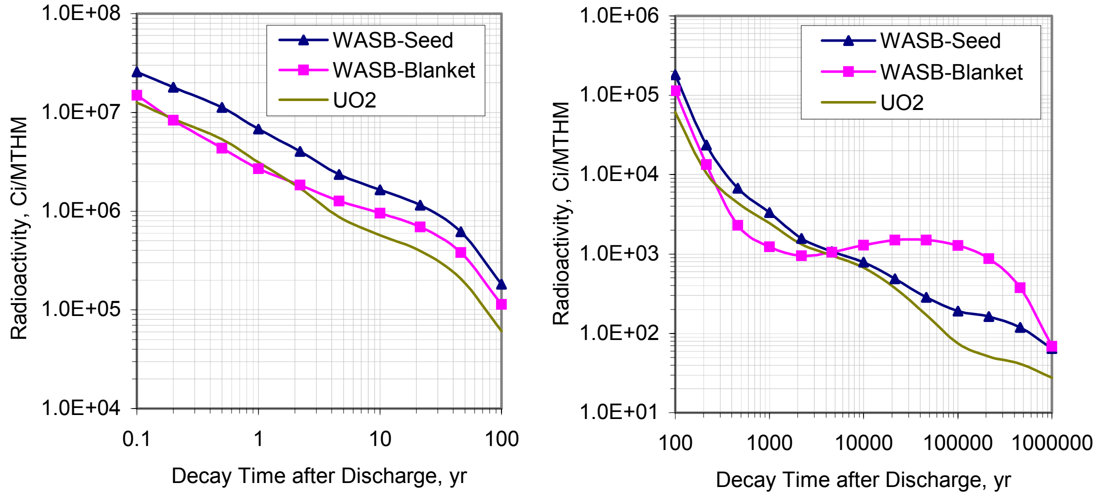


Figure 6.1a Radioactivity of the WASB Spent Fuel in Ci/MTHM

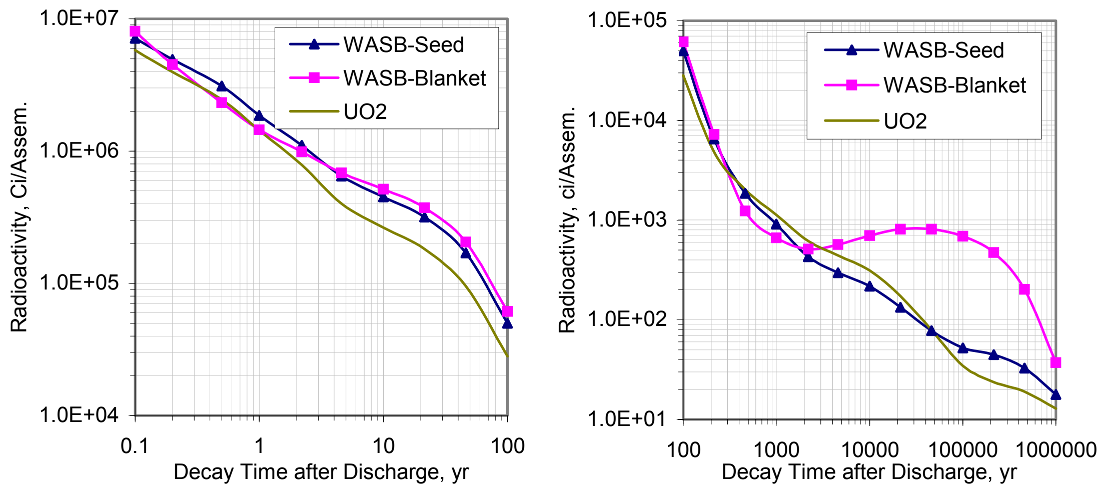


Figure 6.1b Radioactivity of the WASB Spent Fuel in Ci/Assembly

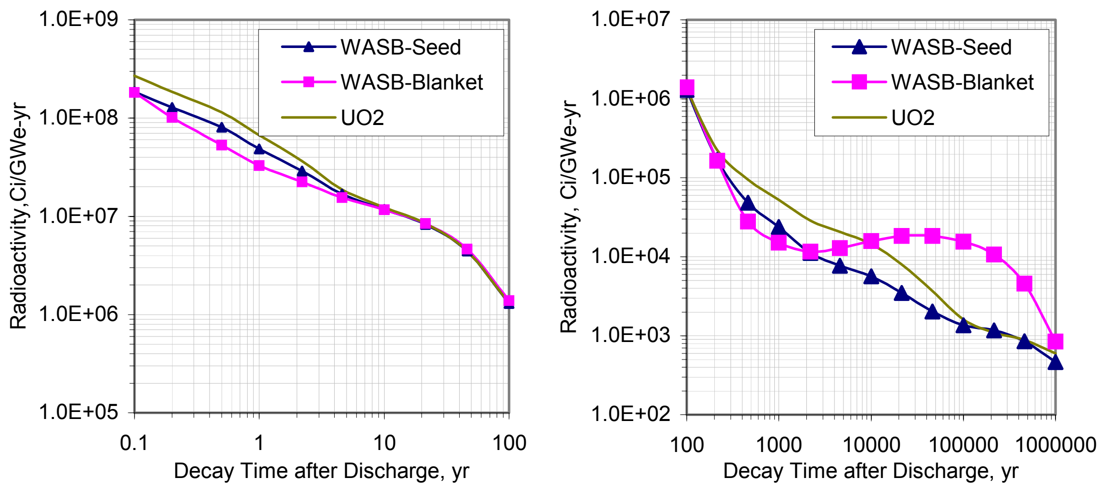


Figure 6.1c Radioactivity of the WASB Spent Fuel in Ci/GWe-yr

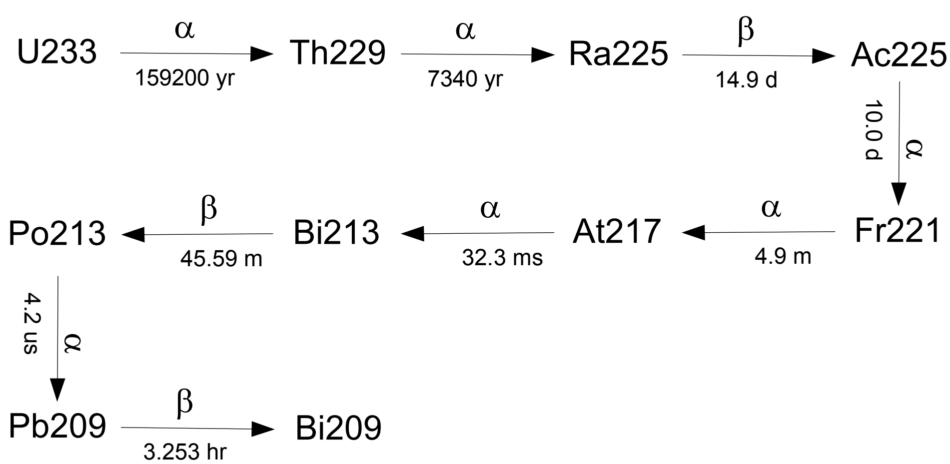


Figure 6.2 Decay Chain of U-233

As mentioned earlier, the heat load of the spent fuel is important for spent fuel storage design. The decay heat of the WASB spent fuel is shown in Figures 6.3a, 6.3b, 6.3c. The comparisons show that the decay heat load per unit heavy metal mass of the seed is higher than that of conventional UO_2 fuel, and the blanket behavior is comparable to the conventional UO_2 in the short term. The decay heat load per unit heavy metal mass of the seed is about twice that of the conventional UO_2 fuel; however, the total heavy metal in the seed assembly is about half of that in the conventional UO_2 assembly. Hence the decay heat load per assembly for the seed is comparable to a conventional UO_2 assembly as shown in

Figure 6.3b. The conventional spent fuel pool with some modification, if necessary, will be suitable for the spent seed if the seed assemblies are stored in the pool without consolidation. Finally, Figure 6.3c shows the seed and blanket decay heat together per GWe-yr as a function of decay time. It is clear that during the first 20,000 years or so, WASB has a lower burden than the UO_2 fuel in terms of decay heat in Watt/GWe-yr. Thereafter, the decay heat of the WASB design is higher than that of the conventional UO_2 design. However, the decay heat in this time frame is only at the level of 200 Watts/GWe-yr and lower which should not be very challenging to dissipate.

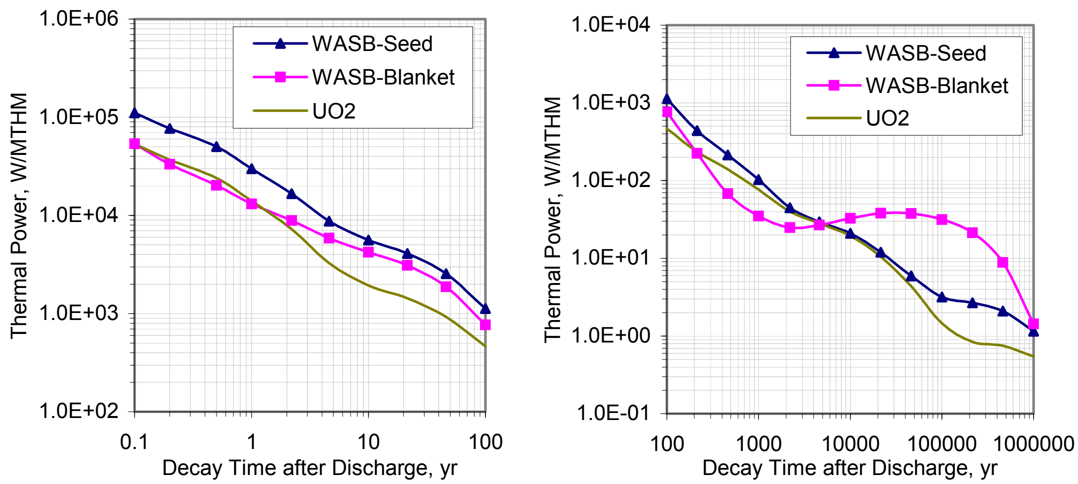


Figure 6.3a Decay Heat of the WASB Spent Fuel in Watts/MTHM

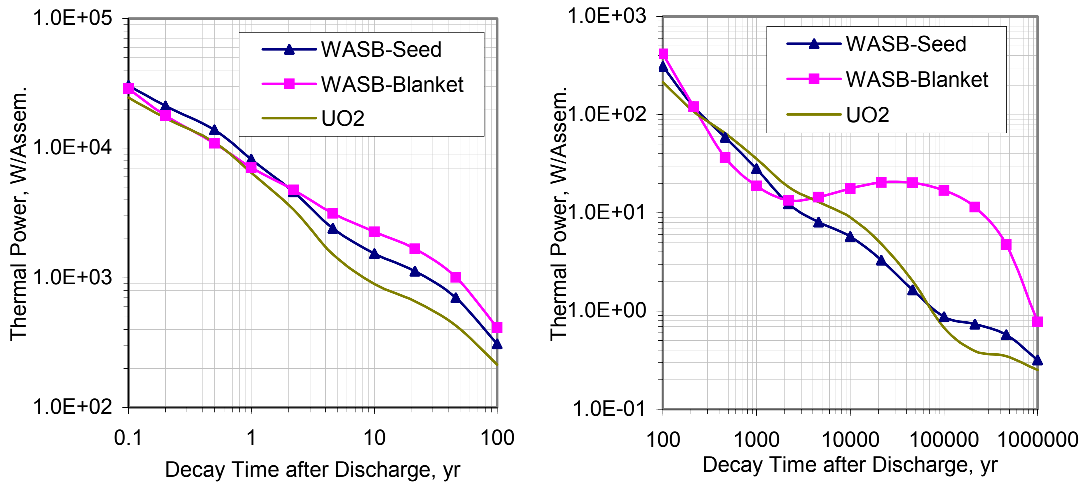


Figure 6.3b Decay Heat of the WASB Spent Fuel in Watts/assembly

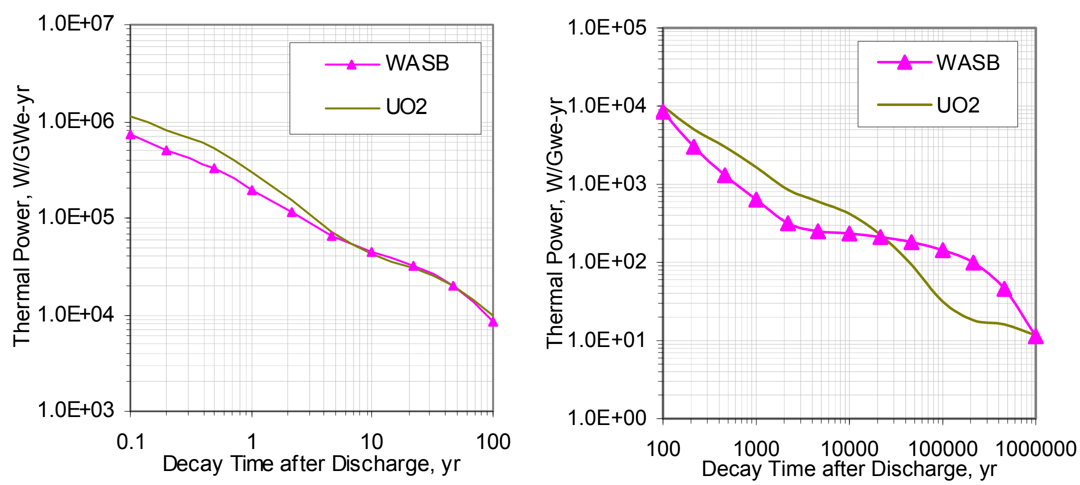


Figure 6.3c Decay Heat of the WASB Spent Fuel in Watts/GWe-yr

6.3 Proliferation Resistance

The proliferation potential of a fuel cycle, or its proliferation resistance, is determined by the quantity and the quality of the weapon usable materials that could be diverted to military use. An additional important factor is the measure of complexity required to separate the fissile component from the normal material flow of the fuel cycle. The assessment of proliferation potential depends to a great extent on the specific proliferation scenario. The scenarios related to the civilian nuclear power fuel cycle include national diversion scenarios, either clandestine or open, and sub-national diversion scenarios. The first is more important for civilian power reactors. In these cases, international treaties and safeguards may not be effective because they can be abrogated or circumvented. In order to separate the development and expansion of nuclear power from the danger of nuclear weapon proliferation, international safeguards are necessary but not sufficient. A decisive barrier to proliferation can be based on inherent properties of the fuel cycle itself. The fuel design should provide assurance that the quantity and quality of the fissile component of the fuel cycle material flow limits the proliferation potential below an acceptable threshold in the context of industrial capabilities and economic realities. For a new fuel cycle, it may be that it has to be better in proliferation resistance than present day conventional PWRs. In this section, the proliferation resistance characteristics of the WASB fuel cycle are discussed.

All transuranic isotopes capable of being assembled into a fast-spectrum critical mass are considered as potentially weapons-usable and therefore of proliferation concern. In addition all uranium isotopes except U-238 will have high fission to capture ratio in fast spectra. It is important to note that the effort required to use any isotope varies with the engineering and scientific skills of a potential proliferator because the isotopes' properties vary (half-life, neutron generation, heat generation, and critical mass). In any case, these skills need to be sophisticated. These materials are initially in a compound form (e.g., an oxide), or as oxide mixtures and will require complex chemical or isotopic separation processes to extract them for explosives use. Table 6.2 summarizes some of the nuclear properties of fissile materials. For comparison, the table also includes two major fertile materials, Th-232 and U-238, which in the presence of neutrons can produce the fissionable isotopes U-233 and Pu-239, respectively. Although Table 6.2 shows that Pu-238 is physically capable of sustaining a fast critical mass, the International Atomic Energy Agency considers plutonium containing more than 81% Pu-238 not weapons-usable because of its high heat generation. Note that Cm-244 is also not suitable for weapons because of its high heat production.

Table 6.2 Nuclear Properties of Fissile and Fertile Nuclear Materials [N-1]

Isotope	Half-life (years)	Spontaneous (Neutrons/ sec-kg)	Decay Heat (Watts/kg)	Critical Mass[a] (kg)
Pa-231	32.8×10^{3}	Nil	1.3	162
Th-232[b]	14.1×10^{9}	Nil	Nil	Infinite
U-233	159×10^{3}	1.23	0.281	16.4
U-235	700×10^{6}	0.364	6×10^{-5}	47.9
U-238[b]	4.5×10^{9}	0.11	8×10^{-6}	Infinite
Np-237	2.1×10^{6}	0.139	0.021	59
Pu-238	88	2.67×10^{6}	560	10
Pu-239	24×10^{3}	21.8	2.0	10.2
Pu-240	6.54×10^{3}	1.03×10^{6}	7.0	36.8
Pu-241	14.7	49.3	6.4	12.9
Pu-242	376×10^{3}	1.73×10^{6}	0.12	89
Am-241	433	1540	115	57
Am-243	7.38×10^{3}	900	6.4	155
Cm-244	18.1	11×10^{9}	2.8×10^{3}	28
Cm-245	8.5×10^{3}	147×10^{3}	5.7	13
Cm-246	4.7×10^{3}	9×10^{9}	10	84
Bk-247	1.4×10^{3}	Nil	36	10
Cf-251	898	Nil	56	9
Cf-252	2.645	2.3×10^{15}	3.8×10^{4}	N/A

a) Bare sphere

b) Not a potentially weapons-usable material

The proliferation potential of a fuel cycle, or its proliferation resistance, is determined by the quantity and quality of the fissile that could be diverted to military use. A decisive barrier to proliferation should be based on inherent properties of the fuel cycle itself in addition to a system of international safeguard measures. Attributes important for determining the effectiveness of the isotopic barrier include:

1. A small critical mass represents a lower barrier than a large critical mass.
2. Spontaneous neutron generation: This can cause pre-initiation of the nuclear explosion. This possibility complicates the design, degrades the yield, and reduces

the reliability of the device. The isotopes with the highest spontaneous rates of neutron generation are Pu-238, PU-240, and Pu-242.

3. Heat-generation rate: Heat produced by nuclear decay of the material complicates nuclear device design. For plutonium, this is strongly dependent on the concentration of Pu-238 and Pu-240.

4. Radiation: The radiation (especially gamma) released by the material itself interferes with the handling, processing, and design of the nuclear device. A lower radiation level represents a lower barrier than a higher level. For plutonium, this is dependent on the concentration of Pu-240 and Pu-242; for U-233, dependent on U-232.

5. Degree of isotopic enrichment: Natural and low-enriched uranium can not be used directly for weapon-making, but they can be converted to weapon-usable material by enrichment or re-enrichment. It is generally agreed that in uranium, U-235 enrichments of less than 20 w/o and U-233 enrichments of less than 12 w/o are non-proliferative [F-1].

Sentell et al have proposed a probabilistic method, which permits integration of all aspects of fissile material proliferation in formulating an overall estimate of relative proliferation risk [S-2].

6.3.1 Plutonium Production

The weight percentages of some important isotopes in discharged seed and blanket fuel are presented in Table 6.3. The weight percentage is relative to the initial heavy metal mass of assemblies. The discharge burnup of the seed is about 145 MWd/kgHM and the blanket is at 88 MWd/kgHM.

Table 6.3 Isotope Weight Percentages in the Discharge Seed and Blanket

Isotope	Discharged Seed, w/o	Discharged Blanket, w/o
Th-232	N/A	77.808
U-232	N/A	0.013
U-233	N/A	1.568
U-234	0.069	0.469
U-235	4.576	0.166
U-236	2.753	0.184
U-238	75.227	10.412
Pu-238	0.135	0.044
Pu-239	0.758	0.197
Pu-240	0.287	0.066
Pu-241	0.262	0.076
Pu-242	0.117	0.070

The Pu production rates and comparisons with conventional PWRs are shown in Table 6.4 and Figure 6.4. The thermodynamic efficiency of the WASB plant was assumed to be 34%. The Pu production rates of the WASB are much less than those of conventional PWRs. The fissile Pu (Pu-239 + Pu-241) annual production rate of the WASB is only 38% of conventional PWRs, and the total Pu annual production rate of the WASB is about 40% of conventional PWRs. As in any uranium-fueled reactor, the seed of the WASB core produces plutonium, but in lesser quantities than in a conventional reactor and with far higher isotopic degradation due to high Pu-238 content. Both characteristics would make the plutonium considerably less desirable for weapons than ordinary reactor-grade plutonium. The presence of U-233 in the blanket can be limited to a concentration below that of proliferation concern. In addition it should be noted that the 20 w/o enriched seed has less Pu production compared to conventional PWRs but its high U-236 content causes more severe reactivity penalties.

Table 6.4 Plutonium Production, kg/GWe-yr

	WASB			High Burnup UO_2 [a)]	A Typical PWR[b)]
	Seed	Blanket	S + B		
Pu-238	6.41	1.78	8.19	12.11	6.05
Pu-239	35.97	7.96	43.93	82.96	121.89
Pu-240	13.62	2.66	16.28	35.06	53.85
Pu-241	12.43	3.07	15.50	27.13	34.86
Pu-242	5.55	2.83	8.38	13.25	15.86
Fissile Pu	48.40	11.03	59.43	110.09	156.75
Total Pu	73.98	18.3	92.28	170.51	232.51

a) 9.75 w/o enriched UO_2 fuel, and B_d = 100 MWd/kgHM

b) 4.5 w/o enriched UO_2 fuel, and B_d = 50 MWd/kgHM

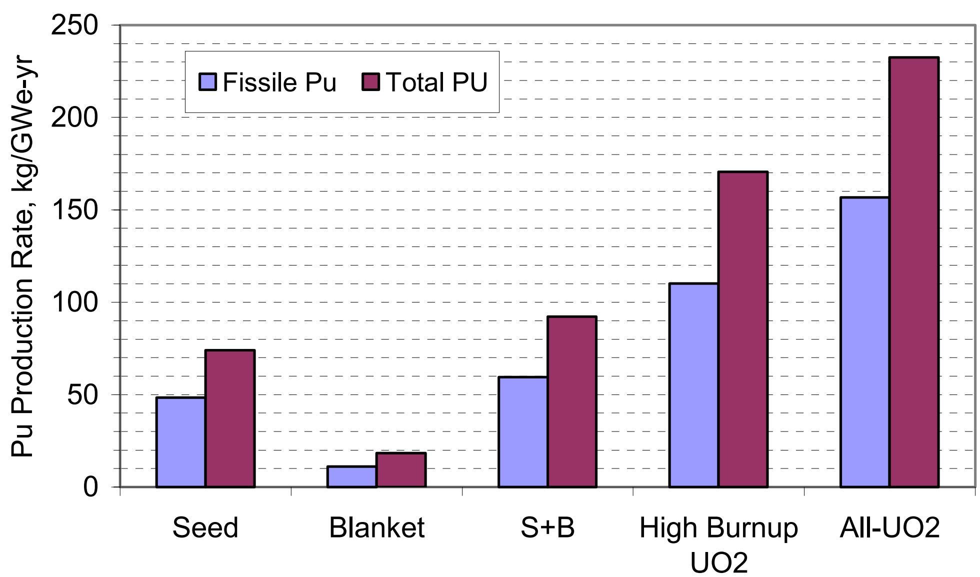


Figure 6.4 Plutonium Production Rates

Isotopic compositions of plutonium for the WASB, high burnup all-UO_2 fuel and a typical all-UO_2 are summarized in Table 6.5. In addition several standard grades of plutonium are also given for comparison in the table.

Table 6.5 Isotopic Composition of Various grades of Plutonium (w/o)

Grade	Pu-238	Pu-239	Pu-240	Pu-241	Pu-242
Super-grade		98.0%	2.0%		
Weapons-grade	0.012%	93.8%	5.8%	0.35%	0.022%
MOX-grade	1.9%	40.4%	32.1%	17.8%	7.8%
A Typical PWR	2.6%	52.4%	23.2%	15.0%	6.8%
High Burnup UO_2	7.1%	48.7%	20.6%	15.9%	7.7%
WASB – Seed	8.7%	48.6%	18.4%	16.8%	7.5%
WASB – Blanket	9.7%	43.5%	14.5%	16.8%	15.5%

6.3.2 Critical Mass, Spontaneous Fission and Heat Generation

From the composition given in Tables 6.2 and 6.5, using a weighted fraction of all isotopes, comparisons of the critical mass of WASB spent fuel with different grades of plutonium are presented in Table 6.6. It can be seen that the seed of the WASB requires nearly the same amount of plutonium as a typical PWR, however the blanket requires 21% more material.

Table 6.6 Critical Mass for Different Plutonium Compositions

Plutonium Source	Critical Mass, kg
Weapons-grade	11.8
A Typical PWR	22.1
High Burnup UO_2	22.2
WASB-seed	21.4
WASB-blanket	26.7

The spontaneous fission source (SFS) defines an important characteristic of weapons material, namely the yield degradation. Neutrons released by spontaneous fission cause pre-initiation, i.e. the start of the explosion before the device has reached its highest supercriticality value, which in turn causes a reduction of the device yield. The relevant data for all considered plutonium compositions are summarized in Table 6.7. The table shows the SFS in neutrons/sec for a critical mass of all considered plutonium compositions.

Table 6.7 Spontaneous Fission Rate for Different Plutonium Compositions, n/s

Isotope	Weapons - grade	A Typical PWR	High Burnup all-UO_2	WASB – Seed	WASB - Blanket
Pu-238	3.20E+02	6.94E+04	1.90E+05	2.32E+05	2.59E+05
Pu-239	2.04E+01	1.14E+01	1.06E+01	1.06E+01	9.48E+00
Pu-240	5.97E+04	2.39E+05	2.12E+05	1.90E+05	1.49E+05
Pu-241	1.73E-01	7.40E+00	7.84E+00	8.28E+00	8.28E+00
Pu-242	3.81E+02	1.18E+05	1.33E+05	1.30E+05	2.68E+05
Total/kg of Pu	6.05E+04	4.26E+05	5.35E+05	5.52E+05	6.77E+05
Total/Critical Mass	7.14E+5	9.43E+06	1.19E+07	1.18E+07	1.81E+07

The total spontaneous fission source for a critical mass of the WASB seed plutonium is 25% higher than that of PWR grade plutonium, and twice higher for the blanket plutonium. The probability that an explosive device, constructed from WASB-Pu, will deliver a nominal yield is small (seed) to negligible (blanket), and the probability of a fizzle yield is relatively high. Thus, the WASB-Pu will produce an unreliable weapon.

An additional barrier to possible diversion of a reactor grade material is the heat emitted by its isotopes. The thermal power produces an increase of the temperature of a device and causes two effects: one is a temperature increase of metallic plutonium which undergoes a metallurgical phase transition at 115 ^{o}C, and the second is an overheating of the high explosive around the plutonium core which may cause the disintegration of this material. The heat produced by different plutonium isotopes is summarized in Table 6.8 and used to estimate the total heat produced by a plutonium metal critical mass in each case.

Table 6.8 Decay Heat Emission for Different Plutonium Compositions, Watts

Isotope	Weapons - grade	A Typical PWR	High Burnup Fuel	WASB – Seed	WASB - Blanket
Pu-238	0.07	14.56	39.76	48.72	54.32
Pu-239	1.88	1.05	0.97	4.96	0.87
Pu-240	0.41	1.62	1.44	6.77	1.02
Pu-241	0.02	0.96	1.02	2.17	1.08
Pu-242	0.00	0.01	0.01	6.68	0.02
Total/kg of Pu	2.37	18.20	43.20	69.29	57.30
Total/Critical Mass	27.97	402.76	957.47	1485.63	1530.18

The total heat produced by the WASB plutonium is much higher than that produced by the PWR grade plutonium. Thus it is reasonable to assume that a weapon device made from the WASB-Pu is more unstable.

6.3.3 Diversion Concern of U-233

U-233 buildup in the blanket presents certain proliferation risks. Forsberg, et al, have developed a combined proliferation limit of 12% fissile uranium (U-233 plus 0.6 times U-235) for cases such as the blanket when both U-233 and U-235 are present. As discussed in Chapter 3, one objective of including 13 v/o UO_2 in the blanket fuel is to denature U-233 to meet the proliferation limit. Figure 6.5 shows this proliferation index as a function of blanket burnup, which is defined as follows:

$$U \ \ proliferation\ index = \frac{{}^{233}U + 0.6\,{}^{235}U}{U^{tot}} \times 100\% . \tag{6.1}$$

The results in Figure 6.5 indicate the present blanket design can meet the requirement of non-proliferation. The margin to the limit decreases as the burnup increases due to buildup of U-233.

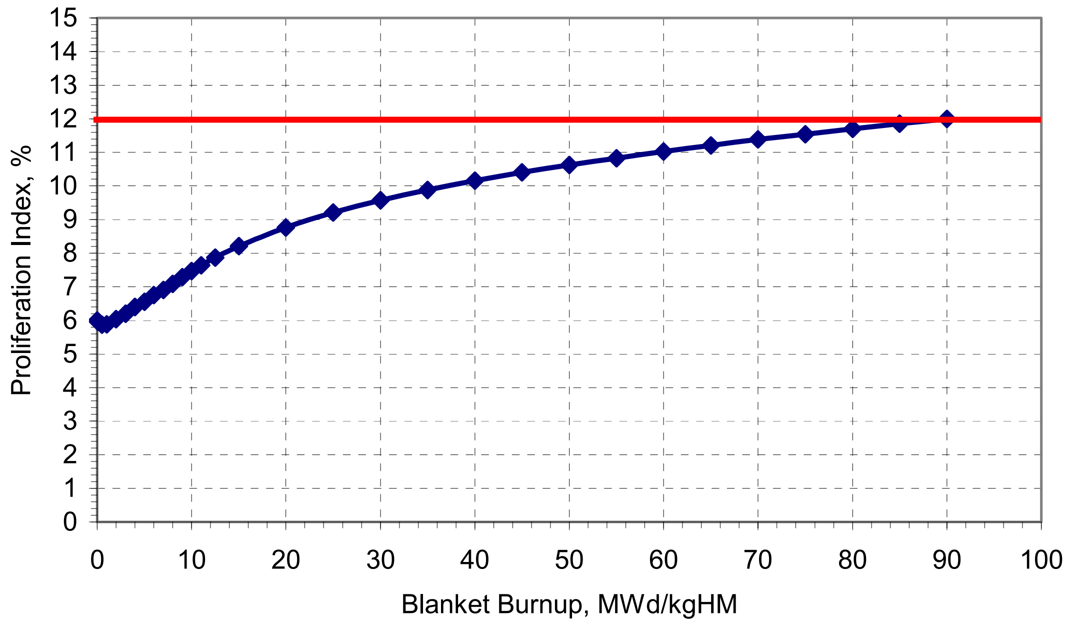


Figure 6.5 Blanket U Proliferation Index

Additional uranium isotopes created during the long in-core residence time of the blanket are U-232, U-234, U-235, and U-236. In principle, all uranium isotopes may be chemically separated from blanket spent fuel and further enriched by standard industrial methods. However, there are several barriers to the diversion of U-233 through this route:

- The contamination of the recycle material by a hard-gama emitter originating in the U-232 chain will require that the reprocessing facility be remotely operated and managed.
- The enrichment of the mixture of separated uranium isotopes will be extremely inefficient due to its isotopic composition. An attempt to separate U-233 from the U-238, U-234 and U-236 isotopes will also remove the fissile U-235 from the resulting enriched stream. Residual amounts of U-234 and U-236 will reduce the criticality of the enriched stream.
- Enrichment in U-233 inherently yields a product with increased U-232 content, exacerbating the gamma problem.

Additionally, the energetic gamma rays (from the decay daughters of U-232) make a nuclear device constructed from U-233 readily detectable to conventional gamma-detectors. 16 kilograms (the minimum required to construct a weapon) of U-233 would become easily detectable less than 5 days after separation, even when shielded by 9 inches of lead, which is the maximum shielding allowable for transport due to weight restrictions for typical shipping containers [L-1]. This detectability characteristic may turn out to be more important than the self-protection aspects, since it makes a device employing U-233 impracticable for covert uses, which has become more of a concern lately due to recent terrorist activities.

It should be stressed that all the technical and organizational difficulties on the diversion path separating and enriching the uranium isotope mixture created in the WASB-blanket may be overcome by a sufficiently dedicated, resourceful and well-funded adversary.

6.4 Seed Spent Fuel Reprocessing

There is an incentive to extend the use of the seed spent fuel and burn it again in LWRs because it contains considerable fissile materials (about 5% of heavy metal). The blanket spent fuel contains about 1.6 w/o U-233 (relative to the initial heavy metal), which would be a good fuel for CANDUs by direct refabrication without removing the fission products. Actually a so-called DUPIC process (Direct Use of Spent Fuel in CANDU) has been under investigation since the early 1990's [Y-1][C-1]. However, this is not the interest of this thesis. This section only focuses on seed spent fuel reprocessing.

Table 6.9 shows the weight percentages of the main fission products and actinides in the spent seed fuel at 10 years after discharge. The burnup calculation was performed using MCODE for a seed assembly, and the discharge burnup is 145 MWd/kgHM. Afterwards the discharged seed spent fuel undergoes 10 years decay, which was simulated using ORIGEN. It can be seen that the U-235 content is about 5 w/o of the total fission products and actinides.

The spent seed fuel might be reused in conventional PWRs by application of the AIROX process. AIROX is a dry process involving irradiated fuel oxidation by O_2. UO_2 fuel pins have small holes implaced via a puncturing machine. The oxidation transforms UO_2 to

U_3O_8 with a volume increase, on the order of 30%, allowing the removal of the volatile and semi-volatile fission products. The cladding ruptures and fuel cracks. The U_3O_8 is reduced to UO_2 by H_2. After several cycles, the spent fuel is pulverized. The remaining fuel is mixed with UO_2 or PuO_2, then resintered and refabricated [R-2].

Different values for fission product removal rates can be found in the literature as shown in Table 6.10. INEEL states that hot cell tests by Atomics International in the 1960s show that a "small" amount of technetium is removed during sintering. However, in later quantitative discussions no forecast is made regarding expected technetium removal. The Scientech forecasts were developed to predict release to fission product waste streams and thus may be biased upwards in some cases to produce a conservative (in terms of waste-handling) estimate [B-3]. KAERI data is the most recent paper on the subject [K-7].

Table 6.9 Seed Spent Fuel Characteristics at 10 Years after Discharge

Fission Products			Actinides		
	Weight Percentage[a)], %	Neutron Absorption Ratio[b)], %		Weight Percentage, %	Neutron Absorption Ratio, %
Kr	0.184	0.139	U-234	0.080	0.221
Zr	1.803	0.194	U-235	5.022	32.839
Mo	1.184	0.753	U-236	2.941	3.487
Tc	0.340	1.017	U-238	76.464	10.116
Ru	0.903	0.366	NP-237	0.271	1.161
Rh	0.142	1.704	PU-238	0.133	0.533
Pd	0.439	0.292	PU-239	0.894	19.313
Ag	0.017	0.206	PU-240	0.327	9.608
Cd	0.033	0.052	PU-241	0.185	3.405
In	0.001	0.030	PU-242	0.110	0.501
I	0.087	0.096	AM-241	0.127	1.816
Xe	2.183	1.066	AM-242M	0.002	0.101
Cs	0.875	1.443	AM-243	0.035	0.232
Ba	0.707	0.044	Total	86.591	83.335
La	0.561	0.113			
Ce	1.087	0.017			
Pr	0.515	0.190			
Nd	1.881	2.533			
Pm	0.004	0.082			
Sm	0.355	3.109			
Eu	0.052	0.827			
Gd	0.057	2.391			
Total	13.409	16.665			

a) relative to total weight of fission products and actinides in the table, data from MCODE and ORIGEN calculations

b) Relative to total neutron absorption of fission products and actinides in the table, data from MCODE and ORIGEN calculations

Table 6.10 Comparison of Fission Product Percent Removal Forecasts During AIROX

Nuclide	AECL [E-3]	INEEL [M-2]	Scientech [S-1]	KAERI [K-7]
^{14}C	100	100	100	100
^{3}H	100	100	100	100
Ag	80	0	0	N/A
Cd	0	75	75	75
Cs	99	90	100	100
I	99	100	100	100
In	0	75	0	N/A
Ir	0	0	75	75
Kr	99	100	100	100
Mo	80	0	0	N/A
Pd	80	0	0	N/A
Rh	80	0	0	N/A
Ru	80	90	100	100
Se	80	0	0	N/A
Tc	80	Discrepancy	0	N/A
Te	99	75	75	75
Xe	100	100	100	100

Three cases were analyzed to investigate the potential of reusing the seed spent fuel in conventional PWRs. The first case is the fuel with all actinides and fission products of seed spent fuel. The second one is the AIROX fuel, where the fission product release rates from KAERI are used. We assume that the release rates for Ag, In, Mo, Pd, Rh, Se, and Tc are zero because the data are not available, thus the fuel reactivity predictions will be conservative. The third case is only the actinides in the fuel. All calculations were performed using CASMO-4 based on a standard 17 × 17 PWR assembly, and all assembly geometrical parameters are the same as the reference core. All actinides and 140 fission products contained in spent seed fuel have been considered in the calculations.

The reactivity analysis results are shown in Figure 6.6. A conventional PWR all-U case is also shown in the figure for comparison. It is clear that the AIROX fuel can not be used in conventional LWRs due to poor reactivity performance. It might be possible that the spent seed fuel be used only in part of the PWR core, thus accumulating further burnup. This case was not addressed here. This is changed if the actinides are separated from the fission products, which requires wet reprocessing such as in the PUREX process [C-2]. As shown in the figure, reactivity-limited burnup would be as good or better than conventional fuel and with a smaller change in reactivity versus burnup. In addition the calculations indicate that it would further improve proliferation resistance if spent seed fuel is reprocessed and reused as shown in Figure 6.7.

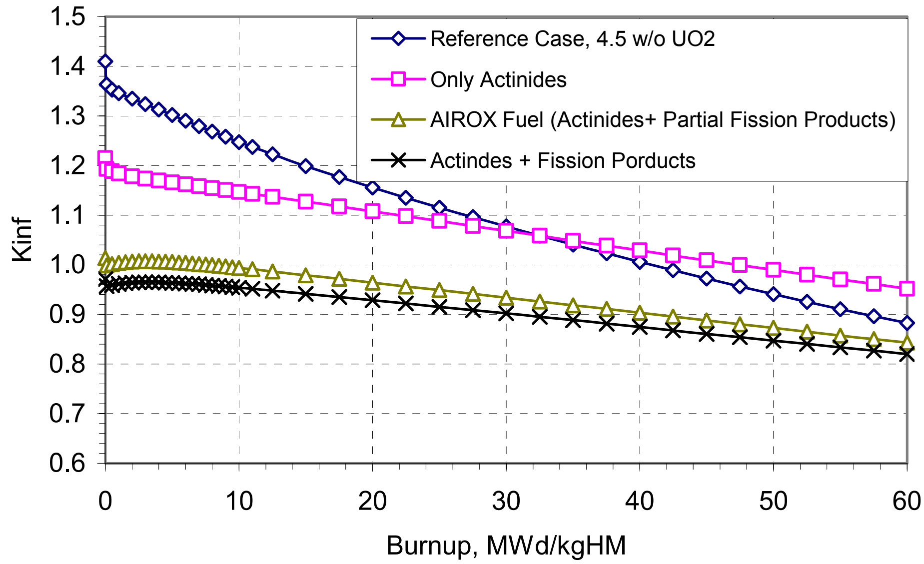


Figure 6.6 Kinf Comparison of Different Refabricated Fuel Compositions

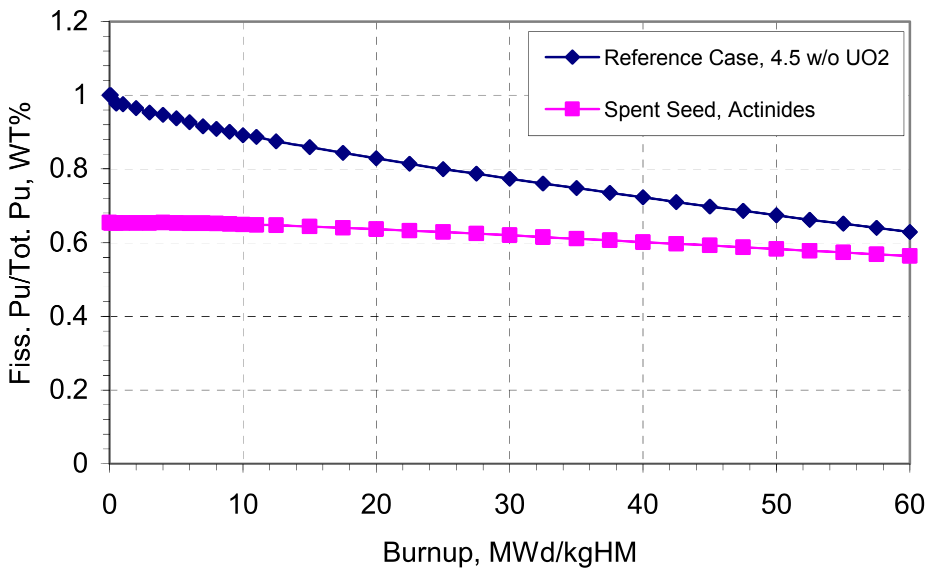


Figure 6.7 Fissile Pu Weight Percentage vs. Burnup

6.5 Chapter Summary

Spent fuel characteristics and proliferation resistance potential of the WASB fuel cycle have been analyzed in this chapter. It can be concluded that the WASB produces 40% less spent fuel assemblies than the conventional UO_2 design and 50% less spent fuel heavy metal in MTHM/GWe-yr. The radioactivity and decay heat of the WASB spent fuel is higher than the conventional UO_2 design per unit heavy metal mass because of its higher burnup. It is shown that during the first 20,000 years or so, WASB has lower decay heat than the all-UO_2 fuel in Watts/GWe-yr. Thereafter, the decay heat of the WASB design is higher than that of the conventional UO_2 design, but only at the level of 200 Watts/GWe-yr and lower.

As in any uranium-fueled reactor, the seed of the WASB core produces plutonium, but in lesser quantities than in a conventional reactor and with far higher isotopic degradation. Both characteristics would make the plutonium considerably less desirable for weapons than ordinary reactor-grade plutonium. The presence of U-233 in the blanket can be limited to a concentration below that of proliferation concern.

Preliminary analyses show that seed spent fuel could be refabricated and burned again in a conventional LWR core if fission products are removed and only the actinides are recycled. The reprocessing and refabrication of blanket spent fuel is not discussed in this chapter.

Chapter 7

FUEL CYCLE COST MODELING

7.1 Introduction

The adoption of a standardized methodology for fuel cycle cost calculations is a prerequisite for a fair comparison among different fuel cycle options. The fuel cycle of the WASB core includes two separate fuel material flows: seed and blanket. The fuel management schemes, inter-refueling intervals and burnups are different for each flow. Therefore there are two streams of cash flows for fuel cycle materials and services during the reactor lifetime. A fuel cycle cost calculation model for the WASB core has been developed based on the OECD' levelized lifetime cost methodology [O-1].

The levelized cost methodology discounts the time series of expenditures and incomes to their present values in a specified base year by applying a discount rate. Applying a discount rate takes into account the time value of money. The discount rate incorporates allowance for financial risks and/or is derived from national macroeconomic analysis; or it may be related to other aspects of the trade-off between costs and benefits for present and future generations. Tax on income and profit charged to the utility and any other overheads that do not influence cost comparisons are excluded. The date selected as the base year for discounting purposes does not affect the levelized cost comparison between different fuel cycles. The absolute values of levelized cost will, however, differ from base year to base year in periods of inflation or deflation. In the present study, the base year for discounting is the year of commissioning. Applied to fuel cycle costs, the levelized lifetime methodology provides costs per unit of electricity generated which are the ratios of the total lifetime expenses to total expected output, expressed in terms of present value equivalent. When this method is applied, the economic merits of different fuel cycles are derived from the comparison of their respective average lifetime levelized costs. Technological and economic assumptions underlying the results are transparent and the method allows for sensitivity analysis.

7.2 Fuel Cycle Cost Modeling

The cash flow for fuel cycle materials and services commences before the reactor starts to generate electricity and continues well after the reactor ceases operation. The exact timing of payments for uranium, fuel fabrication, reprocessing, etc., depends on the fuel cycle chosen and the associated lead and lag times for the fuel cycle components. In order to calculate the overall fuel cycle cost, the magnitude of each component cost and the appropriate point in time that it occurs must be identified. The quantities of fuel required are obtained from reactor neutronics calculations. These quantities of materials and services are adjusted to allow for process losses in the various component stages of the nuclear fuel cycle and then multiplied by the unit costs to obtain the component costs. Table 7.2 lists the notation for all the parameters needed for the calculations.

Table 7.1 Parameter Notation for Fuel Cycle Cost Calculations

<u>General</u>

Discount rate (yr^{-1})	r
Time	t
Base date of monetary unit for calculation of escalation	t_b
Date of fuel loading	t_c
Fuel residence time (yr)	T_r

<u>Material</u>

Mass of natural uranium feed (kg)	M_f
Mass of enriched uranium charged to reactor (kg)	M_p
Mass of uranium in the enrichment plant tails (kg)	M_t
Weight percent of ^{235}U in the uranium feed	x_f (0.711)
Weight percent of ^{235}U charged to reactor	x_p
Weight percent of ^{235}U in the tails	x_t
Conversion factor from kg U to lb U_3O_8 (a lbs U_3O_8 per kg U)	a (2.6)

For each component i of the nuclear fuel cycle:

Total component cost ($)	F_i
Unit cost ($/kg)	P_i
Escalation rate or inflation rate (yr^{-1})	s_i
Material loss (weight percent)	l_i
Total loss factor	f_i
Lead or lag time (yr)*	Δt_i

* Lead time: measured from fuel loading date; Lag time: measured from fuel discharge date

Where
- i = 1: Natural uranium purchase
- i = 2: Conversion into UF_6
- i = 3: Enrichment
- i = 4: Fabrication
- i = 5: Spent fuel storage
- i = 6: Spent fuel disposal

and
- P_1 = Dollars per lb U_3O_8
- P_2 = Dollars per kg U
- P_3 = Dollars per kg SWU
- P_4 = Dollars per kg U fabricated
- P_5 = Dollars per assembly ($/Assembly)
- P_6 = Dollars per kg heavy metal

7.2.1 Component Cost

The component costs for a given fuel batch can be written as follows:

1. Cost of uranium

$$F_1 = M_f \cdot a \cdot f_1 \cdot P_1 \cdot (1+s_1)^{t-t_b} \tag{7.1}$$

where

$$M_f = \frac{(x_p - x_t)}{(x_f - x_t)} \cdot M_p \tag{7.2}$$

$$f_1 = (1+l_2)(1+l_3)(1+l_4) \tag{7.3}$$

Let $F_1' = M_f \cdot a \cdot f_1 \cdot P_1$, then we have

$$F_1 = F_1' \cdot (1+s_1)^{t-t_b} \tag{7.4}$$

For all front-end components: $t = t_c - \Delta t_i$

2. Cost of conversion

$$F_2 = M_f \cdot f_2 \cdot p_2 \cdot (1 + s_2)^{t - t_b} \tag{7.5}$$

where

$$f_2 = (1 + l_2)(1 + l_3)(1 + l_4) \tag{7.6}$$

Let $F_2^{'} = M_f \cdot f_2 \cdot P_2$, then we have

$$F_1 = F_2^{'} \cdot (1 + s_2)^{t - t_b} \tag{7.7}$$

3. Cost of enrichment

$$F_3 = S \cdot f_3 \cdot p_3 \cdot (1 + s_3)^{t - t_b} \tag{7.8}$$

where

$$S = M_p V_p + M_t V_t - M_f V_f \text{, Separative Work Units (SWU)} \tag{7.9}$$

$$M_t = M_f - M_p \tag{7.10}$$

$$V_j = (2x_j - 1)\ln \frac{x_j}{1 - x_j} \tag{7.11}$$

and j is f, p or t as appropriate

$$f_3 = (1 + l_3)(1 + l_4) \tag{7.12}$$

Let $F_3^{'} = S \cdot f_3 \cdot P_3$, then we have

$$F_1 = F_3^{'} \cdot (1 + s_3)^{t - t_b} \tag{7.13}$$

4. Cost of fabrication

$$F_4 = M_p \cdot f_4 \cdot P_4 \cdot (1 + s_4)^{t - t_b} \tag{7.14}$$

where

$$f_4 = (1 + l_4) \tag{7.15}$$

Let $F_4^{'} = M_p \cdot f_4 \cdot P_4$, then we have

$$F_4 = F_4^{'} \cdot (1 + s_4)^{t - t_b} \tag{7.16}$$

5. Cost of spent fuel storage

Assuming zero loss factor for reactor fuel during the irradiation period, we have

$$F_5 = N \cdot P_5 \cdot (1 + s_5)^{t - t_b} \tag{7.17}$$

where

N = Number of discharged assemblies

$t = t_c + T_r + \Delta t_5$: date of back-end components

Let $F_5^{'} = N \cdot P_5$, then we have

$$F_5 = F_5^{'} \cdot (1 + s_5)^{t - t_b} \tag{7.18}$$

6. Cost of spent fuel Disposal

Assuming zero loss factor for reactor fuel during the irradiation period, we have

$$F_6 = N \cdot P_6 \cdot (1 + s_5)^{t - t_b} \tag{7.19}$$

where

N = Number of discharged assemblies

$t = t_c + T_r + \Delta t_6$: date of back-end components

Let $F_6^{'} = N \cdot P_6$, then we have

$$F_6 = F_6^{'} \cdot (1 + s_6)^{t - t_b} \tag{7.20}$$

7.2.2 Discounting and Levelizing of Fuel Cycle Costs

All the component costs during the life of the reactor (L) are discounted back to a selected base date t_0 and added together in order to arrive at a total fuel cost in present value terms. Figure 7.1 shows the flow diagram of fuel cycle cost calculations. Figure 7.2 shows the calculation timeline.

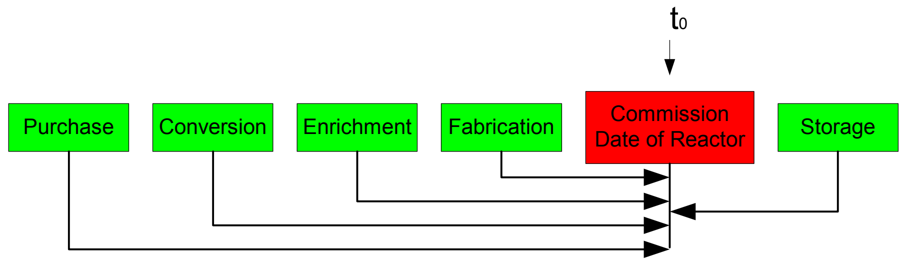


Figure 7.1 Discounting Method

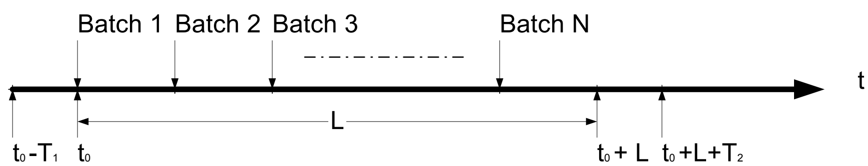


Figure 7.2 Calculation Timeline

The total discounted cost of the nuclear fuel cycle present worth as of the reference date t_0 can be written as:

$$\sum_i \sum_{t=t_0-T_1}^{t_0+L+T_2} \frac{F_i(t)}{(1+r)^{t-t_0}} \tag{7.21}$$

where

t_0 = reference date (commissioning date of reactor)

L = reactor lifetime

T_1 = maximum value of lead time (at front-end)

T_2 = maximum value of lag time (at back-end)

If C is the constant levelized fuel cost per unit of electricity produced by a reactor, the total present worth cost of fuel is also

$$\sum_{t=t_0}^{t_0+L} \frac{C \cdot E(t)}{(1+r)^{t-t_0}} \tag{7.22}$$

where

$E(t) =$ net electrical output at time t, $kWe \cdot hr / yr$

hence

$$\sum_i \sum_{t=t_0-T_1}^{t_0+L+T_2} \frac{F_i(t)}{(1+r)^{t-t_0}} = \sum_{t=t_0}^{t_0+L} \frac{C \cdot E(t)}{(1+r)^{t-t_0}} \tag{7.23}$$

hence

$$C = \frac{\displaystyle\sum_i \sum_{t=t_0-T_1}^{t_0+L+T_2} \frac{F_i(t)}{(1+r)^{t-t_0}}}{\displaystyle\sum_{t=t_0}^{t_0+L} \frac{E(t)}{(1+r)^{t-t_0}}} \tag{7.24}$$

Because electricity is generated more or less continuously during the life of the reactor, a continuous discounting method can be employed. The annual discount rate is then replaced by $r' = \ln(1+r)$, the equivalent continuous discount rate, and the discount factor is replaced by the exponential form:

$$\frac{1}{(1+r)^{t-t_0}} = e^{-r'(t-t_0)} \tag{7.25}$$

Thus the denominator of equation (7.24) can be rewritten in the exponential form and the integration is made over the period in which electricity is generated.

$$\sum_{t=t_0}^{t_0+L} \frac{E(t)}{(1+r)^{t-t_0}} = \int_{t_0}^{t_0+L} e^{-r'(t-t_0)} \cdot E(t)dt \tag{7.26}$$

When the load factor of the plant varies from year to year or from cycle to cycle over the lifetime of reactor, E has different values for each operational year or cycle. In such a case, the above integral will be applied in segments which will be added together in order to arrive at the total discounted electricity output. The procedures mentioned above can be laborious if economic and technical parameters vary with time.

7.2.3 Simplification

If we assume all cycles have the same amount of U refueling and the same refueling cycle length, it follows from Equation (7.21) that:

$$\sum_{i} \sum_{t=t_0-T_1}^{t_0+L+T_2} \frac{F_i(t)}{(1+r)^{t-t_0}} \cong \sum_{i=1}^{4} \frac{F_i^{'} \cdot (1+s_i)^{t_0-t_b-\Delta t_i}}{(1+r)^{-\Delta t_i}} \cdot \frac{1-\left(\frac{1+s_i}{1+r}\right)^{L}}{1-\left(\frac{1+s_i}{1+r}\right)^{T_c}}$$
$$+\frac{F_5^{'} \cdot (1+s_5)^{t_0-t_b+\Delta t_5}}{(1+r)^{\Delta t_5}} \cdot \frac{1-\left(\frac{1+s_5}{1+r}\right)^{L}}{\left(\frac{1+s_5}{1+r}\right)^{-T_c}-1} + \frac{F_6^{'} \cdot (1+s_6)^{t_0-t_b+\Delta t_6}}{(1+r)^{\Delta t_6}} \cdot \frac{1-\left(\frac{1+s_6}{1+r}\right)^{L}}{\left(\frac{1+s_6}{1+r}\right)^{-T_c}-1} \tag{7.27}$$

where:

T_c = Refueling cycle length (yr), time period between successive refuelings.

It should be noted that the two sides of the above equation are strictly equal only for single batch loading. Substituting Equations (7.26) and (7.27) into Equation (7.24), we have

$$C=\frac{\sum_{i=1}^{4} \frac{F_i^{'} \cdot (1+s_i)^{t_0-t_b-\Delta t_i}}{(1+r)^{-\Delta t_i}} \cdot \frac{1-\left(\frac{1+s_i}{1+r}\right)^{L}}{1-\left(\frac{1+s_i}{1+r}\right)^{T_c}} + \frac{F_5^{'} \cdot (1+s_5)^{t_0-t_b+\Delta t_5}}{(1+r)^{\Delta t_5}} \cdot \frac{1-\left(\frac{1+s_5}{1+r}\right)^{L}}{\left(\frac{1+s_5}{1+r}\right)^{-T_c}-1} + \frac{F_6^{'} \cdot (1+s_6)^{t_0-t_b+\Delta t_6}}{(1+r)^{\Delta t_6}} \cdot \frac{1-\left(\frac{1+s_6}{1+r}\right)^{L}}{\left(\frac{1+s_6}{1+r}\right)^{-T_c}-1}}{\int_{t_0}^{t_0+L} e^{-r'(t-t_0)} \cdot E(t)\,dt} \tag{7.28}$$

If $E(t)$ is assumed to be constant over the period of time from 0 to L and equal to E, the denominator of the above equation then becomes

$$\int_{t_0}^{t_0+L} e^{-r'(t-t_0)} \cdot E(t)\,dt = E \int_{t_0}^{t_0+L} e^{-r'(t-t_0)}\,dt = E \cdot \frac{1-e^{-r'L}}{r'} \tag{7.29}$$

E is net electricity output in one year, $kWe \cdot hr / yr$

Substituting Equation (7.29) into Equation (7.28) and setting $s_i = 0$, we have

$$C=\frac{\sum_{i=1}^{4}\frac{F_i^{'}}{(1+r)^{-\Delta t_i}}\cdot\frac{1-\frac{1}{(1+r)^L}}{1-\frac{1}{(1+r)^{T_c}}}+\frac{F_5^{'}}{(1+r)^{\Delta t_5}}\cdot\frac{1-\frac{1}{(1+r)^L}}{(1+r)^{T_c}-1}+\frac{F_6^{'}}{(1+r)^{\Delta t_6}}\cdot\frac{1-\frac{1}{(1+r)^L}}{(1+r)^{T_c}-1}}{E\cdot\frac{1-e^{-r'L}}{r'}}$$

$$=\frac{\sum_{i=1}^{4}F_i^{'}\cdot e^{r'\Delta t_i}\cdot\frac{1-e^{-r'L}}{1-e^{-r'T_c}}+F_5^{'}\cdot e^{-r'\Delta t_5}\cdot\frac{1-e^{-r'L}}{e^{r'T_c}-1}+F_6^{'}\cdot e^{-r'\Delta t_6}\cdot\frac{1-e^{-r'L}}{e^{r'T_c}-1}}{E\cdot\frac{1-e^{-r'L}}{r'}}$$

$$=\frac{\sum_{i=1}^{4}F_i^{'}\cdot e^{r'\Delta t_i}+F_5^{'}\cdot e^{-r'(T_c+\Delta t_5)}+F_6^{'}\cdot e^{-r'(T_c+\Delta t_6)}}{E\cdot\frac{1-e^{-r'T_c}}{r'}}$$

(7.30)

From the above equation we find that the reference date t_0 is not necessary for calculation of the levelized fuel cycle cost, although we selected it as the discount basis time at the beginning of the derivation.

Actually Equation (7.30) can also be written in a different form:

$$C=\frac{\sum_{i=1}^{4}F_i^{'}\cdot e^{r'\Delta t_i}+F_5^{'}\cdot e^{-r'(T_c+\Delta t_5)}+F_6^{'}\cdot e^{-r'(T_c+\Delta t_6)}}{E\cdot\frac{1-e^{-r'T_c}}{r'}}$$

$$=\frac{\sum_{i=1}^{4}\frac{F_i^{'}}{M_p}\cdot e^{r'\Delta t_i}+\frac{F_5^{'}}{M_p}\cdot e^{-r'(T_c+\Delta t_5)}+\frac{F_6^{'}}{M_p}\cdot e^{-r'(T_c+\Delta t_6)}}{\frac{E\cdot T_c}{M_p}\cdot\frac{1-e^{-r'T_c}}{r'T_c}} \tag{7.31}$$

$$=\frac{\sum_{i=1}^{4}\frac{F_i^{'}}{M_p}\cdot e^{r'\Delta t_i}+\frac{F_5^{'}}{M_p}\cdot e^{-r'(T_c+\Delta t_5)}+\frac{F_6^{'}}{M_p}\cdot e^{-r'(T_c+\Delta t_6)}}{B_d\cdot\eta\cdot\frac{1-e^{-r'T_c}}{r'T_c}}$$

where:

$$B_d = \frac{E \cdot T_c}{M_p \cdot \eta}, \text{ discharge burnup}$$

$\eta =$ Thermal efficiency

Equation (7.31) shows how the levelized fuel cycle cost can be calculated using the discharge burnup.

It should be noted that Equation (7.31) can only apply for a single batch core. For a multi-batch core, the initial core is usually loaded with all fresh fuel at the beginning of the reactor life, but the fuel has lower enrichment than the equilibrium core. The core will reach the equilibrium state after several cycles of reloading.

In order to simplify the analysis, assume that the initial core is an equilibrium core. Hence the costs for burned fuel batches can be calculated for an N-batch loading core as follows

$$\begin{aligned}
&\textit{Fuel } \cos t \textit{ of } N-1 \textit{ burned fuel batches} \\
&= \int_0^{(N-1)T_c} C \cdot E(t+T_c) \cdot e^{-r't} dt + \int_0^{(N-2)T_c} C \cdot E(t+2T_c) \cdot e^{-r't} dt + \cdots + \int_0^{T_c} C \cdot E(t+(N-1)T_c) \cdot e^{-r't} dt \\
&= C \cdot E'
\end{aligned} \tag{7.32}$$

where

C	Levelized fuel cycle cost
N	Number of batches in core
$E(t)$	Net electricity output per year from a batch as a function of time t
E	Whole core net electricity output per year
E'	Discounted net electricity output from N-1 burned fuel batches
T_c	Refueling cycle length, years

If we assume the energy generated from a fuel batch is uniform during its life, that is,

$E(t) = E/N$, we have

$$E^{'} = \frac{E}{N}\cdot\left(\int_0^{(N-1)T_c} e^{-r't}dt + \int_0^{(N-2)T_c} e^{-r't}dt + \cdots + \int_0^{T_c} e^{-r't}dt\right) = \frac{E}{N}\cdot\frac{N - \dfrac{1-e^{-r'NT_c}}{1-e^{-r'T_c}}}{r'} \qquad (7.33)$$

Substituting Equation (7.33) into (7.32), we have

$$Fuel\ \cos t\ of\ \ N-1\ burned\ \ fuel\ batches = C\cdot E^{'} = C\cdot\frac{E}{N}\cdot\frac{N - \dfrac{1-e^{-r'NT_c}}{1-e^{-r'T_c}}}{r'} \qquad (7.34)$$

In addition, at the end of the reactor life, there are still N-1 fuel batches which are not burned up. If the cost for those N-1 fuel batches is recoverable, it should be subtracted from total fuel cost. The present value of those M-1 fuel batches is $C\cdot E^{'}\cdot e^{-r'L}$.

So the levelized fuel cycle cost is

$$C = \frac{C\cdot E^{'} + \sum_{i=1}^{4} F_i^{'}\cdot e^{r'\Delta t_i}\cdot\dfrac{1-e^{-r'L}}{1-e^{-r'T_c}} + F_5^{'}\cdot e^{-r'(T_c+\Delta t_5)}\cdot\dfrac{1-e^{-r'L}}{1-e^{-r'T_c}} + F_6^{'}\cdot e^{-r'(T_c+\Delta t_6)}\cdot\dfrac{1-e^{-r'L}}{1-e^{-r'T_c}} - C\cdot E^{'}\cdot e^{-r'L}}{\int_{t_0}^{t_0+L} e^{-r'(t-t_0)}\cdot Edt}$$

$$= \frac{C\cdot E^{'} + \sum_{i=1}^{4} F_i^{'}\cdot e^{r'\Delta t_i}\cdot\dfrac{1-e^{-r'L}}{1-e^{-r'T_c}} + F_5^{'}\cdot e^{-r'(T_c+\Delta t_5)}\cdot\dfrac{1-e^{-r'L}}{1-e^{-r'T_c}} + F_6^{'}\cdot e^{-r'(T_c+\Delta t_6)}\cdot\dfrac{1-e^{-r'L}}{1-e^{-r'T_c}} - C\cdot E^{'}\cdot e^{-r'L}}{E\cdot\dfrac{1-e^{-r'L}}{r'}}$$

(7.35)

From Equation (7.35), we have

$$C = \frac{\sum_{i=1}^{4} F_i^{'}\cdot e^{r'\Delta t_i} + F_5^{'}\cdot e^{-r'(T_c+\Delta t_5)} + F_6^{'}\cdot e^{-r'(T_c+\Delta t_6)}}{E\cdot\dfrac{1-e^{-r'T_c}}{r'}}\cdot\frac{1}{1-\dfrac{E^{'}}{\dfrac{E}{r'}}} \qquad (7.36)$$

Substituting Equation (7.33) into Equation (7.36), we have

$$C = \frac{\sum_{i=1}^{4} F_i^{'} \cdot e^{r'\Delta t_i} + F_5^{'} \cdot e^{-r'(T_c+\Delta t_5)} + F_6^{'} \cdot e^{-r'(T_c+\Delta t_6)}}{E \cdot \frac{1-e^{-r'T_c}}{r'}} \cdot \frac{1}{1-\frac{E^{'}}{\frac{E}{r'}}}$$

$$= \frac{\sum_{i=1}^{4} F_i^{'} \cdot e^{r'\Delta t_i} + F_5^{'} \cdot e^{-r'(T_c+\Delta t_5)} + F_6^{'} \cdot e^{-r'(T_c+\Delta t_6)}}{E \cdot \frac{1-e^{-r'T_c}}{r'}} \cdot \frac{1}{1-\frac{\frac{E}{N} \cdot \frac{N-\frac{1-e^{-r'NT_c}}{1-e^{-r'T_c}}}{r'}}{\frac{E}{r'}}}$$

$$= \frac{\sum_{i=1}^{4} F_i^{'} \cdot e^{r'\Delta t_i} + F_5^{'} \cdot e^{-r'(T_c+\Delta t_5)} + F_6^{'} \cdot e^{-r'(T_c+\Delta t_6)}}{\frac{E}{N} \cdot \frac{1-e^{-r'NT_c}}{r'}}$$

$$= \frac{\sum_{i=1}^{4} F_i^{'} \cdot e^{r'\Delta t_i} + F_5^{'} \cdot e^{-r'(T_c+\Delta t_5)} + F_6^{'} \cdot e^{-r'(T_c+\Delta t_6)}}{\frac{E}{N} \cdot \frac{1-e^{-r'T_b}}{r'}} \tag{7.37}$$

where

$T_b = NT_c$: Fuel Residence time, years

Equation (7.37) provides a simplified way to calculate the levelized fuel cycle cost.

7.2.4 Seed and Blanket Fuel Cycles

For the seed and blanket case such as WASB and SBU, there are two cash flows during the life of the reactor, one is the seed fuel cash flow and the other is the blanket fuel cash flow. The levelized fuel cycle cost of the seed and blanket case is:

$$C=\frac{\left(C_{seed}\cdot E^{'}+\sum_{i=1}^{4}F_i^{'}\cdot e^{r'\Delta t_i}\cdot\frac{1-e^{-r'L}}{1-e^{-r'T_c}}+F_5^{'}\cdot e^{-r'\Delta t_5}\cdot\frac{1-e^{-r'L}}{e^{r'T_c}-1}+F_6^{'}\cdot e^{-r'\Delta t_6}\cdot\frac{1-e^{-r'L}}{e^{r'T_c}-1}-C_{seed}\cdot E^{'}\cdot e^{-r'L}\right)_{seed}}{E\cdot\frac{1-e^{-r'L}}{r'}}$$

$$+\frac{\left(\sum_{i=1}^{4}F_i^{'}\cdot e^{r'\Delta t_i}\cdot\frac{1-e^{-r'L}}{1-e^{-r'T_c}}+F_5^{'}\cdot e^{-r'\Delta t_5}\cdot\frac{1-e^{-r'L}}{e^{r'T_c}-1}+F_6^{'}\cdot e^{-r'\Delta t_6}\cdot\frac{1-e^{-r'L}}{e^{r'T_c}-1}\right)_{blanket}}{E\cdot\frac{1-e^{-r'L}}{r'}}$$

$$=\frac{C_{seed}\cdot E^{'}-C_{seed}\cdot E^{'}\cdot e^{-r'L}}{E\cdot\frac{1-e^{-r'L}}{r'}}+\frac{\sum_{i=1}^{4}F_{i,seed}^{'}\cdot e^{r'\Delta t_i}+F_{5,seed}^{'}\cdot e^{-r'\left(T_{c,seed}+\Delta t_5\right)}+F_{6,seed}^{'}\cdot e^{-r'\left(T_{c,seed}+\Delta t_6\right)}}{E\cdot\frac{1-e^{-r'T_{c,seed}}}{r'}}$$

$$+\frac{\sum_{i=1}^{4}F_{i,blanket}^{'}\cdot e^{r'\Delta t_i}+F_{5,blanket}^{'}\cdot e^{-r'\left(T_{c,blanket}+\Delta t_5\right)}+F_{6,blanket}^{'}\cdot e^{-r'\left(T_{c,blanket}+\Delta t_6\right)}}{E\cdot\frac{1-e^{-r'T_{c,blanket}}}{r'}} \quad (7.38)$$

where

C_{seed} = Seed levelized cost, mills/kWe-hr

$E^{'}=\int_0^{2T_c}E_{seed}\left(t+T_c\right)\cdot e^{-r't}dt+\int_0^{T_c}E_{seed}\left(t+2T_c\right)\cdot e^{-r't}dt$, for a 3-batch seed loading

$E_{seed}(t)$ = One batch seed fuel net electrical output per year at year t

E = Total reactor net electrical output per year, $kWe\cdot hr/yr$

$T_{c,seed}$ = Seed refueling cycle length, years

$T_{c,blanket}$ = Blanket refueling cycle length, years

Assuming the seed energy is uniform during its lifetime in the core and $E_{seed}(t)=\frac{f_{seed}\cdot E}{3}$, we have

$$
\begin{aligned}
E^{'} &= \int_0^{2T_c} E_{seed}\left(t+T_c\right)\cdot e^{-r't}dt + \int_0^{T_c} E_{seed}\left(t+2T_c\right)\cdot e^{-r't}dt \\
&= \frac{f_{seed}\cdot E}{3}\cdot\left(\int_0^{2T_c} e^{-r't}dt + \int_0^{T_c} e^{-r't}dt\right) \\
&= \frac{f_{seed}\cdot E}{3}\cdot\frac{3-\dfrac{1-e^{-r'T_{b,seed}}}{1-e^{-r'T_{c,seed}}}}{r'}
\end{aligned}
\tag{7.39}
$$

where

f_{seed} = Seed power fraction relative to whole core power

$T_{b,seed} = 3T_{c,seed}$, Seed residence time in core, years

From Equation (7.37) we have

$$
C_{seed} = \frac{\sum_{i=1}^{4} F_{i,seed}^{'}\cdot e^{r'\Delta t_i} + F_{5,seed}^{'}\cdot e^{-r'\left(T_{c,seed}+\Delta t_5\right)}}{\dfrac{f_{seed}\cdot E}{3}\cdot\dfrac{1-e^{-r'T_{b,seed}}}{r'}}
\tag{7.40}
$$

Substituting Equations (7.40) and (7.39) into Equation (7.38) we have

$$
\begin{aligned}
C &= \frac{\sum_{i=1}^{4} F_{i,seed}^{'}\cdot e^{r'\Delta t_i} + F_{5,seed}^{'}\cdot e^{-r'\left(T_{c,seed}+\Delta t_5\right)} + F_{6,seed}^{'}\cdot e^{-r'\left(T_{c,seed}+\Delta t_6\right)}}{\dfrac{E}{3}\cdot\dfrac{1-e^{-r'T_{b,seed}}}{r'}} + \frac{\sum_{i=1}^{4} F_{i,blanket}^{'}\cdot e^{r'\Delta t_i} + F_{5,blanket}^{'}\cdot e^{-r'\left(T_{c,blanket}+\Delta t_5\right)} + F_{6,blanket}^{'}\cdot e^{-r'\left(T_{c,blanket}+\Delta t_6\right)}}{E\cdot\dfrac{1-e^{-r'T_{c,blanket}}}{r'}} \\
&= \frac{\sum_{i=1}^{4} \dfrac{F_{i,seed}^{'}}{M_{seed}}\cdot e^{r'\Delta t_i} + \dfrac{F_{5,seed}^{'}}{M_{seed}}\cdot e^{-r'\left(T_{c,seed}+\Delta t_5\right)} + \dfrac{F_{6,seed}^{'}}{M_{seed}}\cdot e^{-r'\left(T_{c,seed}+\Delta t_6\right)}}{\dfrac{f_{seed}\cdot E\cdot T_{b,seed}}{3M_{seed}}\cdot\dfrac{1-e^{-r'T_{b,seed}}}{r'T_{b,seed}}}\cdot f_{seed} + \frac{\sum_{i=1}^{4} \dfrac{F_{i,blanket}^{'}}{M_{blanket}}\cdot e^{r'\Delta t_i} + \dfrac{F_{5,blanket}^{'}}{M_{blanket}}\cdot e^{-r'\left(T_{c,blanket}+\Delta t_5\right)} + \dfrac{F_{6,blanket}^{'}}{M_{blanket}}\cdot e^{-r'\left(T_{c,blanket}+\Delta t_6\right)}}{\dfrac{f_{blanket}\cdot E\cdot T_{c,blanket}}{M_{blanket}}\cdot\dfrac{1-e^{-r'T_{c,blanket}}}{r'T_{c,blanket}}}\cdot f_{blanket} \\
&= \frac{\sum_{i=1}^{4} \dfrac{F_{i,seed}^{'}}{M_{seed}}\cdot e^{r'\Delta t_i} + \dfrac{F_{5,seed}^{'}}{M_{seed}}\cdot e^{-r'\left(T_{c,seed}+\Delta t_5\right)} + \dfrac{F_{6,seed}^{'}}{M_{seed}}\cdot e^{-r'\left(T_{c,seed}+\Delta t_6\right)}}{B_{d,seed}\cdot\eta\cdot\dfrac{1-e^{-r'T_{b,seed}}}{r'T_{b,seed}}}\cdot f_{seed} + \frac{\sum_{i=1}^{4} \dfrac{F_{i,blanket}^{'}}{M_{blanket}}\cdot e^{r'\Delta t_i} + \dfrac{F_{5,blanket}^{'}}{M_{blanket}}\cdot e^{-r'\left(T_{c,blanket}+\Delta t_5\right)} + \dfrac{F_{6,blanket}^{'}}{M_{blanket}}\cdot e^{-r'\left(T_{c,blanket}+\Delta t_6\right)}}{B_{d,blanket}\cdot\eta\cdot\dfrac{1-e^{-r'T_{c,blanket}}}{r'T_{c,blanket}}}\cdot f_{blanket} \\
&= C_{seed}\cdot f_{seed} + C_{blanket}\cdot f_{blanket}
\end{aligned}
\tag{7.41}
$$

Where

M_{seed} = One batch seed heavy metal mass, kg

$M_{blanket}$ = Blanket heavy metal mass, kg

f_{seed} = Seed power fraction

$f_{blanket}$ = Blanket power fraction

$B_{d,seed}$ = Seed discharge burnup

$B_{d,blanket}$ = Blanket discharge burnup

C_{seed} = Seed levelized cost

$C_{blanket}$ = Blanket Levelized cost

η = Thermal efficiency

It can be seen that there are two terms in the above equation: one is the seed cost term; the other is the blanket cost term. The total cost is the sum of the seed and blanket costs weighted by their power fractions in the core. Note that the equation will essentially have the same form as Equation (7.37) if the core only consisted of one type of fuel. Some insights can be extracted from Equation (7.41). The fuel cycle cost decreases as the seed and blanket burnups increase. It can also be reduced if more energy is extracted from the blanket fuel, which means that the power fraction of the blanket increases, hence the power fraction of the seed fuel decreases.

Actually $\frac{F'_{i,seed}}{M_{seed}}$ ($i = 1,2,3,4,5,6$) are the seed component unit costs, but for the blanket one should be careful when calculating $\frac{F'_{i,blanket}}{M_{blanket}}$ because the blanket consists of uranium and thorium, so $F'_{i,blanket}$ must be divided into two terms: $F'_{i,blanket}(U)$ and $F'_{i,blanket}(Th)$. Hence we have

$$\begin{aligned}\frac{F'_{i,blanket}}{M_{blanket}} &= \frac{F'_{i,blanket}(U) + F'_{i,blanket}(Th)}{M_{blanket}} = \frac{F'_{i,blanket}(U)}{M_{blanket}} + \frac{F'_{i,blanket}(Th)}{M_{blanket}} \\ &= \frac{F'_{i,blanket}(U)}{M^U_{blanket}} \cdot \frac{M^U_{blanket}}{M_{blanket}} + \frac{F'_{i,blanket}(Th)}{M^{Th}_{blanket}} \cdot \frac{M^{Th}_{blanket}}{M_{blanket}}\end{aligned} \tag{7.42}$$

Where

$M_{blanket} = M^U_{blanket} + M^{Th}_{blanket}$

$M^U_{blanket}$ = Uranium mass in the blanket, kg

$M^{Th}_{blanket}$ = Thorium mass in the blanket, kg

$$\frac{F'_{i,blanket}(U)}{M^{U}_{blanket}} = \text{Uranium component unit cost, \$/kg}$$

$$\frac{F'_{i,blanket}(Th)}{M^{U}_{blanket}} = \text{Thorium component unit cost, \$/kg}$$

There is another method to derive the levelized fuel cycle cost of the seed and blanket case based on individual seed and blanket's levelized costs (C_{seed} and $C_{blanket}$ as given above):

$$C = \frac{\int_0^L C_{seed} \cdot E_{seed} \cdot e^{-r't} dt + \int_0^L C_{blanket} \cdot E_{blanket} \cdot e^{-r't} dt}{\int_0^L E \cdot e^{-r't} dt} = C_{seed} \cdot \frac{E_{seed}}{E} + C_{blanket} \cdot \frac{E_{blanket}}{E}$$
$$= C_{seed} \cdot f_{seed} + C_{Blanket} \cdot f_{blanket} \tag{7.43}$$

It can be seen that Equation (7.43) is the same as Equation (7.41).

Equation (7.41) is a simplified formula to calculate the levelized fuel cycle cost of the seed and blanket fuel cycle. It should be noted that all analyses are based on the reactor's life history. However the same formula can be obtained from equilibrium cycle analysis.

7.3 WASB Fuel Cycle Cost Analysis

Based on the model developed above, a fuel cycle cost analysis spreadsheet has been developed for the seed and blanket.

7.3.1 Economic Assumptions and Result Analysis

The component unit cost assumptions are summarized in Table 7.2. In order to take into account added fabrication costs associated with high enrichment and annular fuel, the seed fuel fabrication cost is taken to be 500 \$/kg, which doubles the conventional UO_2 fuel fabrication unit cost. 300 \$/kg is used for the blanket fuel fabrication unit cost assuming the ThO_2/UO_2 blanket fuel is more expensive to fabricate. The spent fuel storage cost is assumed to be charged per assembly. This approach ignores one of the seed and blanket core advantages, namely a reduced amount of discharged fuel. The lead time to irradiation for each component is shown in Table 7.3. It should be noted that the lag time to discharge for spent fuel storage is zero in the calculations. In addition, some important assumptions are:

1. Annual continuous discount rate is 8%;
2. Plant thermal efficiency is 33.7%;
3. The refueling period costs are neglected. Thus, the impact of different capacity factors among the various cycle lengths is not accounted for.

Table 7.2 Component Unit Cost

Thorium	50 $/kg
Natural Uranium	50 $/kg
Conversion	6 $/kg
Enrichment	80 $/kgSWU
Fabrication (UO_2 – PWR)	250 $/kg
Fabrication (Seed)	500 $/kg
Fabrication (Blanket)	300 $/kg
Spent Fuel Storage	50000 $/Assembly
Spent Fuel Disposal	500 $/kg

Table 7.3 Lead and Lag Time to Irradiation, months

U_3O_8 (Natural Uranium)	12 (time to irradiation)
Thorium	6 (time to irradiation)
Conversion (UF_6)	6 (time to irradiation)
Enrichment	6 (time to irradiation)
Fabrication	3 (time to irradiation)
Spent Fuel Storage	0 (time after discharge)
Spent Fuel Disposal	90 (time after discharge)

In the fuel cycle cost calculations, the average discharge burnup of the seed fuel is 145 MWd/kgHM and the blanket fuel is at 88 MWd/kgHM. The reference PWR is an all-U PWR case, the enrichment of U-235 in uranium is 4.5 w/o and the discharge burnup is 50 MWd/kgHM. It should be noted that the burnup of the reference case is not calculated by a whole core model, but on a single assembly basis.

Table 7.4 and Figure 7.3 show the fuel cycle cost comparison between the WASB and the reference PWR. It can be seen that, under the above assumptions, the total WASB fuel cycle cost is about 2.2% cheaper than that of the all-U fuel cycle, and is about 8.7% more expensive than the all-U fuel cycle without including the disposal cost. The major contributions to the fuel cycle cost are from the enrichment cost and the uranium ore purchase cost for both cases. The enrichment cost of the WASB is 35% more expensive than that of the all-U case. The WASB is more economic in respect to fabrication cost than the reference case: about 77% of the reference all-U case due to less fuel refueling. The WASB has lower spent fuel storage cost than the all-U case due to reduced spent fuel assembly discharge. Note that the significant advantage of the WASB is the spent fuel disposal cost: it amounts to about 1/3 of the reference all-U case. However unlike the per kg basis used in the Table 7.4 calculation the U.S. DOE charges the fixed value of 1 mill/kWe-hr for spent fuel disposal. The blanket fuel cycle cost is only about 15% of the seed. It should be noted that this analysis ignored differences in burnable poison and soluble boron needs, which affect the fuel cost, also ignored in this analysis are any differences in the outage time associated with refueling.

Table 7.4 Fuel Cycle Cost, mills/kWe-hr

Component	Reference PWR	WASB		
		Seed	Blanket	
			U	Th
Purchase	1.63	1.57	0.13	0.03
Conversion	0.19	0.19	0.01	0
Enrichment	1.53	1.93	0.14	0
Fabrication	0.75	0.34	0.24	
Front End Cost	4.10	4.03	0.55	
Spent Fuel Storage	0.28	0.15	0.02	
Total	**4.38**	**4.76**		
Spent Fuel Disposal	0.72	0.16	0.07	
Total, (+ disposal)	**5.10**	**4.99**		

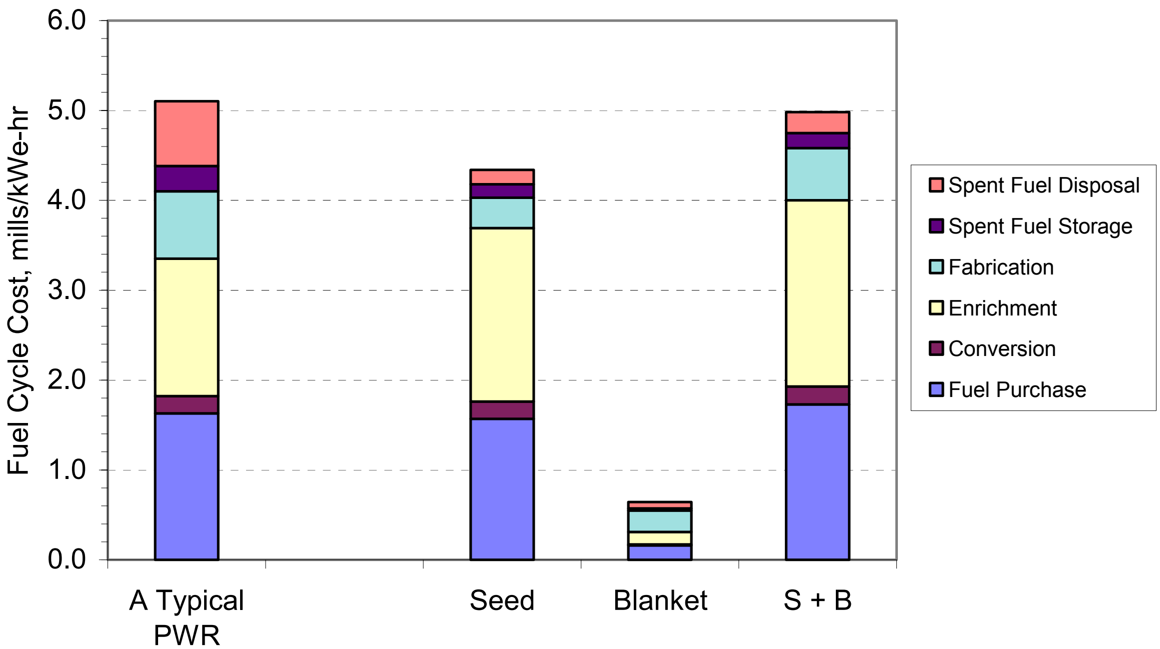


Figure 7.3 Comparison of Fuel Cycle Costs

7.3.2 Sensitivity Analysis

As shown in Equation (7.41), the fuel cycle cost is dependent on parameters such as burnup, cycle length, component unit costs, discount rate, etc. Burnup and cycle length parameters are determined by the neutronic design; hence they are not influenced by economic conditions. However, component unit costs are subject to the nuclear fuel market and technical advancement, and the discount count rate is dependent on the national economy. Hence it is interesting to investigate their effects on the fuel cycle cost.

Figure 7.4 shows the fuel cycle cost of the WASB changes as a function of the uranium ore unit cost. It indicates that the fuel cycle cost is very sensitive to the natural uranium cost. For example, the fuel cycle cost will increase by 1 mill/kWe-hr if the uranium unit cost increases from 50 $/kgU to 80 $/kgU.

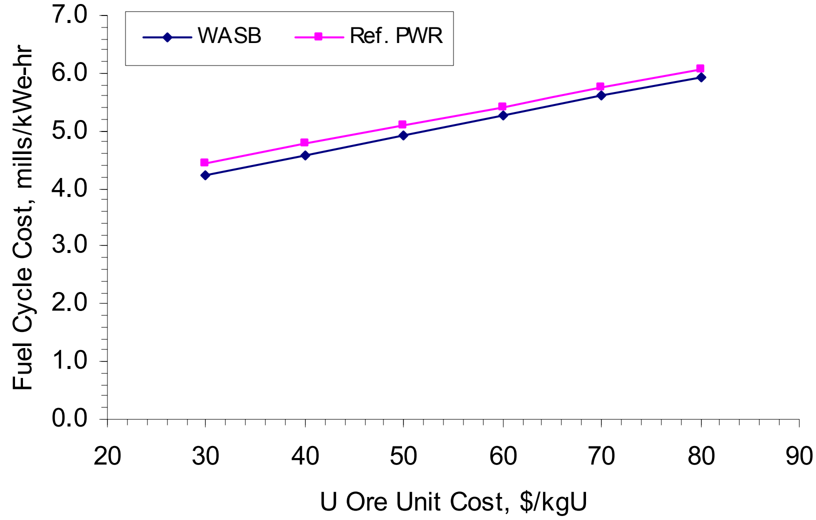


Figure 7.4 Cost versus U Ore Unit Cost

Figure 7.5 shows the impact of varying the SWU unit cost on the fuel cycle cost. The fuel cycle cost increases as the SWU unit cost increases. The slope of the WASB curve is greater than that of the reference case, which means the SWU unit cost has greater impact on the WASB fuel cycle cost because of the greater SWU requirements in the WASB fuel. It can be seen that at 97 $/kgSWU, the WASB fuel cycle cost is equal to the all-U fuel cycle. In the long term it can be predicted that isotopic separation costs should become cheaper. This will further reduce the fuel cycle cost of the WASB, which will make the WASB more competitive.

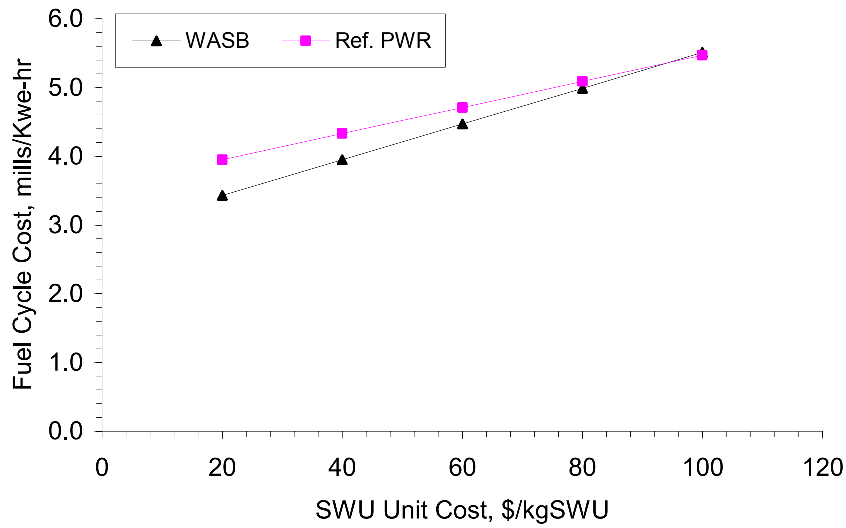


Figure 7.5 Cost versus SWU Unit Cost

The fuel cycle cost versus seed fuel fabrication unit cost is shown in Figure 7.6. The fuel cycle cost will be improved if the seed fuel fabrication cost can be reduced, but the improvement is small because the fabrication cost is only a very small portion of total cost.

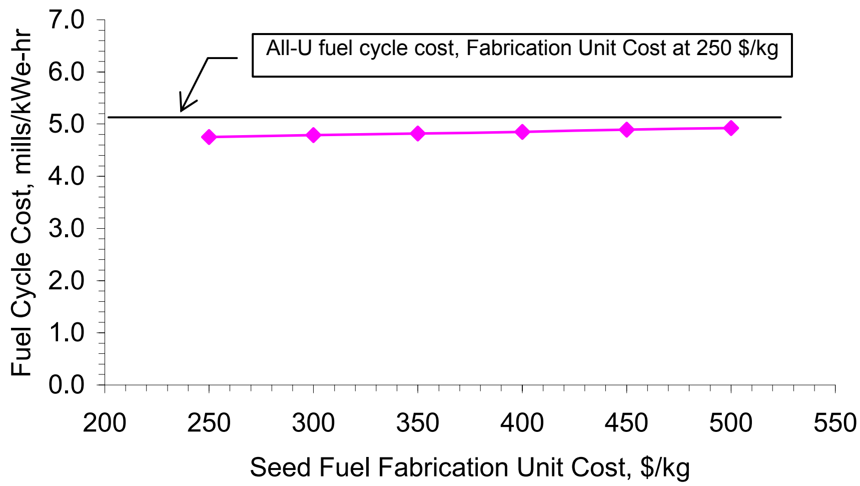


Figure 7.6 Cost versus Seed Fuel Fabrication Unit Cost

Figure 7.7 shows that the waste disposal cost of the conventional all-U case is more sensitive to the disposal unit cost than the WASB fuel cycle because the WASB core has much less waste production. The WASB fuel cycle will be more competitive than the reference case if the disposal unit cost increases. However, the disposal cost is not only dependent on the disposal unit cost, but also on the assumption of the lag time Δt_6 for disposal (measured from discharge) as shown in Equation 7.41. The disposal cost will decrease when the lag time increases.

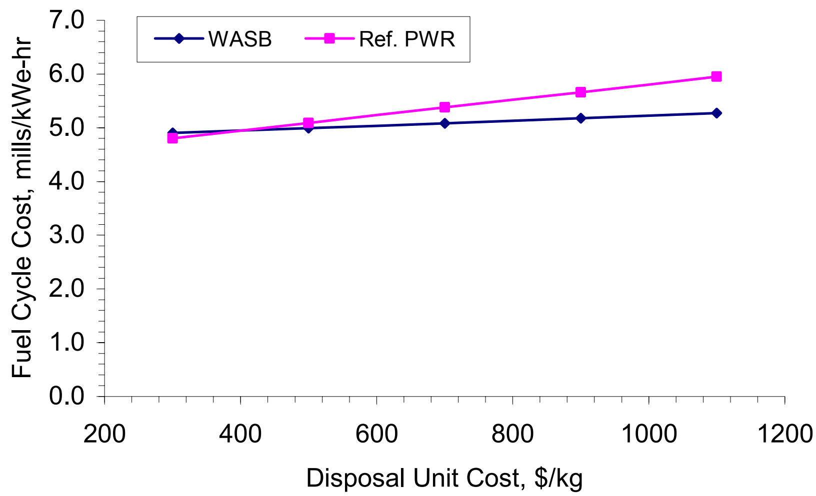


Figure 7.7 Cost versus Spent Fuel Disposal Unit Cost

The discount rate effect on the fuel cycle cost is shown in Figure 7.8. It presents the cost increase as the discount rate increases. The WASB cost increases more quickly than the reference all-U case. At the discount rate of about 9.5% per year the fuel cycle cost of the WASB is equal to that of the reference all-U case. Above this point, the WASB fuel cycle cost is more expensive than the all-U case. Note that the cost here is based on a per kilogram waster disposal fee, not the current 1 mill/kWe-hr fee.

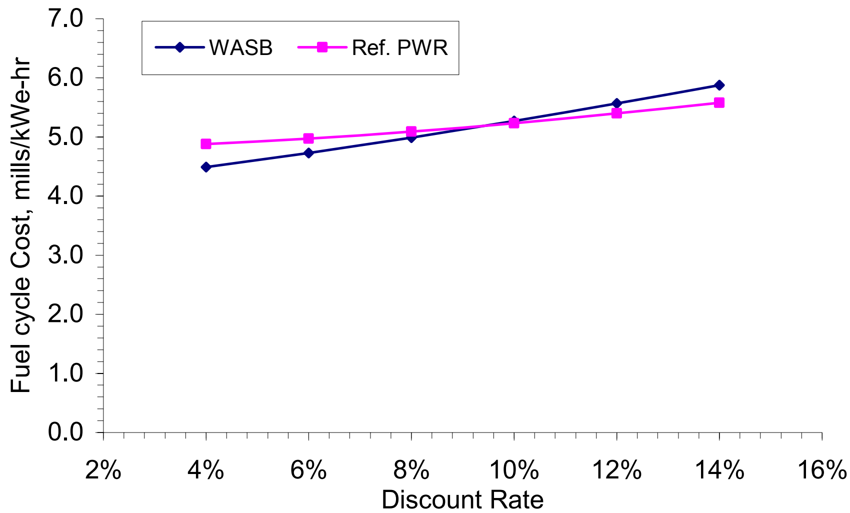


Figure 7.8 Cost versus Discount Rate

7.4 Chapter Summary

A simplified fuel cycle cost calculation model is developed for the WASB fuel cycle economic analysis based on OECD methodology. This model can be also applied to other reactors, where the reactors have multi-batch in-core fuel management schemes.

There are uncertainties in the fuel cycle component unit costs, which will affect the calculated fuel cycle cost calculation results. The results show that the major costs in the fuel cycle cost are the fuel uranium purchase cost and the enrichment cost, which are more than half of the total cost. It can be concluded that the WASB fuel cycle is slightly less expensive than conventional PWRs with respect to reduced fabrication cost, spent fuel storage and disposal costs. However the front end fuel cycle cost of the WASB is 11.7% more expensive than conventional PWRs because of its greater SWU requirements. In the very long term isotopic separation costs should become cheaper and ore more expensive, which will help eliminate this difference. In addition, the fuel cycle cost of the WASB design has two contributors: the seed and the blanket. The seed fuel is much more expensive than the blanket fuel: the blanket fuel cycle cost is only about 13% of the total cost.

Chapter 8

SUMMARY, CONCLUSIONS AND RECOMMENDATIONS

8.1 Summary and Conclusions

In the past several years, there has been renewed interest in thorium fuel as an option for next generation fuel cycles, driven by the incentives of reducing plutonium production and long-term radioactivity of waste. Several design concepts involving thorium have been proposed as once-through, proliferation resistant, high burnup, cycles for a pressurized light water reactor. All these design concepts can be classified into two categories: homogeneously mixed thorium-uranium dioxide fuel and heterogeneous core arrangement of seed and blanket regions. Economic evaluations show that the homogeneously mixed ThO_2-UO_2 fuel is more expensive than the equivalent all-UO_2 fuel in terms of fuel cycle cost. The heterogeneous seed and blanket fuel design concepts improve the thorium fuel utilization, since the thorium fuel will stay in the core for a longer time than the uranium. The work presented here has focused on optimizing a **W**hole **A**ssembly **S**eed and **B**lanket (**WASB**) fuel cycle for PWRs to keep the mechanical design simple and maximize the benefits from inclusion of thorium in the fuel. The major aspect of the WASB approach is that the individual seed and blanket regions each occupy one full-size PWR assembly in an approximate checkerboard array.

8.1.1 Code Validation

WASB core neutronic design and analysis was performed using the Studsvik Core Management System (CMS), which consists of three codes: CASMO-4, TABLES-3 and SIMULATE-3. CASMO-4 can be used to generate 2-group nodal data for SIMULATE-3. The accuracy of the nodal data is dependent upon the accuracy of the CASMO flux distribution, which is used to homogenize and collapse the cross-sections. The most accurate solution may be obtained by modeling the true heterogeneous nature of the lattice in the energy group structure of the nuclear data library.

The CMS code package was benchmarked against MCODE (which couples MCNP and ORIGEN). Based on the benchmark calculations, the agreement between CASMO-4 and MCODE is good, and more than adequate to confirm that CASMO-4, in particular its "colorset" feature, can be used to calculate thorium-fueled seed and blanket assembly arrays for use in PWR cores. SIMULATE always predicts the lowest reactivity values among the three codes. SIMULATE is inconsistent with MCODE at high burnup. Thus the cycle length calculated using SIMULATE will be conservative. In addition, the pin power distribution comparison shows good agreement between SIMULATE and MCODE with the exception of corner pins in an assembly.

8.1.2 Fuel and Core Design

The 193 assembly core of a Westinghouse 4-loop 1150 MWe PWR was selected as the reference plant for this study. Both the seed and blanket assemblies use the standard Westinghouse 17×17 lattice design. The seed and blanket assembly design parameters are summarized in Table 8.1. The overall length of a fuel assembly is 365.76cm. The active fuel length is 335.28 cm. The core loading pattern is devised to meet the requirements of in-core fuel management. There are two separate fuel management streams: a three-batch stream for the seed (18 month cycle length) and a single-batch stream for the blanket, which is to stay in the core for up to 9 seed cycles. Figure 8.1 shows the loading pattern which reduces the flux by placing blanket assemblies at the periphery of the core. The seed assemblies and other blanket assemblies are scattered inside the core in a "checkerboard" scheme. It should be noted that the fresh seed assemblies are located in the core interior as far as possible. The main advantage of this scheme is that it produces a fairly uniform power distribution throughout the core, and therefore leads to lower power peaking and a low-leakage core.

Table 8.1 Seed and Blanket Assembly Optimized Parameters

PARAMETERS	VALUES	
	Seed	Blanket
Fuel Material Composition	UO_2 20 w/o U-235 in U	87 v/o ThO_2 + 13 v/o UO_2 10 w/o U-235 in U
Fuel Density, g/cm^3	10.302	9.403
Fuel Assembly Pitch, cm	21.5	
Fuel Pellet Radius, cm	Void: 0.22, Fuel: 0.385	0.4646
Gas Gap, cm	0.0082	
Cladding Thickness, cm	0.0572	
Pin Pitch, cm	1.26	
Moderator to Fuel Volume Ratio[a]	3.50	1.26

a. Conventional PWR $V_m / V_f = 1.95$

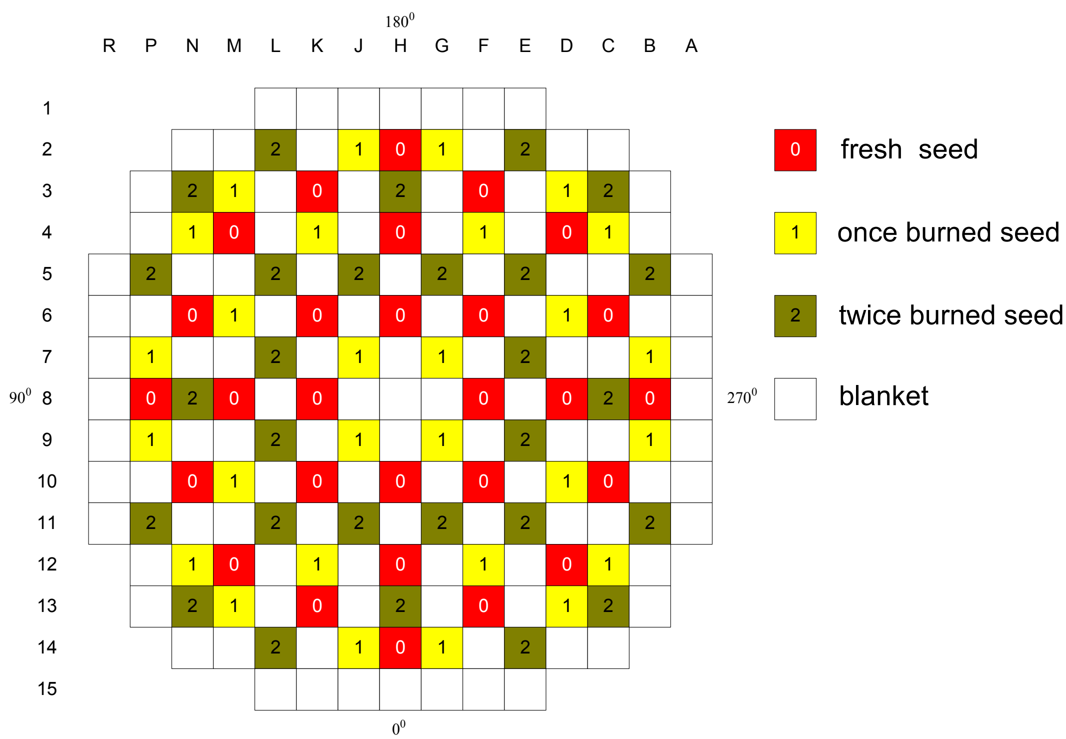


Figure 8.1 The WASB Core Layout and Refueling Scheme

The 20% enriched annular UO_2 seed fuel is a wet lattice to improve neutronic performance and limits the fissile plutonium buildup. Even higher moderator to fuel volume ratios might be achieved by using thinner fuel rods, but a more practical seed fuel pin diameter, of 0.9 cm diameter, was used for the seed fuel because of mechanical and thermal-hydraulic concerns.

The blanket fuel uses a mixture of ThO_2 and UO_2, in which UO_2 serves to denature U-233 bred in the fuel and to share thermal power at the BOL. Detailed analyses were performed to determine the appropriate mixing ratio of ThO_2 and UO_2. The smallest UO_2 proportion needs to be about 13% by volume in order to meet non-proliferation requirements in the blanket fuel. In addition, a tight blanket lattice (fatter fuel rod of 1.06 cm diameter) design leads to better neutronic performance, that is, higher reactivity and a flatter core reactivity curve.

Analyses show that an even distribution of erbium in the seed fuel rods yields a fairly constant reactivity for the cycle, and improves inner assembly power peaking factors. Erbium must be mixed with another material which acts as a void plug, as in the B&W patent cited earlier. By making the plug highly porous it can still serve the function of providing added volume to accommodate gaseous fission products.

The analyses show that the WASB core satisfies the safety and control limits of conventional PWRs. The coefficients of reactivity are comparable with currently operating PWRs, however the MTC of the WASB is more negative than a typical 18-month cycle PWR. In addition the reduction in effective delayed neutron fraction (β_{eff}) requires careful review of the control systems because of their importance to short term power transients.

Whole core analyses show that the total control rod worth of the WASB core is about 1/3 less than those of a typical PWR for the standard arrangement of Ag-In-Cd RCCAs in the core. The use of enriched boron in the control rods can effectively improve the control rod worth. In addition calculations indict that the control rods have higher worth in the seed than those in the blanket. Therefore a new control loading pattern has been designed specifically so that almost all the control rods will be inserted in seed assemblies, however the new pattern requires a redesign of the vessel head of the reactor, which is an added cost in case of retrofitting in existing PWRs.

Obviously all physics aspects of the WASB concept analysis need further refinement. Here only the major issues are addressed:

- Blanket reactivity degradation in later life may require either a gradual increase in the fuel loading of the seed, or earlier refueling of the blanket than 9 seed cycles, or a staggered schedule for blanket replacement.
- The control rod needs to be redesigned because the worth in the WASB core is lower than in a typical PWR due to the hard neutron spectrum in the core, and transient and accident scenarios must be simulated; particular attention should be given to the rod ejection accident.
- Fuel pin performance for very high burnup requires advanced cladding material qualification for long residence periods to withstand water side corrosion and increased fission gas production.

8.1.3 Thermal-hydraulics and Fuel Performance

Thermal-hydraulic performance and fuel behavior of the WASB core have been studied in detail. The thermal-hydraulic analyses show that acceptable MDNBR in the core can be achieved even assuming 12% overpower, increased inlet coolant temperature, and reduced flow rate, if a large local grid loss coefficient is used for the blanket. The results show that the blanket has higher thermal safety margin than the seed. Five variables were quantitatively analyzed to check the sensitivity of MDNBR to those variables: core inlet temperature, coolant flow rate, core power, form loss coefficient, and seed rod diameter.

The more demanding power histories of the seed at BOC and the longer residence time of the blanket bring challenges to the thermal and mechanical performance of fuel rods. The calculated results show the internal pressure of the seed rod is more of a concern than that of the blanket because the large expected fission gas release in the seed fuel leads to an internal pressure of the seed fuel that is greater than the system pressure at EOL. However, further analysis shows that the internal pressure can be effectively reduced if a longer plenum (three times the original size) is used for the seed fuel rods. This will result in higher pressure drop in the core, and may be too difficult to accommodate in a retrofit PWR. The calculated results indicate that cladding materials with improved corrosion resistance such as M5 are required to prevent excessive cladding corrosion.

8.1.4 Spent Fuel Characteristics and Proliferation Resistance

Spent fuel characteristics and proliferation resistance potential of the WASB fuel cycle have been analyzed. It can be concluded that the WASB produces 40% fewer spent fuel assemblies than the conventional UO_2 design and 50% less spent fuel heavy metal in MTHM/GWe-yr. The radioactivity and decay heat of the WASB spent fuel is higher than the conventional UO_2 design in terms of unit heavy metal mass because of higher burnups. It shows that during the first 20,000 years or so, WASB has lower decay heat than the UO_2 fuel in Watts/GWe-yr. Thereafter, the decay heat of the WASB design is higher than that of the conventional UO_2 design, but only at the level of 200 Watts/GWe-yr and lower.

As in any uranium-fueled reactor, the seed of the WASB core produces plutonium, but in lesser quantities than in a conventional reactor and with far higher isotopic degradation. Both characteristics would make the plutonium considerably less desirable for weapons than ordinary reactor-grade plutonium. The presence of U-233 in the blanket can be limited to a concentration below that of proliferation concern. Preliminary analyses show that seed spent fuel could be refabricated and burned again in a conventional LWR core if fission products are removed and only the actinides are recycled. The reprocessing and refabrication of blanket spent fuel is not discussed in this study.

8.1.5 Fuel Cycle Cost Modeling and Analysis

A fuel cycle cost model was developed for the WASB fuel cycle economic analysis based on OECD methodology. This model can be also applied to other reactors, where the reactors have multi in-core fuel management schemes.

There are uncertainties in the fuel cycle component unit costs, which will affect the fuel cycle cost calculation results. The results show that the major costs in the fuel cycle cost are the fuel uranium purchase cost and the enrichment cost, which are more than half of the total cost. The seed fuel cycle is much more expensive than that of the blanket, and the blanket fuel cycle cost is only about 13% of the total cost. It can be concluded that the WASB design would be 2% cheaper than conventional PWRs if the fabrication cost, spent fuel storage costs remain at what they are today and $500 per kg disposal costs are imposed. However the front end fuel cycle cost of the WASB is 11.7% more expensive than conventional PWRs because of its greater SWU requirements. In the very long term isotopic separation costs should become cheaper and the ore more expensive, which will help eliminate this difference.

8.2 Recommendations for Future Work

Although extensive core design and performance assessment of the WASB core have been done in this work, there are some important issues, which require further investigation in the future:

- The present work only evaluates steady-state considerations. To complete the design, the core should be analyzed under transient conditions. In view of the increased rod worth in the fresh seed assemblies and reduced effective delayed neutron fraction in the core indicated in this study, particular attention should be given to the rod ejection accident.
- For safety analysis, an effort should also be made to construct a model of a PWR plant (RELAP5) for the accident analysis of a WASB core. This is important to verify that the transient behavior of the core exhibits adequate safety margin.
- Using a 19 × 19 instead of a 17 × 17 pin array for seed assemblies will probably yield a negligible neutronics effect. However, the lower linear power of the seed fuel will have thermal benefits that may be significant. Using annular fuel with both internal and external cooling for the seed fuel would be a good way to increase the thermal margin to DNB. Pellet temperatures will be lower and so will be the oxide layer thickness, fission gas release and rod internal pressure. These options should be examined.
- A transition core design and fuel management scheme should be worked out to permit changing from conventional to WASB loading. This would permit retrofitting

current PWRs without having to batch-discharge their existing cores which contain tens of millions of dollars of unburned fuel at end of cycle.

- More detailed evaluation of other costs associate with transition is needed: replacement of control rod and possibly the reactor vessel head, use of enriched boron as soluble poison, increasing pumping power to accommodate the higher pressure drop of WASB cores, and the cost of fabrication, transportation and storage facilities for 20 w/o enriched seed fuel.

- The impacts of higher burnup seed and blanket spent fuel on waste management should be further studied. For example the effect on specific limiting nuclides such as Tc-99, I-129, and Np-237. The capacity of the repository should be addressed.

- A way to translate both waste property improvements and proliferation resistant attributes into economic credits should be developed. Otherwise utilities will not have sufficient interest to give serious consideration to using thorium in a seed and blanket core configuration.

- The nuclear fuel reprocessing option is being re-evaluated in the US. If implemented, the use of the fissile material in the spent WASB fuel should be examined.

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Appendix A. WASB Design Evolution

In an effort to meet the design challenges, seed and blanket assemblies and core designs have undergone significant evolution during the four years of the project. The design history of these changes is discussed in the following paragraphs.

1. Assembly Neutronics Analyses (September 1999 – August 2000)

The seed and blanket assembly parameters are shown in Table A.1. The standard PWR assembly values are in the leftmost data column.

Table A.1 Assembly Parameters (for 9/99 – 10/00 Core)

Parameter	Standard PWR	WASB I	
		Seed 001	Blanket 101
Fuel assembly pitch, cm	21.5	21.5	21.5
Fuel material composition	UO_2	UO_2	$(U+Th)O_2$
U enrichment, w/o U-235 in U	3.5 to 5.0	20	12
UO_2 volume fraction, $UO_2/(U+Th)O_2$, %	100	100	10
Number of fuel rods	264	264	288
Fuel pellet outer radius, cm	0.4096	0.377	0.4096
Fuel pellet inner radius, cm	0	0.28	0
Gas gap, cm	0.0082	0.0082	0.0085
Cladding thickness, cm	0.0575	0.03	0.0575
Fuel rod radius, cm	0.475	0.415	0.475
Fuel cell pitch, cm	1.26	1.26	1.26
Moderator/fuel volume ratio	1.948	5.46	1.24
Mass HM per assembly, kgHM	~454	173	469

Neutronic analyses were performed based on a CASMO-4 colorset model as shown in Figure A.1 [T-1].

The proposed WASB core has two separate fuel management streams: a three-batch stream for the seed (18 month cycle length) and a single-batch stream for the blanket, which is to stay in the core for up to 9 seed cycles. CASMO cannot model a reload core, hence a

SIMULATE whole core model is needed in order to investigate and optimize the WASB core design.

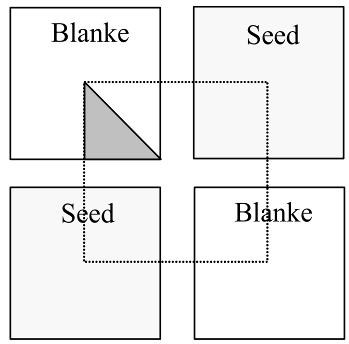

Figure A.1 CASMO Colorset Calculation Model

2. Whole Core Analyses (September 2000 – December 2000)

Two core configurations were investigated in this period as shown in Figure A.2 [T-1]. Configuration A has 96 seed assemblies and 97 blanket assemblies, all control rods are in the blanket assemblies. Configuration B has 97 seed assemblies and 96 blanket assemblies, all control rods are in the seed assemblies. Whole core neutronic analyses were performed using SIMULATE-3.

The seed and blanket design parameters are shown in Table A.2. The major change is in the rod number of a blanket assembly (changed from 288 to 264). Analyses showed that both A and B have high neutron leakage, and the control rod worth in he blanket assemblies is greater than in the seed assemblies.

Table A.2 Seed and Blanket Assembly Parameters (for 9/00 – 12/00 core)

Parameter	WASB II	
	Seed 001	Blanket 102
Fuel assembly pitch, cm	21.5	21.5
Fuel material composition	UO_2	$(U+Th)O_2$
U enrichment, w/o U-235 in U	20	12
UO_2 volume fraction, $UO_2/(U+Th)O_2$, %	100	10
Number of fuel rods	264	264
Fuel pellet outer radius, cm	0.377	0.4096
Fuel pellet inner radius, cm	0.28	0
Gas gap, cm	0.0082	0.0082
Cladding thickness, cm	0.03	0.0575
Fuel rod radius, cm	0.415	0.475
Fuel cell pitch, cm	1.26	1.26
Moderator/fuel volume ratio	5.46	1.948
Mass HM per assembly, kgHM	173	419

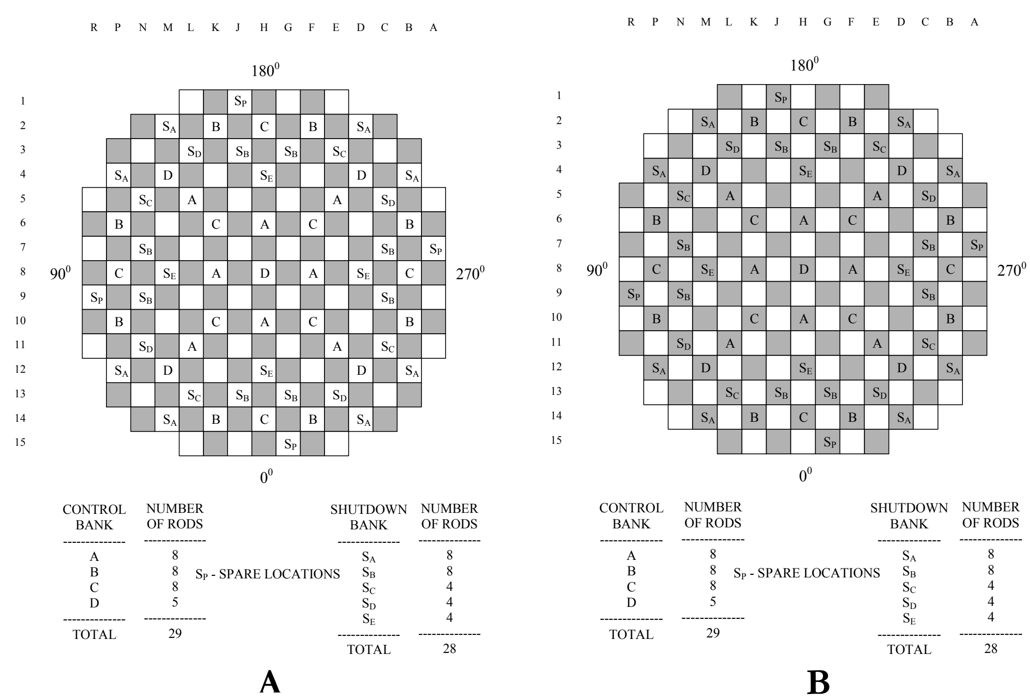

Figure A.2 WASB Core Configuration and Rod Cluster Control Assembly Pattern (Seed Assemblies are Shaded)

3. Whole Core Analyses (January 2001 – December 2001)

This WASB core has 84 seed assemblies and 109 blanket assemblies. The seed and blanket assembly parameters are shown in Table A.3. A seed 002 assembly has more fuel loading than a seed 001 assembly. There are two separate fuel management flows: a standard three-batch scheme was adopted for the seed (18 month cycle length) and a single-batch for the blanket, which is to stay in the core for up to 9 seed cycles.

Core neutronic analyses were performed using SIMULATE-3. Analyses showed that this WASB core has a lower neutron leakage than the core designs shown in Figure A.2, and it can achieve the 18 month cycle length, but the power peaking factors are very high in the core [T-1].

Table A.3 Seed and Blanket Assembly Parameters (for 1/01 – 12/01 core)

Parameter	WASB III	
	Seed 002	Blanket 102
Fuel assembly pitch, cm	21.5	21.5
Fuel material composition	UO_2	$(U+Th)O_2$
U enrichment, w/o U-235 in U	20	12
UO_2 volume fraction, $UO_2/(U+Th)O_2$, %	100	10
Number of fuel rods	264	264
Fuel pellet outer radius, cm	0.385	0.4096
Fuel pellet inner radius, cm	0.22	0
Gas gap, cm	0.0082	0.0082
Cladding thickness, cm	0.0572	0.0575
Fuel rod radius, cm	0.4504	0.475
Fuel cell pitch, cm	1.26	1.26
Moderator/fuel volume ratio	3.50	1.948
Mass HM per assembly, kgHM	275	419

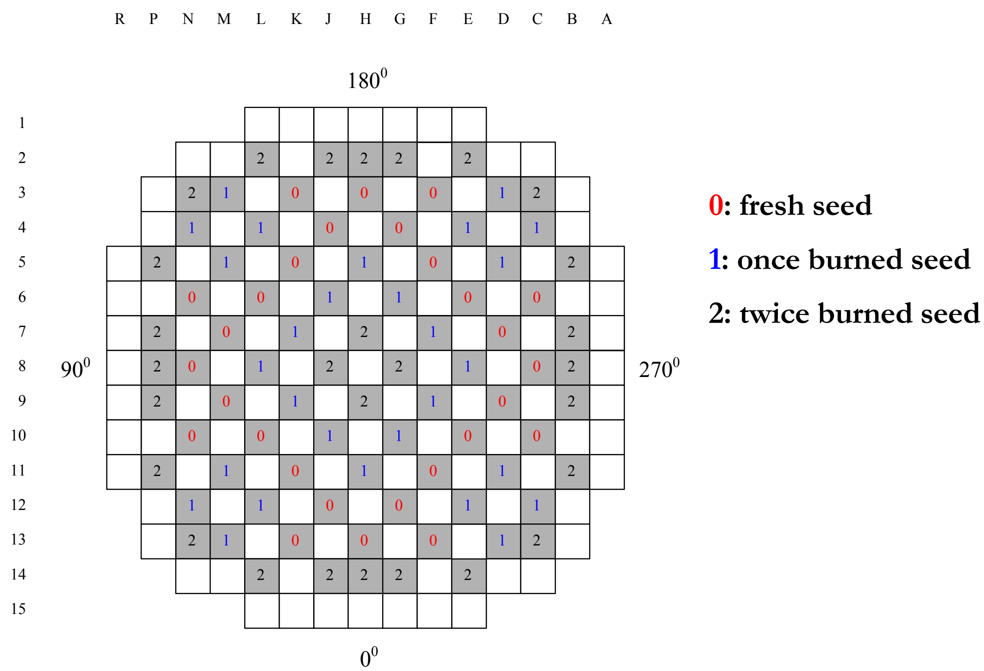

Figure A.3 WASB Core Configuration and Rod Cluster Control Assembly Pattern (Seed Assemblies are Shaded)

4. WASB Core Analyses (December 2001– August 2002)

The core layout is shown in Figure A.4. There are 84 seed assemblies and 109 blanket assemblies in the Core. The seed and blanket assembly design parameters are shown in Table A.4. The major change is the blanket fuel pin diameter, which is fatter than the blanket design in Table A.4. A tight blanket lattice (fatter fuel rod of 1.06 cm diameter) design leads to better neutronic performance, that is, higher reactivity and a flatter core reactivity curve [T-2].

Axial low enrichment regions were used for seed fuel pins in order to reduce axial neutron leakage. In addition, axial poison zoning design was also applied in the seed fuel design. There are no axial low enrichment regions for the blanket fuel pins.

Table A.4 Seed and Blanket Assembly Parameters (for 12/01 – 8/02)

Parameter	WASB IV	
	Seed 002	Blanket 103
Fuel assembly pitch, cm	21.5	21.5
Fuel material composition	UO_2	$(U+Th)O_2$
U enrichment, w/o U-235 in U	20	15
UO_2 volume fraction, $UO_2/(U+Th)O_2$, %	100	10
Number of fuel rods	264	264
Fuel pellet outer radius, cm	0.385	0.4646
Fuel pellet inner radius, cm	0.22	0
Gas gap, cm	0.0082	0.0082
Cladding thickness, cm	0.0572	0.0575
Fuel rod radius, cm	0.4504	0.53
Fuel cell pitch, cm	1.26	1.26
Moderator/fuel volume ratio	3.50	1.26
Mass HM per assembly, kgHM	275	539

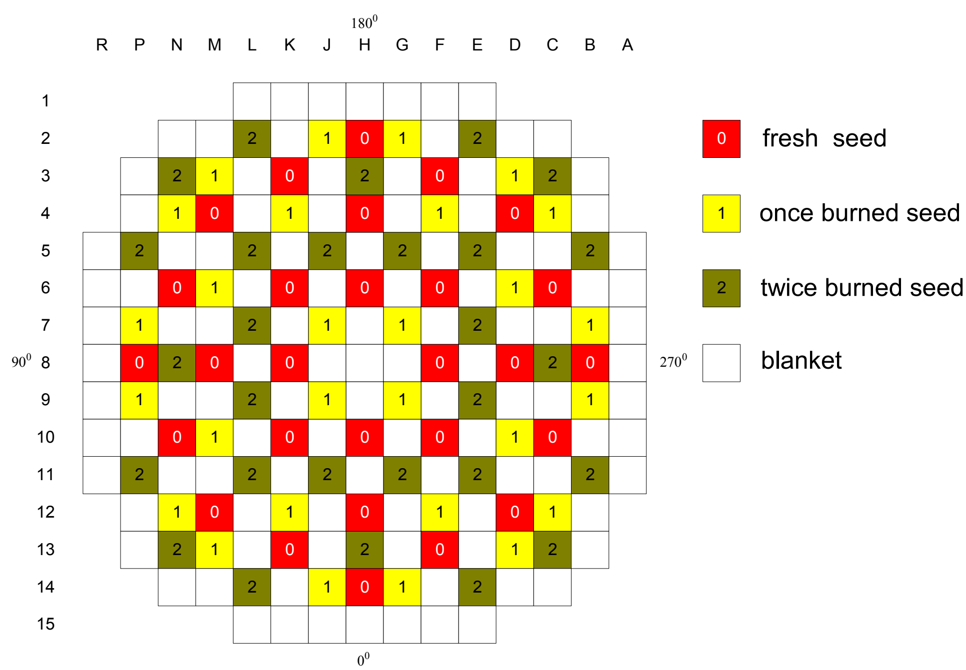

Figure A.4 The WASB Core Layout and Refueling Scheme (Seed Assemblies are Shaded)

5. WASB Core Analyses (September 2002– December 2002)

The core layout is shown in Figure A.5, which is the same as the WASB IV core. The seed and blanket assembly design parameters are shown in Table A.5. The major change is the UO_2 volume fraction in the blanket fuel. The blanket fuel uses a mixture of ThO_2 and UO_2, in which a small amount of UO_2 serves to denature U-233 bred in the fuel and to share thermal power at the BOL. Detailed analyses were performed to determine the appropriate mixing ratio of ThO_2 and UO_2. The smallest UO_2 ratio needs to be about 13% (volume percent) in order to meet non-proliferation requirements in the blanket fuel [T-3].

The thesis work is based on this design.

Table A.5 Seed and Blanket Assembly Parameters

Parameter	WASB V	
	Seed 002	Blanket 104
Fuel assembly pitch, cm	21.5	21.5
Fuel material composition	UO_2	$(U+Th)O_2$
U enrichment, w/o U-235 in U	20	10
UO_2 volume fraction, $UO_2/(U+Th)O_2$, %	100	13
Number of fuel rods	264	264
Fuel pellet outer radius, cm	0.385	0.4646
Fuel pellet inner radius, cm	0.22	0
Gas gap, cm	0.0082	0.0082
Cladding thickness, cm	0.0572	0.0575
Fuel rod radius, cm	0.4504	0.53
Fuel cell pitch, cm	1.26	1.26
Moderator/fuel volume ratio	3.50	1.26
Mass HM per assembly, kgHM	275	541

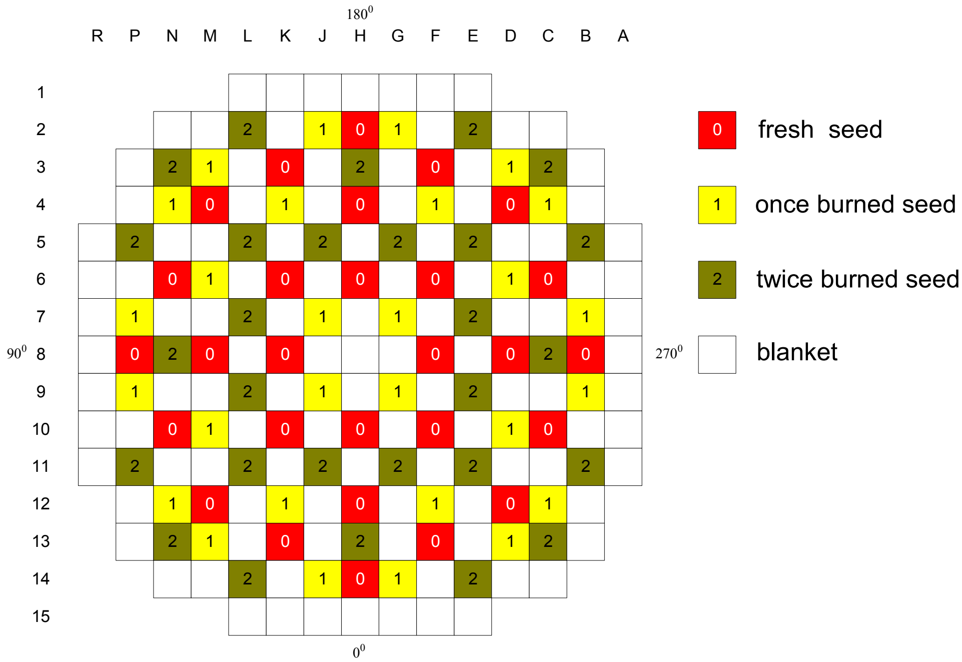

Figure A.5 The WASB Core Layout and Refueling Scheme
(Seed Assemblies are Shaded)

6. A try: 4-batch loading stream for the seed

We also have designed and investigated a WASB VI core design shown in Figure A.6. There are 96 seed assemblies and 97 blanket assemblies in the core. The in-core fuel management is: a 4-batch loading for the seed (18 month cycle length) and a single batch for the blanket, which is to stay in the core for up to 9 seed cycles. We thought this design could increase the seed discharge burnup, but the calculations didn't show that. Its power peaking factors are higher than those of the design of choice shown in Figure A.5. However it has less assembly refueling than the design of choice: 24 seed assemblies will be refueled for each cycle instead of 28 seed assemblies for the design of choice. In addition, the seed fuel assemblies are to stay in the core for six years.

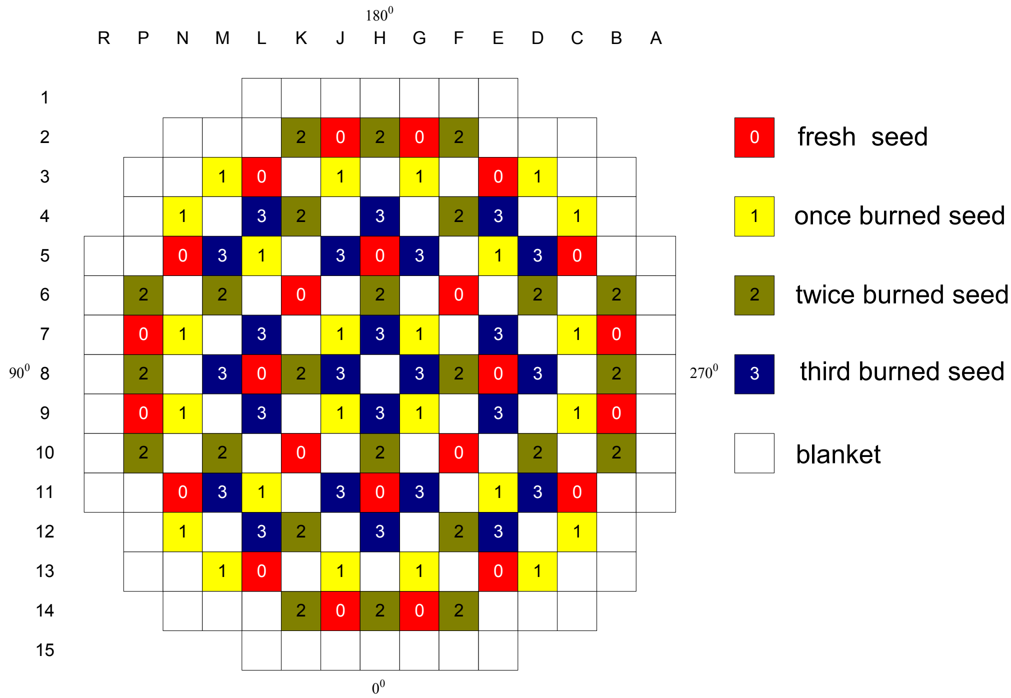

Figure A.6 A New WASB Core Layout and Refueling Scheme
(Seed Assemblies are Shaded)

Appendix B. WASB Core Input Files for the CMS Code Package

As described in the code description for the CMS package in Chapter 1, three types of input files are needed for CASMO-4, TABLES-3, SIMULATE-3, respectively. This appendix includes 18 CASMO-4 input files (9 seed fuel assembly input files 9 blanket fuel assembly input files), one TABLES-3 input file, and 3 SIMULATE-3 input files). The WASB core shown in Figure B.1 is modeled in these input files. The seed and blanket assemblies are shown in Table B.1. The detailed core design discussion can be found in Chapters 3 and 4.

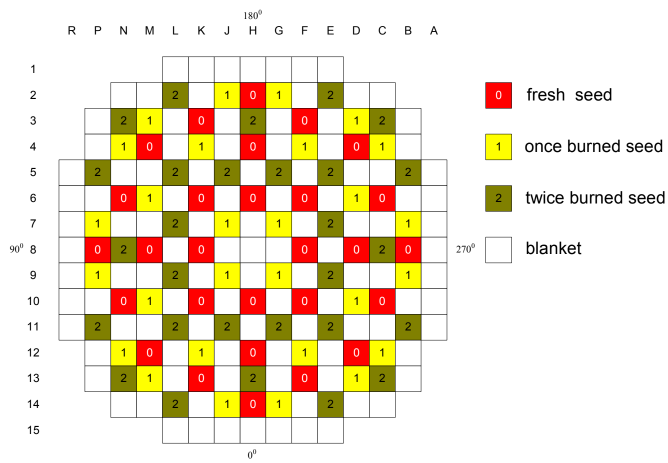

Figure B.1 The WASB Core Layout and Refueling Scheme

Table B.1 Seed and Blanket Assembly Optimized Parameters

PARAMETERS	VALUES	
	Seed	Blanket
Fuel Material Composition	UO_2 20 w/o U-235 in U	87 v/o ThO_2 + 13 v/o UO_2 10 w/o U-235 in U
Fuel Density, g/cm^3	10.302	9.403
Fuel Assembly Pitch, cm	21.5	
Fuel Pellet Radius, cm	Void: 0.22, Fuel: 0.385	0.4646
Gas Gap, cm	0.0082	
Cladding Thickness, cm	0.0572	
Pin Pitch, cm	1.26	
Moderator to Fuel Volume Ratio[a]	3.50	1.26

a. Conventional PWR $V_m / V_f = 1.95$

CASMO Input Files

Seed Assembly:

1. c4.seed_base.inp

```
*       CASMO-4 Input
*       Dean Wang
*       Massachusetts Institute of Technology

TTL,TFU=900,TMO=583.1,BOR=450,VOI=0.0
SIM,'SEED'
MI1,6.550,.7000E-05/302=100.0                                 *ZR4
MI2,7.900,.1800E-04/347=100.0                                *SS347
MI3,8.200,.1800E-04/718=100.0                                *INC718
MI4,8.200,.1800E-04/750=100.0                                *INC750
* MI5,10.16,.2250E-04/47000=80.00,49000=15.00,48000=5.000    *AGINCD
*MI5,2.50/5000=78.2600,6000=21.7400                          *B4C
MI5,5.0/5000=89.1300,6000=10.8700                            *B4C,
double B-10
MI6,2.260,.3250E-05/8000=54.81,14000=37.87,13000=3.440,
5010=.7100,5011=3.170                                        *BORSHIM
MI7,0.1573/5010=100                                          *IFBA
MI8,2.2543/718=73.82,8000=12.33,13000=13.85                  *Pt DET
BOX,6.550,.7000E-05/302=100.0                                *ZR4
FUE,1,10.3024/17.6254,92234=0.141,92238=70.5016,8000=11.732
FUE,2,0.1/0,92238=0,68167=87.434,8000=12.566
PIN,1,.22,.385,.3932,.4504/"AIR","1","AIR","CAN"           *Seed Fuel
PIN,2,.5690,.6147/"COO","BOX"                              *Guide Tube
PIN,5,.1727,.2553,.4166,.5042,.5690,.6147/"AIR","MI3","MI8",
"MI3","COO","BOX"/2,4,6/1,"DET"                            *INSTR THIM
PIN,6,.1727,.2553,.4166,.5042,.5690,.6147/"AIR","MI3","MI8",
"MI3","COO","BOX"/2,4,6/1                                  *INSTR THIM
PIN,7,.4331,.4369,.4839,.5690,.6147/"MI5","AIR","MI2",
"COO","BOX"/1,3,5/1,"AIC"                                  *RCCA
PIN,8,.4331,.4369,.4839,.5690,.6147/"MI5","AIR","MI2",
"COO","BOX"/1,3,5/1,"BTH"                                  *RCCA
SPA,13.99,.1800E-04,,8.154/718=84.59,347=15.41
PRE,155.1296
PDE,129.58,'KWL'
PWR,17,1.260,21.50
DEP,-100,105,110,120,130,140,150,160,170,180,190,200
LPI
   2
   1  1
   1  1  1
   2  1  1  2
   1  1  1  1  1
   1  1  1  1  1  2
   2  1  1  2  1  1  1
   1  1  1  1  1  1  1  1
   1  1  1  1  1  1  1  1  1
*
STA
```

```
TIT *+ TFU BRA
RES,,0,0.5,1,2,3,4,5,6,7,8,9,10,11,15,17,20,25,30,35,40,45,50,55,60,
65,70,75,80,85,90,95,100,105,110,120,130,140,150,160,170,180,190,200
TFU,565.8,1200
STA
TIT *+ TMO BRA
RES,,0,0.5,1,2,3,4,5,6,7,8,9,10,11,15,17,20,25,30,35,40,45,50,55,60,
65,70,75,80,85,90,95,100,105,110,120,130,140,150,160,170,180,190,200
TMO,565.8,600,610
STA
TIT *+ BOR BRA
RES,,0,0.5,1,2,3,4,5,6,7,8,9,10,11,15,17,20,25,30,35,40,45,50,55,60,
65,70,75,80,85,90,95,100,105,110,120,130,140,150,160,170,180,190,200
BOR,0,900,1500,2000
STA
TIT *+ BOR=900, TMO=565.8
RES,,0,0.5,1,2,3,4,5,6,7,8,9,10,11,15,17,20,25,30,35,40,45,50,55,60,
65,70,75,80,85,90,95,100,105,110,120,130,140,150,160,170,180,190,200
TMO,565.8
BOR,900
STA
TIT *+ CRD BRA: AIC NOIFBA
RES,,0,0.5,1,2,3,4,5,6,7,8,9,10,11,15,17,20,25,30,35,40,45,50,55,60,
65,70,75,80,85,90,95,100,105,110,120,130,140,150,160,170,180,190,200
LPI
   2
   1  1
   1  1  1
   7  1  1  7
   1  1  1  1  1
   1  1  1  1  1  7
   7  1  1  7  1  1  1
   1  1  1  1  1  1  1  1
   1  1  1  1  1  1  1  1  1
*
STA
TIT *+ COLD BASE
RES,,0,0.5,1,2,3,4,5,6,7,8,9,10,11,15,17,20,25,30,35,40,45,50,55,60,
65,70,75,80,85,90,95,100,105,110,120,130,140,150,160,170,180,190,200
TFU,565.8
TMO,565.8
BOR,450.0
PDE,0.0001
PRE,155.1296
CNU,'FUE',54135,0
STA
TIT *+ COLD TMO
RES,,0,0.5,1,2,3,4,5,6,7,8,9,10,11,15,17,20,25,30,35,40,45,50,55,60,
65,70,75,80,85,90,95,100,105,110,120,130,140,150,160,170,180,190,200
TFU,565.8
TMO,450,305
BOR,450.0
PDE,0.0001
PRE,155.1296
CNU,'FUE',54135,0
```

```
STA
TIT *+ COLD TFU
RES,,0,0.5,1,2,3,4,5,6,7,8,9,10,11,15,17,20,25,30,35,40,45,50,55,60,
65,70,75,80,85,90,95,100,105,110,120,130,140,150,160,170,180,190,200
TFU,450,305
TMO,565.8
BOR,450.0
PDE,0.0001
PRE,155.1296
CNU,'FUE',54135,0
STA
TIT *+ COLD BOR BASE
RES,,0,0.5,1,2,3,4,5,6,7,8,9,10,11,15,17,20,25,30,35,40,45,50,55,60,
65,70,75,80,85,90,95,100,105,110,120,130,140,150,160,170,180,190,200
TFU,565.8
TMO,565.8
BOR,0,900,1300
PDE,0.0001
PRE,155.1296
CNU,'FUE',54135,0
STA
TIT *+ COLD BOR TMO=450K
RES,,0,0.5,1,2,3,4,5,6,7,8,9,10,11,15,17,20,25,30,35,40,45,50,55,60,
65,70,75,80,85,90,95,100,105,110,120,130,140,150,160,170,180,190,200
TFU,565.8
TMO,450
BOR,0,900,1300
PDE,0.0001
PRE,155.1296
CNU,'FUE',54135,0
STA
TIT *+ COLD BOR TMO=305K
RES,,0,0.5,1,2,3,4,5,6,7,8,9,10,11,15,17,20,25,30,35,40,45,50,55,60,
65,70,75,80,85,90,95,100,105,110,120,130,140,150,160,170,180,190,200
TFU,565.8
TMO,305
BOR,0,900,1300
PDE,0.0001
PRE,155.1296
CNU,'FUE',54135,0
STA
TIT *+ COLD BOR TFU=450K
RES,,0,0.5,1,2,3,4,5,6,7,8,9,10,11,15,17,20,25,30,35,40,45,50,55,60,
65,70,75,80,85,90,95,100,105,110,120,130,140,150,160,170,180,190,200
TFU,450
TMO,565.8
BOR,0,900,1300
PDE,0.0001
PRE,155.1296
CNU,'FUE',54135,0
STA
TIT *+ COLD BOR TFU=305K
RES,,0,0.5,1,2,3,4,5,6,7,8,9,10,11,15,17,20,25,30,35,40,45,50,55,60,
65,70,75,80,85,90,95,100,105,110,120,130,140,150,160,170,180,190,200
TFU,305
```

```
TMO,565.8
BOR,0,900,1300
PDE,0.0001
PRE,155.1296
CNU,'FUE',54135,0
STA
TIT *+ COLD SDC
RES,,0,0.5,1,2,3,4,5,6,7,8,9,10,11,15,17,20,25,30,35,40,45,50,55,60,
65,70,75,80,85,90,95,100,105,110,120,130,140,150,160,170,180,190,200
TFU,565.8
TMO,565.8
BOR,450
PDE,0.0001
PRE,155.1296
CNU,'FUE',54135,0
SDC,200,1991.5,6574.5,8766,26298,43830/'DT'
STA
TIT *+ COLD AIC NOSHIM
RES,,0,0.5,1,2,3,4,5,6,7,8,9,10,11,15,17,20,25,30,35,40,45,50,55,60,
65,70,75,80,85,90,95,100,105,110,120,130,140,150,160,170,180,190,200
TFU,565.8
TMO,565.8
BOR,450
PDE,0.0001
PRE,155.1296
CNU,'FUE',54135,0
LPI
   2
   1  1
   1  1  1
   7  1  1  7
   1  1  1  1  1
   1  1  1  1  1  7
   7  1  1  7  1  1  1
   1  1  1  1  1  1  1  1
   1  1  1  1  1  1  1  1  1
STA
END
```

2. c4.seed_m.inp

```
*       CASMO-4 Input
*       11 Dec. 2000
*       Dean Wang
*       Massachusetts Institute of Technology

TTL,TFU=900,TMO=565.8,BOR=450,VOI=0.0
SIM,'SEED'
MI1,6.550,.7000E-05/302=100.0                                   *ZR4
MI2,7.900,.1800E-04/347=100.0                                   *SS347
MI3,8.200,.1800E-04/718=100.0                                   *INC718
MI4,8.200,.1800E-04/750=100.0                                   *INC750
MI5,10.16,.2250E-04/47000=80.00,49000=15.00,48000=5.000         *AGINCD
MI6,2.260,.3250E-05/8000=54.81,14000=37.87,13000=3.440,
```

```
5010=.7100,5011=3.170                                         *BORSHIM
MI7,0.1573/5010=100                                           *IFBA
MI8,2.2543/718=73.82,8000=12.33,13000=13.85                   *Pt DET
BOX,6.550,.7000E-05/302=100.0                                 *ZR4
FUE,1,10.3024/17.6254,92234=0.141,92238=70.5016,8000=11.732
FUE,2,0.1/0,92238=0,68167=87.434,8000=12.566
PIN,1,.22,.385,.3932,.4504/"AIR","1","AIR","CAN"
PIN,2,.5690,.6147/"COO","BOX"                               *Guide Tube
PIN,5,.1727,.2553,.4166,.5042,.5690,.6147/"AIR","MI3","MI8"
"MI3","COO","BOX"/2,4,6/1,"DET"                             *INSTR THIM
PIN,6,.1727,.2553,.4166,.5042,.5690,.6147/"AIR","MI3","MI8"
"MI3","COO","BOX"/2,4,6/1                                   *INSTR THIM
PIN,7,.4331,.4369,.4839,.5690,.6147/"MI5","AIR","MI2"
"COO","BOX"/1,3,5/1,"AIC"                                      *RCCA
PIN,8,.4331,.4369,.4839,.5690,.6147/"MI5","AIR","MI2"
"COO","BOX"/1,3,5/1,"BTH"                                      *RCCA
SPA,13.99,.1800E-04,,8.154/718=84.59,347=15.41
PRE,155.1296
PDE,129.58,'KWL'
PWR,17,1.260,21.50
DEP,-100,105,110,120,130,140,150,160,170,180,190,200
LPI
   2
   1  1
   1  1  1
   2  1  1  2
   1  1  1  1  1
   1  1  1  1  1  2
   2  1  1  2  1  1  1
   1  1  1  1  1  1  1  1
   1  1  1  1  1  1  1  1  1
*
STA
END
```

3. c4.seed_m1.inp

```
*       CASMO-4 Input
*       11 Dec. 2000
*       Dean Wang
*       Massachusetts Institute of Technology

TTL,TFU=900,TMO=600,BOR=450,VOI=0.0
SIM,'SEED'
MI1,6.550,.7000E-05/302=100.0                                  *ZR4
MI2,7.900,.1800E-04/347=100.0                                  *SS347
MI3,8.200,.1800E-04/718=100.0                                  *INC718
MI4,8.200,.1800E-04/750=100.0                                  *INC750
MI5,10.16,.2250E-04/47000=80.00,49000=15.00,48000=5.000        *AGINCD
MI6,2.260,.3250E-05/8000=54.81,14000=37.87,13000=3.440,
5010=.7100,5011=3.170
*BORSHIM
```

```
MI7,0.1573/5010=100                                         *IFBA
MI8,2.2543/718=73.82,8000=12.33,13000=13.85                 *Pt DET
BOX,6.550,.7000E-05/302=100.0                               *ZR4
FUE,1,10.3024/17.6254,92234=0.141,92238=70.5016,8000=11.732
FUE,2,0.1/0,92238=0,68167=87.434,8000=12.566
PIN,1,.22,.385,.3932,.4504/"AIR","1","AIR","CAN"         *Seed Fuel
PIN,2,.5690,.6147/"COO","BOX"                           *Guide Tube
PIN,5,.1727,.2553,.4166,.5042,.5690,.6147/"AIR","MI3","MI8"
"MI3","COO","BOX"/2,4,6/1,"DET"                         *INSTR THIM
PIN,6,.1727,.2553,.4166,.5042,.5690,.6147/"AIR","MI3","MI8"
"MI3","COO","BOX"/2,4,6/1                               *INSTR THIM
PIN,7,.4331,.4369,.4839,.5690,.6147/"MI5","AIR","MI2"
"COO","BOX"/1,3,5/1,"AIC"                                  *RCCA
PIN,8,.4331,.4369,.4839,.5690,.6147/"MI5","AIR","MI2"
"COO","BOX"/1,3,5/1,"BTH"                                  *RCCA
SPA,13.99,.1800E-04,,8.154/718=84.59,347=15.41
PRE,155.1296
PDE,129.58,'KWL'
PWR,17,1.260,21.50
DEP,-100,105,110,120,130,140,150,160,170,180,190,200
LPI
   2
   1  1
   1  1  1
   2  1  1  2
   1  1  1  1  1
   1  1  1  1  1  2
   2  1  1  2  1  1  1
   1  1  1  1  1  1  1  1
   1  1  1  1  1  1  1  1  1
*
STA
END
```

4. c4.seed_f.inp:

```
*       CASMO-4 Input
*       11 Dec. 2000
*       Dean Wang
*       Massachusetts Institute of Technology

TTL,TFU=565.8,TMO=583.1,BOR=450,VOI=0.0
SIM,'SEED'
MI1,6.550,.7000E-05/302=100.0                               *ZR4
MI2,7.900,.1800E-04/347=100.0                               *SS347
MI3,8.200,.1800E-04/718=100.0                               *INC718
MI4,8.200,.1800E-04/750=100.0                               *INC750
MI5,10.16,.2250E-04/47000=80.00,49000=15.00,48000=5.000     *AGINCD
MI6,2.260,.3250E-05/8000=54.81,14000=37.87,13000=3.440,
5010=.7100,5011=3.170                                      *BORSHIM
MI7,0.1573/5010=100                                         *IFBA
MI8,2.2543/718=73.82,8000=12.33,13000=13.85                *Pt DET
BOX,6.550,.7000E-05/302=100.0                               *ZR4
```

```
FUE,1,10.3024/17.6254,92234=0.141,92238=70.5016,8000=11.732
FUE,2,0.1/0,92238=0,68167=87.434,8000=12.566
PIN,1,.22,.385,.3932,.4504/"AIR","1","AIR","CAN"            *Seed Fuel
PIN,2,.5690,.6147/"COO","BOX"                              *Guide Tube
PIN,5,.1727,.2553,.4166,.5042,.5690,.6147/"AIR","MI3","MI8"
"MI3","COO","BOX"/2,4,6/1,"DET"                            *INSTR THIM
PIN,6,.1727,.2553,.4166,.5042,.5690,.6147/"AIR","MI3","MI8"
"MI3","COO","BOX"/2,4,6/1                                  *INSTR THIM
PIN,7,.4331,.4369,.4839,.5690,.6147/"MI5","AIR","MI2"
"COO","BOX"/1,3,5/1,"AIC"                                      *RCCA
PIN,8,.4331,.4369,.4839,.5690,.6147/"MI5","AIR","MI2"
"COO","BOX"/1,3,5/1,"BTH"                                      *RCCA
SPA,13.99,.1800E-04,,8.154/718=84.59,347=15.41
PRE,155.1296
PDE,129.58,'KWL'
PWR,17,1.260,21.50
DEP,-100,105,110,120,130,140,150,160,170,180,190,200
LPI
   2
   1  1
   1  1  1
   2  1  1  2
   1  1  1  1  1
   1  1  1  1  1  2
   2  1  1  2  1  1  1
   1  1  1  1  1  1  1  1
   1  1  1  1  1  1  1  1  1
*
STA
END
```

5. c4.seed_f1.inp

```
*       CASMO-4 Input
*       11 Dec. 2000
*       Dean Wang
*       Massachusetts Institute of Technology

TTL,TFU=1200,TMO=583.1,BOR=450,VOI=0.0
SIM,'SEED'
MI1,6.550,.7000E-05/302=100.0                                  *ZR4
MI2,7.900,.1800E-04/347=100.0                                  *SS347
MI3,8.200,.1800E-04/718=100.0                                  *INC718
MI4,8.200,.1800E-04/750=100.0                                  *INC750
MI5,10.16,.2250E-04/47000=80.00,49000=15.00,48000=5.000        *AGINCD
MI6,2.260,.3250E-05/8000=54.81,14000=37.87,13000=3.440,
5010=.7100,5011=3.170                                         *BORSHIM
MI7,0.1573/5010=100                                          *IFBA
MI8,2.2543/718=73.82,8000=12.33,13000=13.85                  *Pt DET
BOX,6.550,.7000E-05/302=100.0                                  *ZR4
FUE,1,10.3024/17.6254,92234=0.141,92238=70.5016,8000=11.732
FUE,2,0.1/0,92238=0,68167=87.434,8000=12.566
PIN,1,.22,.385,.3932,.4504/"AIR","1","AIR","CAN"          *Seed Fuel
PIN,2,.5690,.6147/"COO","BOX"                            *Guide Tube
```

```
PIN,5,.1727,.2553,.4166,.5042,.5690,.6147/"AIR","MI3","MI8"
"MI3","COO","BOX"/2,4,6/1,"DET"                              *INSTR THIM
PIN,6,.1727,.2553,.4166,.5042,.5690,.6147/"AIR","MI3","MI8"
"MI3","COO","BOX"/2,4,6/1                                   *INSTR THIM
PIN,7,.4331,.4369,.4839,.5690,.6147/"MI5","AIR","MI2"
"COO","BOX"/1,3,5/1,"AIC"                                      *RCCA
PIN,8,.4331,.4369,.4839,.5690,.6147/"MI5","AIR","MI2"
"COO","BOX"/1,3,5/1,"BTH"                                        *RCCA
SPA,13.99,.1800E-04,,8.154/718=84.59,347=15.41
PRE,155.1296
PDE,129.58,'KWL'
PWR,17,1.260,21.50
DEP,-100,105,110,120,130,140,150,160,170,180,190,200
LPI
   2
   1  1
   1  1  1
   2  1  1  2
   1  1  1  1  1
   1  1  1  1  1  2
   2  1  1  2  1  1  1
   1  1  1  1  1  1  1  1
   1  1  1  1  1  1  1  1  1
*
STA
END
```

6. c4.seed_b.inp:

```
*       CASMO-4 Input
*       11 Dec. 2000
*       Dean Wang
*       Massachusetts Institute of Technology

TTL,TFU=900,TMO=583.1,BOR=0,VOI=0.0
SIM,'SEED'
MI1,6.550,.7000E-05/302=100.0                                   *ZR4
MI2,7.900,.1800E-04/347=100.0                                   *SS347
MI3,8.200,.1800E-04/718=100.0                                   *INC718
MI4,8.200,.1800E-04/750=100.0                                   *INC750
MI5,10.16,.2250E-04/47000=80.00,49000=15.00,48000=5.000         *AGINCD
MI6,2.260,.3250E-05/8000=54.81,14000=37.87,13000=3.440,
5010=.7100,5011=3.170                                          *BORSHIM
MI7,0.1573/5010=100                                             *IFBA
MI8,2.2543/718=73.82,8000=12.33,13000=13.85                     *Pt DET
BOX,6.550,.7000E-05/302=100.0                                   *ZR4
FUE,1,10.3024/17.6254,92234=0.141,92238=70.5016,8000=11.732
FUE,2,0.1/0,92238=0,68167=87.434,8000=12.566
PIN,1,.22,.385,.3932,.4504/"AIR","1","AIR","CAN"            *Seed Fuel
PIN,2,.5690,.6147/"COO","BOX"                               *Guide Tube
PIN,5,.1727,.2553,.4166,.5042,.5690,.6147/"AIR","MI3","MI8"
"MI3","COO","BOX"/2,4,6/1,"DET"                             *INSTR THIM
PIN,6,.1727,.2553,.4166,.5042,.5690,.6147/"AIR","MI3","MI8"
"MI3","COO","BOX"/2,4,6/1                                   *INSTR THIM
```

```
PIN,7,.4331,.4369,.4839,.5690,.6147/"MI5","AIR","MI2"
"COO","BOX"/1,3,5/1,"AIC"                                   *RCCA
PIN,8,.4331,.4369,.4839,.5690,.6147/"MI5","AIR","MI2"
"COO","BOX"/1,3,5/1,"BTH"                                   *RCCA
SPA,13.99,.1800E-04,,8.154/718=84.59,347=15.41
PRE,155.1296
PDE,129.58,'KWL'
PWR,17,1.260,21.50
DEP,-100,105,110,120,130,140,150,160,170,180,190,200
LPI
   2
   1  1
   1  1  1
   2  1  1  2
   1  1  1  1  1
   1  1  1  1  1  2
   2  1  1  2  1  1  1
   1  1  1  1  1  1  1  1
   1  1  1  1  1  1  1  1  1
*
STA
END
```

7. c4.seed_b1.inp

```
*       CASMO-4 Input
*       11 Dec. 2000
*       Dean Wang
*       Massachusetts Institute of Technology

TTL,TFU=900,TMO=583.1,BOR=900,VOI=0.0
SIM,'SEED'
MI1,6.550,.7000E-05/302=100.0                                  *ZR4
MI2,7.900,.1800E-04/347=100.0                                  *SS347
MI3,8.200,.1800E-04/718=100.0                                  *INC718
MI4,8.200,.1800E-04/750=100.0                                  *INC750
MI5,10.16,.2250E-04/47000=80.00,49000=15.00,48000=5.000        *AGINCD
MI6,2.260,.3250E-05/8000=54.81,14000=37.87,13000=3.440,
5010=.7100,5011=3.170                                          BORSHIM
MI7,0.1573/5010=100                                            *IFBA
MI8,2.2543/718=73.82,8000=12.33,13000=13.85                    *Pt DET
BOX,6.550,.7000E-05/302=100.0                                  *ZR4
FUE,1,10.3024/17.6254,92234=0.141,92238=70.5016,8000=11.732
FUE,2,0.1/0,92238=0,68167=87.434,8000=12.566
PIN,1,.22,.385,.3932,.4504/"AIR","1","AIR","CAN"           *Seed Fuel
PIN,2,.5690,.6147/"COO","BOX"                              *Guide Tube
PIN,5,.1727,.2553,.4166,.5042,.5690,.6147/"AIR","MI3","MI8"
"MI3","COO","BOX"/2,4,6/1,"DET"                            *INSTR THIM
PIN,6,.1727,.2553,.4166,.5042,.5690,.6147/"AIR","MI3","MI8"
"MI3","COO","BOX"/2,4,6/1                                  *INSTR THIM
PIN,7,.4331,.4369,.4839,.5690,.6147/"MI5","AIR","MI2"
"COO","BOX"/1,3,5/1,"AIC"                                  *RCCA
PIN,8,.4331,.4369,.4839,.5690,.6147/"MI5","AIR","MI2"
"COO","BOX"/1,3,5/1,"BTH"                                  *RCCA
```

```
SPA,13.99,.1800E-04,,8.154/718=84.59,347=15.41
PRE,155.1296
PDE,129.58,'KWL'
PWR,17,1.260,21.50
DEP,-100,105,110,120,130,140,150,160,170,180,190,200
LPI
   2
   1  1
   1  1  1
   2  1  1  2
   1  1  1  1  1
   1  1  1  1  1  2
   2  1  1  2  1  1  1
   1  1  1  1  1  1  1  1
   1  1  1  1  1  1  1  1  1
*
STA
END
```

8. c4.seed_b2.inp

```
*       CASMO-4 Input
*       11 Dec. 2000
*       Dean Wang
*       Massachusetts Institute of Technology

TTL,TFU=900,TMO=583.1,BOR=1500,VOI=0.0
SIM,'SEED'
MI1,6.550,.7000E-05/302=100.0                                   *ZR4
MI2,7.900,.1800E-04/347=100.0                                   *SS347
MI3,8.200,.1800E-04/718=100.0                                   *INC718
MI4,8.200,.1800E-04/750=100.0                                   *INC750
MI5,10.16,.2250E-04/47000=80.00,49000=15.00,48000=5.000         *AGINCD
MI6,2.260,.3250E-05/8000=54.81,14000=37.87,13000=3.440,
5010=.7100,5011=3.170                                         *BORSHIM
MI7,0.1573/5010=100                                             *IFBA
MI8,2.2543/718=73.82,8000=12.33,13000=13.85                     *Pt DET
BOX,6.550,.7000E-05/302=100.0                                   *ZR4
FUE,1,10.3024/17.6254,92234=0.141,92238=70.5016,8000=11.732
FUE,2,0.1/0,92238=0,68167=87.434,8000=12.566
PIN,1,.22,.385,.3932,.4504/"AIR","1","AIR","CAN"            *Seed Fuel
PIN,2,.5690,.6147/"COO","BOX"                               *Guide Tube
PIN,5,.1727,.2553,.4166,.5042,.5690,.6147/"AIR","MI3","MI8"
"MI3","COO","BOX"/2,4,6/1,"DET"                             *INSTR THIM
PIN,6,.1727,.2553,.4166,.5042,.5690,.6147/"AIR","MI3","MI8"
"MI3","COO","BOX"/2,4,6/1                                   *INSTR THIM
PIN,7,.4331,.4369,.4839,.5690,.6147/"MI5","AIR","MI2"
"COO","BOX"/1,3,5/1,"AIC"                                       *RCCA
PIN,8,.4331,.4369,.4839,.5690,.6147/"MI5","AIR","MI2"
"COO","BOX"/1,3,5/1,"BTH"                                       *RCCA
SPA,13.99,.1800E-04,,8.154/718=84.59,347=15.41
PRE,155.1296
PDE,129.58,'KWL'
```

```
PWR,17,1.260,21.50
DEP,-100,105,110,120,130,140,150,160,170,180,190,200
LPI
   2
   1  1
   1  1  1
   2  1  1  2
   1  1  1  1  1
   1  1  1  1  1  2
   2  1  1  2  1  1  1
   1  1  1  1  1  1  1  1
   1  1  1  1  1  1  1  1  1
*
STA
END
```

9. c4.seed_b3.inp

```
*       CASMO-4 Input
*       11 Dec. 2000
*       Dean Wang
*       Massachusetts Institute of Technology

TTL,TFU=900,TMO=583.1,BOR=2000,VOI=0.0
SIM,'SEED'
MI1,6.550,.7000E-05/302=100.0                                   *ZR4
MI2,7.900,.1800E-04/347=100.0                                   *SS347
MI3,8.200,.1800E-04/718=100.0                                   *INC718
MI4,8.200,.1800E-04/750=100.0                                   *INC750
MI5,10.16,.2250E-04/47000=80.00,49000=15.00,48000=5.000         *AGINCD
MI6,2.260,.3250E-05/8000=54.81,14000=37.87,13000=3.440,
5010=.7100,5011=3.170                                          *BORSHIM
MI7,0.1573/5010=100                                             *IFBA
MI8,2.2543/718=73.82,8000=12.33,13000=13.85                     *Pt DET
BOX,6.550,.7000E-05/302=100.0                                   *ZR4
FUE,1,10.3024/17.6254,92234=0.141,92238=70.5016,8000=11.732
FUE,2,0.1/0,92238=0,68167=87.434,8000=12.566
PIN,1,.22,.385,.3932,.4504/"AIR","1","AIR","CAN"           *Seed Fuel
PIN,2,.5690,.6147/"COO","BOX"                              *Guide Tube
PIN,5,.1727,.2553,.4166,.5042,.5690,.6147/"AIR","MI3","MI8"
"MI3","COO","BOX"/2,4,6/1,"DET"                            *INSTR THIM
PIN,6,.1727,.2553,.4166,.5042,.5690,.6147/"AIR","MI3","MI8"
"MI3","COO","BOX"/2,4,6/1                                  *INSTR THIM
PIN,7,.4331,.4369,.4839,.5690,.6147/"MI5","AIR","MI2"
"COO","BOX"/1,3,5/1,"AIC"                                       *RCCA
PIN,8,.4331,.4369,.4839,.5690,.6147/"MI5","AIR","MI2"
"COO","BOX"/1,3,5/1,"BTH"                                       *RCCA
SPA,13.99,.1800E-04,,8.154/718=84.59,347=15.41
PRE,155.1296
PDE,129.58,'KWL'
PWR,17,1.260,21.50
DEP,-100,105,110,120,130,140,150,160,170,180,190,200
LPI
   2
```

```
   1  1
   1  1  1
   2  1  1  2
   1  1  1  1  1
   1  1  1  1  1  2
   2  1  1  2  1  1  1
   1  1  1  1  1  1  1  1
   1  1  1  1  1  1  1  1  1
*
STA
END
```

Blanket Assembly:

1. c4.blanket_base.inp

```
*       CASMO-4 Input
*       11 Dec. 2000
*       Dean Wang
*       Massachusetts Institute of Technology

TTL,TFU=900,TMO=583.1,BOR=450,VOI=0.0
SIM,'BLANKET'
MI1,6.550,.7000E-05/302=100.0                                  *ZR4
MI2,7.900,.1800E-04/347=100.0                                  *SS347
MI3,8.200,.1800E-04/718=100.0                                  *INC718
MI4,8.200,.1800E-04/750=100.0                                  *INC750
*MI5,10.16,.2250E-04/47000=80.00,49000=15.00,48000=5.000      *AGINCD
*MI5,2.50/5000=78.2600,6000=21.7400                           *B4C
MI5,5.0/5000=89.1300,6000=10.8700                             *B4C
MI6,2.260,.3250E-05/8000=54.81,14000=37.87,13000=3.440,
5010=.7100,5011=3.170                                        *BORSHIM
MI7,0.1573/5010=100                                          *IFBA
MI8,2.2543/718=73.82,8000=12.33,13000=13.85                  *Pt DET
BOX,6.550,.7000E-05/302=100.0                                *ZR4
FUE,1,9.4028/1.2554,90232=75.3660,92234=0.0100,92238=11.2890,8000=12.0
796
PIN,1,.4646,.4728,.53/"1","AIR","CAN"
PIN,2,.5690,.6147/"COO","BOX"                              *Guide Tube
PIN,5,.1727,.2553,.4166,.5042,.5690,.6147/"AIR","MI3","MI8",
"MI3","COO","BOX"/2,4,6/1,"DET"                            *INSTR THIM
PIN,6,.1727,.2553,.4166,.5042,.5690,.6147/"AIR","MI3","MI8",
"MI3","COO","BOX"/2,4,6/1                                  *INSTR THIM
PIN,7,.4331,.4369,.4839,.5690,.6147/"MI5","AIR","MI2",
"COO","BOX"/1,3,5/1,"AIC"                                      *RCCA
PIN,8,.4331,.4369,.4839,.5690,.6147/"MI5","AIR","MI2",
"COO","BOX"/1,3,5/1,"BTH"                                      *RCCA
SPA,13.99,.1800E-04,,8.154/718=84.59,347=15.41
PRE,155.1296
PDE,79.42,'KWL'
PWR,17,1.260,21.50
DEP,-100,105,110,115,120,125,130,135,140,145,150
LPI
```

```
   2
   1  1
   1  1  1
   2  1  1  2
   1  1  1  1  1
   1  1  1  1  1  2
   2  1  1  2  1  1  1
   1  1  1  1  1  1  1  1
   1  1  1  1  1  1  1  1  1
*
STA
TIT *+ TFU BRA
RES,,0,0.5,1,2,3,4,5,6,7,8,9,10,11,15,17,20,25,30,35,40,45,60,65,
70,75,80,85,90,95,100,105,110,115,120,125,130,135,140,145,150
TFU,565.8,1200
STA
TIT *+ TMO BRA
RES,,0,0.5,1,2,3,4,5,6,7,8,9,10,11,15,17,20,25,30,35,40,45,60,65,
70,75,80,85,90,95,100,105,110,115,120,125,130,135,140,145,150
TMO,565.8,600,610
STA
TIT *+ BOR BRA
RES,,0,0.5,1,2,3,4,5,6,7,8,9,10,11,15,17,20,25,30,35,40,45,60,65,
70,75,80,85,90,95,100,105,110,115,120,125,130,135,140,145,150
BOR,0,900,1500,2000
STA
TIT *+ BOR=900, TMO=565.8
RES,,0,0.5,1,2,3,4,5,6,7,8,9,10,11,15,17,20,25,30,35,40,45,60,65,
70,75,80,85,90,95,100,105,110,115,120,125,130,135,140,145,150
TMO,565.8
BOR,900
STA
TIT *+ SDC BRA
RES,,0,0.5,1,2,3,4,5,6,7,8,9,10,11,15,17,20,25,30,35,40,45,60,65,
70,75,80,85,90,95,100,105,110,115,120,125,130,135,140,145,150
SDC,200,1991.5,6574.5,8766,26298,43830/'DT'
STA
TIT *+ CRD BRA: AIC NOIFBA
RES,,0,0.5,1,2,3,4,5,6,7,8,9,10,11,15,17,20,25,30,35,40,45,60,65,
70,75,80,85,90,95,100,105,110,115,120,125,130,135,140,145,150
LPI
   2
   1  1
   1  1  1
   7  1  1  7
   1  1  1  1  1
   1  1  1  1  1  7
   7  1  1  7  1  1  1
   1  1  1  1  1  1  1  1
   1  1  1  1  1  1  1  1  1
STA
TIT *+ COLD BASE
RES,,0,0.5,1,2,3,4,5,6,7,8,9,10,11,15,17,20,25,30,35,40,45,60,65,
70,75,80,85,90,95,100,105,110,115,120,125,130,135,140,145,150
TFU,565.8
```

```
TMO,565.8
BOR,450.0
PDE,0.0001
PRE,155.1296
CNU,'FUE',54135,0
STA
TIT *+ COLD TMO
RES,,0,0.5,1,2,3,4,5,6,7,8,9,10,11,15,17,20,25,30,35,40,45,60,65,
70,75,80,85,90,95,100,105,110,115,120,125,130,135,140,145,150
TFU,565.8
TMO,450,305
BOR,450.0
PDE,0.0001
PRE,155.1296
CNU,'FUE',54135,0
STA
TIT *+ COLD TFU
RES,,0,0.5,1,2,3,4,5,6,7,8,9,10,11,15,17,20,25,30,35,40,45,60,65,
70,75,80,85,90,95,100,105,110,115,120,125,130,135,140,145,150
TFU,450,305
TMO,565.8
BOR,450.0
PDE,0.0001
PRE,155.1296
CNU,'FUE',54135,0
STA
TIT *+ COLD BOR BASE
RES,,0,0.5,1,2,3,4,5,6,7,8,9,10,11,15,17,20,25,30,35,40,45,60,65,
70,75,80,85,90,95,100,105,110,115,120,125,130,135,140,145,150
TFU,565.8
TMO,565.8
BOR,0,900,1300
PDE,0.0001
PRE,155.1296
CNU,'FUE',54135,0
STA
TIT *+ COLD BOR TMO=450K
RES,,0,0.5,1,2,3,4,5,6,7,8,9,10,11,15,17,20,25,30,35,40,45,60,65,
70,75,80,85,90,95,100,105,110,115,120,125,130,135,140,145,150
TFU,565.8
TMO,450
BOR,0,900,1300
PDE,0.0001
PRE,155.1296
CNU,'FUE',54135,0
STA
TIT *+ COLD BOR TMO=305K
RES,,0,0.5,1,2,3,4,5,6,7,8,9,10,11,15,17,20,25,30,35,40,45,60,65,
70,75,80,85,90,95,100,105,110,115,120,125,130,135,140,145,150
TFU,565.8
TMO,305
BOR,0,900,1300
PDE,0.0001
PRE,155.1296
CNU,'FUE',54135,0
```

```
STA
TIT *+ COLD BOR TFU=450K
RES,,0,0.5,1,2,3,4,5,6,7,8,9,10,11,15,17,20,25,30,35,40,45,60,65,
70,75,80,85,90,95,100,105,110,115,120,125,130,135,140,145,150
TFU,450
TMO,565.8
BOR,0,900,1300
PDE,0.0001
PRE,155.1296
CNU,'FUE',54135,0
STA
TIT *+ COLD BOR TFU=305K
RES,,0,0.5,1,2,3,4,5,6,7,8,9,10,11,15,17,20,25,30,35,40,45,60,65,
70,75,80,85,90,95,100,105,110,115,120,125,130,135,140,145,150
TFU,305
TMO,565.8
BOR,0,900,1300
PDE,0.0001
PRE,155.1296
CNU,'FUE',54135,0
STA
TIT *+ COLD SDC
RES,,0,0.5,1,2,3,4,5,6,7,8,9,10,11,15,17,20,25,30,35,40,45,60,65,
70,75,80,85,90,95,100,105,110,115,120,125,130,135,140,145,150
TFU,565.8
TMO,565.8
BOR,450
PDE,0.0001
PRE,155.1296
CNU,'FUE',54135,0
SDC,200,1991.5,6574.5,8766,26298,43830/'DT'
STA
TIT *+ COLD AIC NOSHIM
RES,,0,0.5,1,2,3,4,5,6,7,8,9,10,11,15,17,20,25,30,35,40,45,60,65,
70,75,80,85,90,95,100,105,110,115,120,125,130,135,140,145,150
TFU,565.8
TMO,565.8
BOR,450
PDE,0.0001
PRE,155.1296
CNU,'FUE',54135,0
LPI
   2
   1  1
   1  1  1
   7  1  1  7
   1  1  1  1  1
   1  1  1  1  1  7
   7  1  1  7  1  1  1
   1  1  1  1  1  1  1  1
   1  1  1  1  1  1  1  1  1
STA
END
```

2. c4.blanket_m.inp

```
*       CASMO-4 Input
*       11 Dec. 2000
*       Dean Wang
*       Massachusetts Institute of Technology

TTL,TFU=900,TMO=565.8,BOR=450,VOI=0.0
SIM,'BLANKET'
MI1,6.550,.7000E-05/302=100.0                                  *ZR4
MI2,7.900,.1800E-04/347=100.0                                  *SS347
MI3,8.200,.1800E-04/718=100.0                                  *INC718
MI4,8.200,.1800E-04/750=100.0                                  *INC750
MI5,10.16,.2250E-04/47000=80.00,49000=15.00,48000=5.000        *AGINCD
MI6,2.260,.3250E-05/8000=54.81,14000=37.87,13000=3.440,
5010=.7100,5011=3.170                                        *BORSHIM
MI7,0.1573/5010=100                                           *IFBA
MI8,2.2543/718=73.82,8000=12.33,13000=13.85                    *Pt DET
BOX,6.550,.7000E-05/302=100.0                                  *ZR4
FUE,1,9.4028/1.2554,90232=75.3660,92234=0.0100,92238=11.2890,8000=12.0
796
PIN,1,.4646,.4728,.53/"1","AIR","CAN"
PIN,2,.5690,.6147/"COO","BOX"                           *Guide Tube
PIN,5,.1727,.2553,.4166,.5042,.5690,.6147/"AIR","MI3","MI8"
"MI3","COO","BOX"/2,4,6/1,"DET"                         *INSTR THIM
PIN,6,.1727,.2553,.4166,.5042,.5690,.6147/"AIR","MI3","MI8"
"MI3","COO","BOX"/2,4,6/1                               *INSTR THIM
PIN,7,.4331,.4369,.4839,.5690,.6147/"MI5","AIR","MI2"
"COO","BOX"/1,3,5/1,"AIC"                                      *RCCA
PIN,8,.4331,.4369,.4839,.5690,.6147/"MI5","AIR","MI2"
"COO","BOX"/1,3,5/1,"BTH"                                      *RCCA
SPA,13.99,.1800E-04,,8.154/718=84.59,347=15.41
PRE,155.1296
PDE,79.42,'KWL'
PWR,17,1.260,21.50
DEP,-100,105,110,115,120,125,130,135,140,145,150
LPI
   2
   1  1
   1  1  1
   2  1  1  2
   1  1  1  1  1
   1  1  1  1  1  2
   2  1  1  2  1  1  1
   1  1  1  1  1  1  1  1
   1  1  1  1  1  1  1  1  1
*
STA
END
```

3. c4.blanket_m1.inp

```
*       CASMO-4 Input
*       11 Dec. 2000
*       Dean Wang
```

```
*       Massachusetts Institute of Technology

TTL,TFU=900,TMO=600,BOR=450,VOI=0.0
SIM,'BLANKET'
MI1,6.550,.7000E-05/302=100.0                                  *ZR4
MI2,7.900,.1800E-04/347=100.0                                  *SS347
MI3,8.200,.1800E-04/718=100.0                                  *INC718
MI4,8.200,.1800E-04/750=100.0                                  *INC750
MI5,10.16,.2250E-04/47000=80.00,49000=15.00,48000=5.000        *AGINCD
MI6,2.260,.3250E-05/8000=54.81,14000=37.87,13000=3.440,
5010=.7100,5011=3.170                                         *BORSHIM
MI7,0.1573/5010=100                                            *IFBA
MI8,2.2543/718=73.82,8000=12.33,13000=13.85                    *Pt DET
BOX,6.550,.7000E-05/302=100.0                                  *ZR4
FUE,1,9.4028/1.2554,90232=75.3660,92234=0.0100,92238=11.2890,8000=12.0
796
PIN,1,.4646,.4728,.53/"1","AIR","CAN"
PIN,2,.5690,.6147/"COO","BOX"                           *Guide Tube
PIN,5,.1727,.2553,.4166,.5042,.5690,.6147/"AIR","MI3","MI8"
"MI3","COO","BOX"/2,4,6/1,"DET"                         *INSTR THIM
PIN,6,.1727,.2553,.4166,.5042,.5690,.6147/"AIR","MI3","MI8"
"MI3","COO","BOX"/2,4,6/1                               *INSTR THIM
PIN,7,.4331,.4369,.4839,.5690,.6147/"MI5","AIR","MI2"
"COO","BOX"/1,3,5/1,"AIC"                                      *RCCA
PIN,8,.4331,.4369,.4839,.5690,.6147/"MI5","AIR","MI2"
"COO","BOX"/1,3,5/1,"BTH"                                      *RCCA
SPA,13.99,.1800E-04,,8.154/718=84.59,347=15.41
PRE,155.1296
PDE,79.42,'KWL'
PWR,17,1.260,21.50
DEP,-100,105,110,115,120,125,130,135,140,145,150
LPI
   2
   1  1
   1  1  1
   2  1  1  2
   1  1  1  1  1
   1  1  1  1  1  2
   2  1  1  2  1  1  1
   1  1  1  1  1  1  1  1
   1  1  1  1  1  1  1  1  1
*
STA
END
```

4. c4.blanket_f.inp

```
*       CASMO-4 Input
*       11 Dec. 2000
*       Dean Wang
*       Massachusetts Institute of Technology

TTL,TFU=565.8,TMO=583.1,BOR=450,VOI=0.0
```

```
SIM,'BLANKET'
MI1,6.550,.7000E-05/302=100.0                                  *ZR4
MI2,7.900,.1800E-04/347=100.0                                  *SS347
MI3,8.200,.1800E-04/718=100.0                                  *INC718
MI4,8.200,.1800E-04/750=100.0                                  *INC750
MI5,10.16,.2250E-04/47000=80.00,49000=15.00,48000=5.000        *AGINCD
MI6,2.260,.3250E-05/8000=54.81,14000=37.87,13000=3.440,
5010=.7100,5011=3.170                                         *BORSHIM
MI7,0.1573/5010=100                                            *IFBA
MI8,2.2543/718=73.82,8000=12.33,13000=13.85                    *Pt DET
BOX,6.550,.7000E-05/302=100.0                                  *ZR4
FUE,1,9.4028/1.2554,90232=75.3660,92234=0.0100,92238=11.2890,8000=12.0
796
PIN,1,.4646,.4728,.53/"1","AIR","CAN"
PIN,2,.5690,.6147/"COO","BOX"                           *Guide Tube
PIN,5,.1727,.2553,.4166,.5042,.5690,.6147/"AIR","MI3","MI8"
"MI3","COO","BOX"/2,4,6/1,"DET"                         *INSTR THIM
PIN,6,.1727,.2553,.4166,.5042,.5690,.6147/"AIR","MI3","MI8"
"MI3","COO","BOX"/2,4,6/1                               *INSTR THIM
PIN,7,.4331,.4369,.4839,.5690,.6147/"MI5","AIR","MI2"
"COO","BOX"/1,3,5/1,"AIC"                                   *RCCA
PIN,8,.4331,.4369,.4839,.5690,.6147/"MI5","AIR","MI2"
"COO","BOX"/1,3,5/1,"BTH"                                   *RCCA
SPA,13.99,.1800E-04,,8.154/718=84.59,347=15.41
PRE,155.1296
PDE,79.42,'KWL'
PWR,17,1.260,21.50
DEP,-100,105,110,115,120,125,130,135,140,145,150
LPI
   2
   1  1
   1  1  1
   2  1  1  2
   1  1  1  1  1
   1  1  1  1  1  2
   2  1  1  2  1  1  1
   1  1  1  1  1  1  1  1
   1  1  1  1  1  1  1  1  1
*
STA
END
```

5. c4.blanket_f1.inp

```
*       CASMO-4 Input
*       11 Dec. 2000
*       Dean Wang
*       Massachusetts Institute of Technology

TTL,TFU=1200,TMO=583.1,BOR=450,VOI=0.0
SIM,'BLANKET'
MI1,6.550,.7000E-05/302=100.0                                  *ZR4
MI2,7.900,.1800E-04/347=100.0                                  *SS347
```

```
MI3,8.200,.1800E-04/718=100.0                                   *INC718
MI4,8.200,.1800E-04/750=100.0                                   *INC750
MI5,10.16,.2250E-04/47000=80.00,49000=15.00,48000=5.000         *AGINCD
MI6,2.260,.3250E-05/8000=54.81,14000=37.87,13000=3.440,
5010=.7100,5011=3.170                                         *BORSHIM
MI7,0.1573/5010=100                                             *IFBA
MI8,2.2543/718=73.82,8000=12.33,13000=13.85                     *Pt DET
BOX,6.550,.7000E-05/302=100.0                                   *ZR4
FUE,1,9.4028/1.2554,90232=75.3660,92234=0.0100,92238=11.2890,8000=12.0
796
PIN,1,.4646,.4728,.53/"1","AIR","CAN"
PIN,2,.5690,.6147/"COO","BOX"                            *Guide Tube
PIN,5,.1727,.2553,.4166,.5042,.5690,.6147/"AIR","MI3","MI8"
"MI3","COO","BOX"/2,4,6/1,"DET"                          *INSTR THIM
PIN,6,.1727,.2553,.4166,.5042,.5690,.6147/"AIR","MI3","MI8"
"MI3","COO","BOX"/2,4,6/1                                *INSTR THIM
PIN,7,.4331,.4369,.4839,.5690,.6147/"MI5","AIR","MI2"
"COO","BOX"/1,3,5/1,"AIC"                                       *RCCA
PIN,8,.4331,.4369,.4839,.5690,.6147/"MI5","AIR","MI2"
"COO","BOX"/1,3,5/1,"BTH"                                       *RCCA
SPA,13.99,.1800E-04,,8.154/718=84.59,347=15.41
PRE,155.1296
PDE,79.42,'KWL'
PWR,17,1.260,21.50
DEP,-100,105,110,115,120,125,130,135,140,145,150
LPI
   2
   1  1
   1  1  1
   2  1  1  2
   1  1  1  1  1
   1  1  1  1  1  2
   2  1  1  2  1  1  1
   1  1  1  1  1  1  1  1
   1  1  1  1  1  1  1  1  1
*
STA
END
```

6. c4.blanket_b.inp

```
*       CASMO-4 Input
*       11 Dec. 2000
*       Dean Wang
*       Massachusetts Institute of Technology

TTL,TFU=900,TMO=583.1,BOR=0,VOI=0.0
SIM,'BLANKET'
MI1,6.550,.7000E-05/302=100.0                                   *ZR4
MI2,7.900,.1800E-04/347=100.0                                   *SS347
MI3,8.200,.1800E-04/718=100.0                                   *INC718
MI4,8.200,.1800E-04/750=100.0                                   *INC750
MI5,10.16,.2250E-04/47000=80.00,49000=15.00,48000=5.000         *AGINCD
MI6,2.260,.3250E-05/8000=54.81,14000=37.87,13000=3.440,
```

```
5010=.7100,5011=3.170                                            *BORSHIM
MI7,0.1573/5010=100                                               *IFBA
MI8,2.2543/718=73.82,8000=12.33,13000=13.85                      *Pt DET
BOX,6.550,.7000E-05/302=100.0                                    *ZR4
FUE,1,9.4028/1.2554,90232=75.3660,92234=0.0100,92238=11.2890,8000=12.0
796
PIN,1,.4646,.4728,.53/"1","AIR","CAN"
PIN,2,.5690,.6147/"COO","BOX"                              *Guide Tube
PIN,5,.1727,.2553,.4166,.5042,.5690,.6147/"AIR","MI3","MI8"
"MI3","COO","BOX"/2,4,6/1,"DET"                            *INSTR THIM
PIN,6,.1727,.2553,.4166,.5042,.5690,.6147/"AIR","MI3","MI8"
"MI3","COO","BOX"/2,4,6/1                                  *INSTR THIM
PIN,7,.4331,.4369,.4839,.5690,.6147/"MI5","AIR","MI2"
"COO","BOX"/1,3,5/1,"AIC"                                        *RCCA
PIN,8,.4331,.4369,.4839,.5690,.6147/"MI5","AIR","MI2"
"COO","BOX"/1,3,5/1,"BTH"                                        *RCCA
SPA,13.99,.1800E-04,,8.154/718=84.59,347=15.41
PRE,155.1296
PDE,79.42,'KWL'
PWR,17,1.260,21.50
DEP,-100,105,110,115,120,125,130,135,140,145,150
LPI
   2
   1  1
   1  1  1
   2  1  1  2
   1  1  1  1  1
   1  1  1  1  1  2
   2  1  1  2  1  1  1
   1  1  1  1  1  1  1  1
   1  1  1  1  1  1  1  1  1
*
STA
END
```

7. c4.blanket_b1.inp

```
*       CASMO-4 Input
*       11 Dec. 2000
*       Dean Wang
*       Massachusetts Institute of Technology

TTL,TFU=900,TMO=583.1,BOR=900,VOI=0.0
SIM,'BLANKET'
MI1,6.550,.7000E-05/302=100.0                                    *ZR4
MI2,7.900,.1800E-04/347=100.0                                    *SS347
MI3,8.200,.1800E-04/718=100.0                                    *INC718
MI4,8.200,.1800E-04/750=100.0                                    *INC750
MI5,10.16,.2250E-04/47000=80.00,49000=15.00,48000=5.000          *AGINCD
MI6,2.260,.3250E-05/8000=54.81,14000=37.87,13000=3.440,
5010=.7100,5011=3.170                                          *BORSHIM
MI7,0.1573/5010=100                                               *IFBA
MI8,2.2543/718=73.82,8000=12.33,13000=13.85                      *Pt DET
BOX,6.550,.7000E-05/302=100.0                                    *ZR4
```

```
FUE,1,9.4028/1.2554,90232=75.3660,92234=0.0100,92238=11.2890,8000=12.0
796
PIN,1,.4646,.4728,.53/"1","AIR","CAN"
PIN,2,.5690,.6147/"COO","BOX"                             *Guide Tube
PIN,5,.1727,.2553,.4166,.5042,.5690,.6147/"AIR","MI3","MI8"
"MI3","COO","BOX"/2,4,6/1,"DET"                           *INSTR THIM
PIN,6,.1727,.2553,.4166,.5042,.5690,.6147/"AIR","MI3","MI8"
"MI3","COO","BOX"/2,4,6/1                                 *INSTR THIM
PIN,7,.4331,.4369,.4839,.5690,.6147/"MI5","AIR","MI2"
"COO","BOX"/1,3,5/1,"AIC"                                        *RCCA
PIN,8,.4331,.4369,.4839,.5690,.6147/"MI5","AIR","MI2"
"COO","BOX"/1,3,5/1,"BTH"                                        *RCCA
SPA,13.99,.1800E-04,,8.154/718=84.59,347=15.41
PRE,155.1296
PDE,79.42,'KWL'
PWR,17,1.260,21.50
DEP,-100,105,110,115,120,125,130,135,140,145,150
LPI
   2
   1  1
   1  1  1
   2  1  1  2
   1  1  1  1  1
   1  1  1  1  1  2
   2  1  1  2  1  1  1
   1  1  1  1  1  1  1  1
   1  1  1  1  1  1  1  1  1
*
STA
END
```

8. c4.blanket_b2.inp

```
*       CASMO-4 Input
*       11 Dec. 2000
*       Dean Wang
*       Massachusetts Institute of Technology

TTL,TFU=900,TMO=583.1,BOR=1500,VOI=0.0
SIM,'BLANKET'
MI1,6.550,.7000E-05/302=100.0                                    *ZR4
MI2,7.900,.1800E-04/347=100.0                                    *SS347
MI3,8.200,.1800E-04/718=100.0                                    *INC718
MI4,8.200,.1800E-04/750=100.0                                    *INC750
MI5,10.16,.2250E-04/47000=80.00,49000=15.00,48000=5.000          *AGINCD
MI6,2.260,.3250E-05/8000=54.81,14000=37.87,13000=3.440,
5010=.7100,5011=3.170                                           *BORSHIM
MI7,0.1573/5010=100                                              *IFBA
MI8,2.2543/718=73.82,8000=12.33,13000=13.85                      *Pt DET
BOX,6.550,.7000E-05/302=100.0                                    *ZR4
FUE,1,9.4028/1.2554,90232=75.3660,92234=0.0100,92238=11.2890,8000=12.0
796
PIN,1,.4646,.4728,.53/"1","AIR","CAN"
PIN,2,.5690,.6147/"COO","BOX"                             *Guide Tube
```

```
PIN,5,.1727,.2553,.4166,.5042,.5690,.6147/"AIR","MI3","MI8"
"MI3","COO","BOX"/2,4,6/1,"DET"                         *INSTR THIM
PIN,6,.1727,.2553,.4166,.5042,.5690,.6147/"AIR","MI3","MI8"
"MI3","COO","BOX"/2,4,6/1                               *INSTR THIM
PIN,7,.4331,.4369,.4839,.5690,.6147/"MI5","AIR","MI2"
"COO","BOX"/1,3,5/1,"AIC"                                     *RCCA
PIN,8,.4331,.4369,.4839,.5690,.6147/"MI5","AIR","MI2"
"COO","BOX"/1,3,5/1,"BTH"                                      *RCCA
SPA,13.99,.1800E-04,,8.154/718=84.59,347=15.41
PRE,155.1296
PDE,79.42,'KWL'
PWR,17,1.260,21.50
DEP,-100,105,110,115,120,125,130,135,140,145,150
LPI
   2
   1  1
   1  1  1
   2  1  1  2
   1  1  1  1  1
   1  1  1  1  1  2
   2  1  1  2  1  1  1
   1  1  1  1  1  1  1  1
   1  1  1  1  1  1  1  1  1
*
STA
END
```

9. c4.blanket_b3.inp

```
*       CASMO-4 Input
*       11 Dec. 2000
*       Dean Wang
*       Massachusetts Institute of Technology

TTL,TFU=900,TMO=583.1,BOR=2000,VOI=0.0
SIM,'BLANKET'
MI1,6.550,.7000E-05/302=100.0                                  *ZR4
MI2,7.900,.1800E-04/347=100.0                                  *SS347
MI3,8.200,.1800E-04/718=100.0                                  *INC718
MI4,8.200,.1800E-04/750=100.0                                  *INC750
MI5,10.16,.2250E-04/47000=80.00,49000=15.00,48000=5.000        *AGINCD
MI6,2.260,.3250E-05/8000=54.81,14000=37.87,13000=3.440,
5010=.7100,5011=3.170
*BORSHIM
MI7,0.1573/5010=100                                            *IFBA
MI8,2.2543/718=73.82,8000=12.33,13000=13.85                    *Pt DET
BOX,6.550,.7000E-05/302=100.0                                  *ZR4
*FUE,1,9.3624/1.3220,90232=79.0955,92234=0.0106,92238=7.4914,8000=12.0
805
*FUE,1,9.3624/0.8814,90232=79.0955,92234=0.0071,92238=7.9326,8000=12.0
834
FUE,1,9.4028/1.2554,90232=75.3660,92234=0.0100,92238=11.2890,8000=12.0
796
PIN,1,.4646,.4728,.53/"1","AIR","CAN"
```

```
PIN,2,.5690,.6147/"COO","BOX"                               *Guide Tube
PIN,5,.1727,.2553,.4166,.5042,.5690,.6147/"AIR","MI3","MI8"
"MI3","COO","BOX"/2,4,6/1,"DET"                             *INSTR THIM
PIN,6,.1727,.2553,.4166,.5042,.5690,.6147/"AIR","MI3","MI8"
"MI3","COO","BOX"/2,4,6/1                                   *INSTR THIM
PIN,7,.4331,.4369,.4839,.5690,.6147/"MI5","AIR","MI2"
"COO","BOX"/1,3,5/1,"AIC"                                   *RCCA
PIN,8,.4331,.4369,.4839,.5690,.6147/"MI5","AIR","MI2"
"COO","BOX"/1,3,5/1,"BTH"                                   *RCCA
SPA,13.99,.1800E-04,,8.154/718=84.59,347=15.41
PRE,155.1296
PDE,79.42,'KWL'
PWR,17,1.260,21.50
DEP,-100,105,110,115,120,125,130,135,140,145,150
LPI
   2
   1  1
   1  1  1
   2  1  1  2
   1  1  1  1  1
   1  1  1  1  1  2
   2  1  1  2  1  1  1
   1  1  1  1  1  1  1  1
   1  1  1  1  1  1  1  1  1
*
STA
END
```

TABLES Input Files

```
'COM'   TABLES-3 Input File
'COM'   3-D Model
'COM'  13 Dec. 2000
'COM'  Dean Wang
'COM'  Massachusetts Institute of Technology

'TIT','WASB-BlANKET ASSEMBLY'/
'PWR',0/
'LIB','ADD' 'BLANKET' /
'OPT',4,1/

'COM'      CASMO FILES USED FOR THIS SEGMENT
'CAS','/home2/users/wangda/WASB/c4/blanket/c4.blanket_base.cax' /
'CAS','/home2/users/wangda/WASB/c4/blanket/c4.blanket_f.cax' /
'CAS','/home2/users/wangda/WASB/c4/blanket/c4.blanket_f1.cax' /
'CAS','/home2/users/wangda/WASB/c4/blanket/c4.blanket_m.cax' /
'CAS','/home2/users/wangda/WASB/c4/blanket/c4.blanket_m1.cax' /
'CAS','/home2/users/wangda/WASB/c4/blanket/c4.blanket_b.cax' /
'CAS','/home2/users/wangda/WASB/c4/blanket/c4.blanket_b1.cax' /
'CAS','/home2/users/wangda/WASB/c4/blanket/c4.blanket_b2.cax' /
'CAS','/home2/users/wangda/WASB/c4/blanket/c4.blanket_b3.cax' /

'COM'      VARIABLE LIST


'EXP'   38,1   0,0.5,1,2,3,4,5,6,7,8,9,10,15,20,25,30,35,40,45,60,65,

70,75,80,85,90,95,100,105,110,115,120,125,130,135,140,145,150/
'RES'   20,1
0,1,2,5,6,10,20,30,40,50,60,70,80,90,100,110,120,130,140,150/

'TFU'    3,2   1200,900,565.8/
'TMO'    4,1   583.1,565.8,600,610/
'BOR'    5,1   450,0,900,1500,2000/
'HTFU'   3,2   1200,900,565.8/
'HTMO'   3,2   600,583.1,565.8/
'HBOR'   5,2   0,450,900,1500,2000/
'CRD'    2,1   ' ','AIC'/

'BAS.MAC' 2 'EXP','HTMO',,,'TMO'/
'DEL.MAC' 2 'EXP','TMO'/
'DEL.MAC' 2 'EXP','HBOR',,,'BOR'/
'DEL.MAC' 2 'EXP','BOR'/
'DEL.MAC' 2 'EXP','HTFU',,,'TFU'/
'DEL.MAC' 2 'EXP','TFU'/
'DEL.MAC' 2 'EXP','CRD'/
'SUB.MAC' 'HTMO','TMO','HTMO','TMO'/
'SUB.MAC' 'HBOR','BOR','HBOR','BOR'/
'SUB.MAC' 'HTFU','TFU','HTFU','TFU'/

'BAS.FPD' 2 'EXP','TMO'/
'DEL.FPD' 2 'EXP','BOR'/
```

```
'DEL.FPD' 2 'EXP','TFU'/
'DEL.FPD' 2 'EXP','CRD'/

'COM'    DISCONTINUITY FACTOR DEPENDENCE
'BAS.DFS' 2 'EXP','TMO'/
'DEL.DFS' 2 'EXP','BOR'/
'DEL.DFS' 2 'EXP','TFU'/
'DEL.DFS' 2 'EXP','CRD'/
'PIN.PIN' /
'STA'/
'END'/

'TIT','WASB-SEED ASSEMBLY'/
'LIB','ADD' 'SEED' /
'COM'      CASMO FILES USED FOR THIS SEGMENT
'CAS','/home2/users/wangda/WASB/c4/seed/c4.seed_base.cax' /
'CAS','/home2/users/wangda/WASB/c4/seed/c4.seed_f.cax' /
'CAS','/home2/users/wangda/WASB/c4/seed/c4.seed_f1.cax' /
'CAS','/home2/users/wangda/WASB/c4/seed/c4.seed_m.cax' /
'CAS','/home2/users/wangda/WASB/c4/seed/c4.seed_m1.cax' /
'CAS','/home2/users/wangda/WASB/c4/seed/c4.seed_b.cax' /
'CAS','/home2/users/wangda/WASB/c4/seed/c4.seed_b1.cax' /
'CAS','/home2/users/wangda/WASB/c4/seed/c4.seed_b2.cax' /
'CAS','/home2/users/wangda/WASB/c4/seed/c4.seed_b3.cax' /

'EXP'   39,1
0,0.5,1,2,3,4,5,6,7,8,9,10,15,20,25,30,35,40,45,50,55,60,65,

70,75,80,85,90,100,110,120,130,140,150,160,170,180,190,200/
'RES'   24,1
0,1,3,5,10,20,30,40,50,60,70,80,90,100,110,120,130,140,150,
               160,170,180,190,200/

'TFU'    3,2   1200,900,565.8/
'TMO'    4,1   583.1,565.8,600,610/
'BOR'    5,1   450,0,900,1500,2000/
'HTFU'   3,2   1200,900,565.8/
'HTMO'   3,2   600,583.1,565.8/
'HBOR'   5,2   0,450,900,1500,2000/
'CRD'    2,1   ' ','AIC'/

'BAS.MAC' 2 'EXP','HTMO',,,'TMO'/
'DEL.MAC' 2 'EXP','TMO'/
'DEL.MAC' 2 'EXP','HBOR',,,'BOR'/
'DEL.MAC' 2 'EXP','BOR'/
'DEL.MAC' 2 'EXP','HTFU',,,'TFU'/
'DEL.MAC' 2 'EXP','TFU'/
'DEL.MAC' 2 'EXP','CRD'/
'SUB.MAC' 'HTMO','TMO','HTMO','TMO'/
'SUB.MAC' 'HBOR','BOR','HBOR','BOR'/
'SUB.MAC' 'HTFU','TFU','HTFU','TFU'/

'BAS.FPD' 2 'EXP','TMO'/
'DEL.FPD' 2 'EXP','BOR'/
'DEL.FPD' 2 'EXP','TFU'/
```

```
'DEL.FPD' 2 'EXP','CRD'/

'COM'    DISCONTINUITY FACTOR DEPENDENCE
'BAS.DFS' 2 'EXP','TMO'/
'DEL.DFS' 2 'EXP','BOR'/
'DEL.DFS' 2 'EXP','TFU'/
'DEL.DFS' 2 'EXP','CRD'/
'PIN.PIN' /
'STA'/
'END'/

'TIT','WASB-SEED ASSEMBLY WITH BP'/
'LIB','ADD' 'SEED_BP' /
'COM'      CASMO FILES USED FOR THIS SEGMENT
'CAS','/home2/users/wangda/WASB/c4/seed/seed_bp/c4.seed_base.cax' /
'CAS','/home2/users/wangda/WASB/c4/seed/seed_bp/c4.seed_f.cax' /
'CAS','/home2/users/wangda/WASB/c4/seed/seed_bp/c4.seed_f1.cax' /
'CAS','/home2/users/wangda/WASB/c4/seed/seed_bp/c4.seed_m.cax' /
'CAS','/home2/users/wangda/WASB/c4/seed/seed_bp/c4.seed_m1.cax' /
'CAS','/home2/users/wangda/WASB/c4/seed/seed_bp/c4.seed_b.cax' /
'CAS','/home2/users/wangda/WASB/c4/seed/seed_bp/c4.seed_b1.cax' /
'CAS','/home2/users/wangda/WASB/c4/seed/seed_bp/c4.seed_b2.cax' /
'CAS','/home2/users/wangda/WASB/c4/seed/seed_bp/c4.seed_b3.cax' /

'EXP'   39,1
0,0.5,1,2,3,4,5,6,7,8,9,10,15,20,25,30,35,40,45,50,55,60,65,

70,75,80,85,90,100,110,120,130,140,150,160,170,180,190,200/
'RES'   24,1
0,1,3,5,10,20,30,40,50,60,70,80,90,100,110,120,130,140,150,
               160,170,180,190,200/

'TFU'    3,2   1200,900,565.8/
'TMO'    4,1   583.1,565.8,600,610/
'BOR'    5,1   450,0,900,1500,2000/
'HTFU'   3,2   1200,900,565.8/
'HTMO'   3,2   600,583.1,565.8/
'HBOR'   5,2   0,450,900,1500,2000/
'CRD'    2,1   ' ','AIC'/

'BAS.MAC' 2 'EXP','HTMO',,,'TMO'/
'DEL.MAC' 2 'EXP','TMO'/
'DEL.MAC' 2 'EXP','HBOR',,,'BOR'/
'DEL.MAC' 2 'EXP','BOR'/
'DEL.MAC' 2 'EXP','HTFU',,,'TFU'/
'DEL.MAC' 2 'EXP','TFU'/
'DEL.MAC' 2 'EXP','CRD'/
'SUB.MAC' 'HTMO','TMO','HTMO','TMO'/
'SUB.MAC' 'HBOR','BOR','HBOR','BOR'/
'SUB.MAC' 'HTFU','TFU','HTFU','TFU'/

'BAS.FPD' 2 'EXP','TMO'/
'DEL.FPD' 2 'EXP','BOR'/
'DEL.FPD' 2 'EXP','TFU'/
'DEL.FPD' 2 'EXP','CRD'/
```

```
'COM'    DISCONTINUITY FACTOR DEPENDENCE
'BAS.DFS' 2 'EXP','TMO'/
'DEL.DFS' 2 'EXP','BOR'/
'DEL.DFS' 2 'EXP','TFU'/
'DEL.DFS' 2 'EXP','CRD'/
'PIN.PIN' /
'STA'/
'END'/


'TIT','WASB-SEED_END ASSEMBLY'/
'LIB','ADD' 'SEED_END' /
'COM'      CASMO FILES USED FOR THIS SEGMENT
'CAS','/home2/users/wangda/WASB/c4/seed/seed_end/c4.seed_end_base.cax'
/
'CAS','/home2/users/wangda/WASB/c4/seed/seed_end/c4.seed_end_f.cax' /
'CAS','/home2/users/wangda/WASB/c4/seed/seed_end/c4.seed_end_f1.cax' /
'CAS','/home2/users/wangda/WASB/c4/seed/seed_end/c4.seed_end_m.cax' /
'CAS','/home2/users/wangda/WASB/c4/seed/seed_end/c4.seed_end_m1.cax' /
'CAS','/home2/users/wangda/WASB/c4/seed/seed_end/c4.seed_end_b.cax' /
'CAS','/home2/users/wangda/WASB/c4/seed/seed_end/c4.seed_end_b1.cax' /
'CAS','/home2/users/wangda/WASB/c4/seed/seed_end/c4.seed_end_b2.cax' /
'CAS','/home2/users/wangda/WASB/c4/seed/seed_end/c4.seed_end_b3.cax' /

'EXP'   39,1
0,0.5,1,2,3,4,5,6,7,8,9,10,15,20,25,30,35,40,45,50,55,60,65,

70,75,80,85,90,100,110,120,130,140,150,160,170,180,190,200/
'RES'   24,1
0,1,3,5,10,20,30,40,50,60,70,80,90,100,110,120,130,140,150,
                160,170,180,190,200/

'TFU'    3,2   1200,900,565.8/
'TMO'    4,1   583.1,565.8,600,610/
'BOR'    5,1   450,0,900,1500,2000/
'HTFU'   3,2   1200,900,565.8/
'HTMO'   3,2   600,583.1,565.8/
'HBOR'   5,2   0,450,900,1500,2000/
'CRD'    2,1   ' ','AIC'/

'BAS.MAC' 2 'EXP','HTMO',,,'TMO'/
'DEL.MAC' 2 'EXP','TMO'/
'DEL.MAC' 2 'EXP','HBOR',,,'BOR'/
'DEL.MAC' 2 'EXP','BOR'/
'DEL.MAC' 2 'EXP','HTFU',,,'TFU'/
'DEL.MAC' 2 'EXP','TFU'/
'DEL.MAC' 2 'EXP','CRD'/
'SUB.MAC' 'HTMO','TMO','HTMO','TMO'/
'SUB.MAC' 'HBOR','BOR','HBOR','BOR'/
'SUB.MAC' 'HTFU','TFU','HTFU','TFU'/

'BAS.FPD' 2 'EXP','TMO'/
'DEL.FPD' 2 'EXP','BOR'/
'DEL.FPD' 2 'EXP','TFU'/
```

```
'DEL.FPD' 2 'EXP','CRD'/

'COM'    DISCONTINUITY FACTOR DEPENDENCE
'BAS.DFS' 2 'EXP','TMO'/
'DEL.DFS' 2 'EXP','BOR'/
'DEL.DFS' 2 'EXP','TFU'/
'DEL.DFS' 2 'EXP','CRD'/
'PIN.PIN' /
'STA'/
'END'/


'COM' -------------------- PWR RADIAL REFLECTORS --------------------

'TIT' 'PWR RADIAL REFLECTOR'
'REF' 'RADIAL' 0,155.13/
'LIB' 'ADD' 'RADREF' /

'CAS','/home2/users/wangda/WASB/c4/ref/radial/c4.radref.cax' /
'CAS','/home2/users/wangda/WASB/c4/ref/radial/c4.radref0b.cax' /
'CAS','/home2/users/wangda/WASB/c4/ref/radial/c4.radref9b.cax' /
'CAS','/home2/users/wangda/WASB/c4/ref/radial/c4.radref2kb.cax' /

'COM'  VARIABLE LIST
'BOR'   4,2   0.0 450.0 900.0   2000.0/
'HBOR'  4,2   0.0 450.0 900.0  2000.0/

'COM'  MACROSCOPIC AND DFS DEPENDANCE, NO FPD
'BAS.MAC' 1,,'BOR',,,'HBOR'/
'BAS.DFS' 1,,'BOR',,,'HBOR' /

'STA'/
'END'/

'COM' --------------- PWR BOTTOM AXIAL REFLECTOR ---------------

'TIT' 'PWR BOTTOM AXIAL REFLECTOR'
'REF' 'AXIAL' 0,155.13/
'LIB' 'ADD' 'BOTREF' /

'CAS','/home2/users/wangda/WASB/c4/ref/bottom/c4.botref.cax' /

'COM'  VARIABLE LIST
'BOR'   4,2   0.0 600.0 1200.0  1800.0/
'HBOR'  4,2   0.0 600.0 1200.0 1800.0/

'COM'  MACROSCOPIC AND DFS DEPENDANCE, NO FPD
'BAS.MAC' 1,,'BOR',,,'HBOR'/
'BAS.DFS' 1,,'BOR',,,'HBOR' /

'STA'/
'END'/

'COM' --------------- PWR TOP AXIAL REFLECTOR ---------------
```

```
'TIT' 'PWR TOP AXIAL REFLECTOR'
'REF' 'AXIAL' 0,155.13/
'LIB' 'ADD' 'TOPREF' /

'CAS','/home2/users/wangda/WASB/c4/ref/top/c4.topref.cax' /

'COM'  VARIABLE LIST
'BOR'   4,2   0.0 600.0 1200.0  1800.0/
'HBOR'  4,2   0.0 600.0 1200.0 1800.0/

'COM'  MACROSCOPIC AND DFS DEPENDANCE, NO FPD
'BAS.MAC' 1,,'BOR',,,'HBOR'/
'BAS.DFS' 1,,'BOR',,,'HBOR' /

'STA'/
'END'/
```

SIMULATE Input Files

1. first cycle

```
'DIM.PWR' 15                                    /
'DIM.CAL' 24 2 2 1                               /
'DIM.DEP','EXP','SAM','HTMO','HBOR','HTFU','PIN','EBP'/ * DEPLETION
ARGUMENTS
'DIM.MEM' 12*2                                  /
'COM' 'ERR.CHK' 'PERMIT'  /
'LIB' '/home2/users/wangda/WASB/t3/lib/t3.WASB.lib' /
'TIT.CAS','   ***   WASB: CYCLE 1 CORE FOLLOW MODEL DEPLETION    ***
'/
'TIT.PRO','WASB Whole Core Calculation'/ * TABLES-3 LIBRARIES
'TIT.RUN','cycle 1'/ * CASE NAME
'COR.DAT',21.50364,365.76,104.5,751.53/ * CLD ASY WTH,HT,PD,M FLX/CR
X-A,T
'COR.OPE' 100. 100. 2250                        /
'COR.SYM' 'ROT'                                 /
'COR.TIN' 559.04
'SEG.TFU',0,0,347.38,-5.3799/ * FUEL TEMP FIT VERSUS POWER FOR ALL
FUEL (FROSSTY)
'TAB.TFU',1,0,'EXP',10,'POW',1
      0.    5.0   10.0   15.0   20.0   30.0   35.0   40.0  45.0   50.0
   1  0. -35.45 -52.14 -83.33 -87.25 -82.88 -83.63 -89.22 -88.07 -
76.61 /* FROSSTY
'COM'/*** FUEL: CROSS-SECTION, LOADING, ...
***********************************
'REF.LIB',01,'RADREF',0/ * RADIAL REFLECTOR SEGMENT
'REF.LIB',02,'BOTREF',0/ * BOTTOM AXIAL REFLECTOR SEGMENT
'REF.LIB',03,'TOPREF',0/ * TOP AXIAL REFLECTOR SEGMENT
'SEG.LIB',04,'SEED'/ * SEED INCLUDING Er2O3
'SEG.LIB',05,'BLANKET'/ * BLANKET
'SEG.LIB',06,'SEED_END'/ *
'FUE.ZON',01,1,'SEED01',02,0.,06,15.24,04,350.52,06,365.76,03/
'COM' 'FUE.ZON',01,1,'SEED01',02,0.,04,365.76,03/
'FUE.ZON',02,1,'BLANKET01',02,0.,05,365.76,03/
'FUE.ZON',18,1,'RAREF',02,0.,01,365.76,03/ * RADIAL REFLECTOR
'FUE.TYP',1,
  2  2  1  2  1  1  1  2 18
  2  1  2  1  2  2  1  2 18
  1  2  1  2  1  1  2  2 18
  2  1  2  1  2  2  1  2 18
  1  2  1  2  1  1  2 18 18
  1  2  1  2  1  1  2 18  0
  1  1  2  1  2  2 18 18  0
  2  2  2  2 18 18 18  0  0
 18 18 18 18 18  0  0  0  0/ * LOWER RIGHT QTR CORE
'FUE.BAT' 1
          4  4  1  4  1  3  1  4
          4  3  4  3  4  4  2  4
          1  4  1  4  2  1  4  4
          4  3  4  2  4  4  3  4
          1  4  2  4  1  2  4  0
```

```
          3 4 1 4 2 3 4 0
          1 2 4 3 4 4 0 0
          4 4 4 4 0 0 0 0/
'FUE.GRD', 'ON'   2.82 3.36 'INC'
                 64.87 3.36 'INC'
                117.07 3.36 'INC'
                169.27 3.36 'INC'
                221.46 3.36 'INC'
                273.66 3.36 'INC'
                325.86 3.36 'INC'/ * FUEL GRIDS:1 PER CORE ONLY
'COM'/*** CONTROL RODS
**************************************************
'CRD.GRP',1,
  0  0  0  0 00 00 00 00 00 00 00  0  0  0  0
  0  0 00  9 00  3 00  2 00  3 00  9 00  0  0
  0 00 00 00  6 00  8 00  8 00  7 00 00 00  0
  0  9 00  1 00 00 00  5 00 00 00  1 00  9  0
 00 00  7 00  4 00 00 00 00 00  4 00  6 00 00
 00  3 00 00 00  2 00  4 00  2 00 00 00  3 00
 00 00  8 00 00 00 00 00 00 00 00 00  8 00 00
 00  2 00  5 00  4 00  1 00  4 00  5 00  2 00
 00 00  8 00 00 00 00 00 00 00 00 00  8 00 00
 00  3 00 00 00  2 00  4 00  2 00 00 00  3 00
 00 00  6 00  4 00 00 00 00 00  4 00  7 00 00
  0  9 00  1 00 00 00  5 00 00 00  1 00  9  0
  0 00 00 00  7 00  8 00  8 00  6 00 00 00  0
  0  0 00  9 00  3 00  2 00  3 00  9 00  0  0
  0  0  0  0 00 00 00 00 00 00 00  0  0  0  0/ * CRD BANK LOCS(00 IS
NO BANK)
'CRD.ZON',1,1,'ARO',0,0.0,0,365.76/ * 1=NO CONTROL, 2=AIC CONTROL
'CRD.ZON',2,1,'AIC',0,0.0,0,7.57,10,365.76/ * 2.98*2.54=7.57=INTER LOC
'CRD.DAT',226,1.585/ * 228-1.48IN*1.6ST/IN=226 (365.76-7.57)/226=1.585
'CRD.TYP',1,
 0  0  0  0  1  1  1  1  1  1  1  0  0  0  0
 0  0  1 02  1 02  1 02  1 02  1 02  1  0  0
 0  1  1  1 02  1 02  1 02  1 02  1  1  1  0
 0 02  1 02  1  1  1 02  1  1  1 02  1 02  0
 1  1 02  1 02  1  1  1  1  1 02  1 02  1  1
 1 02  1  1  1 02  1 02  1 02  1  1  1 02  1
 1  1 02  1  1  1  1  1  1  1  1  1 02  1  1
 1 02  1 02  1 02  1 02  1 02  1 02  1 02  1
 1  1 02  1  1  1  1  1  1  1  1  1 02  1  1
 1 02  1  1  1 02  1 02  1 02  1  1  1 02  1
 1  1 02  1 02  1  1  1  1  1 02  1 02  1  1
 0 02  1 02  1  1  1 02  1  1  1 02  1 02  0
 0  1  1  1 02  1 02  1 02  1 02  1  1  1  0
 0  0  1 02  1 02  1 02  1 02  1 02  1  0  0
 0  0  0  0  1  1  1  1  1  1  1  0  0  0  0/ * CONTROL LOCATIONS (02)
'CRD.BNK'  /
'HYD.ITE'                                       /
'ITE.KEF' 1.3 /
'ERR.CHK','PERMIT'/
'COM' 'ITE.BOR' 5000.                                 /
'DEP.CYC' 'CYCLE1' 0.0 01                       /
'PRI.OPT' 60                                    /60 LINES PER PAGE
```

```
'PIN.EDT' 'OFF'                                /
'DEP.STA' 'AVE' 0.                             /
'DEP.FPD' 1                                    /
'WRE' 'WSHUFF1',0.   /
'STA'                                          /
'END'                                          /


'TIT.PRO' 'WASB Whole Core Calculation'                        /
'TIT.RUN' 'WHFP1'                                              /
'TIT.CAS' 'EXP=0.,Pr=100.,CRD=ARO,CB=ITE,XE=1'                 /
'RES' 'WSHUFF1' 0.                /
'COR.OPE' 100. 100. 2250                                       /
'PIN.EDT' 'ON' 'SUMM' '2PIN'                  /
'PRI.STA' 50*' '                                               /
'PRI.STA' '2RPF' '2EXP' '2KIN' 'SRPF' 'SEXP' 'SKIN'             /
'DEP.STA' 'AVE'  0.                                            /
'DEP.FPD'  1                                                   /
'WRE' 'WHFP1' 0.                  /
'STA'                                                          /
'COM' *********************************************************/
'TIT.CAS' 'EXP=0.~14.869,Pr=100.,CRD=ARO,CB=ITE,XE=2'          /
'PRI.STA' '2RPF' '2EXP' '2KINF'  /
'DEP.STA' 'AVE' 0. 0.15 1 2 3 4 5 6 7 -1 21       /
'DEP.FPD'  2                                                   /
'WRE' 'WHFP1' 0.15 1 2 3 4 5         /
'STA'                                                          /
'COM' *********************************************************/
'COM' 'ITE.LIM' 1.,0,0.00001,,,,30,,30
/
'COM' 'ITE.SRC' 'SET' 'EOLEXP',,0.02,,,'KEF' 1.0000,.00001 'MINBOR'
0.0/
'DEP.STA' 'AVE' 22                                             /
'WRE' 'WHFP1' 20000               /
'STA'                                                          /
'END'                                                          /


'COM' CREATE PEAK Sm AND WRITE RESTART FILE FOR SHUFFLING   /
'TIT.PRO' 'WASB'                                         /
'TIT.RUN' 'WPKSM2'                                          /
'TIT.CAS' 'EXP=14.869,Pr=0.,CRD=ARO,CB=ITE,XE=8'            /
'ERR.CHK','PERMIT'/
'RES' 'WHFP1' 20000           /
'COR.OPE' 0.0                                               /
'PRI.STA' 50*' '                                            /
'PRI.STA' '2RPF' '2EXP' '2KIN' 'SRPF' 'SEXP' 'SKIN'         /
'PIN.EDT' 'OFF'                                             /
'DEP.FPD' 8                                                 /
'WRE' 'WPKSM2' 20000          /
'STA'                                                       /
'END'                                                       /
```

2. second cycle

```
'DIM.PWR' 15                                    /
'DIM.CAL' 24 4 2 1                              /
'DIM.DEP','EXP','SAM','HTMO','HBOR','HTFU','PIN','EBP'/ * DEPLETION
ARGUMENTS
'DIM.MEM' 12*2                                  /
'TIT.PRO','WASB Whole Core Calculation'/ * TABLES-3 LIBRARIES
/
'TIT.RUN','cycle 2'/ * CASE NAME
'TIT.CAS','   ***   WASB: CYCLE 2 CORE FOLLOW MODEL DEPLETION    ***
'/
'FUE.LAB' 6  /
 1  1                      01L-01 01K-01 01J-01 01H-01 01G-01 01F-01 01E-01
 2  1               01N-02 01M-02 01L-02 01K-02 01J-02 TYPE01 01G-02 01F-02 01E-02 01D-02 01C-02
 3  1        01P-03 01N-03 01M-03 01L-03 TYPE01 01J-03 01H-03 01G-03 TYPE01 01E-03 01D-03 01C-03 01B-03
 4  1        01P-04 01N-04 TYPE01 01L-04 01K-04 01J-04 TYPE01 01G-04 01F-04 01E-04 TYPE01 01C-04 01B-04
 5  1 01R-05 01P-05 01N-05 01M-05 01L-05 01K-05 01J-05 01H-05 01G-05 01F-05 01E-05 01D-05 01C-05 01B-05 01A-05
 6  1 01R-06 01P-06 TYPE01 01M-06 01L-06 TYPE01 01J-06 TYPE01 01G-06 TYPE01 01E-06 01D-06 TYPE01 01B-06 01A-06
 7  1 01R-07 01P-07 01N-07 01M-07 01L-07 01K-07 01J-07 01H-07 01G-07 01F-07 01E-07 01D-07 01C-07 01B-07 01A-07
 8  1 01R-08 TYPE01 01N-08 TYPE01 01L-08 TYPE01 01J-08 01H-08 01G-08 TYPE01 01E-08 TYPE01 01C-08 TYPE01 01A-08
 9  1 01R-09 01P-09 01N-09 01M-09 01L-09 01K-09 01J-09 01H-09 01G-09 01F-09 01E-09 01D-09 01C-09 01B-09 01A-09
10  1 01R-10 01P-10 TYPE01 01M-10 01L-10 TYPE01 01J-10 TYPE01 01G-10 TYPE01 01E-10 01D-10 TYPE01 01B-10 01A-10
11  1 01R-11 01P-11 01N-11 01M-11 01L-11 01K-11 01J-11 01H-11 01G-11 01F-11 01E-11 01D-11 01C-11 01B-11 01A-11
12  1        01P-12 01N-12 TYPE01 01L-12 01K-12 01J-12 TYPE01 01G-12 01F-12 01E-12 TYPE01 01C-12 01B-12
13  1        01P-13 01N-13 01M-13 01L-13 TYPE01 01J-13 01H-13 01G-13 TYPE01 01E-13 01D-13 01C-13 01B-13
14  1               01N-14 01M-14 01L-14 01K-14 01J-14 TYPE01 01G-14 01F-14 01E-14 01D-14 01C-14
15  1                      01L-15 01K-15 01J-15 01H-15 01G-15 01F-15 01E-15

'FUE.NEW' 'TYPE01' 'D28' 28 1 0 ,, 5 /
'RES' 'WPKSM2' 20000 /
'LIB' '/home2/users/wangda/WASB/t3/lib/t3.WASB.lib' /
'COR.OPE' 100. 100. 2250            /
'COR.TIN' 559.04                    / unit in F
'REF.LIB',01,'RADREF',0/ * RADIAL REFLECTOR SEGMENT
'REF.LIB',02,'BOTREF',0/ * BOTTOM AXIAL REFLECTOR SEGMENT
'REF.LIB',03,'TOPREF',0/ * TOP AXIAL REFLECTOR SEGMENT
'SEG.LIB',04,'SEED'/ * SEED INCLUDING Er2O3
'SEG.LIB',05,'BLANKET'/ * BLANKET
'SEG.LIB',06,'SEED_END'/ *
'FUE.ZON',01,1,'SEED01',02,0.,06,15.24,04,350.52,06,365.76,03/
'COM' 'FUE.ZON',01,1,'SEED01',02,0.,04,365.76,03/
'FUE.ZON',02,1,'BLANKET01',02,0.,05,365.76,03/
'FUE.ZON',18,1,'RAREF',02,0.,01,365.76,03/ * RADIAL REFLECTOR
'FUE.INI' 'CJILAB'                         /
'HYD.ITE'                                  /
'ITE.BOR' 3500.                            /
'DEP.CYC' 'CYCLE2' 0.0 02                  /
'PRI.OPT' 60                               / 60 lines per page
'PRI.STA' 50*' '                           /
'PRI.STA' '2RPF' '2EXP'                    /
'DEP.STA' 'AVE' 0                          /
'DEP.FPD' 1                                /
'ERR.CHK' 'PERMIT'                         /
'WRE' 'WSHUFF2' 0. /
'STA'                                      /
'END'                                      /


'TIT.PRO' 'WASB'                                                /
'TIT.RUN' 'WHFP2'                                                 /
```

```
'TIT.CAS' 'EXP=0.,Pr=100.,CRD=ARO,CB=ITE,XE=1'                 /
'RES' 'WSHUFF2' 0.                  /
'COR.OPE' 100. 100. 2250                                       /
'PIN.EDT' 'ON' 'SUMM' '2PIN'                  /
'BAT.EDT' 'ON' '2RPF' 'QPIN'/
'PRI.STA' 50*' '                                               /
'PRI.STA' '2RPF' '2EXP' '2KIN' 'SRPF' 'SEXP' 'SKIN'             /
'DEP.STA' 'AVE'  0.                                            /
'DEP.FPD'  1                                                   /
'WRE' 'WHFP2' 0.                    /
'STA'                                                          /
'COM' *********************************************************/
'TIT.CAS' 'EXP=0.~14.952,Pr=100.,CRD=ARO,CB=ITE,XE=2'          /
'DEP.STA' 'AVE' 0. 0.15 1 -1 21     /
'DEP.FPD'  2                                                   /
'WRE' 'WHFP2' 0.15 1 2 3 4 5        /
'STA'                                                          /
'COM' *********************************************************/
'ITE.LIM' 1.,0,0.00001,,,,30,,30                               /
'COM' 'ITE.SRC' 'SET' 'EOLEXP',,0.02,,,'KEF' 1.0000,.00001 'MINBOR'
0.0/
'DEP.STA' 'AVE' 22                                         /
'WRE' 'WHFP2' 20000                 /
'STA'                                                          /
'END'                                                          /
```

```
'COM' CREATE PEAK Sm AND WRITE RESTART FILE FOR SHUFFLING   /
'TIT.PRO' 'WASB'                                         /
'TIT.RUN' 'WPKSM3'                                          /
'TIT.CAS' 'EXP=14.869,Pr=0.,CRD=ARO,CB=ITE,XE=8'            /
'RES' 'WHFP2' 20000             /
'COR.OPE' 0.0                                               /
'PRI.STA' 50*' '                                            /
'PRI.STA' '2RPF' '2EXP' '2KIN' 'SRPF' 'SEXP' 'SKIN'         /
'PIN.EDT' 'ON'                                             /
'DEP.FPD' 8                                                 /
'WRE' 'WPKSM3' 20000            /
'STA'                                                       /
'END'                                                       /
```

3. third cycle

```
'DIM.PWR' 15                                  /
'DIM.CAL' 24 4 2 1                            /
'DIM.DEP','EXP','SAM','HTMO','HBOR','HTFU','PIN','EBP'/ * DEPLETION
ARGUMENTS
'DIM.MEM' 12*2                                /
'TIT.PRO','WASB Whole Core Calculation'/ * TABLES-3 LIBRARIES
/
'TIT.RUN','cycle 3'/ * CASE NAME
```

```
'TIT.CAS','   ***   WASB: CYCLE 3 CORE FOLLOW MODEL DEPLETION    ***
'/
'FUE.LAB' 6  /
 1  1                             02L-01 02K-01 02J-01 02H-01 02G-01 02F-01 02E-01
 2  1               02N-02 02M-02 02K-04 02K-02 02H-06 TYPE01 02H-04 02F-02 02F-04 02D-02 02C-02
 3  1        02P-03 02M-03 02K-03 02L-03 TYPE01 02J-03 02J-07 02G-03 TYPE01 02E-03 02F-03 02C-04 02B-03
 4  1        02P-04 02N-06 TYPE01 02L-04 02K-06 02J-04 TYPE01 02G-04 02H-02 02E-04 TYPE01 02C-06 02B-04
 5  1 02R-05 02M-06 02N-05 02M-05 02N-04 02K-05 02J-02 02H-05 02G-02 02F-05 02D-03 02D-05 02C-05 02D-06 02A-05
 6  1 02R-06 02P-06 TYPE01 02P-08 02L-06 TYPE01 02J-06 TYPE01 02G-06 TYPE01 02E-06 02F-06 TYPE01 02B-06 02A-06
 7  1 02R-07 02M-08 02N-07 02M-07 02P-07 02K-07 02M-04 02H-07 02D-04 02F-07 02B-07 02D-07 02C-07 02F-08 02A-07
 8  1 02R-08 TYPE01 02J-09 TYPE01 02L-08 TYPE01 02J-08 02H-08 02G-08 TYPE01 02E-08 TYPE01 02G-07 TYPE01 02A-08
 9  1 02R-09 02K-08 02N-09 02M-09 02P-09 02K-09 02M-12 02H-09 02D-12 02F-09 02B-09 02D-09 02C-09 02D-08 02A-09
10  1 02R-10 02P-10 TYPE01 02K-10 02L-10 TYPE01 02J-10 TYPE01 02G-10 TYPE01 02E-10 02B-08 TYPE01 02B-10 02A-10
11  1 02R-11 02M-10 02N-11 02M-11 02M-13 02K-11 02J-14 02H-11 02G-14 02F-11 02C-12 02D-11 02C-11 02D-10 02A-11
12  1        02P-12 02N-10 TYPE01 02L-12 02H-14 02J-12 TYPE01 02G-12 02F-10 02E-12 TYPE01 02C-10 02B-12
13  1        02P-13 02N-12 02K-13 02L-13 TYPE01 02J-13 02G-09 02G-13 TYPE01 02E-13 02F-13 02D-13 02B-13
14  1               02N-14 02M-14 02K-12 02K-14 02H-12 TYPE01 02H-10 02F-14 02F-12 02D-14 02C-14
15  1                             02L-15 02K-15 02J-15 02H-15 02G-15 02F-15 02E-15

'FUE.NEW' 'TYPE01' 'D28' 28 1 0 ,, 6 /
'RES' 'WPKSM3' 20000 /
'LIB' '/home2/users/wangda/WASB/t3/lib/t3.WASB.lib' /
'COR.OPE' 100. 100. 2250            /
'COR.TIN' 559.04                    / unit in F
'REF.LIB',01,'RADREF',0/ * RADIAL REFLECTOR SEGMENT
'REF.LIB',02,'BOTREF',0/ * BOTTOM AXIAL REFLECTOR SEGMENT
'REF.LIB',03,'TOPREF',0/ * TOP AXIAL REFLECTOR SEGMENT
'SEG.LIB',04,'SEED'/ * SEED INCLUDING Er2O3
'SEG.LIB',07,'SEED_BP'/ * SEED WITH Er2O3
'SEG.LIB',05,'BLANKET'/ * BLANKET
'SEG.LIB',06,'SEED_END'/ *
'FUE.ZON',01,1,'SEED01',02,0.,06,15.24,04,30.48,07,320.04,04,350.52,06
,365.76,03/
'COM' 'FUE.ZON',01,1,'SEED01',02,0.,06,15.24,04,350.52,06,365.76,03/
'COM' 'FUE.ZON',01,1,'SEED01',02,0.,04,365.76,03/
'FUE.ZON',02,1,'BLANKET01',02,0.,05,365.76,03/
'FUE.ZON',18,1,'RAREF',02,0.,01,365.76,03/ * RADIAL REFLECTOR
'FUE.INI' 'CJILAB'                         /
'HYD.ITE'                                  /
'ITE.BOR' 4000.                            /
'DEP.CYC' 'CYCLE3' 0.0 03                  /
'PRI.OPT' 60                               / 60 lines per page
'PRI.STA' 50*' '                           /
'PRI.STA' '2RPF' '2EXP'                    /
'DEP.STA' 'AVE' 0                          /
'DEP.FPD' 1                                /
'ERR.CHK' 'PERMIT'                         /
'WRE' 'WSHUFF3' 0. /
'STA'                                      /
'END'                                      /


'TIT.PRO' 'WASB'                                                   /
'TIT.RUN' 'WHFP3'                                                     /
'TIT.CAS' 'EXP=0.,Pr=100.,CRD=ARO,CB=ITE,XE=1'                        /
'RES' 'WSHUFF3' 0.                  /
'COR.OPE' 100. 100. 2250                                              /
'PIN.EDT' 'ON'                       /
'PRI.STA' 50*' '                                                      /
'PRI.STA' '2RPF' '2EXP' '2KINF'
/
'DEP.STA' 'AVE'  0.                                                   /
```

```
'DEP.FPD'  1                                                    /
'WRE' 'WHFP3' 0.                      /
'STA'                                                           /
'COM' *********************************************************/
'TIT.CAS' 'EXP=0.~14.952,Pr=100.,CRD=ARO,CB=ITE,XE=2'           /
'DEP.STA' 'AVE' 0. 0.15 1 -1 21        /
'DEP.FPD'  2                                                    /
'WRE' 'WHFP3' 0.15 1 2 3 4 5          /
'STA'                                                           /
'COM' *********************************************************/
'ITE.LIM' 1.,0,0.00001,,,,30,,30                                /
'COM' 'ITE.SRC' 'SET' 'EOLEXP',,0.02,,,'KEF' 1.0000,.00001 'MINBOR'
0.0/
'DEP.STA' 'AVE' 22                                              /
'WRE' 'WHFP3' 20000                   /
'STA'                                                           /
'END'                                                           /
```

Appendix C. VIPRE Input File for the WASB Core

A VIPRE input file for the WASB core is presented in this appendix. The detailed model description can be found in Chapter 5.

```
*******************************************************************
*                                                                 *
*         WASB Core Thermal-hydraulics                            *
*                                                                 *
*******************************************************************
1,0,0                                                   *vipre.1
WASB SEED and BLANKET(one eighth core--with BP)     *vipre.2
geom,57,57,16,0,0,0                                     *geom.1
144.,?
0.,0.5          *default sl=0.5                         *geom.2
1,0.14730,1.11415,1.11415,4,3,0.14142,0.49606,4,0.14142,0.49606,?
6,0.14142,0.49606
9,0.14142,0.49606,
2,0.14730,1.11415,1.11415,4,3,0.14142,0.49606,4,0.14142,0.49606,?
12,0.14142,0.49606
15,0.14142,0.49606,
3,0.12599,1.21576,0.83561,2,11,0.07673,0.49606,14,0.07673,0.49606,
4,0.12599,1.21576,0.83561,2,7,0.07673,0.49606,10,0.07673,0.49606,
5,0.12599,1.21576,0.83561,3,6,0.07673,0.49606,9,0.14142,0.49606,?
17,0.21815,0.99213
6,0.12599,1.21576,0.83561,2,7,0.14142,0.49606,17,0.07673,0.86811,
7,0.12599,1.21576,0.83561,2,8,0.07673,0.49606,18,0.14142,0.99213,
8,0.12599,1.21576,0.83561,2,10,0.07673,0.49606,18,0.28283,0.99213,
9,0.14730,1.11415,1.11415,2,11,0.14142,0.49606,17,0.14142,0.86811,
10,0.12599,1.21576,0.83561,2,12,0.14142,0.49606,18,0.14142,0.99213,
11,0.12599,1.21576,0.83561,2,13,0.07673,0.49606,19,0.14142,0.86811,
12,0.12599,1.21576,0.83561,2,16,0.07673,0.49606,20,0.07673,0.99213,
13,0.12599,1.21576,0.83561,2,14,0.07673,0.49606,19,0.28283,0.86811,
14,0.12599,1.21576,0.83561,2,15,0.14142,0.49606,19,0.14142,0.86811,
15,0.14730,1.11415,1.11415,2,16,0.14142,0.49606,20,0.14142,0.86811,
16,0.12599,1.21576,0.83561,1,20,0.21815,0.99213,
17,2.00818,16.00455,14.48397,3,18,0.36929,2.23228,19,0.28283,2.23228,?
21,0.44031,3.12402
18,2.46412,18.10496,18.10496,3,20,0.36929,2.23228,21,0.51110,3.10827,?
22,0.51110,3.10827
19,1.66104,13.87785,11.97713,2,20,0.28283,2.23228,29,0.37161,3.00906,
20,2.00818,16.00455,14.48397,2,22,0.44031,3.12402,30,0.52276,3.00906,
21,7.73453,96.03340,86.52978,2,23,0.95142,4.23228,24,0.51831,4.23228,
22,7.73453,96.03340,86.52978,3,23,0.95142,4.23228,28,0.51831,4.23228,?
30,0.51831,4.23228
23,10.24262,83.03760,73.53399,2,25,0.89437,3.52690,27,0.89437,3.52690,
24,7.73453,96.03340,86.52978,2,25,0.95142,3.52690,26,0.68504,3.52690,
25,5.12131,41.51880,36.76699,
26,3.86727,48.01670,43.26489,
27,5.12131,41.51880,36.76699,1,28,0.95142,3.52690,
28,7.73453,96.03340,86.52978,2,31,0.68504,4.23228,32,0.95142,4.93766,
29,10.24262,83.03760,73.53399,2,30,0.95142,4.23228,33,0.95142,6.34843,
30,7.73453,96.03340,86.52978,2,31,0.68504,4.23228,34,0.95142,6.34843,
31,7.73453,96.03340,86.52978,2,32,0.95142,4.93766,34,0.95142,6.34843,
32,20.48524,166.07521,147.06797,1,35,1.90283,7.05381,
```

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33,15.46907,192.06679,173.05956,2,34,1.90283,6.34843,37,0.95142,8.46457,
34,40.97048,332.15042,294.13594,2,35,1.90283,8.46457,38,1.90283,8.46457,
35,30.93814,384.13359,346.11911,2,36,1.90283,7.05381,39,1.90283,8.46457,
36,20.48524,166.07521,147.06797,1,40,1.90283,7.05381,
37,20.48524,166.07521,147.06797,2,38,1.90283,6.34843,42,1.21780,8.46457,
38,30.93814,384.13359,346.11911,2,39,1.90283,8.46457,43,1.37008,8.46457,
39,40.97048,332.15042,294.13594,2,40,1.90283,8.46457,44,2.43559,8.46457,
40,30.93814,384.13359,346.11911,2,41,1.90283,7.05381,45,1.37008,8.46457,
41,20.48524,166.07521,147.06797,1,46,2.43559,7.05381,
42,20.48524,166.07521,147.06797,2,43,1.90283,6.34843,48,1.21780,8.46457,
43,30.93814,384.13359,346.11911,2,44,1.90283,8.46457,49,1.90283,8.46457,
44,40.97048,332.15042,294.13594,2,45,1.90283,8.46457,50,1.90283,8.46457,
45,30.93814,384.13359,346.11911,2,46,1.90283,8.46457,51,1.90283,8.46457,
46,40.97048,332.15042,294.13594,2,47,2.43559,7.05381,52,1.90283,8.46457,
47,20.48524,166.07521,147.06797,1,53,1.90283,7.05381,
48,20.48524,166.07521,147.06797,2,49,2.43559,6.34843,54,0.95142,8.46457,
49,40.97048,332.15042,294.13594,2,50,1.90283,8.46457,55,1.90283,8.46457,
50,30.93814,384.13359,346.11911,2,51,1.90283,8.46457,56,1.37008,8.46457,
51,40.97048,332.15042,294.13594,2,52,1.90283,8.46457,57,1.90283,8.46457,
52,30.93814,384.13359,346.11911,1,53,1.37008,8.46457,
53,30.93814,384.13359,346.11911,
54,15.46907,192.06679,173.05956,1,55,1.37008,6.34843,
55,30.93814,384.13359,346.11911,1,56,1.37008,8.46457,
56,30.93814,384.13359,346.11911,1,57,1.37008,8.46457,
57,30.93814,384.13359,346.11911,                                *geom.4
prop,0,0,2,1    * internal EPRI functions--use local pressure     *prop.1
rods,1,62,1,2,2,0,0,0,0,0,0  *2 types of materials properties     *rods.1
0.0,0.0,0,0,                                                      *rods.2
-1       * enter chopped cosine                                   *rods.3
1.55      * cosine power shap with 1.55 peak                       *rods.5
* normal rods layout
1,2,2.3854,1,5,0.25,17,0.75,
2,2,2.4202,1,6,0.25,7,0.25,17,0.25,18,0.25
3,2,2.3489,1,7,0.25,8,0.25,18,0.5,
4,2,2.2077,1,8,0.25,18,0.75,
5,2,2.3061,1,5,0.25,9,0.25,17,0.5,
6,2,2.4218,1,1,0.25,5,0.25,6,0.25,9,0.25
7,2,2.4567,1,1,0.25,4,0.25,6,0.25,7,0.25
8,2,2.3489,1,8,0.25,10,0.25,18,0.5,
9,2,2.3235,1,9,0.25,11,0.25,17,0.25,19,0.25
10,2,2.4329,1,1,0.25,3,0.25,9,0.25,11,0.25
11,2,2.3822,1,1,0.25,2,0.25,3,0.25,4,0.25
12,2,2.4583,1,2,0.25,4,0.25,10,0.25,12,0.25
13,2,2.4202,1,10,0.25,12,0.25,18,0.25,20,0.25
14,2,2.4345,1,11,0.25,13,0.25,19,0.5,
15,2,2.4329,1,2,0.25,3,0.25,14,0.25,15,0.25
16,2,2.4218,1,2,0.25,12,0.25,15,0.25,16,0.25
17,2,2.3441,1,13,0.25,19,0.75,
18,2,2.4345,1,13,0.25,14,0.25,19,0.5,
19,2,2.3235,1,14,0.25,15,0.25,19,0.25,20,0.25
20,2,2.3061,1,15,0.25,16,0.25,20,0.5,
21,2,2.3854,1,16,0.25,20,0.75,
22,2,2.1670,1,17,11.75,
23,2,2.1670,1,18,14,
24,2,2.10,1,19,9.75,
25,2,2.1670,1,20,11.75,
26,1,0.9060,1,21,72.25,
27,1,0.9630,1,22,72.25,
28,2,1.7840,1,23,72.25,
29,1,0.9060,1,24,72.25,
30,1,0.7460,1,26,36.125,
31,2,1.7840,1,25,36.125,
```

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32,2,1.7840,1,27,36.125,
33,1,0.9630,1,28,72.25,
34,2,2.1670,1,29,72.25,
35,1,0.9630,1,30,72.25,
36,1,0.9630,1,31,72.25,
37,2,2.0890,1,32,144.5,
38,1,0.8950,1,33,144.5,
39,2,1.1890,1,34,289,
40,1,0.8500,1,35,289,
41,2,1.2500,1,36,144.5,
42,2,2.0240,1,37,144.5,
43,1,0.8580,1,38,289,
44,2,1.6440,1,39,289,
45,1,0.8690,1,40,289,
46,2,2.1550,1,41,144.5,
47,2,1.2660,1,42,144.5,
48,1,0.8650,1,43,289,
49,2,1.9220,1,44,289,
50,1,0.7330,1,45,289,
51,2,1.4500,1,46,289,
52,2,0.8490,1,47,144.5,
53,2,1.9340,1,48,144.5,
54,2,1.3650,1,49,289,
55,1,0.5410,1,50,289,
56,2,0.5530,1,51,289,
57,1,0.3080,1,52,289,
58,1,0.1850,1,53,289,
59,1,0.3950,1,54,144.5,
60,1,0.3160,1,55,289,
61,1,0.1830,1,56,289,
62,1,0.1060,1,57,289,
0
*   blank line for end of rods layout input
2,nucl,0.354646,0.30315,6,0.173228,0.02252   * UO2 seed rods     *rods.62
0,0,0,0,0,2000.0,.97,0.0     *gap conductence--2000, 97% uo2     *rods.63
1,nucl,0.4173,0.3658,6,0.0,0.02252             *bkt.rods         *rods.62
0,1,2,0,0,2000.0,1.0,0.0     *gap conductence--2000, not uo2     *rods.63
*3,dumy,0.484016,0.0,0     *guide tube, no inside flow            *rods.68
1,30,638.256,BLAN,                                               *rods.70
80.33,0.0764,5.356,?
170.33,0.0764,4.902
260.33,0.0764,4.520,?
350.33,0.0764,4.192
440.33,0.0764,3.909,?
530.33,0.0764,3.662
620.33,0.0764,3.444,?
710.33,0.0764,3.251
800.33,0.0764,3.078,?
890.33,0.0764,2.923
980.33,0.0764,2.782,?
1070.33,0.0764,2.655
1160.33,0.0764,2.538,?
1250.33,0.0764,2.432
1340.33,0.0764,2.334,?
1430.33,0.0764,2.243
1520.33,0.0764,2.159,?
1610.33,0.0764,2.082
1700.33,0.0764,2.010,?
1790.33,0.0764,1.942
1880.33,0.0764,1.879,?
1970.33,0.0764,1.820
2060.33,0.0764,1.765,?
```

```
2150.33,0.0764,1.712
2240.33,0.0764,1.663,?
2420.33,0.0767,1.573
2600.33,0.0770,1.492,?
2780.33,0.0772,1.419
2960.33,0.0774,1.352,?
3140.33,0.0776,1.292
2,13,409.0,BCLD                                                   *rods.70
80.33,0.0671,7.330,?                                             *rods.71
260.33,0.0712,8.1158
692.33,0.07904,9.8016,?
1502.33,0.08955,13.2923
1507.73,0.11988,13.3211,?
1543.73,0.14089,13.5166
1579.73,0.14686,13.7172,?
1615.73,0.1717,13.9231
1651.73,0.1949,14.1347,?
1687.73,0.18388,14.3519
1723.73,0.1478,14.5752,?
1759.73,0.112,14.8047
1786.73,0.085,14.9810                                             *rods.71
*
corr,1,1,                                                         *corr.1
epri,epri,epri,none                                               *corr.2
0.2                                                               *corr.3
ditb,thsp,thsp,w-3l,cond,g5.7     *correlation for boiling curve   *corr.6
w-3l                  *dnb analysis by w-3l                       *corr.9
0.043,0.066,0.986   *w-3l input data                              *corr.11
*     no turbulent mixing in flow and energy solution
*MIXX,0,0,0                                                       *mixx.1
*0.8,0.038                                                        *mixx.2
*
drag,1,0,1                                                        *drag.1
0.316,-0.25,0.0,64.0,-1.0,0.0          *axial friction correlation   *drag.2
0.5,0.496        *pitch=0.496,kij=0.5l/p                          *drag.5
*
grid,0,2                                                         *grid.1
0.6,2.0                                                          *grid.2
37,8             * seed and interfere channels                   *grid.4
1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16
17,18,19,20,23,25,27,29,32,34,36,37,39,41,42,44
46,47,48,49,51                                                   *grid.5
16.0,1,32.0,1,48.0,1,64.0,1,80.0,1,96.0,1,?
112.0,1,128.0,1     *grid loc.                                    *grid.6
20,8           *blanket channels                                 *grid.4
21,22,24,26,28,30,31,33,35,38,40,43,45,50,52,53,
54,55,56,57
16.0,2,32.0,2,48.0,2,64.0,2,80.0,2,96.0,2,?                       *grid.6
112.0,2,128.0,2    *grid location                                 *grid.6
0                                                                 *grid.4
* blanket line for terminate of grid input
cont
0.0,0,300,50,3,0,   *iterative solution                           *cont.2
0.0,0.0,0.0,0.0,0.0,0.9,1.5,1.0                                    *cont.3
5,16,0,16,16,0,1,1,0,0,0,1,0,0                                       *cont.6
1000.,0.0,0.0,0.0,0.0,0.0                                         *cont.7
1,2,3,4,8,9,10,12,13,14,15,16,17,18,21,22,   *channel              *cont.8
4,6,7,8,10,11,12,13,15,16,18,19,20,21,22,25, *rods                *cont.10
1,2,3,4,8,9,10,12,13,14,15,16,17,18,21,22    *channel with CHF    *cont.11
*     OPER CONDITION
oper,1,1,-1,1,0,1,0,0,0                                          *oper.1
-1.0,1.3,2.6,0.005,0          *No MDNBR ITERATION=1.3, Fcool=2.6%   *oper.2
```

```
0                                                                  *oper.3
2248.1,562.64,4633.8,68.5,0  *7.079,0  *operation conditions        *oper.5
0          *no forcing functions                                   *oper.12
endd
*
*end of data input
```
0

Appendix D. FRAPCON Input Files for the WASB Fuel

Two FRAPCON input files for the WASB seed and blanket fuel are included in this appendix. The detailed model description can be found in Chapter 5.

Seed Fuel:

```
****************************************************************************
*        frapcon3, steady-state fuel rod analysis code                     *
*----------------------------------------------------------------------    *
*                                                                          *
*         CASE DESCRIPTION: Seed Fuel (Annular)                            *
*                                                                          *
*UNIT     FILE DESCRIPTION                                                 *
*----     ----------------------------------------------Output:            *
*         Output :                                                         *
*    6        STANDARD PRINTER OUTPUT                                      *
*                                                                          *
*         Scratch:                                                         *
*    5        SCRATCH INPUT FILE FROM ECH01                                *
*                                                                          *
*  Input:   FRAPCON3 INPUT FILE (UNIT 55)                                  *
*                                                                          *
****************************************************************************
* GOESINS:
FILE05='nullfile', STATUS='UNKNOWN', FORM='FORMATTED',
         CARRIAGE CONTROL='NONE'
*
* GOESOUTS:
FILE06='seed.out',    STATUS='UNKNOWN', CARRIAGE CONTROL='LIST'
/***************************************************************************

         The Seed Fuel Pin of Dean's Seed-Blanket Design
 $frpcn
 im=149, na=15, nr=15,
 mechan = 2, ngasr = 15,
 $end
 $frpcon
 cpl = 0.4, crdt = 0., crdtr = 0.0, thkcld = 0.57e-3,
 dco = 9.08e-3, pitch = 12.6e-3, crephr=10.
 den = 94.0, dishsd = 0., thkgap= 82e-6, dspg = 5e-3,
 dspgw = 0.9e-3, enrch = 20.,  fgpav = 1.e6,
 hdish = 0.e-4, hplt = 10.e-3, icm = 4, rc=2.2e-3
 icor = 0, idxgas = 1, iplant = -2, iq = 0, jdlpr = 0,fa = 1.0,
 jn = 26,26,26,
 totl = 3.66, roughc = 5e-7, roughf = 6e-7, vs = 8,
 nunits = 0, rsntr = 101.9,
 flux(1) = 16*0.275e17, p2(1) = 15.51e6, tw(1) = 565.95, go(1) = 4219.,
 jst = 52*1,49*2,48*3
 qf(1) =0.,0.52,0.84,0.97,1.000,1.03,1.05,1.07,1.09,1.10,1.11,4*1.12,
 1.11,1.10,1.09,1.07,1.05,1.03,1.00,0.97,0.84,0.52,0.,
 x(1) =0.000,0.07625,0.22875,0.38125,0.53375,0.68625,0.83875,0.99125,
 1.14375,1.29625,1.44875,1.60125,1.75375,1.90625,2.05875,2.21125,
 2.36375,2.51625,2.66875,2.82125,2.97375,3.12625,3.27875,3.43125,3.5075,3.66,
 qf(27) =0.00,0.55,0.90,1.05,1.07,2*1.08,1.07,1.05,2*1.04,4*1.03,
```

```
 2*1.04,1.05,1.07,2*1.08,1.07,1.05,0.90,0.55,0.,
 x(27) = 0.000,0.07625,0.22875,0.38125,0.53375,0.68625,0.83875,0.99125,
 1.14375,1.29625,1.44875,1.60125,1.75375,1.90625,2.05875,2.21125,
 2.36375,2.51625,2.66875,2.82125,2.97375,3.12625,3.27875,3.43125,3.5075,3.66,
 qf(53) = 0.,0.75,0.94,1.03,1.05,2*1.06,1.05,1.03,1.02,2*1.01,2*1.00,
 2*1.01,1.02,1.03,1.05,2*1.06,1.05,1.03,0.94,0.75,0.,
 x(53) =0.000,0.07625,0.22875,0.38125,0.53375,0.68625,0.83875,0.99125,
 1.14375,1.29625,1.44875,1.60125,1.75375,1.90625,2.05875,2.21125,
 2.36375,2.51625,2.66875,2.82125,2.97375,3.12625,3.27875,3.43125,3.5075,3.66,

 time= 1.,2.,5.,10.,20.,30.,40.,50.,63.,70.,80,90,100,110,123,130,140,150,
 160,170,183,190,200,210,220,230,243,250,260,270,280,290,303,310,320,330,
 340,350,363,370,380,390,400,410,423,430,440,450,460,470,483,490,
 500,510,520,530,543,550,560,570,580,590,603,610,620,630,640,650,663,670,
 680,690,700,710,723,730,740,750,760,770,783,790,800,810,820,830,843,850,
 860,870,880,890,903,910,920,930,940,950,963,970,980,
 990,1000,1010,1023,1030,1040,1050,1060,1070,1083,1090,1100,1110,1120,1130,
 1140,1150,1160,1170,1180,1190,1200,1210,1220,1230,1240,1250,1260,1270,1280,
 1290,1300,1310,1320,1330,1340,1350,1360,1370,1380,1390,1400,1410,1420,1430,
 1440,1450,1460,

 qmpy = 52*35,49*27, 48*18
 slim = .05,
 $end
```

Blanket Fuel:

```
*****************************************************************************
*        frapcon3, steady-state fuel rod analysis code                     *
*----------------------------------------------------------------------     *
*                                                                           *
*         CASE DESCRIPTION: Blanket Fuel (Fat Pin)                         *
*                                                                           *
*UNIT     FILE DESCRIPTION                                                  *
*----     ----------------------------------------------Output:            *
*         Output :                                                          *
*    6        STANDARD PRINTER OUTPUT                                       *
*                                                                           *
*         Scratch:                                                          *
*    5        SCRATCH INPUT FILE FROM ECH01                                 *
*                                                                           *
*  Input:   FRAPCON3 INPUT FILE (UNIT 55)                                   *
*                                                                           *
*****************************************************************************
* GOESINS:
FILE05='nullfile', STATUS='UNKNOWN', FORM='FORMATTED',
         CARRIAGE CONTROL='NONE'
*
* GOESOUTS:
FILE06='seed.out',    STATUS='UNKNOWN', CARRIAGE CONTROL='LIST'
/****************************************************************************

         The Blanket Fuel Pin of Dean's Seed-Blanket Design
 $frpcn
 im=393, na=15, nr=15,
 mechan = 2, ngasr = 15,
 $end
 $frpcon
 cpl = 0.40, crdt = 0., crdtr = 0.0, thkcld = 0.57e-3,
```

```
dco = 10.6e-3, pitch = 12.6e-3,
den = 94.0, dishsd = 1.13e-3, thkgap= 82e-6, dspg = 6e-3,
dspgw = 0.9e-3, uofhm=10., enrch = 15.,  fgpav = 1.4e6,
hdish = 3.13e-4, hplt = 13.4e-3, icm = 4, crephr=10.
icor = 0, idxgas = 1, iplant = -2, iq = 0, jdlpr = 0,fa = 1.0,
jn = 26,26,26,
totl = 3.66, roughc = 5e-7, roughf = 6e-7, vs = 8,
nunits = 0, rsntr = 301.9,
flux(1) = 16*0.275e17, p2(1) = 15.51e6, tw(1) = 565.95, go(1) = 4219.0,
jst = 133*1,130*2,130*3
qf(1) =0.,0.52,0.84,0.97,1.000,1.03,1.05,1.07,1.09,1.10,1.11,4*1.12,
1.11,1.10,1.09,1.07,1.05,1.03,1.00,0.97,0.84,0.52,0.,
x(1) =0.000,0.07625,0.22875,0.38125,0.53375,0.68625,0.83875,0.99125,
1.14375,1.29625,1.44875,1.60125,1.75375,1.90625,2.05875,2.21125,
2.36375,2.51625,2.66875,2.82125,2.97375,3.12625,3.27875,3.43125,3.5075,3.66,
qf(27) =0.00,0.55,0.90,1.05,1.07,2*1.08,1.07,1.05,2*1.04,4*1.03,
2*1.04,1.05,1.07,2*1.08,1.07,1.05,0.90,0.55,0.,
x(27) = 0.000,0.07625,0.22875,0.38125,0.53375,0.68625,0.83875,0.99125,
1.14375,1.29625,1.44875,1.60125,1.75375,1.90625,2.05875,2.21125,
2.36375,2.51625,2.66875,2.82125,2.97375,3.12625,3.27875,3.43125,3.5075,3.66,
qf(53) = 0.,0.75,0.94,1.03,1.05,2*1.06,1.05,1.03,1.02,2*1.01,2*1.00,
2*1.01,1.02,1.03,1.05,2*1.06,1.05,1.03,0.94,0.75,0.,
x(53) =0.000,0.07625,0.22875,0.38125,0.53375,0.68625,0.83875,0.99125,
1.14375,1.29625,1.44875,1.60125,1.75375,1.90625,2.05875,2.21125,
2.36375,2.51625,2.66875,2.82125,2.97375,3.12625,3.27875,3.43125,3.5075,3.66,

time= 1.,2.,5.,10.,20.,30.,40.,60.,70.,80,90,100,110,130,140,150,
160,170,180,200,210,220,230,240,250,260,270,280,290,310,320,330,
340,350,370,380,390,400,410,430,440,450,460,470,490,
500,510,520,530,540,550,560,570,580,590,603,610,620,630,640,650,670,
680,700,710,720,730,740,750,760,770,780,790,800,810,820,830,850,
860,870,880,890,903,910,920,930,950,960,970,980,
990,1000,1010,1023,1030,1040,1050,1060,1070,1083,1090,1100,1110,1120,1130,
1140,1150,1160,1170,1180,1190,1210,1220,1230,1240,1250,1260,1270,1280,
1290,1300,1320,1330,1340,1350,1370,1380,1390,1400,1410,1430,
1450,1460,
1470,1480,1500,1510,1520,1530,1540,1550,1560,1570,1580,
1590,1600,1620,1630,1640,1650,1660,1670,1680,1690,1700,1720,1730,
1740,1750,1770,1780,1790,1800,1810,1820,1830,1840,1860,1870,1880,
1890,1900,1920,1930,1940,1950,1960,1970,1980,1990,2000,2020,2030,
2040,2050,2060,2070,2080,2090,2100,2110,2120,2130,2140,2160,2170,2180,
2190,2200,2220,2230,2240,2250,2260,2270,2280,2290,2300,2310,2320,2330,
2340,2350,2360,2370,2380,2390,2400,2410,2420,2430,2450,2460,2470,2480,
2490,2500,2520,2530,2540,2550,2560,2570,2580,2590,2610,2620,2630,
2640,2650,2670,2680,2690,2700,2710,2720,2730,2740,2750,2760,2780,
2790,2800,2810,2820,2840,2850,2860,2870,2880,2890,2910,2920,
2930,
2940,2950,2970,2980,2990,3000,3010,3020,3030,3050,3060,3070,3080,
3090,3100,3120,3130,3140,3150,3160,3170,3190,3200,3210,3220,3230,
3240,3250,3270,3280,3290,3300,3310,3320,3340,3350,3360,3370,3380,
3390,3400,3420,3430,3440,3450,3460,3470,3480,3500,3510,3520,3530,
3540,3550,3560,3580,3590,3600,3610,3620,3630,3640,3650,3660,3670,3680,
3690,3700,3710,3720,3730,3750,3760,3770,3780,3790,3800,3810,3830,
3840,3850,3870,3880,3890,3900,3910,3920,3930,3940,3950,3960,3970,3980,
3990,4000,4010,4020,4030,4050,4060,4070,4080,4090,4100,4120,4130,
4140,4150,4160,4180,4190,4200,4210,4220,4230,4240,4250,4260,4270,4280,
4290,4300,4310,4320,4330,4350,4360,4370,4380,

qmpy = 133*11.355,130*11.355,130*11.355
slim = .05,
$end
```